# PG No. 11 $C_{4\mathrm{h}}$ $4/m$ [ tetragonal ] (lgs basis)

bra: $= \langle s,\uparrow| \, , \langle s,\downarrow|$
ket: $= |s,\uparrow\rangle \, , |s,\downarrow\rangle$

Table 1: (s,s) block.

| No. | multipole | matrix |
|---|---|---|
| 1 | symmetry | $1$ |
| | $\mathbb{Q}_0^{(a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 2 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 3 | symmetry | $y$ |
| | $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$ |
| 4 | symmetry | $x$ |
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$ |

bra: $= \langle s,\uparrow| \, , \langle s,\downarrow|$
ket: $= |p_x,\uparrow\rangle \, , |p_x,\downarrow\rangle \, , |p_y,\uparrow\rangle \, , |p_y,\downarrow\rangle \, , |p_z,\uparrow\rangle \, , |p_z,\downarrow\rangle$

Table 2: (s,p) block.

| No. | multipole | matrix |
|---|---|---|
| 5 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ |
| 6 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 7 | symmetry | $y$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 8 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 9 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 10 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 11 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$ |
| 12 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 13 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 14 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 15 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 16 | symmetry | 1 |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_0^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 17 | symmetry | $z$ |
| | $\mathbb{T}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$ |
| 18 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 19 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 20 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 21 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 22 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 23 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 24 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 25 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 26 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 27 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 28 | symmetry | 1 |
| | $\mathbb{M}_0^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |

bra: $= \langle s,\uparrow| \, , \langle s,\downarrow|$

ket: $= |d_v,\uparrow\rangle \, , |d_v,\downarrow\rangle \, , |d_{xy},\uparrow\rangle \, , |d_{xy},\downarrow\rangle \, , |d_{xz},\uparrow\rangle \, , |d_{xz},\downarrow\rangle \, , |d_{yz},\uparrow\rangle \, , |d_{yz},\downarrow\rangle \, , |d_u,\uparrow\rangle \, , |d_u,\downarrow\rangle$

Table 3: (s,d) block.

| No. | multipole | matrix |
|---|---|---|
| 29 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ |
| 30 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_2^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 31 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 32 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 33 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 34 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 35 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 36 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 37 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 38 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{2}}{4} \\ \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 39 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{10} \end{bmatrix}$ |
| 40 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 41 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 42 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}}{20} \\ \frac{3\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$ |
| 43 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ \frac{3\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$ |
| 44 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{i}{4} & 0 \end{bmatrix}$ |
| 45 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{1}{4} \\ \frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{1}{4} & 0 \end{bmatrix}$ |
| 46 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{10} \end{bmatrix}$ |
| 47 | symmetry | $y$ |
| | $\mathbb{G}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{10}}{20} & 0 \end{bmatrix}$ |
| 48 | symmetry | $x$ |
| | $\mathbb{G}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \end{bmatrix}$ |
| 49 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}_2^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$ |
| 50 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 51 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_2^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 52 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 53 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 54 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 55 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 56 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_2^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 57 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 58 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 59 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}}{10} \end{bmatrix}$ |
| 60 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 61 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 62 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ -\frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$ |
| 63 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} \\ \frac{3\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \end{bmatrix}$ |
| 64 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ -\frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{1}{4} & 0 \end{bmatrix}$ |
| 65 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{i}{4} \\ -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{i}{4} & 0 \end{bmatrix}$ |
| 66 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 67 | symmetry | $y$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} \\ -\frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{10}i}{20} & 0 \end{bmatrix}$ |
| 68 | symmetry | $x$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \end{bmatrix}$ |

bra: $= \langle s,\uparrow| , \langle s,\downarrow|$

ket: $= |f_c,\uparrow\rangle , |f_c,\downarrow\rangle , |f_s,\uparrow\rangle , |f_s,\downarrow\rangle , |f_z^\beta,\uparrow\rangle , |f_z^\beta,\downarrow\rangle , |f_{xyz},\uparrow\rangle , |f_{xyz},\downarrow\rangle , |f_x,\uparrow\rangle , |f_x,\downarrow\rangle , |f_y,\uparrow\rangle , |f_y,\downarrow\rangle , |f_z^\alpha,\uparrow\rangle , |f_z^\alpha,\downarrow\rangle$

Table 4: (s,f) block.

| No. | multipole | matrix |
|---|---|---|
| 69 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ |
| 70 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_3^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 71 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 72 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 73 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 74 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 75 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 76 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{Q}_{3}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 77 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_{3}^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{8} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 78 | symmetry | $\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_{3}^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 \\ \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 79 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}i}{16} & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{16} & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & \frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 80 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 81 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} \\ 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 \end{bmatrix}$ |
| 82 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} & -\frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & \frac{\sqrt{5}}{8} & 0 \end{bmatrix}$ |
| 83 | symmetry | $\frac{\sqrt{21}x^4}{6}-\frac{\sqrt{21}x^2y^2}{2}-\frac{\sqrt{21}x^2z^2}{2}+\frac{\sqrt{21}y^4}{6}-\frac{\sqrt{21}y^2z^2}{2}+\frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{G}_{4}^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{8} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 84 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{24} & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{42} & 0 \\ -\frac{\sqrt{42}i}{24} & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{42} \end{bmatrix}$ |
| 85 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 86 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & \frac{\sqrt{210}}{56} & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 \end{bmatrix}$ |
| 87 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 \\ -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & -\frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 \end{bmatrix}$ |
| 88 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{16} & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{16} & \frac{\sqrt{5}}{8} & 0 \end{bmatrix}$ |
| 89 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{16} & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 \end{bmatrix}$ |
| 90 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{21}}{14} & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & \frac{\sqrt{14}i}{16} & 0 & 0 & -\frac{3\sqrt{21}i}{56} & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 \end{bmatrix}$ |
| 91 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}i}{16} & 0 & 0 & \frac{3\sqrt{21}}{56} & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & -\frac{3\sqrt{21}}{56} & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{112} & \frac{\sqrt{35}}{56} & 0 \end{bmatrix}$ |
| 92 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & \frac{\sqrt{14}}{14} & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{14} \end{bmatrix}$ |
| 93 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{28} & 0 & \frac{\sqrt{70}}{28} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ \frac{\sqrt{70}i}{28} & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 94 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_2^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}}{28} & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ \frac{\sqrt{70}}{28} & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 95 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$ |
| 96 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & \frac{\sqrt{7}}{14} & 0 \end{bmatrix}$ |
| 97 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{T}_3^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$ |
| 98 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}_3^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 99 | symmetry | $\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_3^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 100 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 101 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{T}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 \end{bmatrix}$ |
| 102 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 \end{bmatrix}$ |
| 103 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 104 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 105 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ -\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 106 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 107 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,1}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{16} & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & -\frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 108 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,2}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} \\ 0 & \frac{\sqrt{30}}{16} & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 109 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{6} & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 \end{bmatrix}$ |
| 110 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & \frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} & -\frac{\sqrt{5}i}{8} & 0 \end{bmatrix}$ |
| 111 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 112 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{24} & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & \frac{\sqrt{105}}{42} & 0 \\ -\frac{\sqrt{42}}{24} & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{42} \end{bmatrix}$ |
| 113 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 114 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 \\ \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 \end{bmatrix}$ |
| 115 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & \frac{\sqrt{210}}{56} & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & \frac{\sqrt{210}i}{56} & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 \end{bmatrix}$ |
| 116 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{16} & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{16} & -\frac{\sqrt{5}i}{8} & 0 \end{bmatrix}$ |
| 117 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & \frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} \\ 0 & \frac{\sqrt{2}}{16} & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & \frac{\sqrt{5}}{8} & 0 \end{bmatrix}$ |
| 118 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{56} & 0 & -\frac{\sqrt{21}i}{14} & -\frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & \frac{\sqrt{14}}{16} & 0 & 0 & -\frac{3\sqrt{21}}{56} & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \end{bmatrix}$ |
| 119 | symmetry | $3\sqrt{5}x^2yz-\frac{\sqrt{5}y^3z}{2}-\frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & -\frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 & \frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & \frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{112} & -\frac{\sqrt{35}i}{56} & 0 \end{bmatrix}$ |
| 120 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & -\frac{\sqrt{14}i}{14} & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & -\frac{\sqrt{21}}{14} \end{bmatrix}$ |
| 121 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{28} & 0 & -\frac{\sqrt{70}i}{28} & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ \frac{\sqrt{70}}{28} & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 122 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_2^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{28} & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 123 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \end{bmatrix}$ |
| 124 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & -\frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$ |

bra: $= \langle p_x,\uparrow|, \langle p_x,\downarrow|, \langle p_y,\uparrow|, \langle p_y,\downarrow|, \langle p_z,\uparrow|, \langle p_z,\downarrow|$
ket: $= |p_x,\uparrow\rangle, |p_x,\downarrow\rangle, |p_y,\uparrow\rangle, |p_y,\downarrow\rangle, |p_z,\uparrow\rangle, |p_z,\downarrow\rangle$

Table 5: (p,p) block.

| No. | multipole | matrix |
|---|---|---|
| 125 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 126 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$ |
| 127 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_2^{(a)}(B_g, 1)$ | $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 128 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(a)}(B_g, 2)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 129 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 130 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,2}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 131 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 132 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 133 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 134 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 135 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$ |
| 136 | symmetry | $1$ |
| | $\mathbb{Q}_{0}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 137 | symmetry | $z$ |
| | $\mathbb{G}_{1}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 138 | symmetry | $y$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$ |
| 139 | symmetry | $x$ |
| | $\mathbb{G}_{1,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 140 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 141 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 142 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 143 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 144 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 145 | symmetry | $z$ |
| | $\mathbb{M}_1^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 146 | symmetry | $y$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{1,1}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 147 | symmetry | $x$ |
| | $\mathbb{M}^{(a)}_{1,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 148 | symmetry | $z$ |
| | $\mathbb{M}^{(1,-1;a)}_{1}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 149 | symmetry | $y$ |
| | $\mathbb{M}^{(1,-1;a)}_{1,1}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 150 | symmetry | $x$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 151 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$ |
| 152 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 153 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 154 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$ |
| 155 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$ |
| 156 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 157 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 158 | symmetry | $z$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$ |
| 159 | symmetry | $y$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$ |
| 160 | symmetry | $x$ |
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$ |

bra: $= \langle p_x,\uparrow|\,, \langle p_x,\downarrow|\,, \langle p_y,\uparrow|\,, \langle p_y,\downarrow|\,, \langle p_z,\uparrow|\,, \langle p_z,\downarrow|$
ket: $= |d_v,\uparrow\rangle\,, |d_v,\downarrow\rangle\,, |d_{xy},\uparrow\rangle\,, |d_{xy},\downarrow\rangle\,, |d_{xz},\uparrow\rangle\,, |d_{xz},\downarrow\rangle\,, |d_{yz},\uparrow\rangle\,, |d_{yz},\downarrow\rangle\,, |d_u,\uparrow\rangle\,, |d_u,\downarrow\rangle$

Table 6: (p,d) block.

| No. | multipole | matrix |
|---|---|---|
| 161 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 162 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 163 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \end{bmatrix}$ |
| 164 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} \end{bmatrix}$ |
| 165 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 166 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 167 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{3\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{3\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 168 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \\ 0 & -\frac{3\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \end{bmatrix}$ |
| 169 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$ |
| 170 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 171 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ \frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{10}}{20} & 0 \\ 0 & -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ -\frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 172 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ \frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 173 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ \frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & \frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ \frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 174 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{10}i}{20} & 0 \\ 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 \end{bmatrix}$ |
| 175 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 176 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{2}}{6} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{6} & 0 & -\frac{\sqrt{2}}{12} & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{6} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 177 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{2}}{6} & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{6} & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 \end{bmatrix}$ |
| 178 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 179 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & \frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$ |
| 180 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$ |
| 181 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{60} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ -\frac{\sqrt{15}}{60} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & \frac{\sqrt{15}i}{60} & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ \frac{\sqrt{15}i}{60} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 182 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_3^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{i}{12} & 0 & -\frac{1}{6} & \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{i}{12} & 0 & \frac{1}{6} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & -\frac{1}{12} & 0 & \frac{i}{6} & 0 & 0 & -\frac{i}{6} & 0 & 0 & \frac{\sqrt{3}}{12} \\ \frac{1}{12} & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & \frac{i}{6} & -\frac{\sqrt{3}}{12} & 0 \\ \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 \\ 0 & -\frac{i}{6} & 0 & 0 & -\frac{i}{6} & 0 & -\frac{1}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 183 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{1}{6} & 0 & \frac{i}{12} & 0 & 0 & -\frac{i}{6} & 0 & 0 & \frac{\sqrt{3}}{12} \\ \frac{1}{6} & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & \frac{i}{6} & -\frac{\sqrt{3}}{12} & 0 \\ 0 & \frac{i}{6} & 0 & \frac{1}{12} & -\frac{i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ \frac{i}{6} & 0 & -\frac{1}{12} & 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & -\frac{i}{6} & 0 & 0 & \frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{6} & -\frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 184 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ -\frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & \frac{\sqrt{5}i}{40} & 0 \\ 0 & \frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & -\frac{\sqrt{5}i}{40} \\ 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & -\frac{3\sqrt{5}}{40} \\ -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} & \frac{3\sqrt{5}}{40} & 0 \end{bmatrix}$ |
| 185 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 \\ 0 & \frac{7\sqrt{15}i}{120} & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{40} \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{40} \\ \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & \frac{3\sqrt{5}i}{40} & 0 \end{bmatrix}$ |
| 186 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{5i}{24} & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & -\frac{1}{6} & \frac{\sqrt{3}i}{24} & 0 \\ 0 & \frac{5i}{24} & 0 & 0 & \frac{i}{12} & 0 & \frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}i}{24} \\ 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{6} & \frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 & 0 \\ 0 & -\frac{i}{24} & 0 & -\frac{1}{6} & \frac{i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} \\ -\frac{i}{24} & 0 & \frac{1}{6} & 0 & 0 & -\frac{i}{12} & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 187 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{1}{6} & 0 & -\frac{i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{6} & -\frac{1}{6} & 0 & -\frac{i}{6} & 0 & 0 & 0 \\ \frac{5i}{24} & 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & \frac{1}{12} & \frac{\sqrt{3}i}{24} & 0 \\ 0 & -\frac{5i}{24} & 0 & 0 & -\frac{i}{6} & 0 & -\frac{1}{12} & 0 & 0 & -\frac{\sqrt{3}i}{24} \\ 0 & \frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & \frac{i}{12} & 0 & 0 & -\frac{\sqrt{3}}{8} \\ -\frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{i}{12} & \frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 188 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{15}}{20} \\ -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 189 | symmetry | $x$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{20} & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \end{bmatrix}$ |
| 190 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{10} & \frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 191 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 192 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_2^{(a)}(B_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 193 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2}^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 194 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \end{bmatrix}$ |
| 195 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 196 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} \\ -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{15}}{60} \\ -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & -\frac{\sqrt{15}}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & \frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{15} \end{bmatrix}$ |
| 197 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 198 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_2^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 199 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$ |
| 200 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$ |
| 201 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{30} & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & \frac{\sqrt{5}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & -\frac{\sqrt{15}}{30} & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{5}i}{10} \end{bmatrix}$ |
| 202 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{21} & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\ -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{7}}{28} \\ -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} \end{bmatrix}$ |
| 203 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 204 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{28} \\ -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{28} & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 205 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 206 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & -\frac{\sqrt{3}}{8} \\ \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & \frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 207 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{8} & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 208 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{21}i}{56} & 0 \\ 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}i}{56} \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} \\ -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 \end{bmatrix}$ |
| 209 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{28} & -\frac{\sqrt{21}i}{56} & 0 \\ 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{21}i}{56} \\ 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{56} \\ -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{21}}{56} & 0 \end{bmatrix}$ |
| 210 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_{2}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ \frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ \frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 211 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2}^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{i}{12} & 0 & -\frac{1}{12} & -\frac{i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ \frac{i}{12} & 0 & \frac{1}{12} & 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & \frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ -\frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & -\frac{i}{6} & \frac{\sqrt{3}}{12} & 0 \\ \frac{i}{3} & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & \frac{1}{12} & 0 & 0 \\ 0 & -\frac{i}{3} & 0 & 0 & -\frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 212 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2}^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 & -\frac{i}{6} & 0 & 0 & \frac{\sqrt{3}}{12} \\ -\frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & \frac{i}{6} & -\frac{\sqrt{3}}{12} & 0 \\ 0 & -\frac{i}{12} & 0 & \frac{1}{12} & -\frac{i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ -\frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & \frac{i}{3} & 0 & 0 & -\frac{1}{12} & 0 & -\frac{i}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{3} & \frac{1}{12} & 0 & -\frac{i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 213 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} -\frac{i}{12} & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & -\frac{1}{6} & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & \frac{i}{12} & 0 & 0 & \frac{i}{12} & 0 & \frac{1}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & -\frac{i}{12} & 0 & 0 & \frac{1}{3} & 0 & -\frac{i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{12} & -\frac{1}{3} & 0 & -\frac{i}{12} & 0 & 0 & 0 \\ 0 & -\frac{i}{6} & 0 & -\frac{1}{6} & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 214 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{12} & 0 & 0 & -\frac{1}{12} & 0 & \frac{i}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{12} & \frac{1}{12} & 0 & \frac{i}{3} & 0 & 0 & 0 \\ \frac{i}{12} & 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & \frac{1}{12} & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & -\frac{i}{12} & 0 & 0 & -\frac{i}{6} & 0 & -\frac{1}{12} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{1}{6} & 0 & -\frac{i}{6} & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 \\ -\frac{1}{6} & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 \end{bmatrix}$ |
| 215 | symmetry | 1 |
| | $\mathbb{G}_{0}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} \\ \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{60} \\ \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & \frac{\sqrt{30}}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & \frac{\sqrt{30}i}{30} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{30}i}{30} \end{bmatrix}$ |
| 216 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_{2}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}i}{210} & 0 & -\frac{\sqrt{105}}{210} & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} \\ -\frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 \\ 0 & \frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{35}}{35} \\ -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & -\frac{\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105} & 0 & \frac{2\sqrt{105}}{105} & \frac{3\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105} & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & -\frac{3\sqrt{35}i}{70} \end{bmatrix}$ |
| 217 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{4\sqrt{35}i}{105} & 0 & \frac{\sqrt{35}}{30} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} \\ \frac{4\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{30} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 \\ 0 & \frac{4\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{30} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{105}}{210} \\ -\frac{4\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & -\frac{\sqrt{105}}{210} & 0 \\ \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & \frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 218 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_2^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{35}}{30} & 0 & \frac{4\sqrt{35}i}{105} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{105}}{210} \\ \frac{\sqrt{35}}{30} & 0 & \frac{4\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{105}}{210} & 0 \\ 0 & \frac{\sqrt{35}i}{30} & 0 & \frac{4\sqrt{35}}{105} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} \\ \frac{\sqrt{35}i}{30} & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 219 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{105} & 0 & \frac{\sqrt{35}}{42} & \frac{2\sqrt{105}i}{105} & 0 \\ 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & \frac{4\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{2\sqrt{105}i}{105} \\ 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & -\frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & \frac{4\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{4\sqrt{35}i}{105} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 \end{bmatrix}$ |
| 220 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & \frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{4\sqrt{35}}{105} & \frac{2\sqrt{105}i}{105} & 0 \\ 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & -\frac{2\sqrt{105}i}{105} \\ 0 & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{4\sqrt{35}i}{105} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{105} & \frac{\sqrt{105}}{70} & 0 \end{bmatrix}$ |
| 221 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} \end{bmatrix}$ |
| 222 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 223 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$ |
| 224 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} \end{bmatrix}$ |
| 225 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3}^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 226 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_{3}^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 227 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{3\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & \frac{3\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 228 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{T}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{3\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \end{bmatrix}$ |
| 229 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$ |
| 230 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 231 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}^{(1,-1;a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & \frac{\sqrt{10}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} \\ \frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 232 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}^{(1,-1;a)}_{3}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & \frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{6}i}{12} & 0 \\ \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 233 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3}^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ \frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{6} & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 234 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{60} & -\frac{\sqrt{10}}{20} & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{10}}{20} \\ 0 & \frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \end{bmatrix}$ |
| 235 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{10}}{20} & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{30} & 0 & -\frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 236 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{6}}{12} & 0 \\ 0 & \frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{12} & -\frac{\sqrt{2}i}{6} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{6} & 0 & -\frac{\sqrt{2}i}{12} & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{6} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 237 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,-1;a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{12} & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{2}i}{6} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{6} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 \end{bmatrix}$ |
| 238 | symmetry | $z$ |
| | $\mathbb{T}^{(1,0;a)}_{1}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 239 | symmetry | $x$ |
| | $\mathbb{T}^{(1,0;a)}_{1,1}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{20} & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & \frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & \frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$ |
| 240 | symmetry | $y$ |
| | $\mathbb{T}^{(1,0;a)}_{1,2}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$ |
| 241 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}i}{60} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ \frac{\sqrt{15}i}{60} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & \frac{\sqrt{15}}{60} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ \frac{\sqrt{15}}{60} & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 242 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}_3^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{1}{12} & 0 & \frac{i}{6} & \frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ -\frac{1}{12} & 0 & -\frac{i}{6} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & \frac{i}{12} & 0 & \frac{1}{6} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{i}{12} & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{\sqrt{3}i}{12} & 0 \\ \frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & -\frac{i}{6} & 0 & 0 \\ 0 & -\frac{1}{6} & 0 & 0 & -\frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 243 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_3^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{i}{6} & 0 & \frac{1}{12} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{i}{6} & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{\sqrt{3}i}{12} & 0 \\ 0 & \frac{1}{6} & 0 & -\frac{i}{12} & -\frac{1}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ \frac{1}{6} & 0 & \frac{i}{12} & 0 & 0 & \frac{1}{6} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{6} & \frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 244 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 \\ -\frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & \frac{\sqrt{5}}{40} & 0 \\ 0 & \frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & -\frac{\sqrt{5}}{40} \\ 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & \frac{3\sqrt{5}i}{40} \\ \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & -\frac{3\sqrt{5}i}{40} & 0 \end{bmatrix}$ |
| 245 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & -\frac{\sqrt{5}}{40} & 0 \\ 0 & \frac{7\sqrt{15}}{120} & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{40} \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{40} \\ \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 \end{bmatrix}$ |
| 246 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{5}{24} & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & \frac{i}{6} & \frac{\sqrt{3}}{24} & 0 \\ 0 & \frac{5}{24} & 0 & 0 & \frac{1}{12} & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{3}}{24} \\ 0 & 0 & \frac{1}{6} & 0 & 0 & \frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{6} & -\frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 & 0 \\ 0 & -\frac{1}{24} & 0 & \frac{i}{6} & \frac{1}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} \\ -\frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & -\frac{1}{12} & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 247 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{6} & 0 & 0 & -\frac{i}{6} & 0 & -\frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{6} & \frac{i}{6} & 0 & -\frac{1}{6} & 0 & 0 & 0 \\ \frac{5}{24} & 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & -\frac{i}{12} & \frac{\sqrt{3}}{24} & 0 \\ 0 & -\frac{5}{24} & 0 & 0 & -\frac{1}{6} & 0 & \frac{i}{12} & 0 & 0 & -\frac{\sqrt{3}}{24} \\ 0 & -\frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & \frac{1}{12} & 0 & 0 & \frac{\sqrt{3}i}{8} \\ \frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{1}{12} & -\frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 248 | symmetry | $z$ |
| | $\mathbb{T}_{1}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} \\ -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 249 | symmetry | $x$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{20} & \frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \end{bmatrix}$ |
| 250 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{15}}{20} \\ 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 251 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 252 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(a)}(B_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 253 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2}^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 254 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$ |
| 255 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 256 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}_{2}^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \\ -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}}{60} & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{60} \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & \frac{\sqrt{15}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{15} \end{bmatrix}$ |
| 257 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2}^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 258 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2}^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 259 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \end{bmatrix}$ |
| 260 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & -\frac{\sqrt{5}}{20} & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & \frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$ |
| 261 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{30} & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{5}}{10} \end{bmatrix}$ |
| 262 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{21} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{7}i}{28} \\ \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{7}}{14} \end{bmatrix}$ |
| 263 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & -\frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 264 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{28} & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 265 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 266 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & -\frac{\sqrt{3}}{8} & 0 \\ 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & \frac{\sqrt{3}}{8} \\ 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$ |
| 267 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \\ 0 & \frac{1}{8} & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{8} & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \end{bmatrix}$ |
| 268 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{56} & 0 \\ 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{56} \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} \\ -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 \end{bmatrix}$ |
| 269 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{28} & -\frac{\sqrt{21}}{56} & 0 \\ 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{21}}{56} \\ 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{21}i}{56} \\ \frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & -\frac{\sqrt{21}i}{56} & 0 \end{bmatrix}$ |
| 270 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}_{2}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ -\frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ \frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 271 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_{2}^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{1}{12} & 0 & -\frac{i}{12} & \frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ -\frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & \frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{\sqrt{3}i}{12} & 0 \\ -\frac{1}{3} & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & \frac{i}{12} & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 & \frac{1}{12} & 0 & -\frac{i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 272 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2}^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & \frac{1}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ -\frac{i}{12} & 0 & -\frac{1}{12} & 0 & 0 & 0 & 0 & -\frac{1}{6} & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & \frac{1}{12} & 0 & \frac{i}{12} & \frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ \frac{1}{12} & 0 & -\frac{i}{12} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & -\frac{1}{3} & 0 & 0 & -\frac{i}{12} & 0 & \frac{1}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{i}{12} & 0 & \frac{1}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 273 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{1}{12} & 0 & 0 & 0 & 0 & -\frac{1}{12} & 0 & -\frac{i}{6} & \frac{\sqrt{3}}{12} & 0 \\ 0 & -\frac{1}{12} & 0 & 0 & -\frac{1}{12} & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & \frac{1}{12} & 0 & 0 & \frac{i}{3} & 0 & \frac{1}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{12} & -\frac{i}{3} & 0 & \frac{1}{12} & 0 & 0 & 0 \\ 0 & \frac{1}{6} & 0 & -\frac{i}{6} & -\frac{1}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & 0 & \frac{i}{6} & 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 274 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{12} & 0 & 0 & -\frac{i}{12} & 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{12} & \frac{i}{12} & 0 & -\frac{1}{3} & 0 & 0 & 0 \\ -\frac{1}{12} & 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & \frac{i}{12} & \frac{\sqrt{3}}{12} & 0 \\ 0 & \frac{1}{12} & 0 & 0 & \frac{1}{6} & 0 & -\frac{i}{12} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & \frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 & -\frac{1}{12} & 0 & 0 & 0 \\ -\frac{i}{6} & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 \end{bmatrix}$ |
| 275 | symmetry | 1 |
| | $\mathbb{M}_{0}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} \\ \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{30}i}{60} \\ -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & -\frac{\sqrt{30}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & \frac{\sqrt{30}}{30} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} \end{bmatrix}$ |
| 276 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_{2}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}}{210} & 0 & \frac{\sqrt{105}i}{210} & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} \\ -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 \\ 0 & -\frac{\sqrt{105}i}{210} & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{35}i}{35} \\ \frac{\sqrt{105}i}{210} & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & \frac{\sqrt{35}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{105} & 0 & -\frac{2\sqrt{105}i}{105} & \frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{105} & 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & -\frac{3\sqrt{35}}{70} \end{bmatrix}$ |
| 277 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2}^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{4\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{30} & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} \\ \frac{4\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{30} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ 0 & -\frac{4\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{30} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{105}i}{210} \\ \frac{4\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & \frac{\sqrt{105}i}{210} & 0 \\ \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 278 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2}^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{35}i}{30} & 0 & \frac{4\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{105}i}{210} \\ -\frac{\sqrt{35}i}{30} & 0 & \frac{4\sqrt{35}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & -\frac{\sqrt{105}i}{210} & 0 \\ 0 & \frac{\sqrt{35}}{30} & 0 & -\frac{4\sqrt{35}i}{105} & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} \\ \frac{\sqrt{35}}{30} & 0 & \frac{4\sqrt{35}i}{105} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & -\frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 279 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{42} & \frac{2\sqrt{105}}{105} & 0 \\ 0 & \frac{\sqrt{35}}{105} & 0 & 0 & \frac{4\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{2\sqrt{105}}{105} \\ 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{105} & \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & \frac{4\sqrt{35}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\ \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 \end{bmatrix}$ |
| 280 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{105} & -\frac{\sqrt{35}i}{105} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{4\sqrt{35}i}{105} & \frac{2\sqrt{105}}{105} & 0 \\ 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{4\sqrt{35}i}{105} & 0 & 0 & -\frac{2\sqrt{105}}{105} \\ 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{4\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{105} & -\frac{\sqrt{105}i}{70} & 0 \end{bmatrix}$ |

bra: $= \langle p_x,\uparrow|, \langle p_x,\downarrow|, \langle p_y,\uparrow|, \langle p_y,\downarrow|, \langle p_z,\uparrow|, \langle p_z,\downarrow|$

ket: $= |f_c,\uparrow\rangle, |f_c,\downarrow\rangle, |f_s,\uparrow\rangle, |f_s,\downarrow\rangle, |f_z^{\beta},\uparrow\rangle, |f_z^{\beta},\downarrow\rangle, |f_{xyz},\uparrow\rangle, |f_{xyz},\downarrow\rangle, |f_x,\uparrow\rangle, |f_x,\downarrow\rangle, |f_y,\uparrow\rangle, |f_y,\downarrow\rangle, |f_z^{\alpha},\uparrow\rangle, |f_z^{\alpha},\downarrow\rangle$

Table 7: (p,f) block.

| No. | multipole | matrix |
|---|---|---|
| 281 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{bmatrix}$ |
| 282 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_2^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 283 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 284 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 285 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}^{(a)}_{2,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 286 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{Q}^{(a)}_{4}(A_g,1)$ | $\begin{bmatrix} \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 287 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{Q}^{(a)}_{4}(A_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \end{bmatrix}$ |
| 288 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{Q}^{(a)}_{4}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 289 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 290 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{Q}_4^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 \\ \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 291 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 292 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 \end{bmatrix}$ |
| 293 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 \end{bmatrix}$ |
| 294 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 295 | symmetry | $\frac{\sqrt{21}x^4}{6}-\frac{\sqrt{21}x^2y^2}{2}-\frac{\sqrt{21}x^2z^2}{2}+\frac{\sqrt{21}y^4}{6}-\frac{\sqrt{21}y^2z^2}{2}+\frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{1}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{1}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{i}{8} & 0 \\ 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 \\ -\frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 296 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & \frac{\sqrt{35}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & -\frac{\sqrt{21}}{28} & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 \\ 0 & -\frac{\sqrt{14}}{16} & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 \\ \frac{\sqrt{14}}{16} & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 & 0 \end{bmatrix}$ |
| 297 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 298 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{56} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{28} & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{56} \\ 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{56} & 0 \\ 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\ \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 299 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{56} \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & \frac{\sqrt{105}}{56} & 0 \\ 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{42}}{112} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 \\ \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 300 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 \\ -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{3i}{16} & 0 & 0 & \frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} \\ 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 \end{bmatrix}$ |
| 301 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}}{32} & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{32} & -\frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & \frac{\sqrt{15}i}{16} \\ 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 302 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{3\sqrt{42}}{224} & \frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{224} & -\frac{\sqrt{105}i}{112} & 0 \\ \frac{\sqrt{42}i}{56} & 0 & \frac{3\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}i}{112} \\ 0 & \frac{3\sqrt{42}}{224} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{5\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{3\sqrt{42}}{224} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} & \frac{5\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & \frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{32} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 303 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}}{56} & 0 & \frac{3\sqrt{42}i}{224} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{5\sqrt{70}i}{224} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & \frac{3\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} & -\frac{\sqrt{70}}{56} & 0 & \frac{5\sqrt{70}i}{224} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{112} & 0 \\ -\frac{3\sqrt{42}i}{224} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{112} \\ 0 & 0 & -\frac{\sqrt{42}i}{32} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{32} & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \end{bmatrix}$ |
| 304 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 305 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{42} & \frac{\sqrt{42}}{84} & 0 \\ \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & \frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & \frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{42} & -\frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 306 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{42} & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 307 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & \frac{\sqrt{7}}{84} & -\frac{\sqrt{42}i}{84} & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{7}}{84} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 \\ -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & -\frac{\sqrt{7}}{84} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \end{bmatrix}$ |
| 308 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{7}}{12} & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & -\frac{\sqrt{7}}{12} & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & \frac{\sqrt{42}i}{84} & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & -\frac{\sqrt{42}}{28} & 0 \end{bmatrix}$ |
| 309 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 \\ 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 \\ -\frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 310 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{60} & \frac{\sqrt{35}}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 \\ \frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} \\ 0 & -\frac{\sqrt{210}i}{60} & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 \\ 0 & -\frac{\sqrt{210}}{240} & 0 & -\frac{\sqrt{210}i}{240} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{14}i}{112} & 0 & 0 \\ \frac{\sqrt{210}}{240} & 0 & -\frac{\sqrt{210}i}{240} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 311 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 312 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{\sqrt{105}}{56} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & -\frac{\sqrt{105}}{56} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & \frac{3\sqrt{7}}{56} & 0 \\ \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & -\frac{\sqrt{105}}{140} & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} \\ 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 \\ 0 & \frac{\sqrt{70}}{80} & 0 & -\frac{\sqrt{70}i}{80} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & 0 \\ -\frac{\sqrt{70}}{80} & 0 & -\frac{\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & \frac{3\sqrt{42}}{112} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 313 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} \\ 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{56} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & \frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & -\frac{3\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{70}i}{80} & 0 & -\frac{\sqrt{70}}{80} & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{3\sqrt{42}}{112} & 0 & 0 \\ -\frac{\sqrt{70}i}{80} & 0 & \frac{\sqrt{70}}{80} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & -\frac{3\sqrt{42}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 314 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{40} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 \\ -\frac{\sqrt{10}}{40} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{40} & \frac{\sqrt{6}}{8} & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{i}{16} & 0 \\ -\frac{\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{80} & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{i}{16} \\ 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{160} & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & -\frac{1}{4} & 0 \end{bmatrix}$ |
| 315 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{Q}_{4,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}}{160} & -\frac{\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & -\frac{i}{16} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & \frac{i}{16} \\ 0 & \frac{3\sqrt{10}}{160} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 \\ -\frac{3\sqrt{10}}{160} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{40} & -\frac{\sqrt{6}}{32} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{40} & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & \frac{i}{4} & 0 \end{bmatrix}$ |
| 316 | symmetry | $3\sqrt{5}x^2yz-\frac{\sqrt{5}y^3z}{2}-\frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{70}i}{280} & 0 & -\frac{19\sqrt{70}}{1120} & \frac{11\sqrt{105}i}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{5\sqrt{42}}{224} & -\frac{\sqrt{7}i}{112} & 0 \\ -\frac{3\sqrt{70}i}{280} & 0 & \frac{19\sqrt{70}}{1120} & 0 & 0 & -\frac{11\sqrt{105}i}{560} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{5\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}i}{112} \\ 0 & -\frac{9\sqrt{70}}{1120} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & \frac{5\sqrt{42}}{224} & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{70}}{1120} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{56} & -\frac{5\sqrt{42}}{224} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{56} & -\frac{\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & -\frac{3\sqrt{70}i}{160} & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 317 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{70} & 0 & -\frac{9\sqrt{70}i}{1120} & 0 & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & 0 \\ -\frac{\sqrt{70}}{70} & 0 & -\frac{9\sqrt{70}i}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & 0 & 0 \\ 0 & -\frac{19\sqrt{70}i}{1120} & 0 & -\frac{3\sqrt{70}}{280} & \frac{11\sqrt{105}i}{560} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{112} & 0 \\ -\frac{19\sqrt{70}i}{1120} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{11\sqrt{105}i}{560} & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{112} \\ 0 & 0 & \frac{3\sqrt{70}i}{160} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{7}}{28} \\ 0 & 0 & 0 & -\frac{3\sqrt{70}i}{160} & \frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{224} & \frac{\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 318 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_{2}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 319 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2}^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{84} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{84} & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 \\ 0 & -\frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 \\ \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 320 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_{2}^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{42} & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{84} & 0 \\ 0 & -\frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{210}}{56} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 \\ -\frac{\sqrt{210}i}{56} & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 321 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & \frac{11\sqrt{14}}{168} & \frac{\sqrt{21}i}{21} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{11\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{21} \\ 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 \\ \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 322 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & -\frac{5\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{21} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{11\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 \end{bmatrix}$ |
| 323 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 324 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_3^{(a)}(B_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 325 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 \\ -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 326 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 327 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 \end{bmatrix}$ |
| 328 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 \end{bmatrix}$ |
| 329 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 330 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & \frac{\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & -\frac{\sqrt{210}}{105} & \frac{3\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & -\frac{3\sqrt{35}i}{70} \end{bmatrix}$ |
| 331 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & \frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{21} & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 332 | symmetry | $\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 333 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 \\ \frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & -\frac{3\sqrt{210}}{280} & \frac{\sqrt{35}i}{70} & 0 \\ \frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{35}i}{70} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{35}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{105} & \frac{3\sqrt{35}}{140} & 0 \end{bmatrix}$ |
| 334 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{420} & \frac{\sqrt{35}i}{70} & 0 \\ \frac{3\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & -\frac{\sqrt{35}i}{70} \\ 0 & \frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 \\ -\frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 \end{bmatrix}$ |
| 335 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{84} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{84} & \frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 \\ \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$ |
| 336 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 \\ \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & \frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & -\frac{\sqrt{21}}{28} & 0 \end{bmatrix}$ |
| 337 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 338 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{84} & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & \frac{\sqrt{35}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{21} \end{bmatrix}$ |
| 339 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 340 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{G}_5^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{12} & 0 \\ -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & -\frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 341 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{12} & 0 \\ 0 & -\frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ -\frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 342 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{5\sqrt{14}}{112} & 0 & \frac{13\sqrt{14}i}{336} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 \\ -\frac{5\sqrt{14}}{112} & 0 & \frac{13\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{14}}{84} & \frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{56} & 0 \\ \frac{5\sqrt{14}i}{112} & 0 & \frac{5\sqrt{14}}{84} & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & 0 & \frac{\sqrt{14}i}{48} & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & \frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{48} & -\frac{5\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & -\frac{\sqrt{35}}{56} & 0 \end{bmatrix}$ |
| 343 | symmetry | $x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$ |
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{5\sqrt{14}i}{84} & 0 & -\frac{5\sqrt{14}}{112} & -\frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{336} & \frac{\sqrt{35}i}{56} & 0 \\ \frac{5\sqrt{14}i}{84} & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & -\frac{13\sqrt{14}}{336} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 \\ \frac{13\sqrt{14}}{336} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{48} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{84} & \frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} \\ 0 & \frac{\sqrt{14}i}{48} & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 \end{bmatrix}$ |
| 344 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ -\frac{\sqrt{10}}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{i}{8} & 0 \\ \frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{8} \\ 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & \frac{1}{8} \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{80} & \frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & -\frac{1}{8} & 0 \end{bmatrix}$ |
| 345 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{80} & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & \frac{i}{8} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{i}{8} \\ 0 & -\frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ \frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{20} & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{i}{8} & 0 \end{bmatrix}$ |
| 346 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{30} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ -\frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 347 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{30}}{40} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}i}{12} & 0 \\ \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 348 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_{3}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 349 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{1}{24} & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & -\frac{1}{24} & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & -\frac{\sqrt{15}}{24} & 0 \\ \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & -\frac{i}{24} & 0 & \frac{1}{6} & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 \\ 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 \\ -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{6} & \frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 350 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{G}_3^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & -\frac{1}{24} & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & \frac{i}{6} & 0 & \frac{1}{24} & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{1}{6} & 0 & \frac{i}{24} & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & -\frac{1}{6} & 0 & \frac{i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & \frac{\sqrt{15}}{24} & 0 \\ 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 \\ -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & -\frac{\sqrt{10}}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 351 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{6}i}{96} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{11\sqrt{6}i}{96} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & \frac{i}{16} & 0 \\ \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{48} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{16} \\ 0 & 0 & -\frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 \end{bmatrix}$ |
| 352 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{32} & -\frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{96} & \frac{i}{16} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & \frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{i}{16} \\ 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{11\sqrt{6}}{96} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & \frac{11\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 353 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{96} & \frac{7i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{5\sqrt{10}}{96} & \frac{\sqrt{15}i}{48} & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{96} & 0 & 0 & -\frac{7i}{48} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{5\sqrt{10}}{96} & 0 & 0 & -\frac{\sqrt{15}i}{48} \\ 0 & \frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{24} & 0 & 0 & -\frac{7\sqrt{10}}{96} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 \\ -\frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{24} & \frac{7\sqrt{10}}{96} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 \\ -\frac{3\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & \frac{1}{24} & -\frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{6}i}{32} & 0 & 0 & \frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & \frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 354 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & \frac{i}{24} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & \frac{7\sqrt{10}i}{96} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{24} & \frac{\sqrt{10}}{24} & 0 & \frac{7\sqrt{10}i}{96} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{6}}{24} & \frac{7i}{48} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{10}i}{96} & 0 & \frac{\sqrt{10}}{24} & -\frac{\sqrt{15}i}{48} & 0 \\ -\frac{\sqrt{6}i}{96} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{7i}{48} & 0 & 0 & -\frac{5\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}i}{48} \\ 0 & 0 & -\frac{3\sqrt{6}i}{32} & 0 & 0 & \frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & \frac{\sqrt{10}i}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}i}{32} & -\frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 \end{bmatrix}$ |
| 355 | symmetry | $z$ |
| | $\mathbb{G}_{1}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & \frac{\sqrt{35}}{35} & \frac{\sqrt{210}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}i}{70} \end{bmatrix}$ |
| 356 | symmetry | $y$ |
| | $\mathbb{G}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{35}}{140} & -\frac{\sqrt{210}i}{140} & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{210}}{140} & 0 \end{bmatrix}$ |
| 357 | symmetry | $x$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & -\frac{\sqrt{210}i}{140} & 0 \\ \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \end{bmatrix}$ |
| 358 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{168} & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{5\sqrt{7}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & \frac{5\sqrt{7}}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & -\frac{5\sqrt{7}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & \frac{5\sqrt{42}}{168} & \frac{2\sqrt{7}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & -\frac{2\sqrt{7}i}{21} \end{bmatrix}$ |
| 359 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & \frac{\sqrt{105}}{168} & 0 \\ \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} \\ 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 \\ 0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\ \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 360 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} \\ 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{168} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{168} & 0 \\ 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{42}}{48} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 \\ \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 361 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{13\sqrt{70}i}{672} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{672} & 0 & 0 \\ -\frac{\sqrt{70}}{56} & 0 & -\frac{13\sqrt{70}i}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & -\frac{\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{672} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{42} & -\frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{672} & 0 & -\frac{\sqrt{42}}{42} & -\frac{5\sqrt{7}i}{112} & 0 \\ -\frac{5\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & \frac{5\sqrt{105}i}{336} & 0 & 0 & -\frac{5\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{5\sqrt{7}i}{112} \\ 0 & 0 & -\frac{\sqrt{70}i}{96} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{96} & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{224} & -\frac{\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 362 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{42} & 0 & \frac{5\sqrt{70}}{224} & \frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & -\frac{5\sqrt{42}}{672} & -\frac{5\sqrt{7}i}{112} & 0 \\ \frac{\sqrt{70}i}{42} & 0 & -\frac{5\sqrt{70}}{224} & 0 & 0 & -\frac{5\sqrt{105}i}{336} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & \frac{5\sqrt{42}}{672} & 0 & 0 & \frac{5\sqrt{7}i}{112} \\ 0 & \frac{13\sqrt{70}}{672} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{5\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 \\ -\frac{13\sqrt{70}}{672} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & \frac{5\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & -\frac{5\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & -\frac{\sqrt{70}i}{96} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & \frac{5\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 363 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{11\sqrt{42}}{672} & -\frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{70}}{224} & -\frac{5\sqrt{105}i}{336} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{11\sqrt{42}}{672} & 0 & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & \frac{5\sqrt{105}i}{336} \\ 0 & -\frac{17\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 \\ \frac{17\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} & \frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & -\frac{5\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} \\ 0 & \frac{\sqrt{42}i}{96} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{5\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 \end{bmatrix}$ |
| 364 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{G}_{3,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}}{42} & 0 & -\frac{17\sqrt{42}i}{672} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{224} & 0 & 0 \\ -\frac{\sqrt{42}}{42} & 0 & -\frac{17\sqrt{42}i}{672} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{70}i}{224} & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{42}i}{672} & 0 & -\frac{\sqrt{42}}{56} & -\frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & \frac{5\sqrt{105}i}{336} & 0 \\ -\frac{11\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & \frac{3\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{5\sqrt{105}i}{336} \\ 0 & 0 & -\frac{\sqrt{42}i}{96} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{5\sqrt{70}i}{224} & 0 & 0 & -\frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{96} & \frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}i}{224} & \frac{\sqrt{105}}{84} & 0 \end{bmatrix}$ |
| 365 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \end{bmatrix}$ |
| 366 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}_2^{(a)}(B_g, 1)$ | $\begin{bmatrix} \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 367 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_2^{(a)}(B_g, 2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 368 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 \end{bmatrix}$ |
| 369 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{2,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 370 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{T}^{(a)}_{4}(A_g,1)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 371 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{T}^{(a)}_{4}(A_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} \end{bmatrix}$ |
| 372 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{T}^{(a)}_{4}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 373 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 374 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{T}_4^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 375 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 376 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{T}_{4,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} \\ 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{16} & 0 & 0 \end{bmatrix}$ |
| 377 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{4,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & 0 & \frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \end{bmatrix}$ |
| 378 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{T}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 379 | symmetry | $\frac{\sqrt{21}x^4}{6}-\frac{\sqrt{21}x^2y^2}{2}-\frac{\sqrt{21}x^2z^2}{2}+\frac{\sqrt{21}y^4}{6}-\frac{\sqrt{21}y^2z^2}{2}+\frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & \frac{i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{1}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 \\ -\frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 380 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & -\frac{\sqrt{14}i}{16} & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 \\ \frac{\sqrt{14}i}{16} & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 \end{bmatrix}$ |
| 381 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{8} & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 382 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{105}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & -\frac{\sqrt{105}i}{56} & 0 \\ \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\ 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 \\ 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 \\ \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 383 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{T}_4^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{3\sqrt{7}i}{56} & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\ 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{56} & 0 \\ 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{42}i}{112} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 \\ -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 384 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{32} & 0 & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & \frac{\sqrt{15}}{16} & 0 \\ \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} \\ 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 \end{bmatrix}$ |
| 385 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & \frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{32} & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}}{16} \\ 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 386 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{3\sqrt{42}i}{224} & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{224} & \frac{\sqrt{105}}{112} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & \frac{3\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & -\frac{\sqrt{105}}{112} \\ 0 & \frac{3\sqrt{42}i}{224} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{5\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{3\sqrt{42}i}{224} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} & \frac{5\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{42}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{3\sqrt{7}i}{56} & -\frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{32} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 387 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{T}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{3\sqrt{42}}{224} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{5\sqrt{70}}{224} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & -\frac{3\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & -\frac{\sqrt{70}i}{56} & 0 & -\frac{5\sqrt{70}}{224} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{42}}{224} & 0 & \frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{112} & 0 \\ \frac{3\sqrt{42}}{224} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{112} \\ 0 & 0 & \frac{\sqrt{42}}{32} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{32} & \frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \end{bmatrix}$ |
| 388 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_{2}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 389 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & -\frac{\sqrt{42}i}{84} & 0 \\ \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{42} & \frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 390 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_2^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & -\frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & -\frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 391 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & -\frac{\sqrt{7}i}{84} & -\frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & \frac{\sqrt{7}i}{84} & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{12} & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & \frac{\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \end{bmatrix}$ |
| 392 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & \frac{\sqrt{7}i}{12} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & -\frac{5\sqrt{7}i}{84} & \frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & \frac{5\sqrt{7}i}{84} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & \frac{\sqrt{42}i}{28} & 0 \end{bmatrix}$ |
| 393 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 \\ \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 394 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{60} & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & -\frac{\sqrt{35}i}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 \\ \frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{168} \\ 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 \\ 0 & \frac{\sqrt{210}i}{240} & 0 & -\frac{\sqrt{210}}{240} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 \\ -\frac{\sqrt{210}i}{240} & 0 & -\frac{\sqrt{210}}{240} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 395 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 396 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{T}_4^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{280} & \frac{\sqrt{105}i}{56} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & -\frac{3\sqrt{7}i}{56} & 0 \\ \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & \frac{\sqrt{105}i}{140} & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{70}i}{80} & 0 & -\frac{\sqrt{70}}{80} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{3\sqrt{42}i}{112} & 0 & -\frac{3\sqrt{42}}{112} & 0 & 0 \\ \frac{\sqrt{70}i}{80} & 0 & -\frac{\sqrt{70}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & -\frac{3\sqrt{42}i}{112} & 0 & -\frac{3\sqrt{42}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 397 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{56} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{3\sqrt{7}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{280} & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & \frac{3\sqrt{7}i}{56} & 0 \\ 0 & -\frac{\sqrt{70}}{80} & 0 & \frac{\sqrt{70}i}{80} & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & 0 \\ -\frac{\sqrt{70}}{80} & 0 & -\frac{\sqrt{70}i}{80} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & \frac{3\sqrt{42}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 398 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{6}}{32} & 0 & 0 \\ \frac{\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{10}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & -\frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & -\frac{1}{16} & 0 \\ -\frac{\sqrt{10}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{80} & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{1}{16} \\ 0 & 0 & \frac{3\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{160} & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & \frac{i}{4} & 0 \end{bmatrix}$ |
| 399 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{160} & -\frac{\sqrt{15}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{32} & -\frac{1}{16} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & \frac{1}{16} \\ 0 & -\frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 \\ \frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & \frac{\sqrt{6}i}{32} & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{40} & -\frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & \frac{3\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & \frac{1}{4} & 0 \end{bmatrix}$ |
| 400 | symmetry | $3\sqrt{5}x^2yz-\frac{\sqrt{5}y^3z}{2}-\frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{19\sqrt{70}i}{1120} & \frac{11\sqrt{105}}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{5\sqrt{42}i}{224} & -\frac{\sqrt{7}}{112} & 0 \\ -\frac{3\sqrt{70}}{280} & 0 & -\frac{19\sqrt{70}i}{1120} & 0 & 0 & -\frac{11\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{5\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}}{112} \\ 0 & \frac{9\sqrt{70}i}{1120} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & -\frac{5\sqrt{42}i}{224} & 0 & 0 & 0 & 0 \\ -\frac{9\sqrt{70}i}{1120} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{56} & \frac{5\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & -\frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & -\frac{3\sqrt{70}}{160} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 401 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}i}{70} & 0 & -\frac{9\sqrt{70}}{1120} & 0 & 0 & \frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{224} & 0 & 0 \\ \frac{\sqrt{70}i}{70} & 0 & -\frac{9\sqrt{70}}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & -\frac{5\sqrt{42}}{224} & 0 & 0 & 0 \\ 0 & -\frac{19\sqrt{70}}{1120} & 0 & \frac{3\sqrt{70}i}{280} & \frac{11\sqrt{105}}{560} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{224} & 0 & -\frac{\sqrt{42}i}{56} & \frac{\sqrt{7}}{112} & 0 \\ -\frac{19\sqrt{70}}{1120} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{11\sqrt{105}}{560} & 0 & 0 & -\frac{5\sqrt{42}}{224} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}}{112} \\ 0 & 0 & \frac{3\sqrt{70}}{160} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{160} & -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & -\frac{\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 402 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}^{(1,1;a)}_{2}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 403 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}^{(1,1;a)}_{2}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{84} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{84} & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \\ 0 & -\frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 \\ \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & \frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 404 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}^{(1,1;a)}_{2}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{42} & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{168} & -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & \frac{\sqrt{210}}{56} & 0 & -\frac{\sqrt{210}i}{56} & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 \\ \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 405 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{2,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{11\sqrt{14}i}{168} & -\frac{\sqrt{21}}{21} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{11\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 \\ \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & \frac{\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 406 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & -\frac{5\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{11\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & \frac{\sqrt{21}}{21} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{11\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 \end{bmatrix}$ |
| 407 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 408 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_3^{(a)}(B_g, 1)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 409 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 410 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 411 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 \end{bmatrix}$ |
| 412 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 \end{bmatrix}$ |
| 413 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 414 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & -\frac{\sqrt{35}i}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & \frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & \frac{\sqrt{210}i}{105} & \frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{35}}{70} \end{bmatrix}$ |
| 415 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & \frac{\sqrt{14}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & \frac{\sqrt{14}i}{21} & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 416 | symmetry | $\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 417 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{420} & 0 & 0 \\ -\frac{3\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & \frac{3\sqrt{210}i}{280} & \frac{\sqrt{35}}{70} & 0 \\ \frac{\sqrt{14}}{28} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & \frac{3\sqrt{35}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & -\frac{3\sqrt{35}i}{140} & 0 \end{bmatrix}$ |
| 418 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{28} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & -\frac{\sqrt{210}i}{420} & \frac{\sqrt{35}}{70} & 0 \\ \frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & -\frac{\sqrt{35}}{70} \\ 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 \\ \frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{42} & -\frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 \end{bmatrix}$ |
| 419 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{84} & \frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 \\ -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \end{bmatrix}$ |
| 420 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & \frac{\sqrt{14}i}{24} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & -\frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{42} & 0 \\ \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{21} & \frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$ |
| 421 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 422 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{84} & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & \frac{\sqrt{35}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}}{21} \end{bmatrix}$ |
| 423 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{20} & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 424 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}i}{12} & 0 \\ -\frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ -\frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 425 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & -\frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ -\frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 426 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{5\sqrt{14}i}{112} & 0 & \frac{13\sqrt{14}}{336} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 \\ \frac{5\sqrt{14}i}{112} & 0 & \frac{13\sqrt{14}}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{14}i}{84} & \frac{5\sqrt{21}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{84} & \frac{\sqrt{35}}{56} & 0 \\ \frac{5\sqrt{14}}{112} & 0 & -\frac{5\sqrt{14}i}{84} & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & \frac{\sqrt{14}}{48} & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & -\frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{48} & \frac{5\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & \frac{\sqrt{35}i}{56} & 0 \end{bmatrix}$ |
| 427 | symmetry | $x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{5\sqrt{14}}{84} & 0 & \frac{5\sqrt{14}i}{112} & -\frac{5\sqrt{21}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{336} & \frac{\sqrt{35}}{56} & 0 \\ \frac{5\sqrt{14}}{84} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & \frac{13\sqrt{14}i}{336} & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 \\ -\frac{13\sqrt{14}i}{336} & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{48} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & -\frac{\sqrt{21}i}{84} & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{56} \\ 0 & \frac{\sqrt{14}}{48} & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 & \frac{\sqrt{35}}{56} & 0 \end{bmatrix}$ |
| 428 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ \frac{\sqrt{10}i}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & \frac{1}{8} & 0 \\ \frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & -\frac{1}{8} \\ 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{i}{8} \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{80} & -\frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & \frac{i}{8} & 0 \end{bmatrix}$ |
| 429 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{80} & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & \frac{1}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{1}{8} \\ 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ -\frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & \frac{1}{8} \\ 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & \frac{1}{8} & 0 \end{bmatrix}$ |
| 430 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{30}i}{30} & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 431 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{40} & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & -\frac{\sqrt{3}}{12} & 0 \\ \frac{\sqrt{30}}{30} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 432 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 \end{bmatrix}$ |
| 433 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & \frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & -\frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & -\frac{\sqrt{15}i}{24} & 0 \\ -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & \frac{i}{6} & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & \frac{\sqrt{6}}{24} & 0 & 0 & \frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 \\ 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 \\ -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 434 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_3^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & -\frac{i}{24} & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{1}{6} & 0 & \frac{i}{24} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & \frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & -\frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & \frac{\sqrt{15}i}{24} & 0 \\ 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & -\frac{1}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{48} & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 \\ \frac{\sqrt{6}}{16} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & \frac{1}{6} & 0 & 0 & \frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 \end{bmatrix}$ |
| 435 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{11\sqrt{6}}{96} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & \frac{11\sqrt{6}}{96} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{1}{16} & 0 \\ -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{48} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{1}{16} \\ 0 & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 \end{bmatrix}$ |
| 436 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & \frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & -\frac{1}{16} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & \frac{1}{16} \\ 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & -\frac{11\sqrt{6}i}{96} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & \frac{11\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & \frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 437 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{96} & -\frac{7}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & \frac{5\sqrt{10}i}{96} & -\frac{\sqrt{15}}{48} & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & \frac{7}{48} & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{5\sqrt{10}i}{96} & 0 & 0 & \frac{\sqrt{15}}{48} \\ 0 & \frac{\sqrt{6}i}{96} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & \frac{1}{24} & 0 & 0 & -\frac{7\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 \\ -\frac{\sqrt{6}i}{96} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & -\frac{1}{24} & \frac{7\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 \\ \frac{3\sqrt{6}}{32} & 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & \frac{i}{24} & \frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{6}}{32} & 0 & 0 & -\frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 438 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{1}{24} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{7\sqrt{10}}{96} & 0 & 0 \\ \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{1}{24} & \frac{\sqrt{10}i}{24} & 0 & -\frac{7\sqrt{10}}{96} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{6}i}{24} & -\frac{7}{48} & 0 & 0 & 0 & 0 & \frac{5\sqrt{10}}{96} & 0 & \frac{\sqrt{10}i}{24} & \frac{\sqrt{15}}{48} & 0 \\ \frac{\sqrt{6}}{96} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & \frac{7}{48} & 0 & 0 & \frac{5\sqrt{10}}{96} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{\sqrt{15}}{48} \\ 0 & 0 & \frac{3\sqrt{6}}{32} & 0 & 0 & \frac{i}{6} & 0 & \frac{1}{24} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{32} & -\frac{i}{6} & 0 & \frac{1}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{96} & 0 & 0 \end{bmatrix}$ |
| 439 | symmetry | $z$ |
| | $\mathbb{M}_{1}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{210}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}}{70} \end{bmatrix}$ |
| 440 | symmetry | $y$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{3\sqrt{35}i}{140} & -\frac{\sqrt{210}}{140} & 0 \\ -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & -\frac{\sqrt{210}i}{140} & 0 \end{bmatrix}$ |
| 441 | symmetry | $x$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & -\frac{\sqrt{210}}{140} & 0 \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \end{bmatrix}$ |
| 442 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{105}i}{168} & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & -\frac{5\sqrt{7}i}{84} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & \frac{5\sqrt{7}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & \frac{2\sqrt{7}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & -\frac{2\sqrt{7}}{21} \end{bmatrix}$ |
| 443 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{105}i}{168} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{168} & -\frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & -\frac{\sqrt{105}i}{168} & 0 \\ \frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} \\ 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 \\ 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{42}}{48} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\ -\frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{42}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 444 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} \\ 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{168} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{168} & \frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{168} & 0 \\ 0 & \frac{\sqrt{42}}{48} & 0 & -\frac{\sqrt{42}i}{48} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\ \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 445 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{13\sqrt{70}}{672} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{672} & 0 & 0 \\ \frac{\sqrt{70}i}{56} & 0 & -\frac{13\sqrt{70}}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & \frac{\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{672} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{42} & -\frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{42} & -\frac{5\sqrt{7}}{112} & 0 \\ -\frac{5\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & \frac{5\sqrt{105}}{336} & 0 & 0 & -\frac{5\sqrt{42}}{672} & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & \frac{5\sqrt{7}}{112} \\ 0 & 0 & -\frac{\sqrt{70}}{96} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{5\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{7}i}{28} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{96} & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{224} & \frac{\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 446 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{42} & 0 & -\frac{5\sqrt{70}i}{224} & \frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & \frac{5\sqrt{42}i}{672} & -\frac{5\sqrt{7}}{112} & 0 \\ \frac{\sqrt{70}}{42} & 0 & \frac{5\sqrt{70}i}{224} & 0 & 0 & -\frac{5\sqrt{105}}{336} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & -\frac{5\sqrt{42}i}{672} & 0 & 0 & \frac{5\sqrt{7}}{112} \\ 0 & -\frac{13\sqrt{70}i}{672} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & \frac{5\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 \\ \frac{13\sqrt{70}i}{672} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & -\frac{5\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 \\ \frac{\sqrt{70}}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & -\frac{5\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & -\frac{\sqrt{70}}{96} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{5\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 447 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{11\sqrt{42}i}{672} & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{70}i}{224} & -\frac{5\sqrt{105}}{336} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{11\sqrt{42}i}{672} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{70}i}{224} & 0 & 0 & \frac{5\sqrt{105}}{336} \\ 0 & \frac{17\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{70}i}{224} & 0 & 0 & 0 & 0 \\ -\frac{17\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} & -\frac{3\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & -\frac{5\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \\ 0 & \frac{\sqrt{42}}{96} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{5\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 \end{bmatrix}$ |
| 448 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{42} & 0 & -\frac{17\sqrt{42}}{672} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 \\ \frac{\sqrt{42}i}{42} & 0 & -\frac{17\sqrt{42}}{672} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{56} & \frac{5\sqrt{105}}{336} & 0 \\ -\frac{11\sqrt{42}}{672} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{5\sqrt{105}}{336} \\ 0 & 0 & -\frac{\sqrt{42}}{96} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{5\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{96} & -\frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{224} & -\frac{\sqrt{105}i}{84} & 0 \end{bmatrix}$ |

bra: $=\langle d_v,\uparrow|,\langle d_v,\downarrow|,\langle d_{xy},\uparrow|,\langle d_{xy},\downarrow|,\langle d_{xz},\uparrow|,\langle d_{xz},\downarrow|,\langle d_{yz},\uparrow|,\langle d_{yz},\downarrow|,\langle d_u,\uparrow|,\langle d_u,\downarrow|$

ket: $=|d_v,\uparrow\rangle,|d_v,\downarrow\rangle,|d_{xy},\uparrow\rangle,|d_{xy},\downarrow\rangle,|d_{xz},\uparrow\rangle,|d_{xz},\downarrow\rangle,|d_{yz},\uparrow\rangle,|d_{yz},\downarrow\rangle,|d_u,\uparrow\rangle,|d_u,\downarrow\rangle$

Table 8: (d,d) block.

| No. | multipole | matrix |
|---|---|---|
| 449 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 450 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} \end{bmatrix}$ |
| 451 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 \\ -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 452 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 453 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \end{bmatrix}$ |
| 454 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 \\ \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 455 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A_g,1)$ | $\begin{bmatrix} \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} \end{bmatrix}$ |
| 456 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{Q}_4^{(a)}(A_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{bmatrix}$ |
| 457 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 458 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_4^{(a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 459 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 460 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 461 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 \end{bmatrix}$ |
| 462 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \end{bmatrix}$ |
| 463 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 464 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_{2}^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & \frac{\sqrt{10}}{20} & 0 \\ 0 & \frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 465 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 466 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 467 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 468 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$ |
| 469 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & \frac{\sqrt{15}}{60} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{60} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{60} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & \frac{\sqrt{5}}{20} & 0 \\ 0 & \frac{\sqrt{15}i}{60} & 0 & -\frac{\sqrt{15}}{15} & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{60} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 470 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & -\frac{5\sqrt{21}}{84} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{28} \\ \frac{5\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & \frac{\sqrt{7}}{28} & 0 \\ 0 & -\frac{5\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ -\frac{5\sqrt{21}i}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 471 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 472 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 473 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 474 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 \\ 0 & -\frac{i}{8} & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ \frac{i}{8} & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & -\frac{i}{8} & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 \end{bmatrix}$ |
| 475 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{1}{8} & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & \frac{\sqrt{3}}{8} \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & -\frac{\sqrt{3}}{8} & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{3}i}{8} \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 476 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\ 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & \frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{3\sqrt{21}}{56} \\ -\frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{3\sqrt{21}}{56} & 0 \\ -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & -\frac{\sqrt{21}i}{56} & 0 \\ 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{21}i}{56} \\ 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{14} & 0 & \frac{3\sqrt{21}}{56} & \frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & -\frac{3\sqrt{21}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 477 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & -\frac{\sqrt{21}}{14} & 0 \\ 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{56} \\ \frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{3\sqrt{21}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 \\ 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} \\ 0 & -\frac{\sqrt{21}}{14} & 0 & -\frac{3\sqrt{21}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{21}}{14} & 0 & -\frac{3\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 \end{bmatrix}$ |
| 478 | symmetry | 1 |
| | $\mathbb{Q}_{0}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & -\frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & \frac{\sqrt{5}}{10} \\ \frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & -\frac{\sqrt{5}}{10} & 0 \\ 0 & -\frac{\sqrt{15}i}{30} & 0 & -\frac{\sqrt{15}}{30} & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ -\frac{\sqrt{15}i}{30} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 479 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & \frac{\sqrt{35}}{35} \\ -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & -\frac{\sqrt{35}}{35} & 0 \\ 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} \\ \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 480 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{35}}{70} & -\frac{\sqrt{105}i}{42} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & \frac{\sqrt{105}i}{42} \\ 0 & -\frac{\sqrt{35}}{35} & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} \\ \frac{\sqrt{35}}{35} & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ 0 & \frac{\sqrt{35}i}{35} & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} \\ \frac{\sqrt{35}i}{35} & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 \end{bmatrix}$ |
| 481 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & \frac{3\sqrt{35}}{70} & \frac{\sqrt{105}i}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}i}{70} & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} \\ -\frac{3\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} \\ \frac{3\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 \end{bmatrix}$ |
| 482 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} \\ 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 \\ 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{105}}{42} \\ -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & -\frac{\sqrt{105}}{42} & 0 \\ \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & \frac{2\sqrt{105}i}{105} & 0 \\ 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{2\sqrt{105}i}{105} \\ 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{42} & 0 & -\frac{\sqrt{105}}{42} & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 483 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{105}}{42} \\ 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & -\frac{\sqrt{105}}{42} & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 \\ 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105} \\ 0 & -\frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 & 0 \end{bmatrix}$ |
| 484 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 485 | symmetry | $y$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$ |
| 486 | symmetry | $x$ |
| | $\mathbb{G}_{1,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 487 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 488 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_3^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 489 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 490 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & \frac{3\sqrt{15}i}{40} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{40} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} \\ -\frac{\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ -\frac{3\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{40} & 0 \\ 0 & \frac{3\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{40} & 0 & 0 \end{bmatrix}$ |
| 491 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & -\frac{3\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & \frac{3\sqrt{15}i}{40} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}}{8} \\ -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & \frac{\sqrt{5}}{8} & 0 \\ \frac{3\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & \frac{\sqrt{5}i}{40} & 0 \\ 0 & -\frac{3\sqrt{15}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{40} \\ 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{8} & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & \frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 492 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{1}{8} & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & -\frac{\sqrt{3}}{24} \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & \frac{\sqrt{3}}{24} & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & \frac{\sqrt{3}i}{24} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & -\frac{\sqrt{3}i}{24} \\ 0 & 0 & \frac{i}{4} & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{24} & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 493 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{i}{8} & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{8} & \frac{\sqrt{3}}{6} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} \\ \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 \\ 0 & \frac{i}{8} & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} \\ 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 \end{bmatrix}$ |
| 494 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} \\ -\frac{\sqrt{42}}{28} & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 495 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{21} \\ 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & \frac{\sqrt{42}i}{84} & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\ -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & \frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 496 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_2^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & -\frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\ -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \\ 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & \frac{\sqrt{42}i}{84} & 0 \\ -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{21} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 497 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{2,1}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{14} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & \frac{\sqrt{14}i}{28} & \frac{\sqrt{42}}{84} & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{42} & 0 & -\frac{\sqrt{42}i}{42} & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ \frac{\sqrt{42}}{42} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \end{bmatrix}$ |
| 498 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{T}^{(1,0;a)}_{2,2}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{14} & -\frac{\sqrt{42}}{84} & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & -\frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ \frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & \frac{\sqrt{14}i}{14} & 0 \end{bmatrix}$ |
| 499 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \end{bmatrix}$ |
| 500 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & -\frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} \\ -\frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 \\ 0 & -\frac{\sqrt{105}}{420} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} \\ -\frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 501 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 502 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{T}_4^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & \frac{3\sqrt{105}i}{140} \\ \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & -\frac{3\sqrt{105}i}{140} & 0 \\ 0 & -\frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{140} \\ -\frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{3\sqrt{105}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{3\sqrt{105}i}{140} & 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & -\frac{3\sqrt{105}i}{140} & 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 & 0 \end{bmatrix}$ |
| 503 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{140} \\ -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{3\sqrt{105}}{140} & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{3\sqrt{105}i}{140} \\ -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & \frac{3\sqrt{105}i}{140} & 0 \\ \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{140} & 0 & -\frac{3\sqrt{105}i}{140} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{105}}{140} & 0 & \frac{3\sqrt{105}i}{140} & 0 & 0 & 0 \end{bmatrix}$ |
| 504 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{40} & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} \\ -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & -\frac{\sqrt{5}i}{5} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 \\ 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & -\frac{3\sqrt{5}i}{20} \\ -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} & \frac{3\sqrt{5}i}{20} & 0 \end{bmatrix}$ |
| 505 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{5}i}{40} & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} \\ \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{5}i}{40} & 0 & 0 & \frac{\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \\ 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{40} \\ \frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & \frac{\sqrt{15}i}{40} & 0 \\ -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & -\frac{\sqrt{15}}{40} & 0 \\ 0 & \frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{40} & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{20} \\ \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & \frac{3\sqrt{5}}{20} & 0 \end{bmatrix}$ |
| 506 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{4,1}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{40} & \frac{11\sqrt{35}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ -\frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{40} & 0 & 0 & -\frac{11\sqrt{35}}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 \\ 0 & \frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ -\frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{105}i}{56} & 0 \\ \frac{11\sqrt{35}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & -\frac{\sqrt{35}i}{28} & -\frac{\sqrt{105}}{280} & 0 \\ 0 & -\frac{11\sqrt{35}}{280} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{105}}{280} \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{140} \\ -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{3\sqrt{35}}{140} & 0 \end{bmatrix}$ |
| 507 | symmetry | $-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{40} & 0 & 0 & \frac{11\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{105}i}{140} \\ \frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{40} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{35}}{280} & -\frac{\sqrt{105}i}{140} & 0 \\ 0 & -\frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{35}i}{35} & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\ -\frac{\sqrt{35}}{40} & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ \frac{11\sqrt{35}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{105}}{280} & 0 \\ 0 & -\frac{11\sqrt{35}}{280} & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{105}}{280} \\ 0 & \frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{3\sqrt{35}i}{140} \\ -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & -\frac{3\sqrt{35}i}{140} & 0 \end{bmatrix}$ |
| 508 | symmetry | $z$ |
| | $\mathbb{M}_1^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 509 | symmetry | $y$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{1,1}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 510 | symmetry | $x$ |
| | $\mathbb{M}^{(a)}_{1,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \end{bmatrix}$ |
| 511 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 512 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_3^{(a)}(B_g, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 513 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(B_g, 2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 514 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(a)}(E_g, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 515 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{3,2}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 \\ 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \end{bmatrix}$ |
| 516 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}^{(a)}_{3,1}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \end{bmatrix}$ |
| 517 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 518 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 519 | symmetry | $y$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 520 | symmetry | $x$ |
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \end{bmatrix}$ |
| 521 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{35} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{35} & -\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} \\ -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{35}i}{70} \\ \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & -\frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{105}}{35} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} \end{bmatrix}$ |
| 522 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{21} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{42} & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 523 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{21} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 \\ 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 524 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{35}i}{70} \\ \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & \frac{3\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} \\ 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{70} & -\frac{\sqrt{105}i}{35} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & -\frac{\sqrt{35}}{70} & 0 \\ 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{35}}{70} \\ 0 & -\frac{3\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ \frac{3\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & -\frac{\sqrt{105}i}{70} & 0 \end{bmatrix}$ |
| 525 | symmetry | $x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} \\ \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{35}i}{35} \\ 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & \frac{\sqrt{35}i}{35} & 0 \\ -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & -\frac{\sqrt{35}}{70} & 0 \\ 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{35}}{70} \\ 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{70} & -\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{35} & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\ -\frac{3\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 \end{bmatrix}$ |
| 526 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{21}i}{21} & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{21}}{42} & 0 \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{21} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \end{bmatrix}$ |
| 527 | symmetry | $-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{21}}{42} & 0 \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$ |
| 528 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |
| | $\mathbb{M}_{5}^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 529 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & \frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{42} & 0 & -\frac{\sqrt{7}i}{42} & -\frac{2\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ \frac{\sqrt{7}}{42} & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & \frac{2\sqrt{7}}{21} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 \\ 0 & \frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{2\sqrt{7}}{21} & 0 & 0 & \frac{\sqrt{21}i}{21} \\ -\frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{21} & -\frac{\sqrt{21}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{21} & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{7} \end{bmatrix}$ |
| 530 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 531 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & -\frac{\sqrt{15}i}{30} & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 532 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & \frac{\sqrt{15}i}{30} & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & -\frac{\sqrt{5}i}{10} & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 533 | symmetry | $\frac{15x^4y}{8}-5x^2y^3+\frac{15x^2yz^2}{4}+y^5-5y^3z^2+\frac{15yz^4}{8}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{19\sqrt{7}i}{168} & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{5\sqrt{21}i}{168} \\ \frac{19\sqrt{7}i}{168} & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{84} & \frac{5\sqrt{21}i}{168} & 0 \\ 0 & \frac{\sqrt{7}}{12} & 0 & \frac{2\sqrt{7}i}{21} & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \\ \frac{\sqrt{7}}{12} & 0 & -\frac{2\sqrt{7}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & \frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 \\ \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & \frac{2\sqrt{7}i}{21} & \frac{\sqrt{21}}{28} & 0 \\ 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & -\frac{2\sqrt{7}i}{21} & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & -\frac{5\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{3\sqrt{7}i}{56} \\ \frac{5\sqrt{21}i}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & \frac{3\sqrt{7}i}{56} & 0 \end{bmatrix}$ |
| 534 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{19\sqrt{7}}{168} & 0 & \frac{\sqrt{7}i}{12} & -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} \\ \frac{19\sqrt{7}}{168} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 \\ 0 & \frac{\sqrt{7}i}{12} & 0 & -\frac{2\sqrt{7}}{21} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ -\frac{\sqrt{7}i}{12} & 0 & -\frac{2\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & \frac{\sqrt{21}i}{84} & 0 \\ -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{21} & 0 & -\frac{\sqrt{7}i}{42} & \frac{\sqrt{21}}{28} & 0 \\ 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{2\sqrt{7}}{21} & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & \frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{168} & 0 & -\frac{\sqrt{21}i}{84} & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} \\ -\frac{5\sqrt{21}}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 \end{bmatrix}$ |
| 535 | symmetry | $\frac{3\sqrt{35}x^4y}{8}-\frac{9\sqrt{35}x^2yz^2}{4}+\frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}i}{40} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{15}i}{40} \\ \frac{\sqrt{5}i}{40} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & -\frac{\sqrt{15}i}{40} & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} \\ \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & \frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{3\sqrt{5}i}{40} \\ -\frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & \frac{3\sqrt{5}i}{40} & 0 \end{bmatrix}$ |
| 536 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{40} & 0 & \frac{\sqrt{5}i}{20} & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} \\ \frac{\sqrt{5}}{40} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 \\ 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{15}i}{20} & 0 \\ \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{15}}{20} & 0 \\ 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{40} \\ \frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 \end{bmatrix}$ |
| 537 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{15}i}{15} & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & \frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} \\ -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$ |
| 538 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{15}}{15} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & -\frac{\sqrt{5}}{10} & 0 \\ 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$ |
| 539 | symmetry | $z$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{35} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{35} & \frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{70}}{140} & 0 & -\frac{3\sqrt{70}i}{140} & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ \frac{3\sqrt{70}}{140} & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{210}}{140} & 0 \\ 0 & \frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ -\frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & \frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & -\frac{\sqrt{210}i}{140} & \frac{\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{70}}{35} \end{bmatrix}$ |
| 540 | symmetry | $y$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{210}i}{70} \\ \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{140} & \frac{\sqrt{210}i}{70} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 \\ 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & -\frac{\sqrt{70}i}{35} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & -\frac{\sqrt{70}i}{70} & \frac{\sqrt{210}}{140} & 0 \\ 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & -\frac{\sqrt{210}i}{70} & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{70}i}{70} \\ \frac{\sqrt{210}i}{70} & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & -\frac{\sqrt{70}i}{70} & 0 \end{bmatrix}$ |
| 541 | symmetry | $x$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{210}i}{70} \\ 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & -\frac{\sqrt{210}i}{70} & 0 \\ \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & -\frac{3\sqrt{70}i}{140} & \frac{\sqrt{210}}{140} & 0 \\ 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & \frac{3\sqrt{70}i}{140} & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{70} & 0 & \frac{\sqrt{210}i}{70} & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} \\ -\frac{\sqrt{210}}{70} & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 \end{bmatrix}$ |
| 542 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & -\frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & -\frac{4\sqrt{35}}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\ -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{4\sqrt{35}}{105} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \\ 0 & -\frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ \frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{105} & \frac{\sqrt{105}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{105}i}{42} & \frac{2\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{2\sqrt{35}}{35} \end{bmatrix}$ |
| 543 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{21}i}{21} & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{7}i}{28} \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{21} & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 544 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_3^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{21}i}{21} & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ \frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & -\frac{\sqrt{21}i}{21} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{7}i}{28} \\ \frac{\sqrt{21}i}{21} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{7}i}{28} & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 545 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{19\sqrt{35}i}{420} & 0 & -\frac{\sqrt{35}}{24} & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & 0 & -\frac{\sqrt{105}i}{84} \\ \frac{19\sqrt{35}i}{420} & 0 & -\frac{\sqrt{35}}{24} & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}}{168} & \frac{\sqrt{105}i}{84} & 0 \\ 0 & -\frac{\sqrt{35}}{24} & 0 & \frac{4\sqrt{35}i}{105} & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} \\ -\frac{\sqrt{35}}{24} & 0 & -\frac{4\sqrt{35}i}{105} & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{84} & \frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 \\ -\frac{5\sqrt{35}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{4\sqrt{35}i}{105} & -\frac{\sqrt{105}}{56} & 0 \\ 0 & \frac{5\sqrt{35}}{168} & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{4\sqrt{35}i}{105} & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & -\frac{3\sqrt{35}i}{140} \\ \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{56} & \frac{3\sqrt{35}i}{140} & 0 \end{bmatrix}$ |
| 546 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{19\sqrt{35}}{420} & 0 & -\frac{\sqrt{35}i}{24} & \frac{5\sqrt{35}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} \\ \frac{19\sqrt{35}}{420} & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 \\ 0 & -\frac{\sqrt{35}i}{24} & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{105}i}{168} \\ \frac{\sqrt{35}i}{24} & 0 & -\frac{4\sqrt{35}}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{84} & -\frac{\sqrt{105}i}{168} & 0 \\ \frac{5\sqrt{35}}{168} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{105} & 0 & \frac{\sqrt{35}i}{84} & -\frac{\sqrt{105}}{56} & 0 \\ 0 & -\frac{5\sqrt{35}}{168} & 0 & 0 & -\frac{4\sqrt{35}}{105} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{84} & -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{140} \\ -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{105}}{56} & 0 & 0 & \frac{3\sqrt{35}}{140} & 0 \end{bmatrix}$ |
| 547 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{24} & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & \frac{\sqrt{21}i}{24} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ -\frac{\sqrt{21}i}{24} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & -\frac{3\sqrt{7}i}{56} & 0 \\ -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{21}i}{28} & -\frac{5\sqrt{7}}{56} & 0 \\ 0 & \frac{\sqrt{21}}{168} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{5\sqrt{7}}{56} \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ -\frac{\sqrt{7}}{28} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \end{bmatrix}$ |
| 548 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & \frac{\sqrt{7}i}{28} \\ \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & -\frac{\sqrt{21}}{24} & 0 & \frac{\sqrt{21}i}{21} & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} \\ -\frac{\sqrt{21}}{24} & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{21} & \frac{5\sqrt{7}}{56} & 0 \\ 0 & \frac{\sqrt{21}}{168} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{5\sqrt{7}}{56} \\ 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{21}i}{28} \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & -\frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$ |

bra: $=\langle d_v,\uparrow|,\langle d_v,\downarrow|,\langle d_{xy},\uparrow|,\langle d_{xy},\downarrow|,\langle d_{xz},\uparrow|,\langle d_{xz},\downarrow|,\langle d_{yz},\uparrow|,\langle d_{yz},\downarrow|,\langle d_u,\uparrow|,\langle d_u,\downarrow|$

ket: $=|f_c,\uparrow\rangle,|f_c,\downarrow\rangle,|f_s,\uparrow\rangle,|f_s,\downarrow\rangle,|f_z^\beta,\uparrow\rangle,|f_z^\beta,\downarrow\rangle,|f_{xyz},\uparrow\rangle,|f_{xyz},\downarrow\rangle,|f_x,\uparrow\rangle,|f_x,\downarrow\rangle,|f_y,\uparrow\rangle,|f_y,\downarrow\rangle,|f_z^\alpha,\uparrow\rangle,|f_z^\alpha,\downarrow\rangle$

Table 9: (d,f) block.

| No. | multipole | matrix |
|---|---|---|
| 549 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} \end{bmatrix}$ |
| 550 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 551 | symmetry | $y$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{1,2}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 \end{bmatrix}$ |
| 552 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}^{(a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} \end{bmatrix}$ |
| 553 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 \\ -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 554 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 555 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{240} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{240} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{240} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 556 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{240} & 0 & 0 \\ -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{240} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{240} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{40} \\ 0 & 0 & \frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 \end{bmatrix}$ |
| 557 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 \\ \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} \\ 0 & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \end{bmatrix}$ |
| 558 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 559 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(A_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 560 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{Q}_5^{(a)}(A_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \end{bmatrix}$ |
| 561 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 562 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{Q}_5^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 \\ -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 563 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 564 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |
| | $\mathbb{Q}_{5,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{11\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{11\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 565 | symmetry | $\frac{15x^4y}{8}-5x^2y^3+\frac{15x^2yz^2}{4}+y^5-5y^3z^2+\frac{15yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{5,2}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{11\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{336} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{11\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{336} & 0 & 0 \\ \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} \\ 0 & 0 & \frac{\sqrt{21}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & 0 \end{bmatrix}$ |
| 566 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |
| | $\mathbb{Q}^{(a)}_{5,1}(E_u,2)$ | $\begin{bmatrix} \frac{3\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{15}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 567 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{5,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{16} & 0 & 0 \\ \frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & -\frac{3\sqrt{15}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 \end{bmatrix}$ |
| 568 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{Q}^{(a)}_{5,1}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 \end{bmatrix}$ |
| 569 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{5,2}(E_u,3)$ | $\begin{bmatrix} -\frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{15}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 570 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{Q}^{(1,-1;a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & \frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{\sqrt{70}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & \frac{\sqrt{70}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 571 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & -\frac{\sqrt{42}i}{84} & 0 \\ \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} \\ 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 572 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{84} & 0 \\ \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{42} & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 573 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{70}}{140} \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & -\frac{\sqrt{70}}{140} & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} \\ 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & \frac{\sqrt{105}}{140} & \frac{\sqrt{70}i}{70} & 0 \\ -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{70}i}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 \end{bmatrix}$ |
| 574 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} \\ 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{3\sqrt{70}}{280} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & \frac{3\sqrt{70}}{280} & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{56} & -\frac{\sqrt{70}i}{70} & 0 \\ \frac{\sqrt{7}i}{28} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & \frac{\sqrt{70}i}{70} \\ 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 \\ -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 575 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{168} & 0 \\ 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{56} & \frac{\sqrt{42}i}{42} & 0 \\ \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & \frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 576 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} \\ 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ \frac{\sqrt{105}i}{168} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 \end{bmatrix}$ |
| 577 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{i}{20} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & -\frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & \frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & -\frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{20} & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{20} & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 578 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 \\ -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{210}}{420} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{14}}{42} & 0 & 0 \\ \frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{60} & 0 & \frac{\sqrt{35}i}{60} & 0 & 0 & -\frac{5\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{60} & 0 & \frac{\sqrt{35}i}{60} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{84} & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{60} & 0 & -\frac{\sqrt{35}}{60} & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{60} & 0 & \frac{\sqrt{35}}{60} & 0 & 0 & -\frac{5\sqrt{14}i}{84} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 579 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & \frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & -\frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & \frac{i}{20} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{20} & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{20} & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 580 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & -\frac{\sqrt{5}i}{10} & 0 \\ \frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & \frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 \\ -\frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & \frac{\sqrt{3}}{20} & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & \frac{3\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{40} & \frac{\sqrt{3}}{15} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{6}}{40} & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & -\frac{i}{5} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 581 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 \\ -\frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & -\frac{\sqrt{5}i}{10} & 0 \\ -\frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{40} & \frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & \frac{\sqrt{5}}{20} & 0 \\ \frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{3}}{15} & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & \frac{i}{5} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ -\frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & -\frac{i}{5} & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 582 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |
| | $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & -\frac{\sqrt{35}}{60} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{560} & \frac{\sqrt{35}}{60} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{336} & -\frac{\sqrt{21}}{84} & 0 \\ -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{60} & \frac{13\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} \\ 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & -\frac{\sqrt{35}}{60} & 0 & 0 & -\frac{13\sqrt{14}i}{336} & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 \\ 0 & \frac{17\sqrt{210}}{1680} & 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & -\frac{\sqrt{35}i}{60} & 0 & 0 & \frac{\sqrt{14}}{336} & 0 & -\frac{5\sqrt{14}i}{336} & 0 & 0 \\ -\frac{17\sqrt{210}}{1680} & 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{60} & -\frac{\sqrt{14}}{336} & 0 & -\frac{5\sqrt{14}i}{336} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{11\sqrt{210}}{1680} & \frac{\sqrt{35}i}{120} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{336} & -\frac{\sqrt{21}i}{56} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & \frac{11\sqrt{210}}{1680} & 0 & 0 & -\frac{\sqrt{35}i}{120} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{336} & 0 & 0 & \frac{\sqrt{21}i}{56} \\ 0 & 0 & -\frac{\sqrt{70}i}{80} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{80} & \frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \end{bmatrix}$ |
| 583 | symmetry | $\frac{15x^4y}{8}-5x^2y^3+\frac{15x^2yz^2}{4}+y^5-5y^3z^2+\frac{15yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{60} & 0 & -\frac{\sqrt{35}}{84} & \frac{\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & \frac{\sqrt{35}i}{60} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & -\frac{\sqrt{35}i}{60} & 0 & 0 & -\frac{13\sqrt{14}i}{336} & 0 & 0 & -\frac{5\sqrt{21}}{168} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & \frac{5\sqrt{35}}{168} & 0 & -\frac{\sqrt{35}i}{60} & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}i}{336} & \frac{5\sqrt{21}}{168} & 0 \\ 0 & -\frac{11\sqrt{210}i}{1680} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & -\frac{5\sqrt{14}}{168} & \frac{\sqrt{21}i}{56} & 0 \\ -\frac{11\sqrt{210}i}{1680} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{120} & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{56} \\ 0 & \frac{5\sqrt{210}}{336} & 0 & \frac{17\sqrt{210}i}{1680} & 0 & 0 & \frac{\sqrt{35}i}{60} & 0 & 0 & \frac{5\sqrt{14}}{336} & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 \\ -\frac{5\sqrt{210}}{336} & 0 & \frac{17\sqrt{210}i}{1680} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{60} & -\frac{5\sqrt{14}}{336} & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{105}}{84} & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{80} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 584 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |
| | $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & -\frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{80} & \frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & \frac{\sqrt{15}}{20} & 0 \\ -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 & 0 & -\frac{3i}{40} & 0 & \frac{1}{20} & -\frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{40} \\ 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & -\frac{3i}{40} & 0 & -\frac{1}{20} & 0 & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 \\ 0 & -\frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & -\frac{i}{20} & 0 & 0 & -\frac{\sqrt{10}}{80} & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 \\ \frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 & 0 & \frac{i}{20} & \frac{\sqrt{10}}{80} & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{6}}{16} & \frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{80} & -\frac{\sqrt{15}i}{40} & 0 \\ \frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}i}{40} \\ 0 & 0 & \frac{9\sqrt{2}i}{80} & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}i}{80} & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 \end{bmatrix}$ |
| 585 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 & 0 & \frac{i}{20} & 0 & \frac{1}{20} & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & \frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & -\frac{3}{40} & 0 & -\frac{i}{20} & 0 & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{80} & \frac{3}{40} & 0 & -\frac{i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{80} & -\frac{\sqrt{15}}{40} & 0 \\ 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{40} & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{40} & \frac{\sqrt{15}i}{40} & 0 \\ -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{\sqrt{10}i}{80} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & -\frac{\sqrt{15}i}{40} \\ 0 & -\frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & \frac{i}{20} & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 \\ \frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 & 0 & -\frac{i}{20} & -\frac{3\sqrt{10}}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & 0 \\ \frac{9\sqrt{2}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{3}}{20} & -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}i}{80} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 586 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & -\frac{\sqrt{3}}{10} & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{40} & \frac{\sqrt{3}}{60} & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & -\frac{\sqrt{5}i}{20} & 0 \\ -\frac{\sqrt{2}i}{10} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & \frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{15} & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{10} & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 587 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{40} & -\frac{\sqrt{3}}{15} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 \\ -\frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & -\frac{\sqrt{3}}{15} & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 \\ 0 & -\frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 \\ \frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{2}}{10} & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{40} & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 \end{bmatrix}$ |
| 588 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 589 | symmetry | $x$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & \frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & -\frac{3\sqrt{70}}{140} & 0 \end{bmatrix}$ |
| 590 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 \end{bmatrix}$ |
| 591 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} \\ 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} \\ 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 592 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_3^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}i}{48} & 0 & -\frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{6}}{48} & -\frac{i}{6} & 0 \\ \frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & \frac{i}{6} \\ 0 & \frac{\sqrt{10}}{48} & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ -\frac{\sqrt{10}}{48} & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & -\frac{i}{24} \\ 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{24} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{1}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & -\frac{1}{24} & 0 \\ 0 & \frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{2}}{16} & 0 & 0 \\ \frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 593 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ \frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{48} & 0 & -\frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & \frac{i}{6} & 0 \\ \frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & -\frac{i}{6} \\ 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{1}{24} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & -\frac{1}{24} & 0 \\ \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{24} \\ 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & \frac{i}{24} & 0 \\ 0 & \frac{\sqrt{30}}{48} & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 \\ -\frac{\sqrt{30}}{48} & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 594 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{160} & \frac{\sqrt{15}}{24} & 0 \\ \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{7\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & -\frac{7\sqrt{10}i}{160} & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{3\sqrt{10}}{160} & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & -\frac{3\sqrt{10}}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{96} & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{10}}{160} & \frac{\sqrt{15}i}{240} & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}}{160} & 0 & 0 & -\frac{\sqrt{15}i}{240} \\ 0 & 0 & \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{160} & \frac{\sqrt{5}}{20} & 0 \end{bmatrix}$ |
| 595 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{7\sqrt{10}i}{160} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}i}{160} & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & -\frac{3i}{16} & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}i}{240} & 0 \\ \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & 0 & \frac{7\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{240} \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 \\ \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & -\frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 \end{bmatrix}$ |
| 596 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & \frac{i}{24} \\ 0 & \frac{\sqrt{10}i}{96} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{i}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & \frac{1}{6} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{96} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & -\frac{1}{6} & 0 \\ 0 & -\frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & -\frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & -\frac{\sqrt{6}}{24} & \frac{i}{48} & 0 \\ -\frac{\sqrt{10}i}{96} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}i}{48} & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{48} \\ 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 \\ -\frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & -\frac{\sqrt{30}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{32} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 597 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & -\frac{1}{24} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{96} & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{96} & \frac{1}{24} & 0 \\ \frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & \frac{i}{6} \\ 0 & -\frac{\sqrt{10}i}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & \frac{i}{6} & 0 \\ 0 & \frac{\sqrt{10}}{96} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ -\frac{\sqrt{10}}{96} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & \frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{96} & \frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{32} & \frac{i}{48} & 0 \\ \frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & -\frac{\sqrt{15}i}{48} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & -\frac{i}{48} \\ 0 & 0 & -\frac{\sqrt{30}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{32} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{32} & \frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 598 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{5} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{5} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{5} & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{15} & 0 & 0 & -\frac{i}{10} & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{15} & 0 & 0 & \frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & -\frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 599 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 \\ \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{210}}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{840} & 0 & \frac{\sqrt{210}}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 600 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A_u,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{5} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{5} & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & -\frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{i}{5} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{15} & 0 & 0 & \frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & -\frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & \frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 601 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{120} & 0 & -\frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{30} & 0 \\ \frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{30} \\ 0 & \frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 \\ -\frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & \frac{\sqrt{3}}{10} & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{15} \\ 0 & -\frac{\sqrt{2}i}{60} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{5}i}{15} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{60} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{5}}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{60} & -\frac{\sqrt{3}}{30} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & \frac{\sqrt{5}}{15} & 0 \\ 0 & \frac{\sqrt{6}i}{30} & 0 & \frac{\sqrt{6}}{30} & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ \frac{\sqrt{6}i}{30} & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 602 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 \\ -\frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{30} & 0 \\ -\frac{\sqrt{2}i}{120} & 0 & -\frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{30} \\ 0 & 0 & \frac{\sqrt{2}i}{60} & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{5}}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{60} & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & -\frac{\sqrt{5}}{15} & 0 \\ -\frac{\sqrt{2}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{3}}{30} & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{15} \\ 0 & \frac{\sqrt{2}i}{60} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{5}i}{15} & 0 \\ 0 & -\frac{\sqrt{6}}{30} & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ \frac{\sqrt{6}}{30} & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & \frac{i}{10} & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 603 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{53\sqrt{210}i}{3360} & 0 & 0 & \frac{13\sqrt{35}}{560} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{224} & 0 & 0 & -\frac{\sqrt{21}}{48} \\ 0 & 0 & 0 & \frac{53\sqrt{210}i}{3360} & -\frac{13\sqrt{35}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{224} & \frac{\sqrt{21}}{48} & 0 \\ -\frac{13\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{240} & \frac{3\sqrt{35}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & -\frac{\sqrt{21}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{240} & 0 & 0 & -\frac{3\sqrt{35}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{21}i}{168} \\ 0 & 0 & \frac{\sqrt{70}i}{160} & 0 & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & \frac{5\sqrt{7}}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{160} & \frac{\sqrt{105}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{224} & -\frac{5\sqrt{7}}{112} & 0 \end{bmatrix}$ |
| 604 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |
| | $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} \frac{53\sqrt{210}i}{3360} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{35}i}{560} & 0 & 0 & \frac{3\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{48} \\ 0 & -\frac{53\sqrt{210}i}{3360} & 0 & 0 & -\frac{13\sqrt{35}i}{560} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{224} & 0 & 0 & -\frac{\sqrt{21}i}{48} & 0 \\ 0 & 0 & -\frac{13\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{13\sqrt{210}i}{840} & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{240} & 0 & 0 & \frac{3\sqrt{35}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 \\ -\frac{\sqrt{210}i}{240} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{280} & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{120} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{80} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & -\frac{\sqrt{70}i}{160} & 0 & 0 & -\frac{\sqrt{105}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{112} & 0 \end{bmatrix}$ |
| 605 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{13\sqrt{6}i}{480} & 0 & 0 & -\frac{3}{80} & 0 & \frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}}{240} \\ 0 & 0 & 0 & \frac{13\sqrt{6}i}{480} & \frac{3}{80} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{32} & -\frac{\sqrt{15}}{240} & 0 \\ -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & \frac{1}{10} & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} \\ 0 & \frac{\sqrt{6}i}{40} & 0 & 0 & \frac{i}{10} & 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 \\ 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & -\frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ \frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{i}{10} & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{6}}{48} & -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{3\sqrt{10}}{80} & -\frac{\sqrt{15}i}{120} & 0 \\ \frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}i}{120} \\ 0 & 0 & -\frac{9\sqrt{2}i}{160} & 0 & 0 & \frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & \frac{\sqrt{5}}{16} \\ 0 & 0 & 0 & \frac{9\sqrt{2}i}{160} & -\frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & -\frac{\sqrt{5}}{16} & 0 \end{bmatrix}$ |
| 606 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |
| | $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} \frac{13\sqrt{6}i}{480} & 0 & 0 & 0 & 0 & \frac{3i}{80} & 0 & \frac{1}{10} & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{240} \\ 0 & -\frac{13\sqrt{6}i}{480} & 0 & 0 & \frac{3i}{80} & 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{240} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & \frac{1}{10} & 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{40} & -\frac{1}{10} & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} & \frac{\sqrt{15}}{30} & 0 \\ 0 & \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{20} & -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{20} & \frac{\sqrt{15}i}{120} & 0 \\ \frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{120} \\ 0 & \frac{\sqrt{6}}{60} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & \frac{i}{10} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ -\frac{\sqrt{6}}{60} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & -\frac{i}{10} & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ -\frac{9\sqrt{2}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & -\frac{\sqrt{3}}{10} & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & \frac{9\sqrt{2}i}{160} & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 \end{bmatrix}$ |
| 607 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)$ | $\begin{bmatrix} \frac{37\sqrt{2}i}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{120} & 0 & \frac{\sqrt{3}}{20} & -\frac{\sqrt{30}i}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{24} \\ 0 & -\frac{37\sqrt{2}i}{240} & 0 & 0 & \frac{\sqrt{3}i}{120} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & \frac{\sqrt{30}i}{240} & 0 & 0 & \frac{\sqrt{5}i}{24} & 0 \\ 0 & 0 & -\frac{19\sqrt{2}i}{120} & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}}{60} \\ 0 & 0 & 0 & \frac{19\sqrt{2}i}{120} & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & -\frac{\sqrt{5}}{60} & 0 \\ 0 & -\frac{\sqrt{2}i}{30} & 0 & \frac{\sqrt{2}}{24} & -\frac{\sqrt{3}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{60} & 0 \\ -\frac{\sqrt{2}i}{30} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{60} \\ 0 & \frac{7\sqrt{2}}{120} & 0 & -\frac{\sqrt{2}i}{120} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ -\frac{7\sqrt{2}}{120} & 0 & -\frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} & -\frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & \frac{1}{20} & -\frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & \frac{i}{8} & 0 & -\frac{1}{20} & 0 & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 \end{bmatrix}$ |
| 608 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |
| | $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{37\sqrt{2}i}{240} & 0 & 0 & \frac{\sqrt{3}}{120} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & \frac{\sqrt{30}i}{240} & 0 & 0 & -\frac{\sqrt{5}}{24} \\ 0 & 0 & 0 & -\frac{37\sqrt{2}i}{240} & -\frac{\sqrt{3}}{120} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{240} & \frac{\sqrt{5}}{24} & 0 \\ \frac{19\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{30} & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{60} \\ 0 & -\frac{19\sqrt{2}i}{120} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}i}{60} & 0 \\ 0 & \frac{\sqrt{2}}{120} & 0 & -\frac{7\sqrt{2}i}{120} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 \\ -\frac{\sqrt{2}}{120} & 0 & -\frac{7\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{24} & 0 & \frac{\sqrt{2}}{30} & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{\sqrt{5}i}{60} & 0 \\ -\frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{2}}{30} & 0 & 0 & -\frac{\sqrt{3}i}{60} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{60} \\ 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & -\frac{1}{8} & 0 & \frac{i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{80} & \frac{1}{8} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{80} & -\frac{\sqrt{15}}{24} & 0 \end{bmatrix}$ |
| 609 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 \\ \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & -\frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 \end{bmatrix}$ |
| 610 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & \frac{\sqrt{105}}{140} & 0 \\ \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 \\ \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{70}}{280} & -\frac{\sqrt{105}i}{70} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 \end{bmatrix}$ |
| 611 | symmetry | $y$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & \frac{\sqrt{105}}{140} & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{70} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 612 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{Q}_3^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 \\ \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & \frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 \\ -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & \frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{42} & -\frac{\sqrt{105}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{24} & \frac{\sqrt{70}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 613 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{112} & 0 & \frac{3\sqrt{42}}{112} & \frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{112} & 0 & -\frac{3\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 \\ -\frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 \\ \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} \\ 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & \frac{\sqrt{105}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & -\frac{\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{210}i}{80} & 0 & -\frac{\sqrt{210}}{80} & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 \\ -\frac{\sqrt{210}i}{80} & 0 & \frac{\sqrt{210}}{80} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 614 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_3^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 \\ \frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & -\frac{3\sqrt{42}}{112} & -\frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{3\sqrt{42}}{112} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & -\frac{\sqrt{7}}{56} & 0 \\ \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{120} & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} \\ 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 \\ 0 & -\frac{\sqrt{210}}{80} & 0 & \frac{\sqrt{210}i}{80} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 \\ \frac{\sqrt{210}}{80} & 0 & \frac{\sqrt{210}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 615 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{70}i}{672} & 0 & 0 & -\frac{\sqrt{105}}{168} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{224} & -\frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{672} & \frac{\sqrt{105}}{168} & 0 \\ \frac{\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{24} & -\frac{13\sqrt{70}i}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} \\ 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & \frac{13\sqrt{70}i}{672} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 \\ 0 & -\frac{17\sqrt{42}}{672} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & -\frac{\sqrt{70}}{672} & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 \\ \frac{17\sqrt{42}}{672} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & \frac{\sqrt{70}}{672} & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{84} & 0 & \frac{11\sqrt{42}}{672} & -\frac{\sqrt{7}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{672} & \frac{\sqrt{105}i}{112} & 0 \\ \frac{\sqrt{42}i}{84} & 0 & -\frac{11\sqrt{42}}{672} & 0 & 0 & \frac{\sqrt{7}i}{48} & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & -\frac{\sqrt{70}}{672} & 0 & 0 & -\frac{\sqrt{105}i}{112} \\ 0 & 0 & \frac{\sqrt{14}i}{32} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{32} & -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{224} & 0 & 0 \end{bmatrix}$ |
| 616 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & -\frac{\sqrt{7}}{42} & -\frac{\sqrt{70}i}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} \\ 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{70}i}{672} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & \frac{13\sqrt{70}i}{672} & 0 & 0 & -\frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{224} & \frac{5\sqrt{7}}{84} & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{70}i}{672} & \frac{\sqrt{105}}{84} & 0 \\ 0 & \frac{11\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{84} & -\frac{\sqrt{7}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{672} & 0 & -\frac{\sqrt{70}}{84} & -\frac{\sqrt{105}i}{112} & 0 \\ \frac{11\sqrt{42}i}{672} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{7}i}{48} & 0 & 0 & -\frac{\sqrt{70}i}{672} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{105}i}{112} \\ 0 & \frac{5\sqrt{42}}{168} & 0 & -\frac{17\sqrt{42}i}{672} & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & \frac{\sqrt{70}i}{672} & 0 & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & -\frac{17\sqrt{42}i}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & -\frac{\sqrt{70}}{168} & 0 & \frac{\sqrt{70}i}{672} & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{32} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 617 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{105}}{70} & -\frac{17\sqrt{42}i}{672} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{17\sqrt{42}i}{672} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{224} & \frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{224} & -\frac{\sqrt{7}}{28} & 0 \\ 0 & \frac{23\sqrt{70}i}{1120} & 0 & \frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{56} & \frac{5\sqrt{7}i}{112} & 0 \\ \frac{23\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{80} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & \frac{\sqrt{70}}{140} & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 \\ -\frac{\sqrt{70}}{140} & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{120} & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{160} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{70} & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{160} & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 618 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{17\sqrt{42}i}{672} & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{224} & \frac{\sqrt{105}}{120} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}i}{672} & \frac{3\sqrt{7}}{56} & 0 \\ \frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & \frac{\sqrt{105}}{120} & -\frac{\sqrt{42}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 \\ 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{120} & \frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{23\sqrt{70}}{1120} & -\frac{\sqrt{105}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{224} & \frac{5\sqrt{7}i}{112} & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & \frac{23\sqrt{70}}{1120} & 0 & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & 0 & -\frac{\sqrt{210}i}{160} & 0 & 0 & \frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{160} & -\frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 \end{bmatrix}$ |
| 619 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 620 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_2^{(a)}(B_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 621 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 622 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \end{bmatrix}$ |
| 623 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,2}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 624 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{G}_4^{(a)}(A_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 \\ -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 625 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(A_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 \\ \frac{\sqrt{70}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 626 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_4^{(a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 627 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 \\ \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 628 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_4^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 629 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 630 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 \\ \frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} \\ 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \end{bmatrix}$ |
| 631 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{112} & 0 & 0 \\ -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} \\ 0 & 0 & \frac{3\sqrt{70}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{70}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \end{bmatrix}$ |
| 632 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{112} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{70}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 633 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & \frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & -\frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & \frac{\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{105} & -\frac{\sqrt{70}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{35}}{70} & \frac{\sqrt{210}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{\sqrt{210}i}{70} \end{bmatrix}$ |
| 634 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 635 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 636 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 \end{bmatrix}$ |
| 637 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & -\frac{3\sqrt{70}}{140} & 0 \end{bmatrix}$ |
| 638 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{10}i}{240} & 0 & \frac{11\sqrt{10}}{240} & 0 & 0 \\ \frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{11\sqrt{10}i}{240} & 0 & -\frac{11\sqrt{10}}{240} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{240} & 0 & -\frac{\sqrt{10}i}{240} & 0 & 0 \\ \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{i}{6} & \frac{\sqrt{10}}{240} & 0 & -\frac{\sqrt{10}i}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & -\frac{1}{6} & \frac{\sqrt{10}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{120} \\ 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & \frac{1}{6} & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & -\frac{\sqrt{15}i}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & 0 & -\frac{\sqrt{15}}{120} \\ 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & \frac{\sqrt{15}}{120} & 0 \\ 0 & -\frac{5\sqrt{2}i}{48} & 0 & \frac{5\sqrt{2}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{30}}{80} & \frac{\sqrt{5}i}{15} & 0 \\ -\frac{5\sqrt{2}i}{48} & 0 & -\frac{5\sqrt{2}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & 0 & -\frac{\sqrt{5}i}{15} \end{bmatrix}$ |
| 639 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & -\frac{\sqrt{14}}{336} & 0 & 0 \\ \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & \frac{\sqrt{14}}{336} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{13\sqrt{14}}{336} & 0 & -\frac{13\sqrt{14}i}{336} & 0 & 0 \\ \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & \frac{13\sqrt{14}}{336} & 0 & -\frac{13\sqrt{14}i}{336} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{84} & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}}{168} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & -\frac{\sqrt{21}}{168} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & \frac{\sqrt{21}}{168} & 0 \\ 0 & \frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{70}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & \frac{\sqrt{7}i}{21} & 0 \\ \frac{\sqrt{70}i}{48} & 0 & \frac{\sqrt{70}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{7}i}{21} \end{bmatrix}$ |
| 640 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 641 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{42}}{48} & \frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{42}}{48} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 \\ -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & \frac{\sqrt{105}}{84} & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} \\ 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & \frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 \\ -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{14}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 642 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 \\ \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{42}}{48} & -\frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{56} & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} \\ 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 \\ 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 \\ \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 643 | symmetry | $\frac{\sqrt{35}y^3z}{2}-\frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}i}{96} & 0 & 0 & \frac{1}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}i}{96} & -\frac{1}{8} & 0 \\ -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{32} & \frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{96} & -\frac{i}{16} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & \frac{i}{16} \\ 0 & 0 & \frac{\sqrt{30}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{32} & \frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 644 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{7\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}i}{96} & 0 & 0 & \frac{i}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & \frac{i}{16} & 0 \\ -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{48} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{i}{16} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & -\frac{\sqrt{30}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 645 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}i}{672} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{672} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{13\sqrt{42}i}{672} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{42}i}{672} & \frac{\sqrt{7}}{14} & 0 \\ 0 & -\frac{3\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & -\frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}i}{672} & 0 & \frac{\sqrt{42}}{56} & -\frac{\sqrt{7}i}{112} & 0 \\ -\frac{3\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{5\sqrt{105}i}{336} & 0 & 0 & -\frac{11\sqrt{42}i}{672} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}i}{112} \\ 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{96} & 0 & 0 \\ \frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{96} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & -\frac{\sqrt{210}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{224} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \end{bmatrix}$ |
| 646 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{672} & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{224} & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{672} & \frac{3\sqrt{7}}{56} & 0 \\ -\frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & -\frac{13\sqrt{42}i}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & \frac{13\sqrt{42}i}{672} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & \frac{3\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\ -\frac{3\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & -\frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{70}}{224} & \frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{11\sqrt{42}}{672} & -\frac{\sqrt{7}i}{112} & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & -\frac{5\sqrt{105}i}{336} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{11\sqrt{42}}{672} & 0 & 0 & \frac{\sqrt{7}i}{112} \\ 0 & 0 & -\frac{\sqrt{210}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{224} & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{224} & \frac{\sqrt{21}}{84} & 0 \end{bmatrix}$ |
| 647 | symmetry | $\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{231}i}{616} & 0 & \frac{\sqrt{231}}{616} & -\frac{3\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{616} & 0 & \frac{3\sqrt{385}}{616} & 0 & 0 \\ \frac{\sqrt{231}i}{616} & 0 & -\frac{\sqrt{231}}{616} & 0 & 0 & \frac{3\sqrt{154}i}{308} & 0 & 0 & -\frac{3\sqrt{385}i}{616} & 0 & -\frac{3\sqrt{385}}{616} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{231}i}{462} & 0 & 0 & \frac{\sqrt{154}i}{77} & 0 & 0 & \frac{\sqrt{385}}{154} & 0 & \frac{\sqrt{385}i}{154} & 0 & 0 \\ -\frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{231}i}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{77} & -\frac{\sqrt{385}}{154} & 0 & \frac{\sqrt{385}i}{154} & 0 & 0 & 0 \\ -\frac{\sqrt{231}i}{132} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{308} & 0 & \frac{\sqrt{154}}{77} & -\frac{\sqrt{385}i}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} \\ 0 & \frac{\sqrt{231}i}{132} & 0 & 0 & -\frac{3\sqrt{154}i}{308} & 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & \frac{\sqrt{385}i}{308} & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 \\ 0 & 0 & \frac{\sqrt{231}i}{132} & 0 & 0 & \frac{3\sqrt{154}}{308} & 0 & \frac{\sqrt{154}i}{77} & 0 & 0 & -\frac{\sqrt{385}i}{308} & 0 & 0 & -\frac{\sqrt{2310}}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{231}i}{132} & -\frac{3\sqrt{154}}{308} & 0 & \frac{\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{308} & \frac{\sqrt{2310}}{924} & 0 \\ 0 & -\frac{\sqrt{77}i}{88} & 0 & \frac{\sqrt{77}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{616} & 0 & -\frac{\sqrt{1155}}{616} & \frac{\sqrt{770}i}{308} & 0 \\ -\frac{\sqrt{77}i}{88} & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & -\frac{\sqrt{770}i}{308} \end{bmatrix}$ |
| 648 | symmetry | $-\frac{\sqrt{14}x^6}{8}+\frac{15\sqrt{14}x^4z^2}{8}-\frac{15\sqrt{14}x^2z^4}{8}-\frac{\sqrt{14}y^6}{8}+\frac{15\sqrt{14}y^4z^2}{8}-\frac{15\sqrt{14}y^2z^4}{8}+\frac{\sqrt{14}z^6}{4}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{33}i}{264} & 0 & -\frac{\sqrt{33}}{264} & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\ -\frac{\sqrt{33}i}{264} & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{33}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} \\ 0 & -\frac{\sqrt{33}i}{132} & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{132} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & -\frac{\sqrt{330}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{33}i}{132} & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{44} & \frac{\sqrt{330}}{132} & 0 \\ 0 & \frac{\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{165}}{88} & \frac{\sqrt{110}i}{44} & 0 \\ \frac{\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{88} & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & -\frac{\sqrt{110}i}{44} \end{bmatrix}$ |
| 649 | symmetry | $-\frac{3\sqrt{7}x^5y}{4}+\frac{15\sqrt{7}x^3yz^2}{2}+\frac{3\sqrt{7}xy^5}{4}-\frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & \frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 \\ -\frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & -\frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{264} & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 \\ -\frac{\sqrt{66}i}{264} & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & -\frac{\sqrt{11}}{22} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 650 | symmetry | $-\frac{\sqrt{2310}x^4y^2}{8} + \frac{\sqrt{2310}x^4z^2}{8} + \frac{\sqrt{2310}x^2y^4}{8} - \frac{\sqrt{2310}x^2z^4}{8} - \frac{\sqrt{2310}y^4z^2}{8} + \frac{\sqrt{2310}y^2z^4}{8}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{7\sqrt{5}i}{120} & 0 & -\frac{7\sqrt{5}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & \frac{\sqrt{3}}{24} & -\frac{\sqrt{2}i}{12} & 0 \\ \frac{7\sqrt{5}i}{120} & 0 & \frac{7\sqrt{5}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{3}}{24} & 0 & 0 & \frac{\sqrt{2}i}{12} \\ 0 & -\frac{\sqrt{5}}{15} & 0 & -\frac{\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{15} & 0 & -\frac{\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} \\ 0 & -\frac{\sqrt{5}i}{60} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{60} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{2}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{60} & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & -\frac{\sqrt{2}}{12} & 0 \\ 0 & \frac{\sqrt{15}i}{120} & 0 & \frac{\sqrt{15}}{120} & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 \\ \frac{\sqrt{15}i}{120} & 0 & -\frac{\sqrt{15}}{120} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{i}{8} & 0 & -\frac{1}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 651 | symmetry | $\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{17\sqrt{11}i}{264} & 0 & -\frac{17\sqrt{11}}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{264} & 0 & -\frac{\sqrt{165}}{264} & \frac{\sqrt{110}i}{132} & 0 \\ \frac{17\sqrt{11}i}{264} & 0 & \frac{17\sqrt{11}}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{264} & 0 & \frac{\sqrt{165}}{264} & 0 & 0 & -\frac{\sqrt{110}i}{132} \\ 0 & -\frac{2\sqrt{11}}{33} & 0 & -\frac{2\sqrt{11}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{11}}{33} & 0 & -\frac{2\sqrt{11}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{132} \\ 0 & \frac{\sqrt{11}i}{132} & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & \frac{\sqrt{110}i}{132} & 0 \\ 0 & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & -\frac{\sqrt{66}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & -\frac{\sqrt{110}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{11}i}{132} & \frac{\sqrt{66}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & \frac{\sqrt{110}}{132} & 0 \\ 0 & -\frac{\sqrt{33}i}{264} & 0 & -\frac{\sqrt{33}}{264} & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\ -\frac{\sqrt{33}i}{264} & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 \end{bmatrix}$ |
| 652 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(B_u,3)$ | $\begin{bmatrix} 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 653 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(B_u,4)$ | $\begin{bmatrix} 0 & \frac{\sqrt{55}}{660} & 0 & \frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{55}}{660} & 0 & \frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}i}{660} & 0 & \frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & \frac{\sqrt{22}i}{33} & 0 \\ -\frac{\sqrt{55}i}{660} & 0 & -\frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & \frac{\sqrt{33}}{66} & 0 & 0 & -\frac{\sqrt{22}i}{33} \\ 0 & 0 & -\frac{\sqrt{55}i}{165} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & \frac{\sqrt{22}}{33} \\ 0 & 0 & 0 & \frac{\sqrt{55}i}{165} & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & -\frac{\sqrt{22}}{33} & 0 \\ \frac{\sqrt{55}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{165} & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{33} \\ 0 & -\frac{\sqrt{55}i}{165} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & 0 & \frac{\sqrt{22}i}{33} & 0 \\ 0 & \frac{\sqrt{165}}{330} & 0 & -\frac{\sqrt{165}i}{330} & 0 & 0 & \frac{\sqrt{110}i}{55} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 \\ -\frac{\sqrt{165}}{330} & 0 & -\frac{\sqrt{165}i}{330} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{55} & -\frac{\sqrt{11}}{22} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 \end{bmatrix}$ |
| 654 | symmetry | $\frac{15\sqrt{7}x^2y^3z}{2} - \frac{15\sqrt{7}x^2yz^3}{2} - \frac{3\sqrt{7}y^5z}{4} + \frac{3\sqrt{7}yz^5}{4}$ |
| | $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{5\sqrt{66}i}{528} & 0 & 0 & -\frac{3\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{\sqrt{110}i}{176} & 0 & 0 & -\frac{\sqrt{165}}{264} \\ 0 & 0 & 0 & \frac{5\sqrt{66}i}{528} & \frac{3\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{176} & \frac{\sqrt{165}}{264} & 0 \\ -\frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & \frac{\sqrt{11}}{22} & -\frac{\sqrt{110}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{132} \\ 0 & \frac{\sqrt{66}i}{88} & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & -\frac{\sqrt{165}i}{132} & 0 \\ 0 & -\frac{\sqrt{66}}{88} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 \\ \frac{\sqrt{66}}{88} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & \frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}i}{88} & 0 & \frac{\sqrt{66}}{66} & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 \\ -\frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{132} \\ 0 & 0 & -\frac{3\sqrt{22}i}{176} & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & \frac{\sqrt{55}}{88} \\ 0 & 0 & 0 & \frac{3\sqrt{22}i}{176} & \frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{176} & -\frac{\sqrt{55}}{88} & 0 \end{bmatrix}$ |
| 655 | symmetry | $\frac{3\sqrt{7}x^5z}{4} - \frac{15\sqrt{7}x^3y^2z}{2} + \frac{15\sqrt{7}xy^2z^3}{2} - \frac{3\sqrt{7}xz^5}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{5\sqrt{66}i}{528} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{44} & -\frac{\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{264} \\ 0 & -\frac{5\sqrt{66}i}{528} & 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & \frac{\sqrt{110}i}{176} & 0 & 0 & -\frac{\sqrt{165}i}{264} & 0 \\ 0 & 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & \frac{\sqrt{165}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{66}i}{88} & \frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & -\frac{\sqrt{165}}{132} & 0 \\ 0 & \frac{\sqrt{66}i}{66} & 0 & -\frac{\sqrt{66}}{88} & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & -\frac{\sqrt{165}i}{132} & 0 \\ \frac{\sqrt{66}i}{66} & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & \frac{\sqrt{165}i}{132} \\ 0 & -\frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 \\ \frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & -\frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & 0 \\ -\frac{3\sqrt{22}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{88} & 0 & \frac{\sqrt{33}}{44} & -\frac{\sqrt{330}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} \\ 0 & \frac{3\sqrt{22}i}{176} & 0 & 0 & -\frac{\sqrt{33}i}{88} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 \end{bmatrix}$ |
| 656 | symmetry | $\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$ |
| | $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} \\ 0 & -\frac{i}{32} & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{16} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 \\ \frac{i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & 0 & \frac{3\sqrt{5}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{32} \\ 0 & \frac{\sqrt{3}i}{32} & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{\sqrt{30}i}{32} & 0 \end{bmatrix}$ |
| 657 | symmetry | $\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{32} & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & \frac{\sqrt{10}}{32} \\ 0 & 0 & 0 & -\frac{i}{32} & -\frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{32} & -\frac{\sqrt{10}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{16} & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & \frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & \frac{1}{16} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & 0 & \frac{\sqrt{3}i}{32} & 0 & 0 & \frac{3\sqrt{2}}{32} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{\sqrt{30}}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{32} & -\frac{3\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{32} & -\frac{\sqrt{30}}{32} & 0 \end{bmatrix}$ |
| 658 | symmetry | $\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$ |
| | $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} \frac{17\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & \frac{37\sqrt{330}i}{5280} & 0 & -\frac{\sqrt{330}}{110} & \frac{\sqrt{33}i}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{96} \\ 0 & -\frac{17\sqrt{55}i}{1056} & 0 & 0 & \frac{37\sqrt{330}i}{5280} & 0 & \frac{\sqrt{330}}{110} & 0 & 0 & -\frac{\sqrt{33}i}{96} & 0 & 0 & \frac{\sqrt{22}i}{96} & 0 \\ 0 & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & -\frac{\sqrt{330}}{110} & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & -\frac{\sqrt{22}}{66} \\ 0 & 0 & 0 & \frac{\sqrt{55}i}{66} & \frac{\sqrt{330}}{110} & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & \frac{\sqrt{22}}{66} & 0 \\ 0 & \frac{29\sqrt{55}i}{2640} & 0 & -\frac{\sqrt{55}}{66} & \frac{\sqrt{330}i}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{176} & 0 & -\frac{\sqrt{33}}{66} & \frac{5\sqrt{22}i}{528} & 0 \\ \frac{29\sqrt{55}i}{2640} & 0 & \frac{\sqrt{55}}{66} & 0 & 0 & -\frac{\sqrt{330}i}{240} & 0 & 0 & \frac{\sqrt{33}i}{176} & 0 & \frac{\sqrt{33}}{66} & 0 & 0 & -\frac{5\sqrt{22}i}{528} \\ 0 & -\frac{7\sqrt{55}}{330} & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 \\ \frac{7\sqrt{55}}{330} & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{165} & \frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 \\ \frac{9\sqrt{165}i}{1760} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{160} & 0 & -\frac{\sqrt{110}}{110} & \frac{5\sqrt{11}i}{352} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{1056} \\ 0 & -\frac{9\sqrt{165}i}{1760} & 0 & 0 & \frac{\sqrt{110}i}{160} & 0 & \frac{\sqrt{110}}{110} & 0 & 0 & -\frac{5\sqrt{11}i}{352} & 0 & 0 & \frac{5\sqrt{66}i}{1056} & 0 \end{bmatrix}$ |
| 659 | symmetry | $\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 660 | $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{17\sqrt{55}i}{1056} & 0 & 0 & \frac{37\sqrt{330}}{5280} & 0 & \frac{\sqrt{330}i}{110} & 0 & 0 & -\frac{\sqrt{33}i}{96} & 0 & 0 & -\frac{\sqrt{22}}{96} \\ 0 & 0 & 0 & -\frac{17\sqrt{55}i}{1056} & -\frac{37\sqrt{330}}{5280} & 0 & \frac{\sqrt{330}i}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{96} & \frac{\sqrt{22}}{96} & 0 \\ \frac{\sqrt{55}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{110} & 0 & -\frac{\sqrt{330}}{165} & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} \\ 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & \frac{\sqrt{330}i}{110} & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 \\ 0 & \frac{\sqrt{55}}{66} & 0 & \frac{7\sqrt{55}i}{330} & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 \\ -\frac{\sqrt{55}}{66} & 0 & \frac{7\sqrt{55}i}{330} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{165} & \frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{66} & 0 & -\frac{29\sqrt{55}}{2640} & -\frac{\sqrt{330}i}{240} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & \frac{\sqrt{33}}{176} & \frac{5\sqrt{22}i}{528} & 0 \\ \frac{\sqrt{55}i}{66} & 0 & \frac{29\sqrt{55}}{2640} & 0 & 0 & \frac{\sqrt{330}i}{240} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{176} & 0 & 0 & -\frac{5\sqrt{22}i}{528} \\ 0 & 0 & -\frac{9\sqrt{165}i}{1760} & 0 & 0 & -\frac{\sqrt{110}}{160} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & \frac{5\sqrt{11}i}{352} & 0 & 0 & \frac{5\sqrt{66}}{1056} \\ 0 & 0 & 0 & \frac{9\sqrt{165}i}{1760} & \frac{\sqrt{110}}{160} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{352} & -\frac{5\sqrt{66}}{1056} & 0 \end{bmatrix}$ |
| 661 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\ \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 662 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & \frac{\sqrt{21}i}{42} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 662 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_2^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & \frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 663 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & \frac{\sqrt{21}}{84} & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 664 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{21}}{84} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \end{bmatrix}$ |
| 665 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(A_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{80} & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 \\ -\frac{\sqrt{10}i}{80} & 0 & \frac{\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ -\frac{\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & \frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 \\ -\frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 666 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{G}_4^{(1,0;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{210}i}{1680} & 0 & \frac{17\sqrt{210}}{1680} & 0 & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & -\frac{17\sqrt{210}i}{1680} & 0 & -\frac{17\sqrt{210}}{1680} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & -\frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 \\ -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & \frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{40} & 0 & \frac{\sqrt{21}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} \\ 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & \frac{\sqrt{21}i}{40} & 0 & -\frac{\sqrt{21}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & -\frac{\sqrt{21}}{40} & 0 & \frac{\sqrt{21}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & \frac{\sqrt{21}}{40} & 0 & \frac{\sqrt{21}i}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & \frac{\sqrt{42}i}{80} & 0 & -\frac{\sqrt{42}}{80} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\ \frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{42}}{80} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 667 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{5} & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & -\frac{1}{40} & 0 & -\frac{i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & \frac{1}{40} & 0 & -\frac{i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{40} & 0 & \frac{1}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & -\frac{i}{40} & 0 & -\frac{1}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 668 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{G}_4^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & \frac{\sqrt{105}i}{70} & 0 \\ \frac{3\sqrt{42}i}{560} & 0 & \frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & \frac{3\sqrt{42}}{560} & 0 & \frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{70}}{560} & 0 & \frac{13\sqrt{70}i}{560} & 0 & 0 \\ -\frac{3\sqrt{42}}{560} & 0 & \frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{70}}{560} & 0 & \frac{13\sqrt{70}i}{560} & 0 & 0 & 0 \\ \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{7}}{20} & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{280} \\ 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{3\sqrt{105}i}{280} & 0 \\ 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & -\frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{105}}{280} \\ 0 & 0 & 0 & -\frac{3\sqrt{42}i}{280} & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & -\frac{3\sqrt{105}}{280} & 0 \\ 0 & \frac{3\sqrt{14}i}{80} & 0 & \frac{3\sqrt{14}}{80} & -\frac{\sqrt{21}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{560} & 0 & -\frac{3\sqrt{210}}{560} & 0 & 0 \\ \frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{14}}{80} & 0 & 0 & \frac{\sqrt{21}i}{35} & 0 & 0 & \frac{3\sqrt{210}i}{560} & 0 & \frac{3\sqrt{210}}{560} & 0 & 0 & 0 \end{bmatrix}$ |
| 669 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{42}}{560} & 0 & -\frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{70}}{560} & 0 & \frac{13\sqrt{70}i}{560} & 0 & 0 \\ \frac{3\sqrt{42}}{560} & 0 & -\frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{70}}{560} & 0 & \frac{13\sqrt{70}i}{560} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & -\frac{\sqrt{105}i}{70} & 0 \\ \frac{3\sqrt{42}i}{560} & 0 & \frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{105}}{280} \\ 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} & -\frac{\sqrt{7}}{20} & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & -\frac{3\sqrt{105}}{280} & 0 \\ \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & \frac{\sqrt{7}}{40} & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}i}{280} \\ 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{105}i}{280} & 0 \\ 0 & \frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{14}i}{80} & 0 & 0 & \frac{\sqrt{21}i}{35} & 0 & 0 & -\frac{3\sqrt{210}}{560} & 0 & -\frac{3\sqrt{210}i}{560} & 0 & 0 \\ -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{14}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{35} & \frac{3\sqrt{210}}{560} & 0 & -\frac{3\sqrt{210}i}{560} & 0 & 0 & 0 \end{bmatrix}$ |
| 670 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & 0 & \frac{1}{40} & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{160} & -\frac{1}{40} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{32} & -\frac{\sqrt{15}}{40} & 0 \\ \frac{\sqrt{6}i}{160} & 0 & 0 & 0 & 0 & -\frac{3i}{20} & 0 & -\frac{1}{40} & \frac{\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & -\frac{\sqrt{6}i}{160} & 0 & 0 & -\frac{3i}{20} & 0 & \frac{1}{40} & 0 & 0 & -\frac{\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & \frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & \frac{i}{40} & 0 & 0 & \frac{\sqrt{10}}{160} & 0 & -\frac{3\sqrt{10}i}{40} & 0 & 0 \\ -\frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & -\frac{i}{40} & -\frac{\sqrt{10}}{160} & 0 & -\frac{3\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{6}}{32} & -\frac{i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{160} & \frac{\sqrt{15}i}{80} & 0 \\ \frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{6}}{32} & 0 & 0 & \frac{i}{16} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{160} & 0 & 0 & -\frac{\sqrt{15}i}{80} \\ 0 & 0 & -\frac{9\sqrt{2}i}{160} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{2}i}{160} & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 \end{bmatrix}$ |
| 671 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{160} & 0 & 0 & 0 & 0 & -\frac{i}{40} & 0 & \frac{1}{10} & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} \\ 0 & -\frac{\sqrt{6}i}{160} & 0 & 0 & -\frac{i}{40} & 0 & -\frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{160} & 0 & 0 & -\frac{3}{20} & 0 & \frac{i}{40} & 0 & 0 & -\frac{\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{160} & \frac{3}{20} & 0 & \frac{i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{160} & -\frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{6}i}{32} & 0 & \frac{\sqrt{6}}{20} & -\frac{i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{160} & 0 & -\frac{\sqrt{10}}{20} & -\frac{\sqrt{15}i}{80} & 0 \\ \frac{\sqrt{6}i}{32} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & \frac{i}{16} & 0 & 0 & \frac{\sqrt{10}i}{160} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{15}i}{80} \\ 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{160} & 0 & 0 & -\frac{i}{40} & 0 & 0 & \frac{3\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{160} & 0 & 0 \\ \frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{160} & 0 & 0 & 0 & 0 & \frac{i}{40} & -\frac{3\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{160} & 0 & 0 & 0 \\ -\frac{9\sqrt{2}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{3}}{10} & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{2}i}{160} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 672 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}i}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{3\sqrt{7}}{70} & \frac{19\sqrt{70}i}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ 0 & \frac{\sqrt{42}i}{1120} & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & -\frac{19\sqrt{70}i}{1120} & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & 0 & -\frac{\sqrt{7}}{140} & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & -\frac{23\sqrt{70}i}{1120} & 0 & 0 & \frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{1120} & \frac{\sqrt{7}}{140} & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & 0 & 0 & \frac{23\sqrt{70}i}{1120} & -\frac{\sqrt{105}}{140} & 0 \\ 0 & -\frac{\sqrt{42}i}{1120} & 0 & \frac{\sqrt{42}}{56} & -\frac{\sqrt{7}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{3\sqrt{70}}{280} & \frac{\sqrt{105}i}{560} & 0 \\ -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}i}{80} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}i}{560} \\ 0 & \frac{\sqrt{42}}{140} & 0 & -\frac{29\sqrt{42}i}{1120} & 0 & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \\ -\frac{\sqrt{42}}{140} & 0 & -\frac{29\sqrt{42}i}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & \frac{\sqrt{70}}{140} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & 0 \\ -\frac{9\sqrt{14}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{70} & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{14}i}{160} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & \frac{\sqrt{210}i}{1120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 673 | symmetry | $3\sqrt{5}x^2yz-\frac{\sqrt{5}y^3z}{2}-\frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & 0 & -\frac{\sqrt{7}}{40} & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & -\frac{19\sqrt{70}i}{1120} & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{1120} & \frac{\sqrt{7}}{40} & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & \frac{19\sqrt{70}i}{1120} & -\frac{\sqrt{105}}{56} & 0 \\ \frac{\sqrt{42}i}{1120} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{140} & 0 & \frac{\sqrt{7}}{40} & -\frac{23\sqrt{70}i}{1120} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} \\ 0 & -\frac{\sqrt{42}i}{1120} & 0 & 0 & \frac{\sqrt{7}i}{140} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & \frac{23\sqrt{70}i}{1120} & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 \\ 0 & \frac{29\sqrt{42}}{1120} & 0 & -\frac{\sqrt{42}i}{140} & 0 & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 \\ -\frac{29\sqrt{42}}{1120} & 0 & -\frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{1120} & \frac{\sqrt{7}i}{80} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{224} & \frac{\sqrt{105}i}{560} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{1120} & 0 & 0 & -\frac{\sqrt{7}i}{80} & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}i}{560} \\ 0 & 0 & \frac{9\sqrt{14}i}{160} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{14}i}{160} & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{1120} & 0 & 0 \end{bmatrix}$ |
| 674 | symmetry | 1 |
| | $\mathbb{G}_0^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{210}}{420} & 0 & 0 \\ \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 \\ \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & \frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{70}}{70} & \frac{\sqrt{105}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & -\frac{\sqrt{105}i}{70} \end{bmatrix}$ |
| 675 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 \\ -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 \\ -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{70}}{70} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{105}}{210} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & -\frac{\sqrt{105}}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & \frac{\sqrt{210}}{140} & \frac{\sqrt{35}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{35}i}{35} \end{bmatrix}$ |
| 676 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_2^{(1,1;a)}(B_u, 1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{120} & 0 & -\frac{\sqrt{210}}{120} & -\frac{\sqrt{35}i}{42} & 0 \\ \frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{120} & 0 & \frac{\sqrt{210}}{120} & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & \frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 \\ -\frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & 0 \\ \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{420} \\ 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{420} & 0 \\ 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{420} \\ 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{420} & 0 \\ 0 & -\frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 \\ -\frac{5\sqrt{42}i}{168} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 \end{bmatrix}$ |
| 677 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 \\ -\frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{120} & 0 & -\frac{\sqrt{210}}{120} & -\frac{\sqrt{35}i}{42} & 0 \\ -\frac{\sqrt{14}i}{168} & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{120} & 0 & \frac{\sqrt{210}}{120} & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}}{420} \\ 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & \frac{\sqrt{35}}{420} & 0 \\ -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{420} \\ 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{420} & 0 \\ 0 & \frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 \end{bmatrix}$ |
| 678 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & \frac{\sqrt{35}}{42} & 0 \\ 0 & \frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{105} & 0 \\ \frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{105} \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{120} & 0 & 0 \\ \frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{120} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 \end{bmatrix}$ |
| 679 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 680 | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{105} & -\frac{\sqrt{35}}{42} & 0 \\ \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{120} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 \\ \frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{120} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{280} & \frac{\sqrt{35}i}{105} & 0 \\ -\frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{35}i}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & \frac{\sqrt{105}}{105} & 0 \end{bmatrix}$ |
| | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{G}_4^{(1,1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{2\sqrt{165}i}{165} & 0 & \frac{2\sqrt{165}}{165} & \frac{7\sqrt{110}i}{660} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & \frac{\sqrt{11}}{66} & 0 & 0 \\ \frac{2\sqrt{165}i}{165} & 0 & -\frac{2\sqrt{165}}{165} & 0 & 0 & -\frac{7\sqrt{110}i}{660} & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & -\frac{\sqrt{11}}{66} & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{165}}{660} & 0 & -\frac{7\sqrt{165}i}{660} & 0 & 0 & -\frac{\sqrt{110}i}{330} & 0 & 0 & -\frac{\sqrt{11}}{132} & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 \\ -\frac{7\sqrt{165}}{660} & 0 & -\frac{7\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{330} & \frac{\sqrt{11}}{132} & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 \\ \frac{\sqrt{165}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{165} & 0 & -\frac{\sqrt{110}}{330} & -\frac{5\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} \\ 0 & -\frac{\sqrt{165}i}{220} & 0 & 0 & -\frac{\sqrt{110}i}{165} & 0 & \frac{\sqrt{110}}{330} & 0 & 0 & \frac{5\sqrt{11}i}{132} & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 \\ 0 & 0 & -\frac{\sqrt{165}i}{220} & 0 & 0 & \frac{\sqrt{110}}{165} & 0 & -\frac{\sqrt{110}i}{330} & 0 & 0 & -\frac{5\sqrt{11}i}{132} & 0 & 0 & \frac{\sqrt{66}}{66} \\ 0 & 0 & 0 & \frac{\sqrt{165}i}{220} & -\frac{\sqrt{110}}{165} & 0 & -\frac{\sqrt{110}i}{330} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{132} & -\frac{\sqrt{66}}{66} & 0 \\ 0 & -\frac{\sqrt{55}i}{660} & 0 & \frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{44} & 0 & \frac{\sqrt{33}}{44} & \frac{5\sqrt{22}i}{132} & 0 \\ -\frac{\sqrt{55}i}{660} & 0 & -\frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{44} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & -\frac{5\sqrt{22}i}{132} \end{bmatrix}$ |
| 681 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 682 | $\mathbb{G}_4^{(1,1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{5\sqrt{231}i}{462} & 0 & -\frac{5\sqrt{231}}{462} & -\frac{19\sqrt{154}i}{4620} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{1155} & 0 & \frac{\sqrt{385}}{1155} & 0 & 0 \\ -\frac{5\sqrt{231}i}{462} & 0 & \frac{5\sqrt{231}}{462} & 0 & 0 & \frac{19\sqrt{154}i}{4620} & 0 & 0 & -\frac{\sqrt{385}i}{1155} & 0 & -\frac{\sqrt{385}}{1155} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{231}i}{84} & 0 & 0 & \frac{\sqrt{154}i}{105} & 0 & 0 & -\frac{\sqrt{385}}{420} & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 \\ \frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{231}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{105} & \frac{\sqrt{385}}{420} & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 & 0 \\ -\frac{\sqrt{231}i}{220} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}i}{1155} & 0 & -\frac{\sqrt{154}}{210} & -\frac{5\sqrt{385}i}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{462} \\ 0 & \frac{\sqrt{231}i}{220} & 0 & 0 & -\frac{2\sqrt{154}i}{1155} & 0 & \frac{\sqrt{154}}{210} & 0 & 0 & \frac{5\sqrt{385}i}{924} & 0 & 0 & \frac{\sqrt{2310}i}{462} & 0 \\ 0 & 0 & \frac{\sqrt{231}i}{220} & 0 & 0 & \frac{2\sqrt{154}}{1155} & 0 & -\frac{\sqrt{154}i}{210} & 0 & 0 & -\frac{5\sqrt{385}i}{924} & 0 & 0 & \frac{\sqrt{2310}}{462} \\ 0 & 0 & 0 & -\frac{\sqrt{231}i}{220} & -\frac{2\sqrt{154}}{1155} & 0 & -\frac{\sqrt{154}i}{210} & 0 & 0 & 0 & 0 & \frac{5\sqrt{385}i}{924} & -\frac{\sqrt{2310}}{462} & 0 \\ 0 & \frac{\sqrt{77}i}{660} & 0 & -\frac{\sqrt{77}}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{308} & 0 & \frac{\sqrt{1155}}{308} & \frac{5\sqrt{770}i}{924} & 0 \\ \frac{\sqrt{77}i}{660} & 0 & \frac{\sqrt{77}}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{308} & 0 & -\frac{\sqrt{1155}}{308} & 0 & 0 & -\frac{5\sqrt{770}i}{924} \end{bmatrix}$ |
| 683 | $\mathbb{G}_4^{(1,1;a)}(A_u,3)$ | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ $\begin{bmatrix} 0 & -\frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{11}i}{44} & 0 & 0 & \frac{3\sqrt{66}i}{220} & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 \\ \frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{66}i}{220} & \frac{\sqrt{165}}{660} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{11}}{44} & \frac{3\sqrt{66}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & \frac{\sqrt{165}}{660} & 0 & 0 \\ \frac{3\sqrt{11}i}{44} & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & -\frac{3\sqrt{66}i}{220} & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}i}{110} & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}i}{110} & \frac{\sqrt{66}}{330} & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{11}i}{110} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{330} & 0 & \frac{\sqrt{66}}{330} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{11}i}{110} & 0 & 0 & -\frac{\sqrt{66}i}{330} & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{330} & 0 & -\frac{\sqrt{33}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{33}}{330} & 0 & -\frac{\sqrt{33}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 684 | | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{77}i}{1540} & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}i}{4620} & 0 & -\frac{17\sqrt{1155}}{4620} & -\frac{\sqrt{770}i}{220} & 0 \\ \frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}i}{4620} & 0 & \frac{17\sqrt{1155}}{4620} & 0 & 0 & \frac{\sqrt{770}i}{220} \\ 0 & \frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{420} & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 \\ -\frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{420} & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 & 0 \\ \frac{\sqrt{77}i}{220} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}i}{2310} & 0 & -\frac{\sqrt{462}}{210} & -\frac{\sqrt{1155}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & -\frac{\sqrt{77}i}{220} & 0 & 0 & -\frac{17\sqrt{462}i}{2310} & 0 & \frac{\sqrt{462}}{210} & 0 & 0 & \frac{\sqrt{1155}i}{220} & 0 & 0 & \frac{\sqrt{770}i}{385} & 0 \\ 0 & 0 & \frac{\sqrt{77}i}{220} & 0 & 0 & -\frac{17\sqrt{462}}{2310} & 0 & \frac{\sqrt{462}i}{210} & 0 & 0 & \frac{\sqrt{1155}i}{220} & 0 & 0 & -\frac{\sqrt{770}}{385} \\ 0 & 0 & 0 & -\frac{\sqrt{77}i}{220} & \frac{17\sqrt{462}}{2310} & 0 & \frac{\sqrt{462}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{220} & \frac{\sqrt{770}}{385} & 0 \\ 0 & -\frac{\sqrt{231}i}{165} & 0 & -\frac{\sqrt{231}}{165} & -\frac{3\sqrt{154}i}{220} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{770} & 0 & -\frac{3\sqrt{385}}{770} & 0 & 0 \\ -\frac{\sqrt{231}i}{165} & 0 & \frac{\sqrt{231}}{165} & 0 & 0 & \frac{3\sqrt{154}i}{220} & 0 & 0 & \frac{3\sqrt{385}i}{770} & 0 & \frac{3\sqrt{385}}{770} & 0 & 0 & 0 \end{bmatrix}$ |
| 684 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_4^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{420} & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 \\ \frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{420} & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}i}{1540} & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1155}i}{4620} & 0 & \frac{17\sqrt{1155}}{4620} & \frac{\sqrt{770}i}{220} & 0 \\ \frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1155}i}{4620} & 0 & -\frac{17\sqrt{1155}}{4620} & 0 & 0 & -\frac{\sqrt{770}i}{220} \\ 0 & 0 & -\frac{\sqrt{77}i}{220} & 0 & 0 & -\frac{\sqrt{462}}{210} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & \frac{\sqrt{1155}i}{220} & 0 & 0 & -\frac{\sqrt{770}}{385} \\ 0 & 0 & 0 & \frac{\sqrt{77}i}{220} & \frac{\sqrt{462}}{210} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{220} & \frac{\sqrt{770}}{385} & 0 \\ \frac{\sqrt{77}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{210} & 0 & \frac{17\sqrt{462}}{2310} & \frac{\sqrt{1155}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{385} \\ 0 & -\frac{\sqrt{77}i}{220} & 0 & 0 & \frac{\sqrt{462}i}{210} & 0 & -\frac{17\sqrt{462}}{2310} & 0 & 0 & -\frac{\sqrt{1155}i}{220} & 0 & 0 & -\frac{\sqrt{770}i}{385} & 0 \\ 0 & -\frac{\sqrt{231}}{165} & 0 & \frac{\sqrt{231}i}{165} & 0 & 0 & \frac{3\sqrt{154}i}{220} & 0 & 0 & -\frac{3\sqrt{385}}{770} & 0 & -\frac{3\sqrt{385}i}{770} & 0 & 0 \\ \frac{\sqrt{231}}{165} & 0 & \frac{\sqrt{231}i}{165} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{220} & \frac{3\sqrt{385}}{770} & 0 & -\frac{3\sqrt{385}i}{770} & 0 & 0 & 0 \end{bmatrix}$ |
| 685 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{11}i}{220} & 0 & 0 & \frac{7\sqrt{66}}{660} & 0 & -\frac{3\sqrt{66}i}{440} & 0 & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & \frac{\sqrt{110}}{110} \\ 0 & 0 & 0 & \frac{\sqrt{11}i}{220} & -\frac{7\sqrt{66}}{660} & 0 & -\frac{3\sqrt{66}i}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & -\frac{\sqrt{110}}{110} & 0 \\ \frac{\sqrt{11}i}{220} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{66}i}{440} & 0 & -\frac{\sqrt{66}}{330} & \frac{\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{440} \\ 0 & -\frac{\sqrt{11}i}{220} & 0 & 0 & -\frac{3\sqrt{66}i}{440} & 0 & \frac{\sqrt{66}}{330} & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & -\frac{3\sqrt{110}i}{440} & 0 \\ 0 & \frac{\sqrt{11}}{220} & 0 & -\frac{3\sqrt{11}i}{440} & 0 & 0 & \frac{\sqrt{66}i}{330} & 0 & 0 & \frac{\sqrt{165}}{660} & 0 & -\frac{3\sqrt{165}i}{440} & 0 & 0 \\ -\frac{\sqrt{11}}{220} & 0 & -\frac{3\sqrt{11}i}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{330} & -\frac{\sqrt{165}}{660} & 0 & -\frac{3\sqrt{165}i}{440} & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{11}i}{440} & 0 & -\frac{\sqrt{11}}{44} & -\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{440} & 0 & -\frac{\sqrt{165}}{60} & -\frac{\sqrt{110}i}{55} & 0 \\ -\frac{9\sqrt{11}i}{440} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{3\sqrt{165}i}{440} & 0 & \frac{\sqrt{165}}{60} & 0 & 0 & \frac{\sqrt{110}i}{55} \\ 0 & 0 & -\frac{\sqrt{33}i}{165} & 0 & 0 & \frac{3\sqrt{22}}{110} & 0 & -\frac{9\sqrt{22}i}{440} & 0 & 0 & -\frac{3\sqrt{55}i}{110} & 0 & 0 & \frac{\sqrt{330}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{33}i}{165} & -\frac{3\sqrt{22}}{110} & 0 & -\frac{9\sqrt{22}i}{440} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{110} & -\frac{\sqrt{330}}{132} & 0 \end{bmatrix}$ |
| 686 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{11}i}{220} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{66}i}{660} & 0 & -\frac{3\sqrt{66}}{440} & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{110} \\ 0 & -\frac{\sqrt{11}i}{220} & 0 & 0 & -\frac{7\sqrt{66}i}{660} & 0 & \frac{3\sqrt{66}}{440} & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & 0 & \frac{\sqrt{110}i}{110} & 0 \\ 0 & 0 & \frac{\sqrt{11}i}{220} & 0 & 0 & -\frac{3\sqrt{66}}{440} & 0 & \frac{\sqrt{66}i}{330} & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & \frac{3\sqrt{110}}{440} \\ 0 & 0 & 0 & -\frac{\sqrt{11}i}{220} & \frac{3\sqrt{66}}{440} & 0 & \frac{\sqrt{66}i}{330} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{660} & -\frac{3\sqrt{110}}{440} & 0 \\ 0 & -\frac{\sqrt{11}i}{44} & 0 & -\frac{9\sqrt{11}}{440} & -\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{60} & 0 & \frac{3\sqrt{165}}{440} & \frac{\sqrt{110}i}{55} & 0 \\ -\frac{\sqrt{11}i}{44} & 0 & \frac{9\sqrt{11}}{440} & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{165}i}{60} & 0 & -\frac{3\sqrt{165}}{440} & 0 & 0 & -\frac{\sqrt{110}i}{55} \\ 0 & -\frac{3\sqrt{11}}{440} & 0 & \frac{\sqrt{11}i}{220} & 0 & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & \frac{3\sqrt{165}}{440} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 \\ \frac{3\sqrt{11}}{440} & 0 & \frac{\sqrt{11}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{330} & -\frac{3\sqrt{165}}{440} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{165} & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{110} & 0 & \frac{9\sqrt{22}}{440} & \frac{3\sqrt{55}i}{110} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} \\ 0 & \frac{\sqrt{33}i}{165} & 0 & 0 & \frac{3\sqrt{22}i}{110} & 0 & -\frac{9\sqrt{22}}{440} & 0 & 0 & -\frac{3\sqrt{55}i}{110} & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 \end{bmatrix}$ |
| 687 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & -\frac{29\sqrt{462}i}{4620} & 0 & -\frac{3\sqrt{462}}{440} & -\frac{\sqrt{1155}i}{420} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}i}{1540} & 0 & 0 & -\frac{29\sqrt{462}i}{4620} & 0 & \frac{3\sqrt{462}}{440} & 0 & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & -\frac{3\sqrt{462}}{440} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & \frac{17\sqrt{1155}i}{4620} & 0 & 0 & -\frac{\sqrt{770}}{440} \\ 0 & 0 & 0 & \frac{\sqrt{77}i}{1540} & \frac{3\sqrt{462}}{440} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}i}{4620} & \frac{\sqrt{770}}{440} & 0 \\ 0 & -\frac{3\sqrt{77}i}{220} & 0 & -\frac{\sqrt{77}}{88} & -\frac{\sqrt{462}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & -\frac{\sqrt{1155}}{440} & -\frac{\sqrt{770}i}{385} & 0 \\ -\frac{3\sqrt{77}i}{220} & 0 & \frac{\sqrt{77}}{88} & 0 & 0 & \frac{\sqrt{462}i}{210} & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & \frac{\sqrt{1155}}{440} & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & -\frac{7\sqrt{77}}{440} & 0 & \frac{3\sqrt{77}i}{220} & 0 & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & -\frac{\sqrt{1155}}{440} & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 \\ \frac{7\sqrt{77}}{440} & 0 & \frac{3\sqrt{77}i}{220} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}i}{2310} & \frac{\sqrt{1155}}{440} & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & 0 \\ -\frac{\sqrt{231}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{440} & -\frac{3\sqrt{385}i}{770} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} \\ 0 & \frac{\sqrt{231}i}{165} & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{440} & 0 & 0 & \frac{3\sqrt{385}i}{770} & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 \end{bmatrix}$ |
| 688 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & -\frac{29\sqrt{462}}{4620} & 0 & \frac{3\sqrt{462}i}{440} & 0 & 0 & \frac{\sqrt{1155}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}i}{1540} & \frac{29\sqrt{462}}{4620} & 0 & \frac{3\sqrt{462}i}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{420} & 0 & 0 \\ \frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}i}{440} & 0 & \frac{17\sqrt{462}}{2310} & \frac{17\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{440} \\ 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & \frac{3\sqrt{462}i}{440} & 0 & -\frac{17\sqrt{462}}{2310} & 0 & 0 & -\frac{17\sqrt{1155}i}{4620} & 0 & 0 & -\frac{\sqrt{770}i}{440} & 0 \\ 0 & -\frac{3\sqrt{77}}{220} & 0 & \frac{7\sqrt{77}i}{440} & 0 & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{440} & 0 & 0 \\ \frac{3\sqrt{77}}{220} & 0 & \frac{7\sqrt{77}i}{440} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}i}{2310} & \frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{440} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}i}{88} & 0 & \frac{3\sqrt{77}}{220} & \frac{\sqrt{462}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{440} & 0 & -\frac{\sqrt{1155}}{924} & -\frac{\sqrt{770}i}{385} & 0 \\ \frac{\sqrt{77}i}{88} & 0 & -\frac{3\sqrt{77}}{220} & 0 & 0 & -\frac{\sqrt{462}i}{210} & 0 & 0 & -\frac{\sqrt{1155}i}{440} & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & 0 & \frac{\sqrt{231}i}{165} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{440} & 0 & 0 & -\frac{3\sqrt{385}i}{770} & 0 & 0 & \frac{\sqrt{2310}}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{231}i}{165} & 0 & 0 & -\frac{3\sqrt{154}i}{440} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{770} & -\frac{\sqrt{2310}}{924} & 0 \end{bmatrix}$ |
| 689 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} \end{bmatrix}$ |
| 690 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 691 | symmetry | $y$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{1,2}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 \end{bmatrix}$ |
| 692 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}^{(a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} \end{bmatrix}$ |
| 693 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 694 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_3^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 695 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}i}{240} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}i}{240} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{240} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 696 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{240} & 0 & 0 \\ -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{240} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{240} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} \\ 0 & 0 & \frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 \end{bmatrix}$ |
| 697 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 \\ \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} \\ 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 \end{bmatrix}$ |
| 698 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 699 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(A_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 700 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{T}_5^{(a)}(A_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \end{bmatrix}$ |
| 701 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 702 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{T}_5^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 703 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_5(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 704 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |
| | $\mathbb{T}^{(a)}_{5,1}(E_u,1)$ | $\begin{bmatrix} \frac{11\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{11\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}i}{336} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 705 | symmetry | $\frac{15x^4y}{8}-5x^2y^3+\frac{15x^2yz^2}{4}+y^5-5y^3z^2+\frac{15yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{5,2}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{11\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}i}{336} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{11\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}i}{336} & 0 & 0 \\ \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} \\ 0 & 0 & \frac{\sqrt{21}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & 0 \end{bmatrix}$ |
| 706 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |
| | $\mathbb{T}^{(a)}_{5,1}(E_u,2)$ | $\begin{bmatrix} \frac{3\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 707 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{5,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 \\ \frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & 0 & -\frac{3\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & 0 \end{bmatrix}$ |
| 708 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{T}^{(a)}_{5,1}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 \end{bmatrix}$ |
| 709 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{5,2}(E_u,3)$ | $\begin{bmatrix} -\frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 710 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{T}^{(1,-1;a)}_{3}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{70}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & \frac{\sqrt{70}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 711 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 712 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 713 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{70}i}{140} \\ 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & -\frac{\sqrt{70}i}{140} & 0 \\ \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\ 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{56} & 0 & \frac{\sqrt{105}i}{140} & -\frac{\sqrt{70}}{70} & 0 \\ \frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{56} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{70}}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 \end{bmatrix}$ |
| 714 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} \\ 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & -\frac{3\sqrt{70}i}{280} \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & \frac{3\sqrt{70}i}{280} & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{56} & \frac{\sqrt{70}}{70} & 0 \\ -\frac{\sqrt{7}}{28} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 & -\frac{\sqrt{70}}{70} \\ 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 \\ -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{3\sqrt{14}i}{56} & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 715 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,-1;a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{168} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{168} & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{56} & -\frac{\sqrt{42}}{42} & 0 \\ -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & \frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{168} & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 716 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(1,-1;a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{84} & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} \\ 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{42}}{42} & 0 \\ -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 \end{bmatrix}$ |
| 717 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{10} & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & \frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & \frac{1}{20} & 0 & -\frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & \frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{20} & -\frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{20} & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{20} & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 718 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{T}_5^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 \\ -\frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{42} & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{420} & 0 & \frac{\sqrt{210}i}{420} & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 \\ -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{60} & 0 & -\frac{\sqrt{35}}{60} & 0 & 0 & \frac{5\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{60} & 0 & -\frac{\sqrt{35}}{60} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{84} & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{60} & 0 & -\frac{\sqrt{35}i}{60} & -\frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{60} & 0 & \frac{\sqrt{35}i}{60} & 0 & 0 & \frac{5\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 719 | symmetry | $\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{10} & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & \frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{20} & -\frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{1}{20} & 0 & -\frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{20} & 0 & 0 & -\frac{1}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{20} & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{20} & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 720 | symmetry | $\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$ |
| | $\mathbb{T}_5^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & \frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 \\ -\frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & \frac{\sqrt{3}i}{20} & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & -\frac{3\sqrt{2}}{40} & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{40} & \frac{\sqrt{3}i}{15} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{6}i}{40} & -\frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 \\ \frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{40} & 0 & 0 & \frac{1}{5} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 721 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 \\ -\frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{30}i}{40} & \frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{40} & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & \frac{\sqrt{5}i}{20} & 0 \\ -\frac{3\sqrt{2}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{3}i}{15} & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 \\ 0 & \frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & -\frac{1}{5} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & \frac{1}{5} & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 722 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |
| | $\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & -\frac{\sqrt{35}i}{60} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{14}}{336} & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{560} & \frac{\sqrt{35}i}{60} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{336} & -\frac{\sqrt{21}i}{84} & 0 \\ \frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}}{168} & 0 & \frac{\sqrt{35}i}{60} & -\frac{13\sqrt{14}}{336} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} \\ 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & \frac{5\sqrt{35}}{168} & 0 & -\frac{\sqrt{35}i}{60} & 0 & 0 & \frac{13\sqrt{14}}{336} & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 \\ 0 & \frac{17\sqrt{210}i}{1680} & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & \frac{\sqrt{35}}{60} & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & \frac{5\sqrt{14}}{336} & 0 & 0 \\ -\frac{17\sqrt{210}i}{1680} & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{60} & -\frac{\sqrt{14}i}{336} & 0 & \frac{5\sqrt{14}}{336} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{11\sqrt{210}i}{1680} & -\frac{\sqrt{35}}{120} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{336} & \frac{\sqrt{21}}{56} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & \frac{11\sqrt{210}i}{1680} & 0 & 0 & \frac{\sqrt{35}}{120} & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{336} & 0 & 0 & -\frac{\sqrt{21}}{56} \\ 0 & 0 & \frac{\sqrt{70}}{80} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{80} & \frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \end{bmatrix}$ |
| 723 | symmetry | $\frac{15x^4y}{8}-5x^2y^3+\frac{15x^2yz^2}{4}+y^5-5y^3z^2+\frac{15yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,-1;a)}_{5,2}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{60} & 0 & -\frac{\sqrt{35}i}{84} & -\frac{\sqrt{14}}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & -\frac{\sqrt{35}}{60} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{14}}{336} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{560} & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{60} & 0 & 0 & \frac{13\sqrt{14}}{336} & 0 & 0 & -\frac{5\sqrt{21}i}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{560} & \frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{60} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{336} & \frac{5\sqrt{21}i}{168} & 0 \\ 0 & \frac{11\sqrt{210}}{1680} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{336} & 0 & -\frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{56} & 0 \\ \frac{11\sqrt{210}}{1680} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{120} & 0 & 0 & -\frac{\sqrt{14}}{336} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{56} \\ 0 & \frac{5\sqrt{210}i}{336} & 0 & -\frac{17\sqrt{210}}{1680} & 0 & 0 & -\frac{\sqrt{35}}{60} & 0 & 0 & \frac{5\sqrt{14}i}{336} & 0 & \frac{\sqrt{14}}{336} & 0 & 0 \\ -\frac{5\sqrt{210}i}{336} & 0 & -\frac{17\sqrt{210}}{1680} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{60} & -\frac{5\sqrt{14}i}{336} & 0 & \frac{\sqrt{14}}{336} & 0 & 0 & 0 \\ \frac{\sqrt{70}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & \frac{\sqrt{105}i}{84} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{80} & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 724 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |
| | $\mathbb{T}^{(1,-1;a)}_{5,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & 0 & -\frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{80} & \frac{i}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & \frac{\sqrt{15}i}{20} & 0 \\ \frac{\sqrt{6}}{80} & 0 & 0 & 0 & 0 & \frac{3}{40} & 0 & \frac{i}{20} & \frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & -\frac{\sqrt{6}}{80} & 0 & 0 & \frac{3}{40} & 0 & -\frac{i}{20} & 0 & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}}{40} & 0 \\ 0 & -\frac{\sqrt{6}i}{80} & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & \frac{1}{20} & 0 & 0 & -\frac{\sqrt{10}i}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 \\ \frac{\sqrt{6}i}{80} & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & 0 & 0 & -\frac{1}{20} & \frac{\sqrt{10}i}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{6}i}{16} & -\frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{80} & \frac{\sqrt{15}}{40} & 0 \\ -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}}{40} \\ 0 & 0 & -\frac{9\sqrt{2}}{80} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{2}}{80} & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & 0 \end{bmatrix}$ |
| 725 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{6}}{80} & 0 & 0 & 0 & 0 & -\frac{1}{20} & 0 & \frac{i}{20} & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} \\ 0 & -\frac{\sqrt{6}}{80} & 0 & 0 & -\frac{1}{20} & 0 & -\frac{i}{20} & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & -\frac{3i}{40} & 0 & \frac{1}{20} & 0 & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & \frac{3i}{40} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{80} & -\frac{\sqrt{15}i}{40} & 0 \\ 0 & \frac{\sqrt{6}}{16} & 0 & \frac{\sqrt{6}i}{40} & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{40} & -\frac{\sqrt{15}}{40} & 0 \\ \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & \frac{1}{8} & 0 & 0 & \frac{\sqrt{10}}{80} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & \frac{\sqrt{15}}{40} \\ 0 & -\frac{\sqrt{6}i}{80} & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & -\frac{1}{20} & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 \\ \frac{\sqrt{6}i}{80} & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & 0 & 0 & \frac{1}{20} & -\frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & 0 \\ -\frac{9\sqrt{2}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{3}i}{20} & \frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{2}}{80} & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 726 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |
| | $\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & -\frac{\sqrt{3}i}{10} & -\frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}i}{60} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{40} & \frac{\sqrt{3}i}{60} & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{40} & -\frac{\sqrt{5}i}{20} & 0 \\ 0 & \frac{\sqrt{2}}{10} & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & \frac{\sqrt{5}}{20} & 0 \\ \frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & \frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{15} & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 727 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{40} & -\frac{\sqrt{3}i}{15} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 \\ \frac{\sqrt{2}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & -\frac{\sqrt{3}i}{15} & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & \frac{\sqrt{3}i}{15} & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 \\ 0 & -\frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 \\ \frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{2}i}{10} & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 \\ \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \end{bmatrix}$ |
| 728 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 729 | symmetry | $x$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{1,1}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & \frac{3\sqrt{70}i}{140} & 0 \end{bmatrix}$ |
| 730 | symmetry | $y$ |
| | $\mathbb{T}^{(1,0;a)}_{1,2}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 \end{bmatrix}$ |
| 731 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} \\ 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \\ 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & 0 \end{bmatrix}$ |
| 732 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}_3^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{48} & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & -\frac{1}{6} & 0 \\ \frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & \frac{1}{6} \\ 0 & -\frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ \frac{\sqrt{10}i}{48} & 0 & \frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & -\frac{1}{24} \\ 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{1}{24} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & \frac{i}{24} & 0 \\ 0 & \frac{\sqrt{30}}{48} & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 \\ \frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 733 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3}^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}i}{48} & 0 & -\frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ -\frac{\sqrt{10}i}{48} & 0 & -\frac{\sqrt{10}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{48} & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & \frac{1}{6} & 0 \\ \frac{\sqrt{10}}{48} & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{1}{6} \\ 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & \frac{i}{24} & 0 \\ \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & \frac{1}{24} \\ 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & \frac{1}{24} & 0 \\ 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{2}}{16} & 0 & 0 \\ \frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 734 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{96} & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{160} & -\frac{\sqrt{15}i}{24} & 0 \\ \frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & \frac{7\sqrt{10}}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & -\frac{7\sqrt{10}}{160} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{6}i}{96} & -\frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}i}{160} & \frac{\sqrt{15}}{240} & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{3}{16} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}}{240} \\ 0 & 0 & \frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & \frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & -\frac{\sqrt{5}i}{20} & 0 \end{bmatrix}$ |
| 735 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{3,2}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & \frac{3\sqrt{10}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & -\frac{7\sqrt{10}}{160} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}}{160} & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & \frac{7\sqrt{10}}{160} & 0 & 0 & -\frac{\sqrt{15}}{240} & 0 \\ \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{3}{16} & 0 & 0 & \frac{7\sqrt{10}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{240} \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & 0 & 0 \\ \frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} \\ 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 \end{bmatrix}$ |
| 736 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{1}{24} \\ 0 & \frac{\sqrt{10}}{96} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & \frac{1}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & -\frac{i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{96} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{96} & \frac{i}{6} & 0 \\ 0 & -\frac{\sqrt{10}}{96} & 0 & \frac{\sqrt{10}i}{24} & -\frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & \frac{\sqrt{6}i}{24} & \frac{1}{48} & 0 \\ -\frac{\sqrt{10}}{96} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & \frac{\sqrt{15}}{48} & 0 & 0 & -\frac{\sqrt{6}}{32} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{1}{48} \\ 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 \\ \frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{30}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{32} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 737 | symmetry | $\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & \frac{i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{96} & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{96} & -\frac{i}{24} & 0 \\ \frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & -\frac{\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{1}{6} \\ 0 & -\frac{\sqrt{10}}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}}{96} & 0 & 0 & \frac{1}{6} & 0 \\ 0 & -\frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 \\ \frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{96} & \frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{32} & \frac{1}{48} & 0 \\ \frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{96} & 0 & 0 & -\frac{\sqrt{15}}{48} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{1}{48} \\ 0 & 0 & -\frac{\sqrt{30}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{32} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{32} & -\frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 738 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |
| | $\mathbb{T}_{5}^{(1,0;a)}(A_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 \\ -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & \frac{1}{5} & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{15} & 0 & 0 & 0 & 0 & -\frac{1}{10} & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & -\frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & -\frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{15} & \frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 739 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{840} & 0 & \frac{\sqrt{210}}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{840} & 0 & \frac{\sqrt{210}}{840} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{840} & 0 & -\frac{\sqrt{210}i}{840} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 740 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{T}_5^{(1,0;a)}(A_u,3)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & \frac{1}{5} & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & -\frac{1}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & \frac{1}{5} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & -\frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{15} & \frac{i}{10} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{15} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{15} & 0 & 0 & \frac{1}{10} & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 741 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{120} & -\frac{\sqrt{5}}{30} & 0 \\ \frac{\sqrt{2}}{120} & 0 & -\frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}}{30} \\ 0 & -\frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 \\ \frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & -\frac{\sqrt{3}i}{10} & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{15} \\ 0 & -\frac{\sqrt{2}}{60} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{5}}{15} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{60} & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{5}i}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{60} & \frac{\sqrt{3}i}{30} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & -\frac{\sqrt{5}i}{15} & 0 \\ 0 & \frac{\sqrt{6}}{30} & 0 & -\frac{\sqrt{6}i}{30} & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ \frac{\sqrt{6}}{30} & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & \frac{1}{10} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 742 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{T}_5^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 \\ \frac{\sqrt{2}i}{120} & 0 & \frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{120} & 0 & -\frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{30}i}{120} & -\frac{\sqrt{5}}{30} & 0 \\ -\frac{\sqrt{2}}{120} & 0 & \frac{\sqrt{2}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}}{30} \\ 0 & 0 & \frac{\sqrt{2}}{60} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{5}i}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{60} & -\frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{5}i}{15} & 0 \\ -\frac{\sqrt{2}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{3}i}{30} & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{15} \\ 0 & \frac{\sqrt{2}}{60} & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{5}}{15} & 0 \\ 0 & \frac{\sqrt{6}i}{30} & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & -\frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ -\frac{\sqrt{6}i}{30} & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & \frac{1}{10} & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 743 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{5,1}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{53\sqrt{210}}{3360} & 0 & 0 & -\frac{13\sqrt{35}i}{560} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{224} & 0 & 0 & \frac{\sqrt{21}i}{48} \\ 0 & 0 & 0 & \frac{53\sqrt{210}}{3360} & \frac{13\sqrt{35}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} & -\frac{\sqrt{21}i}{48} & 0 \\ -\frac{13\sqrt{210}}{840} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{210}}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{240} & \frac{3\sqrt{35}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & -\frac{\sqrt{21}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{240} & 0 & 0 & -\frac{3\sqrt{35}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}}{168} \\ 0 & 0 & \frac{\sqrt{70}}{160} & 0 & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{160} & -\frac{\sqrt{105}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{224} & \frac{5\sqrt{7}i}{112} & 0 \end{bmatrix}$ |
| 744 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |
| | $\mathbb{T}^{(1,0;a)}_{5,2}(E_u,1)$ | $\begin{bmatrix} \frac{53\sqrt{210}}{3360} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{35}}{560} & 0 & 0 & \frac{3\sqrt{14}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{48} \\ 0 & -\frac{53\sqrt{210}}{3360} & 0 & 0 & -\frac{13\sqrt{35}}{560} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} & 0 & 0 & -\frac{\sqrt{21}}{48} & 0 \\ 0 & 0 & -\frac{13\sqrt{210}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{13\sqrt{210}}{840} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{240} & 0 & 0 & \frac{3\sqrt{35}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{21}}{168} & 0 \\ -\frac{\sqrt{210}}{240} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{280} & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{120} & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{70}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{112} \\ 0 & -\frac{\sqrt{70}}{160} & 0 & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}}{112} & 0 \end{bmatrix}$ |
| 745 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{5,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{13\sqrt{6}}{480} & 0 & 0 & \frac{3i}{80} & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}i}{240} \\ 0 & 0 & 0 & \frac{13\sqrt{6}}{480} & -\frac{3i}{80} & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} & \frac{\sqrt{15}i}{240} & 0 \\ -\frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & -\frac{i}{10} & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} \\ 0 & \frac{\sqrt{6}}{40} & 0 & 0 & \frac{1}{10} & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & \frac{\sqrt{15}}{30} & 0 \\ 0 & \frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{60} & 0 & 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ -\frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{1}{10} & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{6}i}{48} & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & \frac{3\sqrt{10}i}{80} & -\frac{\sqrt{15}}{120} & 0 \\ \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{1}{8} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & \frac{\sqrt{15}}{120} \\ 0 & 0 & -\frac{9\sqrt{2}}{160} & 0 & 0 & -\frac{\sqrt{3}i}{80} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & 0 & 0 & \frac{9\sqrt{2}}{160} & \frac{\sqrt{3}i}{80} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{160} & \frac{\sqrt{5}i}{16} & 0 \end{bmatrix}$ |
| 746 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |
| | $\mathbb{T}^{(1,0;a)}_{5,2}(E_u,2)$ | $\begin{bmatrix} \frac{13\sqrt{6}}{480} & 0 & 0 & 0 & 0 & \frac{3}{80} & 0 & -\frac{i}{10} & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{240} \\ 0 & -\frac{13\sqrt{6}}{480} & 0 & 0 & \frac{3}{80} & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & \frac{\sqrt{15}}{240} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & -\frac{i}{10} & 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & \frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{40} & \frac{i}{10} & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & -\frac{\sqrt{15}i}{30} & 0 \\ 0 & \frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{20} & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & \frac{\sqrt{10}i}{20} & \frac{\sqrt{15}}{120} & 0 \\ \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & \frac{1}{8} & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{120} \\ 0 & -\frac{\sqrt{6}i}{60} & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & \frac{1}{10} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\ \frac{\sqrt{6}i}{60} & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & -\frac{1}{10} & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ -\frac{9\sqrt{2}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{3}i}{10} & \frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{16} \\ 0 & \frac{9\sqrt{2}}{160} & 0 & 0 & \frac{\sqrt{3}}{80} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & -\frac{\sqrt{5}}{16} & 0 \end{bmatrix}$ |
| 747 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{5,1}(E_u,3)$ | $\begin{bmatrix} \frac{37\sqrt{2}}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{120} & 0 & -\frac{\sqrt{3}i}{20} & -\frac{\sqrt{30}}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{24} \\ 0 & -\frac{37\sqrt{2}}{240} & 0 & 0 & \frac{\sqrt{3}}{120} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & \frac{\sqrt{30}}{240} & 0 & 0 & \frac{\sqrt{5}}{24} & 0 \\ 0 & 0 & -\frac{19\sqrt{2}}{120} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}i}{60} \\ 0 & 0 & 0 & \frac{19\sqrt{2}}{120} & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & \frac{\sqrt{5}i}{60} & 0 \\ 0 & -\frac{\sqrt{2}}{30} & 0 & -\frac{\sqrt{2}i}{24} & -\frac{\sqrt{3}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & -\frac{\sqrt{5}}{60} & 0 \\ -\frac{\sqrt{2}}{30} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & \frac{\sqrt{5}}{60} \\ 0 & -\frac{7\sqrt{2}i}{120} & 0 & -\frac{\sqrt{2}}{120} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 \\ \frac{7\sqrt{2}i}{120} & 0 & -\frac{\sqrt{2}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{80} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & -\frac{i}{20} & -\frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} \\ 0 & \frac{\sqrt{6}}{80} & 0 & 0 & \frac{1}{8} & 0 & \frac{i}{20} & 0 & 0 & \frac{\sqrt{10}}{80} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 \end{bmatrix}$ |
| 748 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |
| | $\mathbb{T}^{(1,0;a)}_{5,2}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{37\sqrt{2}}{240} & 0 & 0 & -\frac{\sqrt{3}i}{120} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & \frac{\sqrt{30}}{240} & 0 & 0 & \frac{\sqrt{5}i}{24} \\ 0 & 0 & 0 & -\frac{37\sqrt{2}}{240} & \frac{\sqrt{3}i}{120} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{240} & -\frac{\sqrt{5}i}{24} & 0 \\ \frac{19\sqrt{2}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{3}i}{30} & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{60} \\ 0 & -\frac{19\sqrt{2}}{120} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}}{60} & 0 \\ 0 & -\frac{\sqrt{2}i}{120} & 0 & -\frac{7\sqrt{2}}{120} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 \\ \frac{\sqrt{2}i}{120} & 0 & -\frac{7\sqrt{2}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & -\frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{24} & 0 & -\frac{\sqrt{2}i}{30} & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}}{60} & 0 \\ -\frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{30} & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{60} \\ 0 & 0 & \frac{\sqrt{6}}{80} & 0 & 0 & \frac{i}{8} & 0 & \frac{1}{20} & 0 & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & -\frac{i}{8} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{80} & \frac{\sqrt{15}i}{24} & 0 \end{bmatrix}$ |
| 749 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & -\frac{\sqrt{105}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 \end{bmatrix}$ |
| 750 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & \frac{\sqrt{105}i}{140} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 \\ \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{70}i}{280} & \frac{\sqrt{105}}{70} & 0 \\ \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 \end{bmatrix}$ |
| 751 | symmetry | $y$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{1,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} \\ 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{105}}{140} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & \frac{\sqrt{105}i}{140} & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{70} & 0 \\ \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 752 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{T}_{3}^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 \\ \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 \\ \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & \frac{\sqrt{105}i}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & -\frac{\sqrt{105}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & \frac{\sqrt{7}i}{24} & -\frac{\sqrt{70}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 753 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & \frac{3\sqrt{42}i}{112} & -\frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & \frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 \\ -\frac{\sqrt{70}i}{560} & 0 & -\frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{120} & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{105}}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}i}{56} \\ 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & \frac{\sqrt{105}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & -\frac{\sqrt{7}i}{56} & 0 \\ 0 & \frac{\sqrt{210}}{80} & 0 & -\frac{\sqrt{210}i}{80} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 \\ \frac{\sqrt{210}}{80} & 0 & \frac{\sqrt{210}i}{80} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 754 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_3^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 \\ \frac{\sqrt{70}i}{560} & 0 & \frac{\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{112} & 0 & -\frac{3\sqrt{42}i}{112} & \frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{3\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}i}{56} \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & -\frac{\sqrt{7}i}{56} & 0 \\ -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{120} & \frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} \\ 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{210}i}{80} & 0 & -\frac{\sqrt{210}}{80} & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 \\ \frac{\sqrt{210}i}{80} & 0 & -\frac{\sqrt{210}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 755 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,1;a)}_{3,1}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{70}}{672} & 0 & 0 & -\frac{\sqrt{105}i}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & -\frac{\sqrt{7}i}{24} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{672} & \frac{\sqrt{105}i}{168} & 0 \\ -\frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & -\frac{\sqrt{7}i}{24} & \frac{13\sqrt{70}}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} \\ 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & -\frac{13\sqrt{70}}{672} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 \\ 0 & -\frac{17\sqrt{42}i}{672} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & -\frac{\sqrt{70}i}{672} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 \\ \frac{17\sqrt{42}i}{672} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{24} & \frac{\sqrt{70}i}{672} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{11\sqrt{42}i}{672} & \frac{\sqrt{7}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{672} & -\frac{\sqrt{105}}{112} & 0 \\ -\frac{\sqrt{42}}{84} & 0 & -\frac{11\sqrt{42}i}{672} & 0 & 0 & -\frac{\sqrt{7}}{48} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{672} & 0 & 0 & \frac{\sqrt{105}}{112} \\ 0 & 0 & -\frac{\sqrt{14}}{32} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{32} & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{224} & 0 & 0 \end{bmatrix}$ |
| 756 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |
| | $\mathbb{T}^{(1,1;a)}_{3,2}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & -\frac{\sqrt{7}i}{42} & \frac{\sqrt{70}}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} \\ 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & -\frac{\sqrt{70}}{672} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & -\frac{13\sqrt{70}}{672} & 0 & 0 & -\frac{\sqrt{105}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{224} & \frac{5\sqrt{7}i}{84} & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & 0 & 0 & \frac{13\sqrt{70}}{672} & \frac{\sqrt{105}i}{84} & 0 \\ 0 & -\frac{11\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{84} & \frac{\sqrt{7}}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{672} & 0 & -\frac{\sqrt{70}i}{84} & \frac{\sqrt{105}}{112} & 0 \\ -\frac{11\sqrt{42}}{672} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{7}}{48} & 0 & 0 & \frac{\sqrt{70}}{672} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & -\frac{\sqrt{105}}{112} \\ 0 & \frac{5\sqrt{42}i}{168} & 0 & \frac{17\sqrt{42}}{672} & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & -\frac{\sqrt{70}}{672} & 0 & 0 \\ -\frac{5\sqrt{42}i}{168} & 0 & \frac{17\sqrt{42}}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & -\frac{\sqrt{70}i}{168} & 0 & -\frac{\sqrt{70}}{672} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{42} & -\frac{\sqrt{210}}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{32} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{210}}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 757 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,1;a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{105}i}{70} & \frac{17\sqrt{42}}{672} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & -\frac{\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}}{120} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{17\sqrt{42}}{672} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & \frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & -\frac{23\sqrt{70}}{1120} & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{42}i}{56} & -\frac{5\sqrt{7}}{112} & 0 \\ -\frac{23\sqrt{70}}{1120} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{80} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{5\sqrt{7}}{112} \\ 0 & \frac{\sqrt{70}i}{140} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 \\ -\frac{\sqrt{70}i}{140} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{120} & \frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{160} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{70} & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{160} & 0 & 0 & \frac{3\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{5\sqrt{14}}{224} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 758 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |
| | $\mathbb{T}^{(1,1;a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & -\frac{17\sqrt{42}}{672} & 0 & 0 & -\frac{3\sqrt{7}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & \frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{672} & \frac{3\sqrt{7}i}{56} & 0 \\ -\frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{120} & \frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & -\frac{\sqrt{105}i}{120} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & \frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{105}}{120} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{120} & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{23\sqrt{70}i}{1120} & \frac{\sqrt{105}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{224} & -\frac{5\sqrt{7}}{112} & 0 \\ \frac{\sqrt{70}}{56} & 0 & \frac{23\sqrt{70}i}{1120} & 0 & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & \frac{5\sqrt{7}}{112} \\ 0 & 0 & \frac{\sqrt{210}}{160} & 0 & 0 & \frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{160} & -\frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{224} & 0 & 0 \end{bmatrix}$ |
| 759 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(A_u)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 760 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(a)}(B_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 761 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 762 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \end{bmatrix}$ |
| 763 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,2}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 764 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{M}_4^{(a)}(A_u, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 765 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(A_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 \\ -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 766 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_4^{(a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 767 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(B_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\ 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 \\ -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 768 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_4^{(a)}(B_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 769 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{4,1}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 770 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{M}^{(a)}_{4,2}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 \\ -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} \\ 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 \end{bmatrix}$ |
| 771 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{4,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}i}{112} & 0 & 0 \\ \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{56} \\ 0 & 0 & -\frac{3\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \end{bmatrix}$ |
| 772 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{M}^{(a)}_{4,2}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}i}{112} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 773 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & \frac{\sqrt{105}}{420} & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & \frac{\sqrt{105}i}{420} & 0 & \frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & \frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & \frac{\sqrt{70}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{2\sqrt{105}}{105} & 0 & 0 & -\frac{\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{105} & \frac{\sqrt{70}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{35}i}{70} & \frac{\sqrt{210}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & -\frac{\sqrt{210}}{70} \end{bmatrix}$ |
| 774 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 775 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 776 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 \end{bmatrix}$ |
| 777 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & -\frac{\sqrt{210}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{3\sqrt{70}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & \frac{3\sqrt{70}i}{140} & 0 \end{bmatrix}$ |
| 778 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & -\frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{10}}{240} & 0 & -\frac{11\sqrt{10}i}{240} & 0 & 0 \\ \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{1}{6} & 0 & 0 & -\frac{11\sqrt{10}}{240} & 0 & \frac{11\sqrt{10}i}{240} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{10}i}{240} & 0 & -\frac{\sqrt{10}}{240} & 0 & 0 \\ -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & \frac{1}{6} & -\frac{\sqrt{10}i}{240} & 0 & -\frac{\sqrt{10}}{240} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & \frac{i}{6} & \frac{\sqrt{10}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{120} \\ 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 & -\frac{\sqrt{15}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 & \frac{\sqrt{15}i}{120} \\ 0 & 0 & 0 & 0 & -\frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & -\frac{\sqrt{15}i}{120} & 0 \\ 0 & -\frac{5\sqrt{2}}{48} & 0 & -\frac{5\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{30}i}{80} & \frac{\sqrt{5}}{15} & 0 \\ -\frac{5\sqrt{2}}{48} & 0 & \frac{5\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 & -\frac{\sqrt{5}}{15} \end{bmatrix}$ |
| 779 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{210}i}{336} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{336} & 0 & \frac{\sqrt{14}i}{336} & 0 & 0 \\ \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}}{336} & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{13\sqrt{14}i}{336} & 0 & -\frac{13\sqrt{14}}{336} & 0 & 0 \\ -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & -\frac{13\sqrt{14}i}{336} & 0 & -\frac{13\sqrt{14}}{336} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & -\frac{\sqrt{35}i}{84} & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & \frac{\sqrt{21}i}{168} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & -\frac{\sqrt{21}i}{168} & 0 \\ 0 & \frac{\sqrt{70}}{48} & 0 & \frac{\sqrt{70}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{42}i}{112} & \frac{\sqrt{7}}{21} & 0 \\ \frac{\sqrt{70}}{48} & 0 & -\frac{\sqrt{70}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{7}}{21} \end{bmatrix}$ |
| 780 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 781 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{42}i}{48} & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{48} & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 \\ \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} \\ 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{56} & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{56} & 0 \\ 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & 0 \\ -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 782 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_4^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 \\ -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{48} & 0 & -\frac{\sqrt{42}i}{48} & -\frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{56} & 0 \\ -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{168} & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} \\ 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}}{56} & 0 \\ 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 \\ -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 783 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}}{96} & 0 & 0 & -\frac{i}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{32} & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}}{96} & \frac{i}{8} & 0 \\ -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & -\frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{32} & \frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{96} & -\frac{1}{16} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & 0 & -\frac{\sqrt{15}}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & \frac{1}{16} \\ 0 & 0 & \frac{\sqrt{30}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{32} & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{32} & -\frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$ |
| 784 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{7\sqrt{6}}{96} & 0 & 0 & 0 & 0 & \frac{1}{8} \\ 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}}{96} & 0 & 0 & \frac{1}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{30}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{32} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \end{bmatrix}$ |
| 785 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & \frac{\sqrt{42}}{672} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} \\ 0 & -\frac{\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{672} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & -\frac{13\sqrt{42}}{672} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{42}}{672} & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & -\frac{3\sqrt{70}}{224} & 0 & -\frac{\sqrt{70}i}{56} & -\frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{672} & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{112} & 0 \\ -\frac{3\sqrt{70}}{224} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{5\sqrt{105}}{336} & 0 & 0 & -\frac{11\sqrt{42}}{672} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{112} \\ 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 & 0 \\ \frac{\sqrt{210}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & -\frac{\sqrt{210}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{224} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \end{bmatrix}$ |
| 786 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{672} & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{224} & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{672} & -\frac{3\sqrt{7}i}{56} & 0 \\ -\frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & -\frac{13\sqrt{42}}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & \frac{\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{13\sqrt{42}}{672} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & -\frac{3\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}i}{96} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 \\ \frac{3\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & \frac{\sqrt{42}i}{96} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{70}i}{224} & \frac{5\sqrt{105}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{11\sqrt{42}i}{672} & -\frac{\sqrt{7}}{112} & 0 \\ -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{70}i}{224} & 0 & 0 & -\frac{5\sqrt{105}}{336} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{11\sqrt{42}i}{672} & 0 & 0 & \frac{\sqrt{7}}{112} \\ 0 & 0 & -\frac{\sqrt{210}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{224} & -\frac{\sqrt{21}i}{84} & 0 \end{bmatrix}$ |
| 787 | symmetry | $\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{231}}{616} & 0 & -\frac{\sqrt{231}i}{616} & -\frac{3\sqrt{154}}{308} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{616} & 0 & -\frac{3\sqrt{385}i}{616} & 0 & 0 \\ \frac{\sqrt{231}}{616} & 0 & \frac{\sqrt{231}i}{616} & 0 & 0 & \frac{3\sqrt{154}}{308} & 0 & 0 & -\frac{3\sqrt{385}}{616} & 0 & \frac{3\sqrt{385}i}{616} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{231}}{462} & 0 & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & -\frac{\sqrt{385}i}{154} & 0 & \frac{\sqrt{385}}{154} & 0 & 0 \\ \frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{231}}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & \frac{\sqrt{385}i}{154} & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 \\ -\frac{\sqrt{231}}{132} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{308} & 0 & -\frac{\sqrt{154}i}{77} & -\frac{\sqrt{385}}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{924} \\ 0 & \frac{\sqrt{231}}{132} & 0 & 0 & -\frac{3\sqrt{154}}{308} & 0 & \frac{\sqrt{154}i}{77} & 0 & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & -\frac{\sqrt{2310}}{924} & 0 \\ 0 & 0 & \frac{\sqrt{231}}{132} & 0 & 0 & -\frac{3\sqrt{154}i}{308} & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & \frac{\sqrt{2310}i}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{132} & \frac{3\sqrt{154}i}{308} & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{308} & -\frac{\sqrt{2310}i}{924} & 0 \\ 0 & -\frac{\sqrt{77}}{88} & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{1155}i}{616} & \frac{\sqrt{770}}{308} & 0 \\ -\frac{\sqrt{77}}{88} & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & 0 & -\frac{\sqrt{770}}{308} \end{bmatrix}$ |
| 788 | symmetry | $-\frac{\sqrt{14}x^6}{8}+\frac{15\sqrt{14}x^4z^2}{8}-\frac{15\sqrt{14}x^2z^4}{8}-\frac{\sqrt{14}y^6}{8}+\frac{15\sqrt{14}y^4z^2}{8}-\frac{15\sqrt{14}y^2z^4}{8}+\frac{\sqrt{14}z^6}{4}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{33}}{264} & 0 & \frac{\sqrt{33}i}{264} & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\ -\frac{\sqrt{33}}{264} & 0 & -\frac{\sqrt{33}i}{264} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} \\ 0 & -\frac{\sqrt{33}}{132} & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{132} & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & \frac{\sqrt{330}i}{132} \\ 0 & 0 & 0 & \frac{\sqrt{33}}{132} & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & -\frac{\sqrt{330}i}{132} & 0 \\ 0 & \frac{\sqrt{11}}{88} & 0 & \frac{\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{165}i}{88} & \frac{\sqrt{110}}{44} & 0 \\ \frac{\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & -\frac{\sqrt{165}i}{88} & 0 & 0 & -\frac{\sqrt{110}}{44} \end{bmatrix}$ |
| 789 | symmetry | $-\frac{3\sqrt{7}x^5y}{4}+\frac{15\sqrt{7}x^3yz^2}{2}+\frac{3\sqrt{7}xy^5}{4}-\frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\ \frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{264} & 0 & \frac{\sqrt{66}i}{264} & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 \\ -\frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & \frac{\sqrt{11}i}{22} & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 790 | symmetry | $-\frac{\sqrt{2310}x^4y^2}{8} + \frac{\sqrt{2310}x^4z^2}{8} + \frac{\sqrt{2310}x^2y^4}{8} - \frac{\sqrt{2310}x^2z^4}{8} - \frac{\sqrt{2310}y^4z^2}{8} + \frac{\sqrt{2310}y^2z^4}{8}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{7\sqrt{5}}{120} & 0 & \frac{7\sqrt{5}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{3}i}{24} & -\frac{\sqrt{2}}{12} & 0 \\ \frac{7\sqrt{5}}{120} & 0 & -\frac{7\sqrt{5}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & \frac{\sqrt{2}}{12} \\ 0 & \frac{\sqrt{5}i}{15} & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{15} & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} \\ 0 & -\frac{\sqrt{5}}{60} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{60} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{60} & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & \frac{\sqrt{2}i}{12} & 0 \\ 0 & \frac{\sqrt{15}}{120} & 0 & -\frac{\sqrt{15}i}{120} & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & -\frac{i}{8} & 0 & 0 \\ \frac{\sqrt{15}}{120} & 0 & \frac{\sqrt{15}i}{120} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{1}{8} & 0 & \frac{i}{8} & 0 & 0 & 0 \end{bmatrix}$ |
| 791 | symmetry | $\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{17\sqrt{11}}{264} & 0 & \frac{17\sqrt{11}i}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & \frac{\sqrt{165}i}{264} & \frac{\sqrt{110}}{132} & 0 \\ \frac{17\sqrt{11}}{264} & 0 & -\frac{17\sqrt{11}i}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{264} & 0 & 0 & -\frac{\sqrt{110}}{132} \\ 0 & \frac{2\sqrt{11}i}{33} & 0 & -\frac{2\sqrt{11}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{11}i}{33} & 0 & -\frac{2\sqrt{11}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{11}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{132} & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{132} \\ 0 & \frac{\sqrt{11}}{132} & 0 & 0 & -\frac{\sqrt{66}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{110}}{132} & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{132} & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{110}i}{132} \\ 0 & 0 & 0 & \frac{\sqrt{11}}{132} & -\frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & -\frac{\sqrt{110}i}{132} & 0 \\ 0 & -\frac{\sqrt{33}}{264} & 0 & \frac{\sqrt{33}i}{264} & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\ -\frac{\sqrt{33}}{264} & 0 & -\frac{\sqrt{33}i}{264} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 \end{bmatrix}$ |
| 792 | $\mathbb{M}_6^{(1,-1;a)}(B_u,3)$ | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ $\begin{bmatrix} 0 & -\frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 793 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(B_u,4)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{55}i}{660} & 0 & \frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}i}{660} & 0 & \frac{\sqrt{55}}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{660} & 0 & -\frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & \frac{\sqrt{33}i}{66} & \frac{\sqrt{22}}{33} & 0 \\ -\frac{\sqrt{55}}{660} & 0 & \frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & -\frac{\sqrt{22}}{33} \\ 0 & 0 & -\frac{\sqrt{55}}{165} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & -\frac{\sqrt{22}i}{33} \\ 0 & 0 & 0 & \frac{\sqrt{55}}{165} & 0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & \frac{\sqrt{22}i}{33} & 0 \\ \frac{\sqrt{55}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{165} & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{33} \\ 0 & -\frac{\sqrt{55}}{165} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & \frac{\sqrt{22}}{33} & 0 \\ 0 & -\frac{\sqrt{165}i}{330} & 0 & -\frac{\sqrt{165}}{330} & 0 & 0 & \frac{\sqrt{110}}{55} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & \frac{\sqrt{11}}{22} & 0 & 0 \\ \frac{\sqrt{165}i}{330} & 0 & -\frac{\sqrt{165}}{330} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{55} & \frac{\sqrt{11}i}{22} & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 \end{bmatrix}$ |
| 794 | symmetry | $\frac{15\sqrt{7}x^2y^3z}{2} - \frac{15\sqrt{7}x^2yz^3}{2} - \frac{3\sqrt{7}y^5z}{4} + \frac{3\sqrt{7}yz^5}{4}$ |
| | $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{5\sqrt{66}}{528} & 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & -\frac{\sqrt{110}}{176} & 0 & 0 & \frac{\sqrt{165}i}{264} \\ 0 & 0 & 0 & \frac{5\sqrt{66}}{528} & -\frac{3\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{176} & -\frac{\sqrt{165}i}{264} & 0 \\ -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{11}i}{22} & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} \\ 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 \\ 0 & \frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 \\ -\frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & -\frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{88} & 0 & -\frac{\sqrt{66}i}{66} & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & \frac{\sqrt{165}}{132} & 0 \\ -\frac{\sqrt{66}}{88} & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} \\ 0 & 0 & -\frac{3\sqrt{22}}{176} & 0 & 0 & \frac{\sqrt{33}i}{88} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & -\frac{\sqrt{55}i}{88} \\ 0 & 0 & 0 & \frac{3\sqrt{22}}{176} & -\frac{\sqrt{33}i}{88} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & \frac{\sqrt{55}i}{88} & 0 \end{bmatrix}$ |
| 795 | symmetry | $\frac{3\sqrt{7}x^5z}{4} - \frac{15\sqrt{7}x^3y^2z}{2} + \frac{15\sqrt{7}xy^2z^3}{2} - \frac{3\sqrt{7}xz^5}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{5\sqrt{66}}{528} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{88} & 0 & \frac{\sqrt{11}i}{44} & -\frac{\sqrt{110}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} \\ 0 & -\frac{5\sqrt{66}}{528} & 0 & 0 & \frac{3\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & \frac{\sqrt{110}}{176} & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & -\frac{\sqrt{165}i}{132} \\ 0 & 0 & 0 & \frac{\sqrt{66}}{88} & -\frac{\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & \frac{\sqrt{165}i}{132} & 0 \\ 0 & \frac{\sqrt{66}}{66} & 0 & \frac{\sqrt{66}i}{88} & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & -\frac{\sqrt{165}}{132} & 0 \\ \frac{\sqrt{66}}{66} & 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & \frac{\sqrt{165}}{132} \\ 0 & \frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\ -\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ -\frac{3\sqrt{22}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{44} & -\frac{\sqrt{330}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} \\ 0 & \frac{3\sqrt{22}}{176} & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 \end{bmatrix}$ |
| 796 | symmetry | $\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$ |
| | $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{1}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & -\frac{\sqrt{15}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{32} \\ 0 & -\frac{1}{32} & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{16} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & 0 & \frac{\sqrt{10}}{16} & 0 \\ \frac{1}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{32} & 0 & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{32} \\ 0 & \frac{\sqrt{3}}{32} & 0 & 0 & -\frac{3\sqrt{2}}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{32} & 0 & 0 & \frac{\sqrt{30}}{32} & 0 \end{bmatrix}$ |
| 797 | symmetry | $\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{32} & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & 0 & -\frac{\sqrt{10}i}{32} \\ 0 & 0 & 0 & -\frac{1}{32} & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{32} & \frac{\sqrt{10}i}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{16} & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & \frac{\sqrt{10}}{16} & 0 \\ 0 & 0 & -\frac{i}{16} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & -\frac{\sqrt{10}}{16} \\ 0 & 0 & \frac{\sqrt{3}}{32} & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & -\frac{\sqrt{30}i}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{32} & \frac{3\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{32} & \frac{\sqrt{30}i}{32} & 0 \end{bmatrix}$ |
| 798 | symmetry | $\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$ |
| | $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} \frac{17\sqrt{55}}{1056} & 0 & 0 & 0 & 0 & \frac{37\sqrt{330}}{5280} & 0 & \frac{\sqrt{330}i}{110} & \frac{\sqrt{33}}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{96} \\ 0 & -\frac{17\sqrt{55}}{1056} & 0 & 0 & \frac{37\sqrt{330}}{5280} & 0 & -\frac{\sqrt{330}i}{110} & 0 & 0 & -\frac{\sqrt{33}}{96} & 0 & 0 & \frac{\sqrt{22}}{96} & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & \frac{\sqrt{330}i}{110} & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & \frac{\sqrt{22}i}{66} \\ 0 & 0 & 0 & \frac{\sqrt{55}}{66} & -\frac{\sqrt{330}i}{110} & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{66} & -\frac{\sqrt{22}i}{66} & 0 \\ 0 & \frac{29\sqrt{55}}{2640} & 0 & \frac{\sqrt{55}i}{66} & \frac{\sqrt{330}}{240} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{176} & 0 & \frac{\sqrt{33}i}{66} & \frac{5\sqrt{22}}{528} & 0 \\ \frac{29\sqrt{55}}{2640} & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & -\frac{\sqrt{330}}{240} & 0 & 0 & \frac{\sqrt{33}}{176} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & -\frac{5\sqrt{22}}{528} \\ 0 & \frac{7\sqrt{55}i}{330} & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 \\ -\frac{7\sqrt{55}i}{330} & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & -\frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 \\ \frac{9\sqrt{165}}{1760} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{160} & 0 & \frac{\sqrt{110}i}{110} & \frac{5\sqrt{11}}{352} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{1056} \\ 0 & -\frac{9\sqrt{165}}{1760} & 0 & 0 & \frac{\sqrt{110}}{160} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & -\frac{5\sqrt{11}}{352} & 0 & 0 & \frac{5\sqrt{66}}{1056} & 0 \end{bmatrix}$ |
| 799 | symmetry | $\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & \frac{17\sqrt{55}}{1056} & 0 & 0 & -\frac{37\sqrt{330}i}{5280} & 0 & \frac{\sqrt{330}}{110} & 0 & 0 & -\frac{\sqrt{33}}{96} & 0 & 0 & \frac{\sqrt{22}i}{96} \\ 0 & 0 & 0 & -\frac{17\sqrt{55}}{1056} & \frac{37\sqrt{330}i}{5280} & 0 & \frac{\sqrt{330}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{96} & -\frac{\sqrt{22}i}{96} & 0 \\ \frac{\sqrt{55}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{110} & 0 & \frac{\sqrt{330}i}{165} & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{66} \\ 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & \frac{\sqrt{330}}{110} & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & \frac{\sqrt{33}}{66} & 0 & 0 & -\frac{\sqrt{22}}{66} & 0 \\ 0 & -\frac{\sqrt{55}i}{66} & 0 & \frac{7\sqrt{55}}{330} & 0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 \\ \frac{\sqrt{55}i}{66} & 0 & \frac{7\sqrt{55}}{330} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & -\frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}}{66} & 0 & \frac{29\sqrt{55}i}{2640} & -\frac{\sqrt{330}}{240} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{176} & \frac{5\sqrt{22}}{528} & 0 \\ \frac{\sqrt{55}}{66} & 0 & -\frac{29\sqrt{55}i}{2640} & 0 & 0 & \frac{\sqrt{330}}{240} & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & \frac{\sqrt{33}i}{176} & 0 & 0 & -\frac{5\sqrt{22}}{528} \\ 0 & 0 & -\frac{9\sqrt{165}}{1760} & 0 & 0 & \frac{\sqrt{110}i}{160} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 & \frac{5\sqrt{11}}{352} & 0 & 0 & -\frac{5\sqrt{66}i}{1056} \\ 0 & 0 & 0 & \frac{9\sqrt{165}}{1760} & -\frac{\sqrt{110}i}{160} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{352} & \frac{5\sqrt{66}i}{1056} & 0 \end{bmatrix}$ |
| 800 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}_{2}^{(1,0;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 \\ -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 801 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 802 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_2^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & -\frac{\sqrt{21}}{42} & 0 \\ \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 803 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & \frac{\sqrt{21}i}{84} & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & \frac{\sqrt{21}}{42} & 0 \\ \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 804 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{21}i}{84} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{14}i}{56} & \frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \end{bmatrix}$ |
| 805 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{10}}{80} & 0 & -\frac{\sqrt{10}i}{80} & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 \\ \frac{\sqrt{10}}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{80} & 0 & \frac{\sqrt{10}}{80} & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\ -\frac{\sqrt{10}i}{80} & 0 & \frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} \\ 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & \frac{\sqrt{15}i}{40} & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 \\ 0 & \frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 \\ \frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 \end{bmatrix}$ |
| 806 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |
| | $\mathbb{M}_4^{(1,0;a)}(A_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{14}i}{112} & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{17\sqrt{210}}{1680} & 0 & \frac{17\sqrt{210}i}{1680} & 0 & 0 \\ \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & \frac{17\sqrt{210}}{1680} & 0 & -\frac{17\sqrt{210}i}{1680} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & \frac{\sqrt{210}}{560} & 0 & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{30} & \frac{\sqrt{210}i}{560} & 0 & \frac{\sqrt{210}}{560} & 0 & 0 & 0 \\ \frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{40} & 0 & \frac{\sqrt{21}i}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & -\frac{\sqrt{21}}{40} & 0 & -\frac{\sqrt{21}i}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & -\frac{\sqrt{21}i}{40} & 0 & -\frac{\sqrt{21}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{20} & \frac{\sqrt{21}i}{40} & 0 & -\frac{\sqrt{21}}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 \\ 0 & -\frac{\sqrt{42}}{80} & 0 & -\frac{\sqrt{42}i}{80} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\ -\frac{\sqrt{42}}{80} & 0 & \frac{\sqrt{42}i}{80} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 \end{bmatrix}$ |
| 807 | symmetry | $\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(A_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{5} & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & -\frac{i}{40} & 0 & \frac{1}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{10} & \frac{i}{40} & 0 & \frac{1}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{40} & 0 & \frac{i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & 0 & \frac{1}{40} & 0 & -\frac{i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{40} & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{40} & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 808 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{M}_4^{(1,0;a)}(B_u,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{42}}{560} & 0 & -\frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & -\frac{\sqrt{105}}{70} & 0 \\ -\frac{3\sqrt{42}}{560} & 0 & \frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & \frac{3\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{70}i}{560} & 0 & -\frac{13\sqrt{70}}{560} & 0 & 0 \\ -\frac{3\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{70}i}{560} & 0 & -\frac{13\sqrt{70}}{560} & 0 & 0 & 0 \\ -\frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & \frac{\sqrt{7}i}{20} & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{280} \\ 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & \frac{3\sqrt{105}}{280} & 0 \\ 0 & 0 & -\frac{3\sqrt{42}}{280} & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & \frac{3\sqrt{105}i}{280} \\ 0 & 0 & 0 & \frac{3\sqrt{42}}{280} & \frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & -\frac{3\sqrt{105}i}{280} & 0 \\ 0 & -\frac{3\sqrt{14}}{80} & 0 & \frac{3\sqrt{14}i}{80} & \frac{\sqrt{21}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{560} & 0 & -\frac{3\sqrt{210}i}{560} & 0 & 0 \\ -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{14}i}{80} & 0 & 0 & -\frac{\sqrt{21}}{35} & 0 & 0 & -\frac{3\sqrt{210}}{560} & 0 & \frac{3\sqrt{210}i}{560} & 0 & 0 & 0 \end{bmatrix}$ |
| 809 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{42}i}{560} & 0 & \frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{70}i}{560} & 0 & -\frac{13\sqrt{70}}{560} & 0 & 0 \\ \frac{3\sqrt{42}i}{560} & 0 & \frac{3\sqrt{42}}{560} & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{70}i}{560} & 0 & -\frac{13\sqrt{70}}{560} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{42}}{560} & 0 & -\frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{560} & 0 & \frac{\sqrt{70}i}{560} & \frac{\sqrt{105}}{70} & 0 \\ -\frac{3\sqrt{42}}{560} & 0 & \frac{3\sqrt{42}i}{560} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{560} & 0 & -\frac{\sqrt{70}i}{560} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & \frac{3\sqrt{105}i}{280} \\ 0 & 0 & 0 & -\frac{3\sqrt{42}}{280} & -\frac{\sqrt{7}i}{20} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & -\frac{3\sqrt{105}i}{280} & 0 \\ -\frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & \frac{\sqrt{7}i}{40} & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{280} \\ 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & -\frac{\sqrt{7}i}{40} & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{105}}{280} & 0 \\ 0 & \frac{3\sqrt{14}i}{80} & 0 & \frac{3\sqrt{14}}{80} & 0 & 0 & -\frac{\sqrt{21}}{35} & 0 & 0 & -\frac{3\sqrt{210}i}{560} & 0 & \frac{3\sqrt{210}}{560} & 0 & 0 \\ -\frac{3\sqrt{14}i}{80} & 0 & \frac{3\sqrt{14}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{35} & \frac{3\sqrt{210}i}{560} & 0 & \frac{3\sqrt{210}}{560} & 0 & 0 & 0 \end{bmatrix}$ |
| 810 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{160} & 0 & 0 & \frac{i}{40} & 0 & -\frac{1}{10} & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & \frac{\sqrt{15}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{160} & -\frac{i}{40} & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} & -\frac{\sqrt{15}i}{40} & 0 \\ -\frac{\sqrt{6}}{160} & 0 & 0 & 0 & 0 & \frac{3}{20} & 0 & -\frac{i}{40} & -\frac{\sqrt{10}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} \\ 0 & \frac{\sqrt{6}}{160} & 0 & 0 & \frac{3}{20} & 0 & \frac{i}{40} & 0 & 0 & \frac{\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 \\ 0 & \frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & -\frac{1}{40} & 0 & 0 & \frac{\sqrt{10}i}{160} & 0 & \frac{3\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & 0 & 0 & \frac{1}{40} & -\frac{\sqrt{10}i}{160} & 0 & \frac{3\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{6}i}{32} & \frac{1}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{160} & -\frac{\sqrt{15}}{80} & 0 \\ -\frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{1}{16} & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}}{80} \\ 0 & 0 & \frac{9\sqrt{2}}{160} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}}{160} & \frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 \end{bmatrix}$ |
| 811 | symmetry | $-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{6}}{160} & 0 & 0 & 0 & 0 & \frac{1}{40} & 0 & \frac{i}{10} & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} \\ 0 & \frac{\sqrt{6}}{160} & 0 & 0 & \frac{1}{40} & 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{160} & 0 & 0 & -\frac{3i}{20} & 0 & -\frac{1}{40} & 0 & 0 & \frac{\sqrt{10}}{160} & 0 & 0 & \frac{\sqrt{15}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{160} & \frac{3i}{20} & 0 & -\frac{1}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{160} & -\frac{\sqrt{15}i}{20} & 0 \\ 0 & -\frac{\sqrt{6}}{32} & 0 & \frac{\sqrt{6}i}{20} & \frac{1}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{160} & 0 & -\frac{\sqrt{10}i}{20} & \frac{\sqrt{15}}{80} & 0 \\ -\frac{\sqrt{6}}{32} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & -\frac{1}{16} & 0 & 0 & -\frac{\sqrt{10}}{160} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{15}}{80} \\ 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{6}}{160} & 0 & 0 & \frac{1}{40} & 0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{160} & 0 & 0 \\ \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{6}}{160} & 0 & 0 & 0 & 0 & -\frac{1}{40} & -\frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{160} & 0 & 0 & 0 \\ \frac{9\sqrt{2}}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{3}i}{10} & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}}{160} & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 812 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{42}}{1120} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{3\sqrt{7}i}{70} & -\frac{19\sqrt{70}}{1120} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & -\frac{\sqrt{42}}{1120} & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & \frac{19\sqrt{70}}{1120} & 0 & 0 & \frac{\sqrt{105}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{1120} & 0 & 0 & -\frac{\sqrt{7}i}{140} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & \frac{23\sqrt{70}}{1120} & 0 & 0 & \frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{1120} & \frac{\sqrt{7}i}{140} & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{70}}{1120} & -\frac{\sqrt{105}i}{140} & 0 \\ 0 & \frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{42}i}{56} & \frac{\sqrt{7}}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{3\sqrt{70}i}{280} & -\frac{\sqrt{105}}{560} & 0 \\ \frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}}{80} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & \frac{\sqrt{105}}{560} \\ 0 & \frac{\sqrt{42}i}{140} & 0 & \frac{29\sqrt{42}}{1120} & 0 & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \\ -\frac{\sqrt{42}i}{140} & 0 & \frac{29\sqrt{42}}{1120} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & \frac{\sqrt{70}i}{140} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 & 0 \\ \frac{9\sqrt{14}}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{70} & \frac{\sqrt{210}}{1120} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{14}}{160} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{70} & 0 & 0 & -\frac{\sqrt{210}}{1120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 813 | symmetry | $3\sqrt{5}x^2yz-\frac{\sqrt{5}y^3z}{2}-\frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{1120} & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & \frac{19\sqrt{70}}{1120} & 0 & 0 & \frac{\sqrt{105}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{1120} & \frac{\sqrt{7}i}{40} & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{70}}{1120} & -\frac{\sqrt{105}i}{56} & 0 \\ -\frac{\sqrt{42}}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{140} & 0 & \frac{\sqrt{7}i}{40} & \frac{23\sqrt{70}}{1120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & \frac{\sqrt{42}}{1120} & 0 & 0 & -\frac{\sqrt{7}}{140} & 0 & -\frac{\sqrt{7}i}{40} & 0 & 0 & -\frac{23\sqrt{70}}{1120} & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 \\ 0 & \frac{29\sqrt{42}i}{1120} & 0 & \frac{\sqrt{42}}{140} & 0 & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{140} & 0 & 0 \\ -\frac{29\sqrt{42}i}{1120} & 0 & \frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{1120} & -\frac{\sqrt{7}}{80} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{224} & -\frac{\sqrt{105}}{560} & 0 \\ \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & 0 & \frac{\sqrt{7}}{80} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 & \frac{\sqrt{105}}{560} \\ 0 & 0 & -\frac{9\sqrt{14}}{160} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & \frac{\sqrt{210}}{1120} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{14}}{160} & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{1120} & 0 & 0 \end{bmatrix}$ |
| 814 | symmetry | 1 |
| | $\mathbb{M}_0^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 \\ \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 \\ -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & \frac{\sqrt{35}i}{70} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & -\frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & -\frac{\sqrt{70}i}{70} & \frac{\sqrt{105}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{105}}{70} \end{bmatrix}$ |
| 815 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(A_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{84} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 \\ -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 \\ \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{105}}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{105}i}{210} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & \frac{\sqrt{105}i}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & -\frac{\sqrt{210}i}{140} & \frac{\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{35}}{35} \end{bmatrix}$ |
| 816 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(B_u, 1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{120} & 0 & \frac{\sqrt{210}i}{120} & -\frac{\sqrt{35}}{42} & 0 \\ \frac{\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{120} & 0 & -\frac{\sqrt{210}i}{120} & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{280} & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{280} & 0 & 0 \\ \frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 \\ \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{420} \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{420} & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{420} \\ 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{420} & 0 \\ 0 & -\frac{5\sqrt{42}}{168} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 \end{bmatrix}$ |
| 817 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 \\ \frac{\sqrt{14}i}{168} & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{168} & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{120} & 0 & \frac{\sqrt{210}i}{120} & -\frac{\sqrt{35}}{42} & 0 \\ -\frac{\sqrt{14}}{168} & 0 & \frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{120} & 0 & -\frac{\sqrt{210}i}{120} & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}i}{420} \\ 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{35}i}{420} & 0 \\ -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{420} \\ 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{420} & 0 \\ 0 & -\frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 \\ \frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 \end{bmatrix}$ |
| 818 | symmetry | $\sqrt{3}xz$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & \frac{\sqrt{14}}{42} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & \frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{210}i}{168} & \frac{\sqrt{35}}{105} & 0 \\ \frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}}{105} \\ 0 & \frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{120} & 0 & 0 \\ -\frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{120} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 \end{bmatrix}$ |
| 819 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{42} & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & \frac{\sqrt{35}i}{42} & 0 \\ \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ 0 & \frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}i}{120} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 \\ -\frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{210}i}{120} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{280} & \frac{\sqrt{35}}{105} & 0 \\ -\frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{35}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{105}i}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & -\frac{\sqrt{105}i}{105} & 0 \end{bmatrix}$ |
| 820 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |
| | $\mathbb{M}_4^{(1,1;a)}(A_u,1)$ | $\begin{bmatrix} 0 & \frac{2\sqrt{165}}{165} & 0 & -\frac{2\sqrt{165}i}{165} & \frac{7\sqrt{110}}{660} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{66} & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 \\ \frac{2\sqrt{165}}{165} & 0 & \frac{2\sqrt{165}i}{165} & 0 & 0 & -\frac{7\sqrt{110}}{660} & 0 & 0 & -\frac{\sqrt{11}}{66} & 0 & \frac{\sqrt{11}i}{66} & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{165}i}{660} & 0 & -\frac{7\sqrt{165}}{660} & 0 & 0 & -\frac{\sqrt{110}}{330} & 0 & 0 & \frac{\sqrt{11}i}{132} & 0 & -\frac{\sqrt{11}}{132} & 0 & 0 \\ \frac{7\sqrt{165}i}{660} & 0 & -\frac{7\sqrt{165}}{660} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{330} & -\frac{\sqrt{11}i}{132} & 0 & -\frac{\sqrt{11}}{132} & 0 & 0 & 0 \\ \frac{\sqrt{165}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{165} & 0 & \frac{\sqrt{110}i}{330} & -\frac{5\sqrt{11}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} \\ 0 & -\frac{\sqrt{165}}{220} & 0 & 0 & -\frac{\sqrt{110}}{165} & 0 & -\frac{\sqrt{110}i}{330} & 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & \frac{\sqrt{66}}{66} & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{220} & 0 & 0 & -\frac{\sqrt{110}i}{165} & 0 & -\frac{\sqrt{110}}{330} & 0 & 0 & -\frac{5\sqrt{11}}{132} & 0 & 0 & -\frac{\sqrt{66}i}{66} \\ 0 & 0 & 0 & \frac{\sqrt{165}}{220} & \frac{\sqrt{110}i}{165} & 0 & -\frac{\sqrt{110}}{330} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{132} & \frac{\sqrt{66}i}{66} & 0 \\ 0 & -\frac{\sqrt{55}}{660} & 0 & -\frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & -\frac{\sqrt{33}i}{44} & \frac{5\sqrt{22}}{132} & 0 \\ -\frac{\sqrt{55}}{660} & 0 & \frac{\sqrt{55}i}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & -\frac{5\sqrt{22}}{132} \end{bmatrix}$ |
| 821 | symmetry | $-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(A_u,2)$ | $\begin{bmatrix} 0 & -\frac{5\sqrt{231}}{462} & 0 & \frac{5\sqrt{231}i}{462} & -\frac{19\sqrt{154}}{4620} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{1155} & 0 & -\frac{\sqrt{385}i}{1155} & 0 & 0 \\ -\frac{5\sqrt{231}}{462} & 0 & -\frac{5\sqrt{231}i}{462} & 0 & 0 & \frac{19\sqrt{154}}{4620} & 0 & 0 & -\frac{\sqrt{385}}{1155} & 0 & \frac{\sqrt{385}i}{1155} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{231}}{84} & 0 & 0 & \frac{\sqrt{154}}{105} & 0 & 0 & \frac{\sqrt{385}i}{420} & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 \\ -\frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{231}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{105} & -\frac{\sqrt{385}i}{420} & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 & 0 \\ -\frac{\sqrt{231}}{220} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{1155} & 0 & \frac{\sqrt{154}i}{210} & -\frac{5\sqrt{385}}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} \\ 0 & \frac{\sqrt{231}}{220} & 0 & 0 & -\frac{2\sqrt{154}}{1155} & 0 & -\frac{\sqrt{154}i}{210} & 0 & 0 & \frac{5\sqrt{385}}{924} & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 \\ 0 & 0 & \frac{\sqrt{231}}{220} & 0 & 0 & -\frac{2\sqrt{154}i}{1155} & 0 & -\frac{\sqrt{154}}{210} & 0 & 0 & -\frac{5\sqrt{385}}{924} & 0 & 0 & -\frac{\sqrt{2310}i}{462} \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{220} & \frac{2\sqrt{154}i}{1155} & 0 & -\frac{\sqrt{154}}{210} & 0 & 0 & 0 & 0 & \frac{5\sqrt{385}}{924} & \frac{\sqrt{2310}i}{462} & 0 \\ 0 & \frac{\sqrt{77}}{660} & 0 & \frac{\sqrt{77}i}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & -\frac{\sqrt{1155}i}{308} & \frac{5\sqrt{770}}{924} & 0 \\ \frac{\sqrt{77}}{660} & 0 & -\frac{\sqrt{77}i}{660} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{1155}i}{308} & 0 & 0 & -\frac{5\sqrt{770}}{924} \end{bmatrix}$ |
| 822 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_4^{(1,1;a)}(A_u,3)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{11}}{44} & 0 & 0 & \frac{3\sqrt{66}}{220} & 0 & 0 & \frac{\sqrt{165}i}{660} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 \\ -\frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{66}}{220} & -\frac{\sqrt{165}i}{660} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{11}}{44} & 0 & -\frac{3\sqrt{11}i}{44} & \frac{3\sqrt{66}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 \\ \frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{11}i}{44} & 0 & 0 & -\frac{3\sqrt{66}}{220} & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & \frac{\sqrt{165}i}{660} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{110} & 0 & 0 & \frac{\sqrt{66}i}{330} & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}}{110} & -\frac{\sqrt{66}i}{330} & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{11}}{110} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{11}}{110} & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & \frac{\sqrt{66}i}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}i}{330} & 0 & -\frac{\sqrt{33}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{330} & 0 & -\frac{\sqrt{33}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 823 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(B_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}}{4620} & 0 & \frac{17\sqrt{1155}i}{4620} & -\frac{\sqrt{770}}{220} & 0 \\ \frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}}{4620} & 0 & -\frac{17\sqrt{1155}i}{4620} & 0 & 0 & \frac{\sqrt{770}}{220} \\ 0 & -\frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{420} & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 \\ \frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{420} & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 & 0 \\ \frac{\sqrt{77}}{220} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}}{2310} & 0 & \frac{\sqrt{462}i}{210} & -\frac{\sqrt{1155}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{385} \\ 0 & -\frac{\sqrt{77}}{220} & 0 & 0 & -\frac{17\sqrt{462}}{2310} & 0 & -\frac{\sqrt{462}i}{210} & 0 & 0 & \frac{\sqrt{1155}}{220} & 0 & 0 & \frac{\sqrt{770}}{385} & 0 \\ 0 & 0 & \frac{\sqrt{77}}{220} & 0 & 0 & \frac{17\sqrt{462}i}{2310} & 0 & \frac{\sqrt{462}}{210} & 0 & 0 & \frac{\sqrt{1155}}{220} & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & 0 & 0 & -\frac{\sqrt{77}}{220} & -\frac{17\sqrt{462}i}{2310} & 0 & \frac{\sqrt{462}}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{220} & -\frac{\sqrt{770}i}{385} & 0 \\ 0 & -\frac{\sqrt{231}}{165} & 0 & \frac{\sqrt{231}i}{165} & -\frac{3\sqrt{154}}{220} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{770} & 0 & \frac{3\sqrt{385}i}{770} & 0 & 0 \\ -\frac{\sqrt{231}}{165} & 0 & -\frac{\sqrt{231}i}{165} & 0 & 0 & \frac{3\sqrt{154}}{220} & 0 & 0 & \frac{3\sqrt{385}}{770} & 0 & -\frac{3\sqrt{385}i}{770} & 0 & 0 & 0 \end{bmatrix}$ |
| 824 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_4^{(1,1;a)}(B_u,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{77}i}{1540} & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{420} & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 \\ -\frac{\sqrt{77}i}{1540} & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{420} & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1155}}{4620} & 0 & -\frac{17\sqrt{1155}i}{4620} & \frac{\sqrt{770}}{220} & 0 \\ \frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{77}i}{1540} & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1155}}{4620} & 0 & \frac{17\sqrt{1155}i}{4620} & 0 & 0 & -\frac{\sqrt{770}}{220} \\ 0 & 0 & -\frac{\sqrt{77}}{220} & 0 & 0 & \frac{\sqrt{462}i}{210} & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & \frac{\sqrt{1155}}{220} & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & 0 & 0 & \frac{\sqrt{77}}{220} & -\frac{\sqrt{462}i}{210} & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{220} & -\frac{\sqrt{770}i}{385} & 0 \\ \frac{\sqrt{77}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{210} & 0 & -\frac{17\sqrt{462}i}{2310} & \frac{\sqrt{1155}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{385} \\ 0 & -\frac{\sqrt{77}}{220} & 0 & 0 & \frac{\sqrt{462}}{210} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & -\frac{\sqrt{1155}}{220} & 0 & 0 & -\frac{\sqrt{770}}{385} & 0 \\ 0 & \frac{\sqrt{231}i}{165} & 0 & \frac{\sqrt{231}}{165} & 0 & 0 & \frac{3\sqrt{154}}{220} & 0 & 0 & \frac{3\sqrt{385}i}{770} & 0 & -\frac{3\sqrt{385}}{770} & 0 & 0 \\ -\frac{\sqrt{231}i}{165} & 0 & \frac{\sqrt{231}}{165} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{220} & -\frac{3\sqrt{385}i}{770} & 0 & -\frac{3\sqrt{385}}{770} & 0 & 0 & 0 \end{bmatrix}$ |
| 825 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{11}}{220} & 0 & 0 & -\frac{7\sqrt{66}i}{660} & 0 & -\frac{3\sqrt{66}}{440} & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & -\frac{\sqrt{110}i}{110} \\ 0 & 0 & 0 & \frac{\sqrt{11}}{220} & \frac{7\sqrt{66}i}{660} & 0 & -\frac{3\sqrt{66}}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & \frac{\sqrt{110}i}{110} & 0 \\ \frac{\sqrt{11}}{220} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{66}}{440} & 0 & \frac{\sqrt{66}i}{330} & \frac{\sqrt{165}}{660} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{440} \\ 0 & -\frac{\sqrt{11}}{220} & 0 & 0 & -\frac{3\sqrt{66}}{440} & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & -\frac{3\sqrt{110}}{440} & 0 \\ 0 & -\frac{\sqrt{11}i}{220} & 0 & -\frac{3\sqrt{11}}{440} & 0 & 0 & \frac{\sqrt{66}}{330} & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & -\frac{3\sqrt{165}}{440} & 0 & 0 \\ \frac{\sqrt{11}i}{220} & 0 & -\frac{3\sqrt{11}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{330} & \frac{\sqrt{165}i}{660} & 0 & -\frac{3\sqrt{165}}{440} & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{11}}{440} & 0 & \frac{\sqrt{11}i}{44} & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{440} & 0 & \frac{\sqrt{165}i}{60} & -\frac{\sqrt{110}}{55} & 0 \\ -\frac{9\sqrt{11}}{440} & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & -\frac{3\sqrt{165}}{440} & 0 & -\frac{\sqrt{165}i}{60} & 0 & 0 & \frac{\sqrt{110}}{55} \\ 0 & 0 & -\frac{\sqrt{33}}{165} & 0 & 0 & -\frac{3\sqrt{22}i}{110} & 0 & -\frac{9\sqrt{22}}{440} & 0 & 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}i}{132} \\ 0 & 0 & 0 & \frac{\sqrt{33}}{165} & \frac{3\sqrt{22}i}{110} & 0 & -\frac{9\sqrt{22}}{440} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{110} & \frac{\sqrt{330}i}{132} & 0 \end{bmatrix}$ |
| 826 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{11}}{220} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{66}}{660} & 0 & \frac{3\sqrt{66}i}{440} & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{110} \\ 0 & -\frac{\sqrt{11}}{220} & 0 & 0 & -\frac{7\sqrt{66}}{660} & 0 & -\frac{3\sqrt{66}i}{440} & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{110}}{110} & 0 \\ 0 & 0 & \frac{\sqrt{11}}{220} & 0 & 0 & \frac{3\sqrt{66}i}{440} & 0 & \frac{\sqrt{66}}{330} & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & -\frac{3\sqrt{110}i}{440} \\ 0 & 0 & 0 & -\frac{\sqrt{11}}{220} & -\frac{3\sqrt{66}i}{440} & 0 & \frac{\sqrt{66}}{330} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{660} & \frac{3\sqrt{110}i}{440} & 0 \\ 0 & -\frac{\sqrt{11}}{44} & 0 & \frac{9\sqrt{11}i}{440} & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{60} & 0 & -\frac{3\sqrt{165}i}{440} & \frac{\sqrt{110}}{55} & 0 \\ -\frac{\sqrt{11}}{44} & 0 & -\frac{9\sqrt{11}i}{440} & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & \frac{\sqrt{165}}{60} & 0 & \frac{3\sqrt{165}i}{440} & 0 & 0 & -\frac{\sqrt{110}}{55} \\ 0 & \frac{3\sqrt{11}i}{440} & 0 & \frac{\sqrt{11}}{220} & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & -\frac{3\sqrt{165}i}{440} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 \\ -\frac{3\sqrt{11}i}{440} & 0 & \frac{\sqrt{11}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{330} & \frac{3\sqrt{165}i}{440} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{165} & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{110} & 0 & -\frac{9\sqrt{22}i}{440} & \frac{3\sqrt{55}}{110} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} \\ 0 & \frac{\sqrt{33}}{165} & 0 & 0 & \frac{3\sqrt{22}}{110} & 0 & \frac{9\sqrt{22}i}{440} & 0 & 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 \end{bmatrix}$ |
| 827 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & -\frac{29\sqrt{462}}{4620} & 0 & \frac{3\sqrt{462}i}{440} & -\frac{\sqrt{1155}}{420} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{1540} & 0 & 0 & -\frac{29\sqrt{462}}{4620} & 0 & -\frac{3\sqrt{462}i}{440} & 0 & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & \frac{3\sqrt{462}i}{440} & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & \frac{17\sqrt{1155}}{4620} & 0 & 0 & \frac{\sqrt{770}i}{440} \\ 0 & 0 & 0 & \frac{\sqrt{77}}{1540} & -\frac{3\sqrt{462}i}{440} & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1155}}{4620} & -\frac{\sqrt{770}i}{440} & 0 \\ 0 & -\frac{3\sqrt{77}}{220} & 0 & \frac{\sqrt{77}i}{88} & -\frac{\sqrt{462}}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 & \frac{\sqrt{1155}i}{440} & -\frac{\sqrt{770}}{385} & 0 \\ -\frac{3\sqrt{77}}{220} & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & \frac{\sqrt{462}}{210} & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{440} & 0 & 0 & \frac{\sqrt{770}}{385} \\ 0 & \frac{7\sqrt{77}i}{440} & 0 & \frac{3\sqrt{77}}{220} & 0 & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & \frac{\sqrt{1155}i}{440} & 0 & -\frac{\sqrt{1155}}{924} & 0 & 0 \\ -\frac{7\sqrt{77}i}{440} & 0 & \frac{3\sqrt{77}}{220} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}}{2310} & -\frac{\sqrt{1155}i}{440} & 0 & -\frac{\sqrt{1155}}{924} & 0 & 0 & 0 \\ -\frac{\sqrt{231}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}i}{440} & -\frac{3\sqrt{385}}{770} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} \\ 0 & \frac{\sqrt{231}}{165} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{440} & 0 & 0 & \frac{3\sqrt{385}}{770} & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 \end{bmatrix}$ |
| 828 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & \frac{29\sqrt{462}i}{4620} & 0 & \frac{3\sqrt{462}}{440} & 0 & 0 & \frac{\sqrt{1155}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}}{1540} & -\frac{29\sqrt{462}i}{4620} & 0 & \frac{3\sqrt{462}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{420} & 0 & 0 \\ \frac{\sqrt{77}}{1540} & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{440} & 0 & -\frac{17\sqrt{462}i}{2310} & \frac{17\sqrt{1155}}{4620} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{440} \\ 0 & -\frac{\sqrt{77}}{1540} & 0 & 0 & \frac{3\sqrt{462}}{440} & 0 & \frac{17\sqrt{462}i}{2310} & 0 & 0 & -\frac{17\sqrt{1155}}{4620} & 0 & 0 & -\frac{\sqrt{770}}{440} & 0 \\ 0 & \frac{3\sqrt{77}i}{220} & 0 & \frac{7\sqrt{77}}{440} & 0 & 0 & \frac{17\sqrt{462}}{2310} & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & -\frac{\sqrt{1155}}{440} & 0 & 0 \\ -\frac{3\sqrt{77}i}{220} & 0 & \frac{7\sqrt{77}}{440} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{462}}{2310} & -\frac{\sqrt{1155}i}{924} & 0 & -\frac{\sqrt{1155}}{440} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{88} & 0 & -\frac{3\sqrt{77}i}{220} & \frac{\sqrt{462}}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{440} & 0 & \frac{\sqrt{1155}i}{924} & -\frac{\sqrt{770}}{385} & 0 \\ \frac{\sqrt{77}}{88} & 0 & \frac{3\sqrt{77}i}{220} & 0 & 0 & -\frac{\sqrt{462}}{210} & 0 & 0 & -\frac{\sqrt{1155}}{440} & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & \frac{\sqrt{770}}{385} \\ 0 & 0 & \frac{\sqrt{231}}{165} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{440} & 0 & 0 & -\frac{3\sqrt{385}}{770} & 0 & 0 & -\frac{\sqrt{2310}i}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{165} & 0 & 0 & -\frac{3\sqrt{154}}{440} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{770} & \frac{\sqrt{2310}i}{924} & 0 \end{bmatrix}$ |

bra: $= \langle f_c,\uparrow|\,,\langle f_c,\downarrow|\,,\langle f_s,\uparrow|\,,\langle f_s,\downarrow|\,,\langle f_z^\beta,\uparrow|\,,\langle f_z^\beta,\downarrow|\,,\langle f_{xyz},\uparrow|\,,\langle f_{xyz},\downarrow|\,,\langle f_x,\uparrow|\,,\langle f_x,\downarrow|\,,\langle f_y,\uparrow|\,,\langle f_y,\downarrow|\,,\langle f_z^\alpha,\uparrow|\,,\langle f_z^\alpha,\downarrow|$

ket: $= |f_c,\uparrow\rangle\,,|f_c,\downarrow\rangle\,,|f_s,\uparrow\rangle\,,|f_s,\downarrow\rangle\,,|f_z^\beta,\uparrow\rangle\,,|f_z^\beta,\downarrow\rangle\,,|f_{xyz},\uparrow\rangle\,,|f_{xyz},\downarrow\rangle\,,|f_x,\uparrow\rangle\,,|f_x,\downarrow\rangle\,,|f_y,\uparrow\rangle\,,|f_y,\downarrow\rangle\,,|f_z^\alpha,\uparrow\rangle\,,|f_z^\alpha,\downarrow\rangle$

Table 10: (f,f) block.

| No. | multipole | matrix |
|---|---|---|
| 829 | symmetry | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_0^{(a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \end{bmatrix}$ |
| 830 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(A_g)$ | $\begin{bmatrix} -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} \end{bmatrix}$ |
| 831 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 832 | $\mathbb{Q}_2^{(a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 833 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 834 | $\mathbb{Q}^{(a)}_{2,1}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 \\ -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{2,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 835 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A_g,1)$ | $\begin{bmatrix} \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} \end{bmatrix}$ |
| 836 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A_g,2)$ | $\begin{pmatrix} \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} \end{pmatrix}$ |
| 837 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 838 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 839 | symmetry | $-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(B_g,2)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} \\ 0 & 0 & \frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 \\ -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 840 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 841 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} \\ 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 \\ -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} \\ 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 \end{pmatrix}$ |
| 842 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{88} \\ 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & 0 & 0 \\ -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} \\ 0 & 0 & -\frac{3\sqrt{154}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{154}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 \end{pmatrix}$ |
| 843 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{154}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{154}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 844 | symmetry | $\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{616} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{616} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{616} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{616} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{462} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{462} \end{bmatrix}$ |
| 845 | symmetry | $-\frac{\sqrt{14}x^6}{8}+\frac{15\sqrt{14}x^4z^2}{8}-\frac{15\sqrt{14}x^2z^4}{8}-\frac{\sqrt{14}y^6}{8}+\frac{15\sqrt{14}y^4z^2}{8}-\frac{15\sqrt{14}y^2z^4}{8}+\frac{\sqrt{14}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{66} \end{bmatrix}$ |
| 846 | symmetry | $-\frac{3\sqrt{7}x^5y}{4}+\frac{15\sqrt{7}x^3yz^2}{2}+\frac{3\sqrt{7}xy^5}{4}-\frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 847 | symmetry | $-\frac{\sqrt{2310}x^4y^2}{8}+\frac{\sqrt{2310}x^4z^2}{8}+\frac{\sqrt{2310}x^2y^4}{8}-\frac{\sqrt{2310}x^2z^4}{8}-\frac{\sqrt{2310}y^4z^2}{8}+\frac{\sqrt{2310}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(B_g,1)$ | $\begin{bmatrix} \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 848 | symmetry | $\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(B_g,2)$ | $\begin{bmatrix} \frac{\sqrt{11}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{11}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 849 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(B_g,3)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 850 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(B_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} \\ 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 \\ \frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 851 | symmetry | $\frac{3\sqrt{7}x^5z}{4} - \frac{15\sqrt{7}x^3y^2z}{2} + \frac{15\sqrt{7}xy^2z^3}{2} - \frac{3\sqrt{7}xz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 852 | symmetry | $\frac{15\sqrt{7}x^2y^3z}{2} - \frac{15\sqrt{7}x^2yz^3}{2} - \frac{3\sqrt{7}y^5z}{4} + \frac{3\sqrt{7}yz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,2}^{(a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} \\ 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{11}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 \\ -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} \\ 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & 0 \end{pmatrix}$ |
| 853 | symmetry | $\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{32} \\ 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} \\ 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 & 0 \end{bmatrix}$ |
| 854 | symmetry | $\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}^{(a)}_{6,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{32} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 855 | symmetry | $\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{352} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{352} \\ 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{32} & 0 & 0 \\ \frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} \\ 0 & 0 & -\frac{9\sqrt{22}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{22}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 & 0 \end{bmatrix}$ |
| 856 | symmetry | $\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,2}^{(a)}(E_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}}{352} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}}{352} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{17\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{22}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{22}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{1056} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 857 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(A_g)$ | $\begin{pmatrix} 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 \\ -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{14}}{56} & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \end{pmatrix}$ |
| 858 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \end{pmatrix}$ |
| 859 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \end{pmatrix}$ |
| 860 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 861 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)$ | $\begin{matrix} 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 \end{matrix}$ |
| 862 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} \\ 0 & 0 & 0 & -\frac{i}{6} & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 \\ -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} \\ 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 \\ 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{6} & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{6} & \frac{\sqrt{6}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & \frac{i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 \\ 0 & -\frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 \end{pmatrix}$ |
| 863 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,2)$ | $\begin{pmatrix} 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{24} & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} \\ 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 \\ 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 \\ \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & -\frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 & \frac{\sqrt{14}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{168} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{24} & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & \frac{\sqrt{210}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & \frac{\sqrt{14}}{24} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 \\ 0 & \frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 \\ -\frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 \end{pmatrix}$ |
| 864 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 865 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{42}}{168} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{42}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 \\ 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 \\ -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & \frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 \\ 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \end{pmatrix}$ |
| 866 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}}{168} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{42}}{168} & 0 \\ 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & \frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 \\ -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 \\ 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{168} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \end{pmatrix}$ |
| 867 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 \\ -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} \\ 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 \\ \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & \frac{i}{8} & 0 \\ 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 \\ 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 \\ 0 & \frac{i}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 \\ \frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} \\ 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 \end{pmatrix}$ |
| 868 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & \frac{\sqrt{6}i}{48} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} \\ 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{1}{8} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & -\frac{1}{8} & 0 \\ 0 & 0 & 0 & -\frac{1}{8} & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{10}i}{16} \\ 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 869 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & -\frac{\sqrt{42}i}{48} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{42}i}{48} \\ 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & -\frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & \frac{3\sqrt{7}}{56} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & \frac{3\sqrt{7}}{56} & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & -\frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & -\frac{\sqrt{70}i}{112} & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & \frac{\sqrt{70}i}{112} \\ 0 & -\frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{56} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{7}}{56} & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 870 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{42}i}{48} & 0 \\ \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{42}i}{48} \\ 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & -\frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & \frac{3\sqrt{7}i}{56} & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{7}i}{56} & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 \\ \frac{3\sqrt{7}i}{56} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} \\ 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{48} & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 \end{pmatrix}$ |
| 871 | symmetry | $\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{154}i}{616} & 0 & 0 & \frac{\sqrt{231}}{1848} & 0 & -\frac{\sqrt{231}i}{308} & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & \frac{\sqrt{385}}{88} \\ 0 & 0 & 0 & \frac{\sqrt{154}i}{616} & -\frac{\sqrt{231}}{1848} & 0 & -\frac{\sqrt{231}i}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & -\frac{\sqrt{385}}{88} & 0 \\ \frac{\sqrt{154}i}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{1848} & 0 & -\frac{\sqrt{231}}{308} & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{88} \\ 0 & -\frac{\sqrt{154}i}{616} & 0 & 0 & -\frac{\sqrt{231}i}{1848} & 0 & \frac{\sqrt{231}}{308} & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & \frac{\sqrt{385}i}{88} & 0 \\ 0 & -\frac{\sqrt{231}}{1848} & 0 & \frac{\sqrt{231}i}{1848} & 0 & 0 & \frac{\sqrt{154}i}{154} & 0 & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{5\sqrt{385}i}{616} & 0 & 0 \\ \frac{\sqrt{231}}{1848} & 0 & \frac{\sqrt{231}i}{1848} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{154} & -\frac{5\sqrt{385}}{616} & 0 & \frac{5\sqrt{385}i}{616} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{308} & 0 & \frac{\sqrt{231}}{308} & -\frac{\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{308} & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 \\ \frac{\sqrt{231}i}{308} & 0 & -\frac{\sqrt{231}}{308} & 0 & 0 & \frac{\sqrt{154}i}{154} & 0 & 0 & \frac{\sqrt{385}i}{308} & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & -\frac{5\sqrt{385}}{616} & 0 & -\frac{\sqrt{385}i}{308} & 0 & 0 & -\frac{5\sqrt{154}i}{616} & 0 & 0 & -\frac{5\sqrt{231}}{1848} \\ 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & \frac{5\sqrt{385}}{616} & 0 & -\frac{\sqrt{385}i}{308} & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{616} & \frac{5\sqrt{231}}{1848} & 0 \\ -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}i}{616} & 0 & \frac{\sqrt{385}}{308} & \frac{5\sqrt{154}i}{616} & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}i}{1848} \\ 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & -\frac{5\sqrt{385}i}{616} & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & -\frac{5\sqrt{154}i}{616} & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 \\ 0 & -\frac{\sqrt{385}}{88} & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & -\frac{5\sqrt{231}i}{1848} & 0 & 0 \\ \frac{\sqrt{385}}{88} & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & -\frac{5\sqrt{231}i}{1848} & 0 & 0 & 0 \end{bmatrix}$ |
| 872 | symmetry | $-\frac{\sqrt{14}x^6}{8}+\frac{15\sqrt{14}x^4z^2}{8}-\frac{15\sqrt{14}x^2z^4}{8}-\frac{\sqrt{14}y^6}{8}+\frac{15\sqrt{14}y^4z^2}{8}-\frac{15\sqrt{14}y^2z^4}{8}+\frac{\sqrt{14}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A_g,2)$ | $\begin{matrix} 0 & 0 & -\frac{\sqrt{22}i}{88} & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{132} & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & -\frac{\sqrt{55}}{88} \\ 0 & 0 & 0 & \frac{\sqrt{22}i}{88} & \frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{264} & \frac{\sqrt{55}}{88} & 0 \\ \frac{\sqrt{22}i}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{88} & 0 & -\frac{\sqrt{33}}{132} & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} \\ 0 & -\frac{\sqrt{22}i}{88} & 0 & 0 & \frac{\sqrt{33}i}{88} & 0 & \frac{\sqrt{33}}{132} & 0 & 0 & \frac{\sqrt{330}i}{264} & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 \\ 0 & \frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{88} & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\ -\frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{33}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & -\frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}i}{132} & 0 & \frac{\sqrt{33}}{132} & -\frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & \frac{\sqrt{55}}{44} & 0 & 0 \\ \frac{\sqrt{33}i}{132} & 0 & -\frac{\sqrt{33}}{132} & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}i}{264} & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & -\frac{5\sqrt{22}i}{88} & 0 & 0 & -\frac{5\sqrt{33}}{264} \\ 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & \frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{88} & \frac{5\sqrt{33}}{264} & 0 \\ \frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{44} & \frac{5\sqrt{22}i}{88} & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{264} \\ 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & -\frac{5\sqrt{22}i}{88} & 0 & 0 & \frac{5\sqrt{33}i}{264} & 0 \\ 0 & \frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & -\frac{5\sqrt{33}i}{264} & 0 & 0 \\ -\frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{264} & 0 & -\frac{5\sqrt{33}i}{264} & 0 & 0 & 0 \end{matrix}$ |
| 873 | symmetry | $-\frac{3\sqrt{7}x^5y}{4}+\frac{15\sqrt{7}x^3yz^2}{2}+\frac{3\sqrt{7}xy^5}{4}-\frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A_g,3)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{264} & \frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{264} & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{264} & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & -\frac{\sqrt{110}}{44} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{264} & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{66} & \frac{\sqrt{110}}{44} & 0 \\ 0 & \frac{\sqrt{66}i}{264} & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 \\ \frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 \\ -\frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} & -\frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{110}i}{44} & 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{110}i}{44} & 0 & -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 874 | symmetry | $-\frac{\sqrt{2310}x^4y^2}{8}+\frac{\sqrt{2310}x^4z^2}{8}+\frac{\sqrt{2310}x^2y^4}{8}-\frac{\sqrt{2310}x^2z^4}{8}-\frac{\sqrt{2310}y^4z^2}{8}+\frac{\sqrt{2310}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{24} & 0 & -\frac{\sqrt{5}i}{12} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{3}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{24} & 0 & -\frac{\sqrt{5}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{3}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{24} & 0 & \frac{\sqrt{5}}{12} & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{24} & 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 \\ 0 & \frac{\sqrt{5}}{24} & 0 & \frac{\sqrt{5}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 \\ -\frac{\sqrt{5}}{24} & 0 & \frac{\sqrt{5}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{12} & 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{3}}{12} & \frac{\sqrt{2}i}{6} & 0 \\ \frac{\sqrt{5}i}{12} & 0 & \frac{\sqrt{5}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{6} \\ 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & -\frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{24} & 0 \\ -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{24} \\ 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{24} & 0 \\ 0 & -\frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & -\frac{\sqrt{5}}{24} & 0 & -\frac{\sqrt{5}i}{24} & 0 & 0 \\ \frac{\sqrt{3}}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & \frac{\sqrt{5}}{24} & 0 & -\frac{\sqrt{5}i}{24} & 0 & 0 & 0 \end{pmatrix}$ |
| 875 | symmetry | $\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(B_g,2)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{11}}{264} & 0 & -\frac{7\sqrt{11}i}{132} & 0 & 0 & -\frac{\sqrt{110}i}{132} & 0 & 0 & -\frac{\sqrt{165}}{264} \\ 0 & 0 & 0 & 0 & \frac{19\sqrt{11}}{264} & 0 & -\frac{7\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{132} & \frac{\sqrt{165}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{11}i}{264} & 0 & \frac{7\sqrt{11}}{132} & \frac{\sqrt{110}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{264} \\ 0 & 0 & 0 & 0 & -\frac{19\sqrt{11}i}{264} & 0 & -\frac{7\sqrt{11}}{132} & 0 & 0 & -\frac{\sqrt{110}i}{132} & 0 & 0 & \frac{\sqrt{165}i}{264} & 0 \\ 0 & \frac{19\sqrt{11}}{264} & 0 & \frac{19\sqrt{11}i}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{264} & 0 & 0 \\ -\frac{19\sqrt{11}}{264} & 0 & \frac{19\sqrt{11}i}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{264} & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{11}i}{132} & 0 & -\frac{7\sqrt{11}}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & \frac{\sqrt{165}}{132} & -\frac{\sqrt{110}i}{66} & 0 \\ \frac{7\sqrt{11}i}{132} & 0 & \frac{7\sqrt{11}}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{110}i}{66} \\ 0 & 0 & -\frac{\sqrt{110}i}{132} & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{264} \\ 0 & 0 & 0 & \frac{\sqrt{110}i}{132} & \frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{264} & 0 \\ \frac{\sqrt{110}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{264} & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{264} \\ 0 & -\frac{\sqrt{110}i}{132} & 0 & 0 & \frac{\sqrt{165}i}{264} & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{264} & 0 \\ 0 & \frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{264} & 0 & 0 & \frac{\sqrt{110}i}{66} & 0 & 0 & \frac{5\sqrt{11}}{264} & 0 & \frac{5\sqrt{11}i}{264} & 0 & 0 \\ -\frac{\sqrt{165}}{264} & 0 & -\frac{\sqrt{165}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{66} & -\frac{5\sqrt{11}}{264} & 0 & \frac{5\sqrt{11}i}{264} & 0 & 0 & 0 \end{matrix}$ |
| 876 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(B_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 877 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(B_g,4)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{132} & 0 & \frac{\sqrt{55}}{132} & -\frac{\sqrt{22}i}{33} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{132} & 0 & -\frac{\sqrt{55}}{132} & 0 & 0 & \frac{\sqrt{22}i}{33} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{132} & 0 & \frac{\sqrt{55}i}{132} & 0 & 0 & -\frac{\sqrt{22}i}{33} & 0 & 0 & -\frac{\sqrt{33}}{66} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{132} & 0 & \frac{\sqrt{55}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{33} & \frac{\sqrt{33}}{66} & 0 \\ 0 & -\frac{\sqrt{55}i}{132} & 0 & \frac{\sqrt{55}}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & -\frac{\sqrt{33}}{33} & \frac{2\sqrt{22}i}{33} & 0 \\ -\frac{\sqrt{55}i}{132} & 0 & -\frac{\sqrt{55}}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & -\frac{2\sqrt{22}i}{33} \\ 0 & -\frac{\sqrt{55}}{132} & 0 & -\frac{\sqrt{55}i}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 \\ \frac{\sqrt{55}}{132} & 0 & -\frac{\sqrt{55}i}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 \\ \frac{\sqrt{22}i}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{66} \\ 0 & -\frac{\sqrt{22}i}{33} & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & \frac{\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{66} & 0 \\ 0 & 0 & \frac{\sqrt{22}i}{33} & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{66} \\ 0 & 0 & 0 & -\frac{\sqrt{22}i}{33} & -\frac{\sqrt{33}}{33} & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{66} & 0 \\ 0 & \frac{\sqrt{33}i}{66} & 0 & \frac{\sqrt{33}}{66} & -\frac{2\sqrt{22}i}{33} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{66} & 0 & \frac{\sqrt{55}}{66} & 0 & 0 \\ \frac{\sqrt{33}i}{66} & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & \frac{2\sqrt{22}i}{33} & 0 & 0 & -\frac{\sqrt{55}i}{66} & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & 0 \end{matrix}$ |
| 878 | symmetry | $\frac{3\sqrt{7}x^5z}{4} - \frac{15\sqrt{7}x^3y^2z}{2} + \frac{15\sqrt{7}xy^2z^3}{2} - \frac{3\sqrt{7}xz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{11}i}{176} & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & -\frac{5\sqrt{165}i}{528} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{11}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & \frac{\sqrt{165}}{66} & 0 & -\frac{5\sqrt{165}i}{528} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{11}i}{176} & 0 & 0 & -\frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{176} & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 \\ -\frac{\sqrt{11}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{176} & 0 & 0 & -\frac{3\sqrt{165}i}{176} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{176} \\ 0 & 0 & \frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 & 0 & \frac{\sqrt{165}}{66} \\ 0 & 0 & 0 & -\frac{\sqrt{66}i}{176} & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{176} & -\frac{\sqrt{165}}{66} & 0 \\ \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} \\ 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 \\ 0 & \frac{\sqrt{165}}{66} & 0 & \frac{3\sqrt{165}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{176} & 0 & 0 \\ -\frac{\sqrt{165}}{66} & 0 & \frac{3\sqrt{165}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{176} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{165}i}{528} & 0 & 0 & -\frac{3\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{176} & 0 & 0 & \frac{5\sqrt{66}i}{528} & 0 \\ \frac{5\sqrt{165}i}{528} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 & 0 & -\frac{5\sqrt{11}i}{176} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{528} \\ 0 & 0 & -\frac{3\sqrt{110}i}{176} & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & -\frac{5\sqrt{66}i}{528} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{110}i}{176} & \frac{\sqrt{165}}{66} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{528} & 0 & 0 \end{pmatrix}$ |
| 879 | symmetry | $\frac{15\sqrt{7}x^2y^3z}{2} - \frac{15\sqrt{7}x^2yz^3}{2} - \frac{3\sqrt{7}y^5z}{4} + \frac{3\sqrt{7}yz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{11}}{176} & -\frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{176} & -\frac{3\sqrt{110}i}{176} & 0 \\ 0 & 0 & \frac{\sqrt{11}}{176} & 0 & 0 & \frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{176} & 0 & 0 & \frac{3\sqrt{110}i}{176} \\ 0 & \frac{\sqrt{11}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & \frac{5\sqrt{165}}{528} & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 \\ -\frac{\sqrt{11}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & -\frac{5\sqrt{165}}{528} & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & 0 \\ \frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{44} & -\frac{3\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} \\ 0 & -\frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 \\ 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & \frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{165}}{528} & \frac{3\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{176} & \frac{5\sqrt{66}i}{528} & 0 \\ 0 & 0 & \frac{5\sqrt{165}}{528} & 0 & 0 & -\frac{3\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{176} & 0 & 0 & -\frac{5\sqrt{66}i}{528} \\ 0 & -\frac{3\sqrt{165}}{176} & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{176} & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{165}}{176} & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{176} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{110}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{66} & 0 & -\frac{\sqrt{165}}{132} & -\frac{5\sqrt{66}i}{528} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{110}i}{176} & 0 & 0 & \frac{\sqrt{165}i}{66} & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & \frac{5\sqrt{66}i}{528} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 880 | symmetry | $\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{64} & \frac{3i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{64} & \frac{\sqrt{15}i}{32} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{64} & 0 & 0 & -\frac{3i}{32} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{64} & 0 & 0 & -\frac{\sqrt{15}i}{32} \\ 0 & \frac{\sqrt{6}}{64} & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{16} & -\frac{\sqrt{10}}{64} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{16} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & -\frac{\sqrt{10}}{16} \\ 0 & 0 & 0 & \frac{i}{16} & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & \frac{\sqrt{10}}{16} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{64} & \frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{64} & \frac{5i}{32} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{64} & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{64} & 0 & 0 & -\frac{5i}{32} \\ 0 & \frac{3\sqrt{10}}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & \frac{5\sqrt{6}}{64} & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & -\frac{5\sqrt{6}}{64} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & -\frac{5i}{32} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & \frac{5i}{32} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 881 | symmetry | $\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}i}{64} & 0 & 0 & \frac{i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{64} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{64} & 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & 0 & \frac{\sqrt{10}i}{64} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{64} & 0 & 0 & -\frac{3i}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{64} & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 \\ \frac{\sqrt{6}i}{64} & 0 & 0 & 0 & 0 & \frac{3i}{32} & 0 & 0 & -\frac{3\sqrt{10}i}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{32} \\ 0 & 0 & \frac{3i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{32} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 \\ -\frac{i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} \\ 0 & \frac{i}{16} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{64} & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{64} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{10}i}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & -\frac{5\sqrt{6}i}{64} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{64} & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{64} & 0 & 0 & -\frac{5i}{32} & 0 \\ -\frac{\sqrt{10}i}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & \frac{5\sqrt{6}i}{64} & 0 & 0 & 0 & 0 & \frac{5i}{32} \\ 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & \frac{5i}{32} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{5i}{32} & 0 & 0 \end{pmatrix}$ |
| 882 | symmetry | $\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{330}}{2112} & \frac{19\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & \frac{2\sqrt{22}i}{33} & 0 & -\frac{47\sqrt{22}}{2112} & -\frac{9\sqrt{33}i}{352} & 0 \\ 0 & 0 & \frac{\sqrt{330}}{2112} & 0 & 0 & -\frac{19\sqrt{55}i}{1056} & 0 & 0 & \frac{2\sqrt{22}i}{33} & 0 & \frac{47\sqrt{22}}{2112} & 0 & 0 & \frac{9\sqrt{33}i}{352} \\ 0 & \frac{\sqrt{330}}{2112} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{528} & 0 & 0 & -\frac{91\sqrt{22}}{2112} & 0 & -\frac{\sqrt{22}i}{33} & 0 & 0 \\ -\frac{\sqrt{330}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{55}i}{528} & \frac{91\sqrt{22}}{2112} & 0 & -\frac{\sqrt{22}i}{33} & 0 & 0 & 0 \\ -\frac{19\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{528} & -\frac{7\sqrt{33}i}{352} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{33} \\ 0 & \frac{19\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{528} & 0 & 0 & \frac{7\sqrt{33}i}{352} & 0 & 0 & -\frac{\sqrt{22}i}{33} & 0 \\ 0 & 0 & \frac{7\sqrt{55}i}{528} & 0 & 0 & -\frac{\sqrt{330}}{528} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{176} & 0 & 0 & \frac{\sqrt{22}}{48} \\ 0 & 0 & 0 & -\frac{7\sqrt{55}i}{528} & \frac{\sqrt{330}}{528} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{176} & -\frac{\sqrt{22}}{48} & 0 \\ 0 & -\frac{2\sqrt{22}i}{33} & 0 & \frac{91\sqrt{22}}{2112} & \frac{7\sqrt{33}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{330}}{2112} & \frac{5\sqrt{55}i}{1056} & 0 \\ -\frac{2\sqrt{22}i}{33} & 0 & -\frac{91\sqrt{22}}{2112} & 0 & 0 & -\frac{7\sqrt{33}i}{352} & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}}{2112} & 0 & 0 & -\frac{5\sqrt{55}i}{1056} \\ 0 & \frac{47\sqrt{22}}{2112} & 0 & \frac{\sqrt{22}i}{33} & 0 & 0 & -\frac{\sqrt{33}i}{176} & 0 & 0 & \frac{5\sqrt{330}}{2112} & 0 & 0 & 0 & 0 \\ -\frac{47\sqrt{22}}{2112} & 0 & \frac{\sqrt{22}i}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{176} & -\frac{5\sqrt{330}}{2112} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{33}i}{352} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{33} & 0 & -\frac{\sqrt{22}}{48} & -\frac{5\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{33}i}{352} & 0 & 0 & \frac{\sqrt{22}i}{33} & 0 & \frac{\sqrt{22}}{48} & 0 & 0 & \frac{5\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 883 | symmetry | $\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 884 | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{330}i}{2112} & 0 & 0 & -\frac{7\sqrt{55}i}{528} & 0 & 0 & -\frac{\sqrt{22}}{33} & 0 & -\frac{91\sqrt{22}i}{2112} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}i}{2112} & 0 & 0 & 0 & 0 & \frac{7\sqrt{55}i}{528} & \frac{\sqrt{22}}{33} & 0 & -\frac{91\sqrt{22}i}{2112} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{2112} & 0 & 0 & -\frac{19\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & -\frac{47\sqrt{22}i}{2112} & 0 & \frac{2\sqrt{22}}{33} & -\frac{9\sqrt{33}i}{352} & 0 \\ \frac{\sqrt{330}i}{2112} & 0 & 0 & 0 & 0 & \frac{19\sqrt{55}i}{1056} & 0 & 0 & -\frac{47\sqrt{22}i}{2112} & 0 & -\frac{2\sqrt{22}}{33} & 0 & 0 & \frac{9\sqrt{33}i}{352} \\ 0 & 0 & \frac{19\sqrt{55}i}{1056} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{528} & 0 & 0 & -\frac{7\sqrt{33}i}{352} & 0 & 0 & -\frac{\sqrt{22}}{33} \\ 0 & 0 & 0 & -\frac{19\sqrt{55}i}{1056} & 0 & 0 & \frac{\sqrt{330}i}{528} & 0 & 0 & 0 & 0 & \frac{7\sqrt{33}i}{352} & \frac{\sqrt{22}}{33} & 0 \\ \frac{7\sqrt{55}i}{528} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{528} & 0 & 0 & -\frac{\sqrt{33}i}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{48} \\ 0 & -\frac{7\sqrt{55}i}{528} & 0 & 0 & -\frac{\sqrt{330}i}{528} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{176} & 0 & 0 & -\frac{\sqrt{22}i}{48} & 0 \\ 0 & \frac{\sqrt{22}}{33} & 0 & \frac{47\sqrt{22}i}{2112} & 0 & 0 & \frac{\sqrt{33}i}{176} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & 0 \\ -\frac{\sqrt{22}}{33} & 0 & \frac{47\sqrt{22}i}{2112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{176} & 0 & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & 0 & 0 \\ 0 & \frac{91\sqrt{22}i}{2112} & 0 & -\frac{2\sqrt{22}}{33} & \frac{7\sqrt{33}i}{352} & 0 & 0 & 0 & 0 & \frac{5\sqrt{330}i}{2112} & 0 & 0 & -\frac{5\sqrt{55}i}{1056} & 0 \\ \frac{91\sqrt{22}i}{2112} & 0 & \frac{2\sqrt{22}}{33} & 0 & 0 & -\frac{7\sqrt{33}i}{352} & 0 & 0 & \frac{5\sqrt{330}i}{2112} & 0 & 0 & 0 & 0 & \frac{5\sqrt{55}i}{1056} \\ 0 & 0 & \frac{9\sqrt{33}i}{352} & 0 & 0 & \frac{\sqrt{22}}{33} & 0 & \frac{\sqrt{22}i}{48} & 0 & 0 & \frac{5\sqrt{55}i}{1056} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{33}i}{352} & -\frac{\sqrt{22}}{33} & 0 & \frac{\sqrt{22}i}{48} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{55}i}{1056} & 0 & 0 \end{pmatrix}$ |
| | symmetry | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_0^{(1,1;a)}(A_g)$ | $\begin{pmatrix} 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \end{pmatrix}$ |
| 885 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{42}}{42} & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{42} & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{42} & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{42} & 0 & \frac{\sqrt{42}}{42} & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 886 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{84} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{84} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 \\ 0 & \frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 \\ -\frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & -\frac{\sqrt{35}i}{42} & 0 \\ -\frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} \\ 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 \\ 0 & \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 \\ -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 887 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{84} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{210}}{84} & 0 \\ 0 & -\frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & \frac{\sqrt{35}i}{42} & 0 \\ -\frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & -\frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 \\ \frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} \\ 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & \frac{\sqrt{210}}{84} & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 \\ 0 & \frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 \\ \frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 \end{pmatrix}$ |
| 888 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 889 | $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} \\ 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{14}i}{21} & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{14}i}{21} \\ 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 \\ 0 & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 \\ \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 \end{bmatrix}$ |
| 890 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & -\frac{\sqrt{110}i}{264} & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}}{132} \\ 0 & 0 & 0 & \frac{\sqrt{110}i}{264} & -\frac{\sqrt{165}}{66} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{88} & -\frac{\sqrt{11}}{132} & 0 \\ \frac{\sqrt{110}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & -\frac{\sqrt{165}}{132} & -\frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{132} \\ 0 & -\frac{\sqrt{110}i}{264} & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}i}{132} & 0 \\ 0 & -\frac{\sqrt{165}}{66} & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & \frac{\sqrt{110}i}{66} & 0 & 0 & -\frac{\sqrt{11}}{33} & 0 & -\frac{\sqrt{11}i}{33} & 0 & 0 \\ \frac{\sqrt{165}}{66} & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{66} & \frac{\sqrt{11}}{33} & 0 & -\frac{\sqrt{11}i}{33} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}i}{132} & 0 & \frac{\sqrt{165}}{132} & -\frac{\sqrt{110}i}{66} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{132} & 0 & -\frac{5\sqrt{11}}{132} & 0 & 0 \\ \frac{\sqrt{165}i}{132} & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & \frac{\sqrt{110}i}{66} & 0 & 0 & \frac{5\sqrt{11}i}{132} & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}}{33} & 0 & -\frac{5\sqrt{11}i}{132} & 0 & 0 & -\frac{5\sqrt{110}i}{264} & 0 & 0 & \frac{\sqrt{165}}{132} \\ 0 & 0 & 0 & -\frac{\sqrt{66}i}{88} & -\frac{\sqrt{11}}{33} & 0 & -\frac{5\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & \frac{5\sqrt{110}i}{264} & -\frac{\sqrt{165}}{132} & 0 \\ \frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{33} & 0 & \frac{5\sqrt{11}}{132} & \frac{5\sqrt{110}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{132} \\ 0 & -\frac{\sqrt{66}i}{88} & 0 & 0 & \frac{\sqrt{11}i}{33} & 0 & -\frac{5\sqrt{11}}{132} & 0 & 0 & -\frac{5\sqrt{110}i}{264} & 0 & 0 & -\frac{\sqrt{165}i}{132} & 0 \\ 0 & -\frac{\sqrt{11}}{132} & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & \frac{\sqrt{165}i}{132} & 0 & 0 \\ \frac{\sqrt{11}}{132} & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & \frac{\sqrt{165}i}{132} & 0 & 0 & 0 \end{pmatrix}$ |
| 891 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{5\sqrt{154}i}{1848} & 0 & 0 & -\frac{2\sqrt{231}}{231} & 0 & \frac{13\sqrt{231}i}{924} & 0 & 0 & \frac{\sqrt{2310}i}{440} & 0 & 0 & -\frac{\sqrt{385}}{660} \\ 0 & 0 & 0 & \frac{5\sqrt{154}i}{1848} & \frac{2\sqrt{231}}{231} & 0 & \frac{13\sqrt{231}i}{924} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{440} & \frac{\sqrt{385}}{660} & 0 \\ \frac{5\sqrt{154}i}{1848} & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}i}{231} & 0 & \frac{13\sqrt{231}}{924} & \frac{\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{660} \\ 0 & -\frac{5\sqrt{154}i}{1848} & 0 & 0 & \frac{2\sqrt{231}i}{231} & 0 & -\frac{13\sqrt{231}}{924} & 0 & 0 & -\frac{\sqrt{2310}i}{440} & 0 & 0 & -\frac{\sqrt{385}i}{660} & 0 \\ 0 & \frac{2\sqrt{231}}{231} & 0 & -\frac{2\sqrt{231}i}{231} & 0 & 0 & \frac{5\sqrt{154}i}{462} & 0 & 0 & -\frac{13\sqrt{385}}{2310} & 0 & -\frac{13\sqrt{385}i}{2310} & 0 & 0 \\ -\frac{2\sqrt{231}}{231} & 0 & -\frac{2\sqrt{231}i}{231} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{462} & \frac{13\sqrt{385}}{2310} & 0 & -\frac{13\sqrt{385}i}{2310} & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{231}i}{924} & 0 & -\frac{13\sqrt{231}}{924} & -\frac{5\sqrt{154}i}{462} & 0 & 0 & 0 & 0 & \frac{19\sqrt{385}i}{4620} & 0 & -\frac{19\sqrt{385}}{4620} & 0 & 0 \\ -\frac{13\sqrt{231}i}{924} & 0 & \frac{13\sqrt{231}}{924} & 0 & 0 & \frac{5\sqrt{154}i}{462} & 0 & 0 & \frac{19\sqrt{385}i}{4620} & 0 & \frac{19\sqrt{385}}{4620} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}i}{440} & 0 & 0 & \frac{13\sqrt{385}}{2310} & 0 & -\frac{19\sqrt{385}i}{4620} & 0 & 0 & -\frac{25\sqrt{154}i}{1848} & 0 & 0 & \frac{5\sqrt{231}}{924} \\ 0 & 0 & 0 & \frac{\sqrt{2310}i}{440} & -\frac{13\sqrt{385}}{2310} & 0 & -\frac{19\sqrt{385}i}{4620} & 0 & 0 & 0 & 0 & \frac{25\sqrt{154}i}{1848} & -\frac{5\sqrt{231}}{924} & 0 \\ -\frac{\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}i}{2310} & 0 & \frac{19\sqrt{385}}{4620} & \frac{25\sqrt{154}i}{1848} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{924} \\ 0 & \frac{\sqrt{2310}i}{440} & 0 & 0 & \frac{13\sqrt{385}i}{2310} & 0 & -\frac{19\sqrt{385}}{4620} & 0 & 0 & -\frac{25\sqrt{154}i}{1848} & 0 & 0 & -\frac{5\sqrt{231}i}{924} & 0 \\ 0 & \frac{\sqrt{385}}{660} & 0 & \frac{\sqrt{385}i}{660} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{924} & 0 & \frac{5\sqrt{231}i}{924} & 0 & 0 \\ -\frac{\sqrt{385}}{660} & 0 & \frac{\sqrt{385}i}{660} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{924} & 0 & \frac{5\sqrt{231}i}{924} & 0 & 0 & 0 \end{bmatrix}$ |
| 892 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{11}}{44} & \frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{330} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{44} & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & -\frac{3\sqrt{110}i}{220} & 0 & 0 & -\frac{\sqrt{165}i}{330} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} & 0 & -\frac{3\sqrt{11}i}{44} & 0 & 0 & -\frac{3\sqrt{110}i}{220} & 0 & 0 & \frac{\sqrt{165}}{330} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & -\frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{220} & -\frac{\sqrt{165}}{330} & 0 \\ 0 & -\frac{3\sqrt{11}i}{44} & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & \frac{\sqrt{165}}{660} & 0 & 0 \\ -\frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{660} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{660} & 0 & \frac{\sqrt{165}i}{660} & 0 & 0 \\ \frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & \frac{\sqrt{165}i}{660} & 0 & 0 & 0 \\ -\frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{660} & 0 & -\frac{\sqrt{165}}{660} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & \frac{\sqrt{165}i}{660} & 0 & \frac{\sqrt{165}}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & -\frac{\sqrt{165}}{660} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{110}i}{220} & \frac{\sqrt{165}}{660} & 0 & -\frac{\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}i}{330} & 0 & -\frac{\sqrt{165}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{165}i}{330} & 0 & \frac{\sqrt{165}}{330} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 893 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & -\frac{\sqrt{1155}}{165} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{220} & \frac{\sqrt{1155}}{165} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & -\frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{165} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & \frac{\sqrt{77}}{308} & 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{1155}i}{165} & 0 \\ 0 & \frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 \\ -\frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{77}i}{308} & 0 & \frac{\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & \frac{23\sqrt{1155}}{4620} & \frac{\sqrt{770}i}{110} & 0 \\ -\frac{\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & -\frac{23\sqrt{1155}}{4620} & 0 & 0 & -\frac{\sqrt{770}i}{110} \\ 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & -\frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} \\ 0 & 0 & 0 & -\frac{\sqrt{770}i}{220} & -\frac{\sqrt{1155}}{924} & 0 & -\frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 \\ -\frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & -\frac{23\sqrt{1155}}{4620} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} \\ 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & \frac{23\sqrt{1155}}{4620} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} & 0 \\ 0 & \frac{\sqrt{1155}}{165} & 0 & -\frac{\sqrt{1155}i}{165} & 0 & 0 & -\frac{\sqrt{770}i}{110} & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & \frac{\sqrt{77}i}{154} & 0 & 0 \\ -\frac{\sqrt{1155}}{165} & 0 & -\frac{\sqrt{1155}i}{165} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & -\frac{\sqrt{77}}{154} & 0 & \frac{\sqrt{77}i}{154} & 0 & 0 & 0 \end{bmatrix}$ |
| 894 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & -\frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{165} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & \frac{\sqrt{77}}{308} & 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{1155}i}{165} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{308} & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & -\frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{1155}}{165} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{308} & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{220} & -\frac{\sqrt{1155}}{165} & 0 \\ 0 & \frac{\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & \frac{23\sqrt{1155}}{4620} & \frac{\sqrt{770}i}{110} & 0 \\ \frac{\sqrt{77}i}{308} & 0 & \frac{\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & -\frac{23\sqrt{1155}}{4620} & 0 & 0 & -\frac{\sqrt{770}i}{110} \\ 0 & \frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 \\ -\frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & 0 \\ \frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1155}i}{4620} & 0 & -\frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} \\ 0 & -\frac{\sqrt{770}i}{220} & 0 & 0 & -\frac{23\sqrt{1155}i}{4620} & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} & 0 \\ 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & -\frac{23\sqrt{1155}}{4620} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} \\ 0 & 0 & 0 & -\frac{\sqrt{770}i}{220} & \frac{23\sqrt{1155}}{4620} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 \\ 0 & -\frac{\sqrt{1155}i}{165} & 0 & -\frac{\sqrt{1155}}{165} & -\frac{\sqrt{770}i}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 \\ -\frac{\sqrt{1155}i}{165} & 0 & \frac{\sqrt{1155}}{165} & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 \end{bmatrix}$ |
| 895 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{3\sqrt{110}}{220} & 0 & -\frac{\sqrt{110}i}{440} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & \frac{3\sqrt{110}}{220} & 0 & -\frac{\sqrt{110}i}{440} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}i}{264} & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & \frac{7\sqrt{110}i}{440} & 0 & 0 & \frac{\sqrt{165}i}{165} & 0 \\ -\frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & \frac{7\sqrt{110}i}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{165} \\ 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{7\sqrt{165}i}{660} & 0 & 0 & \frac{3\sqrt{110}}{220} \\ 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & \frac{7\sqrt{165}i}{660} & -\frac{3\sqrt{110}}{220} & 0 \\ -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{165}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{55} \\ 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{60} & 0 & 0 & -\frac{\sqrt{110}i}{55} & 0 \\ 0 & \frac{3\sqrt{110}}{220} & 0 & -\frac{7\sqrt{110}i}{440} & 0 & 0 & -\frac{\sqrt{165}i}{60} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 \\ -\frac{3\sqrt{110}}{220} & 0 & -\frac{7\sqrt{110}i}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{60} & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}i}{440} & 0 & 0 & \frac{7\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 \\ \frac{\sqrt{110}i}{440} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}i}{660} & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} \\ 0 & 0 & -\frac{\sqrt{165}i}{165} & 0 & 0 & -\frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{110}i}{55} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{165}i}{165} & \frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{110}i}{55} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 \end{pmatrix}$ |
| 896 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{66}}{264} & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}}{440} & -\frac{\sqrt{165}i}{165} & 0 \\ 0 & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & \frac{7\sqrt{110}}{440} & 0 & 0 & \frac{\sqrt{165}i}{165} \\ 0 & \frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & \frac{\sqrt{110}}{440} & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 \\ -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & -\frac{\sqrt{110}}{440} & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & 0 \\ \frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & \frac{7\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{220} \\ 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & -\frac{7\sqrt{165}i}{660} & 0 & 0 & -\frac{3\sqrt{110}i}{220} & 0 \\ 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{60} & 0 & 0 & -\frac{\sqrt{110}}{55} \\ 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & \frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{60} & \frac{\sqrt{110}}{55} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{110}}{440} & -\frac{7\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & -\frac{\sqrt{11}i}{22} & 0 \\ 0 & 0 & \frac{\sqrt{110}}{440} & 0 & 0 & \frac{7\sqrt{165}i}{660} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & \frac{\sqrt{11}i}{22} \\ 0 & \frac{7\sqrt{110}}{440} & 0 & -\frac{3\sqrt{110}i}{220} & 0 & 0 & -\frac{\sqrt{165}i}{60} & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 \\ -\frac{7\sqrt{110}}{440} & 0 & -\frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{60} & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{165}i}{165} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{220} & 0 & \frac{\sqrt{110}}{55} & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{165}i}{165} & 0 & 0 & \frac{3\sqrt{110}i}{220} & 0 & -\frac{\sqrt{110}}{55} & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 897 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{462}}{1848} & -\frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 & \frac{3\sqrt{770}}{440} & \frac{\sqrt{1155}i}{165} & 0 \\ 0 & 0 & \frac{\sqrt{462}}{1848} & 0 & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 & -\frac{3\sqrt{770}}{440} & 0 & 0 & -\frac{\sqrt{1155}i}{165} \\ 0 & \frac{\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & \frac{3\sqrt{770}}{440} & 0 & -\frac{\sqrt{770}i}{220} & 0 & 0 \\ -\frac{\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{308} & -\frac{3\sqrt{770}}{440} & 0 & -\frac{\sqrt{770}i}{220} & 0 & 0 & 0 \\ \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{462} & \frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{220} \\ 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{462} & 0 & 0 & -\frac{23\sqrt{1155}i}{4620} & 0 & 0 & -\frac{\sqrt{770}i}{220} & 0 \\ 0 & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & -\frac{\sqrt{462}}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & \frac{\sqrt{462}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 \\ 0 & -\frac{\sqrt{770}i}{110} & 0 & -\frac{3\sqrt{770}}{440} & -\frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & -\frac{\sqrt{77}i}{154} & 0 \\ -\frac{\sqrt{770}i}{110} & 0 & \frac{3\sqrt{770}}{440} & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & \frac{\sqrt{77}i}{154} \\ 0 & -\frac{3\sqrt{770}}{440} & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{770}}{440} & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{924} & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1155}i}{165} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}i}{165} & 0 & 0 & \frac{\sqrt{770}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 898 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 899 | symmetry | $z$ |

$\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)$

$$\begin{pmatrix}
0 & 0 & 0 & -\frac{\sqrt{462}i}{1848} & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & -\frac{\sqrt{770}}{220} & 0 & \frac{3\sqrt{770}i}{440} & 0 & 0 \\
0 & 0 & -\frac{\sqrt{462}i}{1848} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{308} & \frac{\sqrt{770}}{220} & 0 & \frac{3\sqrt{770}i}{440} & 0 & 0 & 0 \\
0 & \frac{\sqrt{462}i}{1848} & 0 & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}i}{440} & 0 & \frac{\sqrt{770}}{110} & \frac{\sqrt{1155}i}{165} & 0 \\
\frac{\sqrt{462}i}{1848} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & \frac{3\sqrt{770}i}{440} & 0 & -\frac{\sqrt{770}}{110} & 0 & 0 & -\frac{\sqrt{1155}i}{165} \\
0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & 0 & -\frac{\sqrt{770}}{220} \\
0 & 0 & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & \frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1155}i}{4620} & \frac{\sqrt{770}}{220} & 0 \\
\frac{\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{770}}{220} & 0 & -\frac{3\sqrt{770}i}{440} & 0 & 0 & -\frac{\sqrt{1155}i}{924} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{1848} & 0 & 0 \\
-\frac{\sqrt{770}}{220} & 0 & -\frac{3\sqrt{770}i}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & -\frac{5\sqrt{462}i}{1848} & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{770}i}{440} & 0 & -\frac{\sqrt{770}}{110} & -\frac{23\sqrt{1155}i}{4620} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{1848} & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 \\
-\frac{3\sqrt{770}i}{440} & 0 & \frac{\sqrt{770}}{110} & 0 & 0 & \frac{23\sqrt{1155}i}{4620} & 0 & 0 & \frac{5\sqrt{462}i}{1848} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} \\
0 & 0 & -\frac{\sqrt{1155}i}{165} & 0 & 0 & \frac{\sqrt{770}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{1155}i}{165} & -\frac{\sqrt{770}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & 0
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(A_g)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \end{pmatrix}$ |
| 900 | symmetry | $y$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 901 | symmetry | $x$ |

$\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$

$$\begin{pmatrix}
0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\
-\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,2}^{(1,0;a)}(E_g)$ | $\begin{matrix} 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \end{matrix}$ |
| 902 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$ |
| 903 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(B_g,1)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & -\frac{i}{6} & -\frac{\sqrt{6}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 \\ 0 & \frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 \\ -\frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & -\frac{i}{6} & 0 \\ -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{i}{6} \\ 0 & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} \\ 0 & 0 & 0 & -\frac{i}{6} & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 \\ -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} \\ 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 \\ 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & -\frac{i}{6} & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 \end{matrix}$ |
| 904 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & -\frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & \frac{i}{6} & \frac{\sqrt{6}}{24} & 0 \\ 0 & \frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & -\frac{i}{6} & 0 \\ \frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{i}{6} \\ 0 & \frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 \\ -\frac{\sqrt{10}}{24} & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 \\ \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} \\ 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 \\ 0 & 0 & \frac{i}{6} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{24} \\ 0 & 0 & 0 & -\frac{i}{6} & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 \\ \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{i}{6} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 \end{pmatrix}$ |
| 905 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 \\ 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{i}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} \\ 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 \\ -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 \\ -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} \\ 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 \end{pmatrix}$ |
| 906 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{1}{8} & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & \frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & \frac{1}{8} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 \\ -\frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & \frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{24} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{\sqrt{6}i}{48} & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & -\frac{\sqrt{6}i}{48} \\ 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 907 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 908 | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & \frac{1}{24} & -\frac{\sqrt{6}i}{16} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & -\frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & \frac{\sqrt{6}i}{16} \\ 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & -\frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 \\ -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{48} & \frac{1}{24} & 0 & -\frac{i}{6} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{6} \\ 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{i}{6} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{1}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{48} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & \frac{1}{24} & 0 \\ 0 & \frac{i}{6} & 0 & \frac{1}{24} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & \frac{\sqrt{10}i}{48} & 0 \\ \frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & -\frac{\sqrt{10}i}{48} \\ 0 & -\frac{1}{24} & 0 & \frac{i}{6} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 \\ \frac{1}{24} & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & \frac{1}{24} & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{i}{6} & 0 & -\frac{1}{24} & 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & -\frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{48} & \frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & -\frac{1}{6} & -\frac{\sqrt{6}i}{16} & 0 \\ \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & \frac{i}{24} & 0 & \frac{1}{6} & 0 & 0 & \frac{\sqrt{6}i}{16} \\ 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{1}{6} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & \frac{1}{6} & 0 \\ \frac{\sqrt{10}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{i}{24} \\ 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & \frac{i}{24} & 0 \\ 0 & \frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 \\ -\frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 \\ 0 & \frac{i}{24} & 0 & \frac{1}{6} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 \\ \frac{i}{24} & 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{48} \\ 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & \frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & \frac{\sqrt{10}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & -\frac{1}{6} & 0 & -\frac{i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{48} & 0 & 0 \end{pmatrix}$ |
| 909 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A_g,1)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{10}}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & \frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 910 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 911 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A_g,3)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & \frac{\sqrt{10}}{20} & 0 \\ 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 912 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}}{30} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & -\frac{\sqrt{30}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{2}}{24} & -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 \\ 0 & \frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ -\frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{24} & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & \frac{\sqrt{5}i}{15} & 0 \\ -\frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & -\frac{\sqrt{5}i}{15} \\ 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 \\ -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} \\ 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 \\ 0 & -\frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 \\ \frac{\sqrt{30}}{30} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{15} & -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 \end{pmatrix}$ |
| 913 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & \frac{\sqrt{2}}{24} & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}}{30} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & -\frac{\sqrt{30}}{30} & 0 \\ 0 & -\frac{\sqrt{2}i}{24} & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{120} & -\frac{\sqrt{5}i}{15} & 0 \\ -\frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & \frac{\sqrt{5}i}{15} \\ 0 & -\frac{\sqrt{2}}{24} & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ \frac{\sqrt{2}}{24} & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} \\ 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & -\frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 \\ 0 & -\frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{30}}{30} & \frac{\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & \frac{\sqrt{2}}{12} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 \end{pmatrix}$ |
| 914 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 915 | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{96} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & \frac{5\sqrt{21}i}{96} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{224} & 0 & 0 & -\frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{96} & 0 & 0 & -\frac{\sqrt{14}i}{32} & 0 \\ -\frac{\sqrt{35}i}{224} & 0 & 0 & 0 & 0 & \frac{11\sqrt{210}i}{672} & 0 & 0 & \frac{\sqrt{21}i}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{32} \\ 0 & 0 & \frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{11\sqrt{210}i}{672} & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 \\ \frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} \\ 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}i}{24} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{96} & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}i}{224} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{96} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{5\sqrt{35}i}{224} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{96} & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{224} & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 \\ -\frac{5\sqrt{21}i}{96} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & -\frac{5\sqrt{35}i}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{672} \\ 0 & 0 & \frac{\sqrt{14}i}{32} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & \frac{\sqrt{210}i}{672} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{32} & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & 0 \end{pmatrix}$ |
| | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{35}}{224} & -\frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{96} & \frac{\sqrt{14}i}{32} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{224} & 0 & 0 & \frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{96} & 0 & 0 & -\frac{\sqrt{14}i}{32} \\ 0 & \frac{\sqrt{35}}{224} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & -\frac{5\sqrt{21}}{96} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & \frac{5\sqrt{21}}{96} & 0 & 0 & 0 & 0 & 0 \\ \frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{56} & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{21}}{24} \\ 0 & 0 & 0 & \frac{5\sqrt{210}i}{336} & \frac{\sqrt{35}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & -\frac{\sqrt{21}}{24} & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}}{96} & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{224} & -\frac{\sqrt{210}i}{672} & 0 \\ 0 & 0 & -\frac{5\sqrt{21}}{96} & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}}{224} & 0 & 0 & \frac{\sqrt{210}i}{672} \\ 0 & \frac{\sqrt{21}}{96} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & \frac{5\sqrt{35}}{224} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & -\frac{5\sqrt{35}}{224} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & \frac{\sqrt{210}i}{672} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{24} & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 916 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{32} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & -\frac{13\sqrt{15}i}{480} & 0 & 0 \\ 0 & 0 & \frac{i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & -\frac{\sqrt{15}}{15} & 0 & -\frac{13\sqrt{15}i}{480} & 0 & 0 & 0 \\ 0 & -\frac{i}{32} & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{160} & 0 & 0 & \frac{9\sqrt{10}i}{160} & 0 \\ -\frac{i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{3\sqrt{15}i}{160} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{160} \\ 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{160} & \frac{\sqrt{15}}{15} & 0 \\ \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{120} \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}i}{120} & 0 \\ 0 & -\frac{\sqrt{15}}{15} & 0 & \frac{3\sqrt{15}i}{160} & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & \frac{5i}{32} & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & \frac{3\sqrt{15}i}{160} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & \frac{5i}{32} & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{15}i}{480} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{5i}{32} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 \\ \frac{13\sqrt{15}i}{480} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & -\frac{5i}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{96} \\ 0 & 0 & -\frac{9\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & \frac{\sqrt{15}i}{120} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{10}i}{160} & -\frac{\sqrt{15}}{15} & 0 & \frac{\sqrt{15}i}{120} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 \end{bmatrix}$ |
| 917 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)$ | $\begin{matrix} 0 & 0 & 0 & -\frac{1}{32} & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{160} & -\frac{9\sqrt{10}i}{160} & 0 \\ 0 & 0 & \frac{1}{32} & 0 & 0 & \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{160} & 0 & 0 & \frac{9\sqrt{10}i}{160} \\ 0 & \frac{1}{32} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & \frac{13\sqrt{15}}{480} & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 \\ -\frac{1}{32} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{48} & -\frac{13\sqrt{15}}{480} & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} \\ 0 & -\frac{\sqrt{6}i}{32} & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & -\frac{\sqrt{15}}{120} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{48} & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & \frac{\sqrt{15}}{120} & 0 \\ 0 & 0 & 0 & -\frac{13\sqrt{15}}{480} & \frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{32} & -\frac{\sqrt{6}i}{96} & 0 \\ 0 & 0 & \frac{13\sqrt{15}}{480} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & \frac{5}{32} & 0 & 0 & \frac{\sqrt{6}i}{96} \\ 0 & -\frac{3\sqrt{15}}{160} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & 0 & \frac{5}{32} & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{160} & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{80} & -\frac{5}{32} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{10}i}{160} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{15}}{120} & \frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{10}i}{160} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & -\frac{\sqrt{15}}{120} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & 0 & 0 & 0 \end{matrix}$ |
| 918 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}}{48} & \frac{7\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & \frac{13\sqrt{5}}{240} & \frac{\sqrt{30}i}{80} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{48} & 0 & 0 & -\frac{7\sqrt{2}i}{48} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & -\frac{13\sqrt{5}}{240} & 0 & 0 & -\frac{\sqrt{30}i}{80} \\ 0 & \frac{\sqrt{3}}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & -\frac{7\sqrt{5}}{240} & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 \\ -\frac{\sqrt{3}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & \frac{7\sqrt{5}}{240} & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 \\ -\frac{7\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} \\ 0 & \frac{7\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{12} & 0 \\ 0 & \frac{\sqrt{5}i}{15} & 0 & \frac{7\sqrt{5}}{240} & -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{48} & -\frac{\sqrt{2}i}{48} & 0 \\ \frac{\sqrt{5}i}{15} & 0 & -\frac{7\sqrt{5}}{240} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{48} & 0 & 0 & \frac{\sqrt{2}i}{48} \\ 0 & -\frac{13\sqrt{5}}{240} & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{48} & 0 & 0 & 0 & 0 \\ \frac{13\sqrt{5}}{240} & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{48} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & -\frac{\sqrt{5}}{12} & \frac{\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & \frac{\sqrt{5}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{48} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 919 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{48} & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & -\frac{7\sqrt{5}i}{240} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & -\frac{\sqrt{5}}{30} & 0 & -\frac{7\sqrt{5}i}{240} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{48} & 0 & 0 & -\frac{7\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & \frac{13\sqrt{5}i}{240} & 0 & -\frac{\sqrt{5}}{15} & \frac{\sqrt{30}i}{80} & 0 \\ \frac{\sqrt{3}i}{48} & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}i}{48} & 0 & 0 & \frac{13\sqrt{5}i}{240} & 0 & \frac{\sqrt{5}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{80} \\ 0 & 0 & \frac{7\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & \frac{\sqrt{5}}{30} \\ 0 & 0 & 0 & -\frac{7\sqrt{2}i}{48} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & -\frac{\sqrt{5}}{30} & 0 \\ \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{12} \\ 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{12} & 0 \\ 0 & -\frac{\sqrt{5}}{30} & 0 & -\frac{13\sqrt{5}i}{240} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{48} & 0 & 0 \\ \frac{\sqrt{5}}{30} & 0 & -\frac{13\sqrt{5}i}{240} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{48} & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{5}i}{240} & 0 & \frac{\sqrt{5}}{15} & -\frac{\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{48} & 0 & 0 & \frac{\sqrt{2}i}{48} & 0 \\ \frac{7\sqrt{5}i}{240} & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & 0 & \frac{5\sqrt{3}i}{48} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{48} \\ 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 & \frac{\sqrt{5}i}{12} & 0 & 0 & -\frac{\sqrt{2}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{80} & \frac{\sqrt{5}}{30} & 0 & \frac{\sqrt{5}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{48} & 0 & 0 \end{pmatrix}$ |
| 920 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(A_g)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{42}i}{168} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{5\sqrt{42}i}{168} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{168} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \end{pmatrix}$ |
| 921 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 922 | $\mathbb{T}_2^{(1,0;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 \\ \frac{5\sqrt{14}i}{168} & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & \frac{\sqrt{35}}{21} & 0 \\ -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{35}}{21} \\ 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{14}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{14}i}{84} & 0 \\ -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} \\ 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{21} & \frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 923 | $\mathbb{T}_2^{(1,0;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & 0 \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{35}}{21} & 0 \\ -\frac{5\sqrt{14}}{168} & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & \frac{\sqrt{35}}{21} \\ 0 & \frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 \\ -\frac{5\sqrt{14}i}{168} & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{210}i}{168} & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} \\ 0 & \frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & -\frac{\sqrt{14}i}{84} \\ 0 & 0 & 0 & \frac{\sqrt{35}}{42} & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{14}i}{84} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 \\ \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 \\ \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{35}}{42} & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{42} & 0 \\ 0 & \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{14}}{84} & 0 \\ \frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} \\ 0 & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & -\frac{\sqrt{35}i}{42} & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{42} & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 \end{pmatrix}$ |
| 924 | symmetry | $\sqrt{3}xz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{2,2}(E_g)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 \\ -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & \frac{\sqrt{35}i}{84} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}i}{42} \\ 0 & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & \frac{\sqrt{35}i}{42} & 0 \\ \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 \\ 0 & -\frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \\ \frac{\sqrt{35}i}{84} & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{5\sqrt{21}i}{84} & -\frac{\sqrt{14}}{84} & 0 \\ \frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{14}}{84} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & -\frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & \frac{\sqrt{21}i}{21} & 0 \end{pmatrix}$ |
| 925 | symmetry | $\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{66}i}{88} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & -\frac{\sqrt{66}i}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 \\ 0 & -\frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & \frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 \\ \frac{\sqrt{110}i}{88} & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & -\frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{88} & 0 & -\frac{\sqrt{110}i}{88} & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 \\ \frac{\sqrt{110}}{88} & 0 & \frac{\sqrt{110}i}{88} & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{66}i}{88} & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} \\ 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & -\frac{\sqrt{66}i}{88} & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 \\ \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} \\ 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 \\ 0 & \frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\ -\frac{\sqrt{66}i}{88} & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{88} & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \end{pmatrix}$ |
| 926 | symmetry | $-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{616} & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & -\frac{\sqrt{385}}{110} & 0 & 0 & -\frac{\sqrt{2310}i}{440} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{616} & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{110} & \frac{\sqrt{2310}i}{440} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & -\frac{5\sqrt{154}i}{616} & -\frac{\sqrt{385}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{440} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & \frac{5\sqrt{154}i}{616} & 0 & 0 & \frac{\sqrt{385}}{110} & 0 & 0 & \frac{\sqrt{2310}}{440} & 0 \\ 0 & -\frac{5\sqrt{154}i}{616} & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & \frac{\sqrt{231}}{66} & 0 & 0 & \frac{9\sqrt{2310}i}{3080} & 0 & -\frac{9\sqrt{2310}}{3080} & 0 & 0 \\ \frac{5\sqrt{154}i}{616} & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{66} & -\frac{9\sqrt{2310}i}{3080} & 0 & -\frac{9\sqrt{2310}}{3080} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{154}}{616} & 0 & -\frac{5\sqrt{154}i}{616} & \frac{\sqrt{231}}{66} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2310}}{9240} & 0 & \frac{13\sqrt{2310}i}{9240} & 0 & 0 \\ \frac{5\sqrt{154}}{616} & 0 & \frac{5\sqrt{154}i}{616} & 0 & 0 & -\frac{\sqrt{231}}{66} & 0 & 0 & \frac{13\sqrt{2310}}{9240} & 0 & -\frac{13\sqrt{2310}i}{9240} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{385}}{110} & 0 & 0 & \frac{9\sqrt{2310}i}{3080} & 0 & \frac{13\sqrt{2310}}{9240} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{616} \\ 0 & 0 & 0 & \frac{\sqrt{385}}{110} & -\frac{9\sqrt{2310}i}{3080} & 0 & \frac{13\sqrt{2310}}{9240} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{616} & 0 \\ -\frac{\sqrt{385}}{110} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2310}}{3080} & 0 & \frac{13\sqrt{2310}i}{9240} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} \\ 0 & \frac{\sqrt{385}}{110} & 0 & 0 & -\frac{9\sqrt{2310}}{3080} & 0 & -\frac{13\sqrt{2310}i}{9240} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 \\ 0 & -\frac{\sqrt{2310}i}{440} & 0 & \frac{\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{616} & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 \\ \frac{\sqrt{2310}i}{440} & 0 & \frac{\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{616} & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 \end{pmatrix}$ |
| 927 | symmetry | $\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_g,3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{55} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{220} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{55} & 0 & 0 & \frac{3\sqrt{110}}{220} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{55} & 0 & 0 & \frac{3\sqrt{110}i}{220} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{55} & -\frac{3\sqrt{110}i}{220} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{220} & 0 & -\frac{\sqrt{110}i}{220} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & -\frac{\sqrt{110}}{220} & 0 & \frac{\sqrt{110}i}{220} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & -\frac{\sqrt{110}i}{220} & 0 & \frac{\sqrt{110}}{220} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & \frac{\sqrt{110}i}{220} & 0 & \frac{\sqrt{110}}{220} & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{55} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{220} & 0 & -\frac{\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{55} & 0 & 0 & -\frac{\sqrt{110}}{220} & 0 & \frac{\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{165}}{55} & 0 & 0 & -\frac{\sqrt{110}i}{220} & 0 & \frac{\sqrt{110}}{220} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{55} & \frac{\sqrt{110}i}{220} & 0 & \frac{\sqrt{110}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{110}}{220} & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{110}}{220} & 0 & -\frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 928 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{616} & 0 & -\frac{3\sqrt{462}}{616} & 0 & 0 & -\frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{3\sqrt{770}i}{440} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{462}i}{616} & 0 & -\frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}}{770} & -\frac{3\sqrt{770}i}{440} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{616} & 0 & -\frac{3\sqrt{462}i}{616} & \frac{3\sqrt{1155}}{770} & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{440} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{616} & 0 & \frac{3\sqrt{462}i}{616} & 0 & 0 & -\frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{3\sqrt{770}}{440} & 0 \\ 0 & -\frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 \\ \frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{462}}{616} & 0 & -\frac{3\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & \frac{19\sqrt{770}i}{3080} & -\frac{\sqrt{1155}}{770} & 0 \\ -\frac{3\sqrt{462}}{616} & 0 & \frac{3\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & -\frac{19\sqrt{770}i}{3080} & 0 & 0 & \frac{\sqrt{1155}}{770} \\ 0 & 0 & \frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & -\frac{19\sqrt{770}}{3080} & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & \frac{3\sqrt{462}i}{616} \\ 0 & 0 & 0 & -\frac{3\sqrt{1155}}{770} & -\frac{\sqrt{770}i}{616} & 0 & -\frac{19\sqrt{770}}{3080} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & -\frac{3\sqrt{462}i}{616} & 0 \\ -\frac{3\sqrt{1155}}{770} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & \frac{19\sqrt{770}i}{3080} & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{616} \\ 0 & \frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & -\frac{19\sqrt{770}i}{3080} & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & -\frac{3\sqrt{462}}{616} & 0 \\ 0 & \frac{3\sqrt{770}i}{440} & 0 & \frac{3\sqrt{770}}{440} & 0 & 0 & -\frac{\sqrt{1155}}{770} & 0 & 0 & \frac{3\sqrt{462}i}{616} & 0 & -\frac{3\sqrt{462}}{616} & 0 & 0 \\ -\frac{3\sqrt{770}i}{440} & 0 & \frac{3\sqrt{770}}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{770} & -\frac{3\sqrt{462}i}{616} & 0 & -\frac{3\sqrt{462}}{616} & 0 & 0 & 0 \end{bmatrix}$ |
| 929 | symmetry | $-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{616} & 0 & -\frac{3\sqrt{462}i}{616} & \frac{3\sqrt{1155}}{770} & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{440} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{616} & 0 & \frac{3\sqrt{462}i}{616} & 0 & 0 & -\frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{3\sqrt{770}}{440} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & \frac{3\sqrt{1155}}{770} & 0 & 0 & -\frac{3\sqrt{770}i}{440} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1155}}{770} & \frac{3\sqrt{770}i}{440} & 0 \\ 0 & \frac{3\sqrt{462}}{616} & 0 & \frac{3\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & \frac{19\sqrt{770}i}{3080} & -\frac{\sqrt{1155}}{770} & 0 \\ \frac{3\sqrt{462}}{616} & 0 & -\frac{3\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & -\frac{19\sqrt{770}i}{3080} & 0 & 0 & \frac{\sqrt{1155}}{770} \\ 0 & -\frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 \\ \frac{3\sqrt{462}i}{616} & 0 & \frac{3\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 \\ \frac{3\sqrt{1155}}{770} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & -\frac{\sqrt{770}i}{616} & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{616} \\ 0 & -\frac{3\sqrt{1155}}{770} & 0 & 0 & -\frac{19\sqrt{770}}{3080} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & -\frac{3\sqrt{462}}{616} & 0 \\ 0 & 0 & \frac{3\sqrt{1155}}{770} & 0 & 0 & \frac{19\sqrt{770}i}{3080} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & -\frac{3\sqrt{462}i}{616} \\ 0 & 0 & 0 & -\frac{3\sqrt{1155}}{770} & -\frac{19\sqrt{770}i}{3080} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & \frac{3\sqrt{462}i}{616} & 0 \\ 0 & \frac{3\sqrt{770}}{440} & 0 & -\frac{3\sqrt{770}i}{440} & -\frac{\sqrt{1155}}{770} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{616} & 0 & -\frac{3\sqrt{462}i}{616} & 0 & 0 \\ \frac{3\sqrt{770}}{440} & 0 & \frac{3\sqrt{770}i}{440} & 0 & 0 & \frac{\sqrt{1155}}{770} & 0 & 0 & -\frac{3\sqrt{462}}{616} & 0 & \frac{3\sqrt{462}i}{616} & 0 & 0 & 0 \end{bmatrix}$ |
| 930 | symmetry | $-\frac{\sqrt{35}x^3z}{2}+\frac{\sqrt{35}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,1}(E_g,1)$ | $\begin{pmatrix} 0 & -\frac{3\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & \frac{7\sqrt{165}i}{440} & 0 & -\frac{\sqrt{165}}{220} & 0 & 0 \\ \frac{3\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{176} & -\frac{7\sqrt{165}i}{440} & 0 & -\frac{\sqrt{165}}{220} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}i}{88} & \frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{110} & 0 & \frac{\sqrt{165}i}{88} & -\frac{9\sqrt{110}}{880} & 0 \\ 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & \frac{\sqrt{165}}{110} & 0 & -\frac{\sqrt{165}i}{88} & 0 & 0 & \frac{9\sqrt{110}}{880} \\ 0 & 0 & \frac{\sqrt{66}}{176} & 0 & 0 & \frac{3\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{88} & 0 & 0 & \frac{7\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{165}i}{220} \\ 0 & 0 & 0 & -\frac{\sqrt{66}}{176} & -\frac{3\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{88} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}}{880} & \frac{\sqrt{165}i}{220} & 0 \\ -\frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{88} & 0 & \frac{\sqrt{11}i}{11} & -\frac{\sqrt{110}}{880} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{440} \\ 0 & \frac{\sqrt{66}}{176} & 0 & 0 & -\frac{\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & \frac{\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{165}}{440} & 0 \\ 0 & \frac{7\sqrt{165}i}{440} & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & -\frac{\sqrt{110}}{880} & 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 \\ -\frac{7\sqrt{165}i}{440} & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{880} & -\frac{3\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{165}}{220} & 0 & \frac{\sqrt{165}i}{88} & \frac{7\sqrt{110}}{880} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & -\frac{5\sqrt{11}i}{88} & -\frac{\sqrt{66}}{176} & 0 \\ -\frac{\sqrt{165}}{220} & 0 & -\frac{\sqrt{165}i}{88} & 0 & 0 & -\frac{7\sqrt{110}}{880} & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & \frac{5\sqrt{11}i}{88} & 0 & 0 & \frac{\sqrt{66}}{176} \\ 0 & 0 & -\frac{9\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{165}i}{220} & 0 & -\frac{\sqrt{165}}{440} & 0 & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & -\frac{3\sqrt{11}i}{44} \\ 0 & 0 & 0 & \frac{9\sqrt{110}}{880} & \frac{\sqrt{165}i}{220} & 0 & -\frac{\sqrt{165}}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{176} & \frac{3\sqrt{11}i}{44} & 0 \end{pmatrix}$ |
| 931 | symmetry | $\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{11}}{88} & 0 & 0 & \frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{165}i}{110} & \frac{9\sqrt{110}}{880} & 0 \\ \frac{3\sqrt{11}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & -\frac{\sqrt{165}i}{110} & 0 & 0 & -\frac{9\sqrt{110}}{880} \\ 0 & 0 & 0 & \frac{3\sqrt{11}}{88} & 0 & 0 & \frac{\sqrt{66}}{176} & 0 & 0 & -\frac{\sqrt{165}i}{220} & 0 & \frac{7\sqrt{165}}{440} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{176} & \frac{\sqrt{165}i}{220} & 0 & \frac{7\sqrt{165}}{440} & 0 & 0 & 0 \\ \frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & -\frac{\sqrt{11}i}{88} & -\frac{7\sqrt{110}}{880} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{220} \\ 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & \frac{\sqrt{11}i}{88} & 0 & 0 & \frac{7\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{165}}{220} & 0 \\ 0 & 0 & \frac{\sqrt{66}}{176} & 0 & 0 & -\frac{\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & -\frac{\sqrt{110}}{880} & 0 & 0 & \frac{\sqrt{165}i}{440} \\ 0 & 0 & 0 & -\frac{\sqrt{66}}{176} & \frac{\sqrt{11}i}{88} & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{880} & -\frac{\sqrt{165}i}{440} & 0 \\ 0 & \frac{\sqrt{165}}{88} & 0 & -\frac{\sqrt{165}i}{220} & -\frac{7\sqrt{110}}{880} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{44} & -\frac{\sqrt{66}}{176} & 0 \\ \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{165}i}{220} & 0 & 0 & \frac{7\sqrt{110}}{880} & 0 & 0 & \frac{5\sqrt{11}}{88} & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & \frac{\sqrt{66}}{176} \\ 0 & \frac{\sqrt{165}i}{110} & 0 & \frac{7\sqrt{165}}{440} & 0 & 0 & -\frac{\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & -\frac{3\sqrt{11}}{88} & 0 & 0 \\ -\frac{\sqrt{165}i}{110} & 0 & \frac{7\sqrt{165}}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{880} & \frac{\sqrt{11}i}{44} & 0 & -\frac{3\sqrt{11}}{88} & 0 & 0 & 0 \\ \frac{9\sqrt{110}}{880} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{220} & 0 & \frac{\sqrt{165}i}{440} & -\frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} \\ 0 & -\frac{9\sqrt{110}}{880} & 0 & 0 & -\frac{\sqrt{165}}{220} & 0 & -\frac{\sqrt{165}i}{440} & 0 & 0 & \frac{\sqrt{66}}{176} & 0 & 0 & \frac{3\sqrt{11}}{44} & 0 \end{bmatrix}$ |
| 932 | symmetry | $3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,1}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{3080} & 0 & \frac{\sqrt{1155}i}{385} & -\frac{9\sqrt{770}}{880} & 0 \\ \frac{3\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{3080} & 0 & -\frac{\sqrt{1155}i}{385} & 0 & 0 & \frac{9\sqrt{770}}{880} \\ 0 & 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{1155}i}{140} & 0 & -\frac{13\sqrt{1155}}{3080} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{1232} & \frac{\sqrt{1155}i}{140} & 0 & -\frac{13\sqrt{1155}}{3080} & 0 & 0 & 0 \\ \frac{\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & \frac{\sqrt{77}i}{88} & \frac{17\sqrt{770}}{6160} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1540} \\ 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & -\frac{17\sqrt{770}}{6160} & 0 & 0 & \frac{\sqrt{1155}}{1540} & 0 \\ 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{616} \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{1232} & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{1232} & -\frac{\sqrt{1155}i}{616} & 0 \\ 0 & \frac{\sqrt{1155}}{3080} & 0 & -\frac{\sqrt{1155}i}{140} & \frac{17\sqrt{770}}{6160} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{616} & 0 & -\frac{5\sqrt{77}i}{308} & -\frac{\sqrt{462}}{1232} & 0 \\ \frac{\sqrt{1155}}{3080} & 0 & \frac{\sqrt{1155}i}{140} & 0 & 0 & -\frac{17\sqrt{770}}{6160} & 0 & 0 & -\frac{3\sqrt{77}}{616} & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & \frac{\sqrt{462}}{1232} \\ 0 & \frac{\sqrt{1155}i}{385} & 0 & -\frac{13\sqrt{1155}}{3080} & 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & \frac{5\sqrt{77}}{616} & 0 & 0 \\ -\frac{\sqrt{1155}i}{385} & 0 & -\frac{13\sqrt{1155}}{3080} & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{1232} & \frac{5\sqrt{77}i}{308} & 0 & \frac{5\sqrt{77}}{616} & 0 & 0 & 0 \\ -\frac{9\sqrt{770}}{880} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1540} & 0 & \frac{\sqrt{1155}i}{616} & -\frac{\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{308} \\ 0 & \frac{9\sqrt{770}}{880} & 0 & 0 & \frac{\sqrt{1155}}{1540} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 \end{pmatrix}$ |
| 933 | symmetry | $-\frac{\sqrt{5}x^3z}{2}+3\sqrt{5}xy^2z-\frac{\sqrt{5}xz^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{77}i}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & \frac{13\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{1155}}{140} & 0 & 0 \\ -\frac{3\sqrt{77}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{1232} & -\frac{13\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{1155}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{77}i}{616} & -\frac{\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{385} & 0 & -\frac{\sqrt{1155}i}{3080} & -\frac{9\sqrt{770}}{880} & 0 \\ 0 & 0 & -\frac{3\sqrt{77}i}{616} & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{1155}}{385} & 0 & \frac{\sqrt{1155}i}{3080} & 0 & 0 & \frac{9\sqrt{770}}{880} \\ 0 & 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & \frac{17\sqrt{770}}{6160} & 0 & 0 & -\frac{\sqrt{1155}i}{1540} \\ 0 & 0 & 0 & \frac{\sqrt{462}}{1232} & \frac{\sqrt{77}i}{44} & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{770}}{6160} & \frac{\sqrt{1155}i}{1540} & 0 \\ \frac{\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{616} \\ 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{616} & 0 \\ 0 & \frac{13\sqrt{1155}i}{3080} & 0 & -\frac{\sqrt{1155}}{385} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{5\sqrt{77}i}{616} & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 \\ -\frac{13\sqrt{1155}i}{3080} & 0 & -\frac{\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{5\sqrt{77}i}{616} & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{140} & 0 & -\frac{\sqrt{1155}i}{3080} & \frac{17\sqrt{770}}{6160} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & -\frac{3\sqrt{77}i}{616} & \frac{\sqrt{462}}{1232} & 0 \\ \frac{\sqrt{1155}}{140} & 0 & \frac{\sqrt{1155}i}{3080} & 0 & 0 & -\frac{17\sqrt{770}}{6160} & 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & \frac{3\sqrt{77}i}{616} & 0 & 0 & -\frac{\sqrt{462}}{1232} \\ 0 & 0 & -\frac{9\sqrt{770}}{880} & 0 & 0 & -\frac{\sqrt{1155}i}{1540} & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & \frac{\sqrt{462}}{1232} & 0 & 0 & \frac{3\sqrt{77}i}{308} \\ 0 & 0 & 0 & \frac{9\sqrt{770}}{880} & \frac{\sqrt{1155}i}{1540} & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{1232} & -\frac{3\sqrt{77}i}{308} & 0 \end{pmatrix}$ |
| 934 | symmetry | $\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{264} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & \frac{\sqrt{55}i}{88} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{264} & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & -\frac{\sqrt{55}i}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{264} & 0 & \frac{\sqrt{33}i}{44} & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{264} & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 \\ 0 & -\frac{5\sqrt{33}i}{264} & 0 & -\frac{5\sqrt{33}}{264} & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\ \frac{5\sqrt{33}i}{264} & 0 & -\frac{5\sqrt{33}}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & -\frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{44} & 0 & \frac{\sqrt{33}i}{44} & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 \\ -\frac{\sqrt{33}}{44} & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{264} \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & -\frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{264} & 0 \\ \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} \\ 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 \\ 0 & \frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{264} & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 \\ -\frac{\sqrt{55}i}{88} & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{264} & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 \end{pmatrix}$ |
| 935 | symmetry | $-\frac{\sqrt{14}x^6}{8}+\frac{15\sqrt{14}x^4z^2}{8}-\frac{15\sqrt{14}x^2z^4}{8}-\frac{\sqrt{14}y^6}{8}+\frac{15\sqrt{14}y^4z^2}{8}-\frac{15\sqrt{14}y^2z^4}{8}+\frac{\sqrt{14}z^6}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{231}}{924} & 0 & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & -\frac{\sqrt{385}i}{616} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{231}}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & \frac{\sqrt{385}i}{616} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{616} & 0 & -\frac{\sqrt{231}i}{924} & -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{616} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{231}i}{924} & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & \frac{\sqrt{385}}{616} & 0 \\ 0 & \frac{3\sqrt{231}i}{616} & 0 & \frac{3\sqrt{231}}{616} & 0 & 0 & -\frac{\sqrt{154}}{154} & 0 & 0 & -\frac{5\sqrt{385}i}{616} & 0 & \frac{5\sqrt{385}}{616} & 0 & 0 \\ -\frac{3\sqrt{231}i}{616} & 0 & \frac{3\sqrt{231}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{154} & \frac{5\sqrt{385}i}{616} & 0 & \frac{5\sqrt{385}}{616} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}}{924} & 0 & -\frac{\sqrt{231}i}{924} & -\frac{\sqrt{154}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{308} & 0 & -\frac{\sqrt{385}i}{308} & 0 & 0 \\ \frac{\sqrt{231}}{924} & 0 & \frac{\sqrt{231}i}{924} & 0 & 0 & \frac{\sqrt{154}}{154} & 0 & 0 & -\frac{\sqrt{385}}{308} & 0 & \frac{\sqrt{385}i}{308} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & -\frac{5\sqrt{385}i}{616} & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}i}{264} \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & \frac{5\sqrt{385}i}{616} & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{264} & 0 \\ -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0 & \frac{5\sqrt{385}}{616} & 0 & -\frac{\sqrt{385}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{264} \\ 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{\sqrt{385}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{264} & 0 \\ 0 & -\frac{\sqrt{385}i}{616} & 0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}i}{264} & 0 & \frac{5\sqrt{231}}{264} & 0 & 0 \\ \frac{\sqrt{385}i}{616} & 0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{264} & 0 & \frac{5\sqrt{231}}{264} & 0 & 0 & 0 \end{pmatrix}$ |
| 936 | symmetry | $-\frac{3\sqrt{7}x^5y}{4}+\frac{15\sqrt{7}x^3yz^2}{2}+\frac{3\sqrt{7}xy^5}{4}-\frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{168} & 0 & \frac{\sqrt{462}i}{168} & \frac{\sqrt{1155}}{231} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{462}i}{168} & 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{\sqrt{770}i}{308} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{231} & \frac{\sqrt{770}i}{308} & 0 \\ 0 & -\frac{\sqrt{462}}{168} & 0 & \frac{\sqrt{462}i}{168} & \frac{2\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{616} & 0 & -\frac{3\sqrt{770}i}{616} & 0 & 0 \\ -\frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{462}i}{168} & 0 & 0 & -\frac{2\sqrt{77}}{77} & 0 & 0 & -\frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{770}i}{616} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{2\sqrt{77}}{77} & 0 & 0 & -\frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 \\ -\frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{77} & \frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 \\ \frac{\sqrt{1155}}{231} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{616} & 0 & -\frac{3\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{231} & \frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{770}}{308} & 0 & -\frac{\sqrt{770}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{770}}{308} & 0 & \frac{\sqrt{770}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 937 | symmetry | $-\frac{\sqrt{2310}x^4y^2}{8}+\frac{\sqrt{2310}x^4z^2}{8}+\frac{\sqrt{2310}x^2y^4}{8}-\frac{\sqrt{2310}x^2z^4}{8}-\frac{\sqrt{2310}y^4z^2}{8}+\frac{\sqrt{2310}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & -\frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{56} & \frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{168} & \frac{\sqrt{21}i}{56} & 0 \\ -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{168} & 0 & \frac{\sqrt{35}i}{84} & -\frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} \\ 0 & \frac{\sqrt{210}}{56} & 0 & 0 & \frac{\sqrt{35}}{168} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 \\ 0 & -\frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 \\ \frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{35}}{168} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{84} & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{21}i}{84} & \frac{\sqrt{14}}{42} & 0 \\ \frac{\sqrt{35}}{84} & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{\sqrt{14}}{42} \\ 0 & 0 & -\frac{\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{5\sqrt{35}i}{168} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{168} & \frac{\sqrt{21}i}{24} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & -\frac{5\sqrt{35}i}{168} & 0 \\ \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & \frac{\sqrt{21}i}{84} & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{168} \\ 0 & -\frac{\sqrt{14}}{168} & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{5\sqrt{35}}{168} & 0 \\ 0 & -\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & \frac{5\sqrt{35}i}{168} & 0 & -\frac{5\sqrt{35}}{168} & 0 & 0 \\ \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & -\frac{5\sqrt{35}i}{168} & 0 & -\frac{5\sqrt{35}}{168} & 0 & 0 & 0 \end{bmatrix}$ |
| 938 | symmetry | $\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{462}}{56} & 0 & 0 & -\frac{19\sqrt{77}i}{1848} & 0 & \frac{\sqrt{77}}{132} & 0 & 0 & -\frac{\sqrt{770}}{1848} & 0 & 0 & \frac{\sqrt{1155}i}{616} \\ 0 & 0 & 0 & \frac{\sqrt{462}}{56} & \frac{19\sqrt{77}i}{1848} & 0 & \frac{\sqrt{77}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{1848} & -\frac{\sqrt{1155}i}{616} & 0 \\ -\frac{\sqrt{462}}{56} & 0 & 0 & 0 & 0 & \frac{19\sqrt{77}}{1848} & 0 & \frac{\sqrt{77}i}{132} & \frac{\sqrt{770}}{1848} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{616} \\ 0 & \frac{\sqrt{462}}{56} & 0 & 0 & \frac{19\sqrt{77}}{1848} & 0 & -\frac{\sqrt{77}i}{132} & 0 & 0 & -\frac{\sqrt{770}}{1848} & 0 & 0 & \frac{\sqrt{1155}}{616} & 0 \\ 0 & -\frac{19\sqrt{77}i}{1848} & 0 & \frac{19\sqrt{77}}{1848} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 \\ \frac{19\sqrt{77}i}{1848} & 0 & \frac{19\sqrt{77}}{1848} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{132} & 0 & \frac{\sqrt{77}i}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{924} & -\frac{\sqrt{770}}{462} & 0 \\ \frac{\sqrt{77}}{132} & 0 & -\frac{\sqrt{77}i}{132} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & \frac{\sqrt{770}}{462} \\ 0 & 0 & \frac{\sqrt{770}}{1848} & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & -\frac{25\sqrt{77}i}{1848} \\ 0 & 0 & 0 & -\frac{\sqrt{770}}{1848} & -\frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & \frac{25\sqrt{77}i}{1848} & 0 \\ -\frac{\sqrt{770}}{1848} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & -\frac{\sqrt{1155}i}{924} & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & \frac{25\sqrt{77}}{1848} \\ 0 & \frac{\sqrt{770}}{1848} & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & \frac{\sqrt{1155}i}{924} & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & \frac{25\sqrt{77}}{1848} & 0 \\ 0 & \frac{\sqrt{1155}i}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & -\frac{\sqrt{770}}{462} & 0 & 0 & -\frac{25\sqrt{77}i}{1848} & 0 & \frac{25\sqrt{77}}{1848} & 0 & 0 \\ -\frac{\sqrt{1155}i}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{462} & \frac{25\sqrt{77}i}{1848} & 0 & \frac{25\sqrt{77}}{1848} & 0 & 0 & 0 \end{bmatrix}$ |
| 939 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(B_g,3)$ | $\begin{bmatrix} \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 940 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(B_g,4)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{924} & 0 & \frac{\sqrt{385}i}{924} & -\frac{\sqrt{154}}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{924} & 0 & -\frac{\sqrt{385}i}{924} & 0 & 0 & \frac{\sqrt{154}}{462} & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{924} & 0 & -\frac{\sqrt{385}}{924} & 0 & 0 & -\frac{\sqrt{154}}{462} & 0 & 0 & \frac{\sqrt{231}i}{154} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{924} & 0 & -\frac{\sqrt{385}}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{462} & -\frac{\sqrt{231}i}{154} & 0 \\ 0 & -\frac{\sqrt{385}}{924} & 0 & -\frac{\sqrt{385}i}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & \frac{\sqrt{231}i}{231} & \frac{2\sqrt{154}}{231} & 0 \\ -\frac{\sqrt{385}}{924} & 0 & \frac{\sqrt{385}i}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & -\frac{\sqrt{231}i}{231} & 0 & 0 & -\frac{2\sqrt{154}}{231} \\ 0 & \frac{\sqrt{385}i}{924} & 0 & -\frac{\sqrt{385}}{924} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{66} & 0 & \frac{\sqrt{231}}{66} & 0 & 0 \\ -\frac{\sqrt{385}i}{924} & 0 & -\frac{\sqrt{385}}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{66} & 0 & \frac{\sqrt{231}}{66} & 0 & 0 & 0 \\ -\frac{\sqrt{154}}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & \frac{\sqrt{231}i}{66} & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{462} \\ 0 & \frac{\sqrt{154}}{462} & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & -\frac{\sqrt{231}i}{66} & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 & -\frac{5\sqrt{385}}{462} & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{462} & 0 & 0 & \frac{\sqrt{231}i}{231} & 0 & \frac{\sqrt{231}}{66} & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 & -\frac{5\sqrt{385}i}{462} \\ 0 & 0 & 0 & \frac{\sqrt{154}}{462} & -\frac{\sqrt{231}i}{231} & 0 & \frac{\sqrt{231}}{66} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & \frac{5\sqrt{385}i}{462} & 0 \\ 0 & -\frac{\sqrt{231}}{154} & 0 & \frac{\sqrt{231}i}{154} & \frac{2\sqrt{154}}{231} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{462} & 0 & -\frac{5\sqrt{385}i}{462} & 0 & 0 \\ -\frac{\sqrt{231}}{154} & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & -\frac{2\sqrt{154}}{231} & 0 & 0 & -\frac{5\sqrt{385}}{462} & 0 & \frac{5\sqrt{385}i}{462} & 0 & 0 & 0 \end{matrix}$ |
| 941 | symmetry | $\frac{3\sqrt{7}x^5z}{4}-\frac{15\sqrt{7}x^3y^2z}{2}+\frac{15\sqrt{7}xy^2z^3}{2}-\frac{3\sqrt{7}xz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{77}i}{616} & 0 & \frac{\sqrt{77}}{112} & 0 & 0 & -\frac{\sqrt{462}}{132} & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 & -\frac{\sqrt{1155}}{528} & 0 & 0 \\ -\frac{5\sqrt{77}i}{616} & 0 & \frac{\sqrt{77}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{132} & \frac{\sqrt{1155}i}{264} & 0 & -\frac{\sqrt{1155}}{528} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{112} & 0 & -\frac{3\sqrt{77}i}{308} & -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1232} & 0 & \frac{\sqrt{1155}i}{308} & \frac{3\sqrt{770}}{1232} & 0 \\ \frac{\sqrt{77}}{112} & 0 & \frac{3\sqrt{77}i}{308} & 0 & 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{1232} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & 0 & -\frac{3\sqrt{770}}{1232} \\ 0 & 0 & -\frac{9\sqrt{462}}{1232} & 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{924} \\ 0 & 0 & 0 & \frac{9\sqrt{462}}{1232} & \frac{5\sqrt{77}i}{308} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{\sqrt{1155}i}{924} & 0 \\ -\frac{\sqrt{462}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{308} & 0 & \frac{2\sqrt{77}i}{77} & \frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1155}}{924} \\ 0 & \frac{\sqrt{462}}{132} & 0 & 0 & -\frac{\sqrt{77}}{308} & 0 & -\frac{2\sqrt{77}i}{77} & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 & 0 & -\frac{5\sqrt{1155}}{924} & 0 \\ 0 & -\frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{1232} & 0 & 0 & \frac{\sqrt{770}}{308} & 0 & 0 & -\frac{5\sqrt{77}i}{616} & 0 & -\frac{25\sqrt{77}}{1232} & 0 & 0 \\ \frac{\sqrt{1155}i}{264} & 0 & \frac{\sqrt{1155}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & \frac{5\sqrt{77}i}{616} & 0 & -\frac{25\sqrt{77}}{1232} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{528} & 0 & \frac{\sqrt{1155}i}{308} & \frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & -\frac{25\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{77}i}{308} & \frac{5\sqrt{462}}{3696} & 0 \\ -\frac{\sqrt{1155}}{528} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & -\frac{25\sqrt{77}}{1232} & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & -\frac{5\sqrt{462}}{3696} \\ 0 & 0 & \frac{3\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & -\frac{5\sqrt{1155}}{924} & 0 & 0 & \frac{5\sqrt{462}}{3696} & 0 & 0 & \frac{5\sqrt{77}i}{308} \\ 0 & 0 & 0 & -\frac{3\sqrt{770}}{1232} & -\frac{\sqrt{1155}i}{924} & 0 & -\frac{5\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{3696} & -\frac{5\sqrt{77}i}{308} & 0 \end{pmatrix}$ |
| 942 | symmetry | $\frac{15\sqrt{7}x^2y^3z}{2} - \frac{15\sqrt{7}x^2yz^3}{2} - \frac{3\sqrt{7}y^5z}{4} + \frac{3\sqrt{7}yz^5}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{6,2}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{77}}{308} & 0 & -\frac{\sqrt{77}i}{112} & -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{1155}i}{1232} & -\frac{3\sqrt{770}}{1232} & 0 \\ \frac{3\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{112} & 0 & 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & -\frac{\sqrt{1155}i}{1232} & 0 & 0 & \frac{3\sqrt{770}}{1232} \\ 0 & -\frac{\sqrt{77}i}{112} & 0 & -\frac{5\sqrt{77}}{616} & 0 & 0 & \frac{\sqrt{462}}{132} & 0 & 0 & -\frac{\sqrt{1155}i}{528} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 \\ \frac{\sqrt{77}i}{112} & 0 & -\frac{5\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{132} & \frac{\sqrt{1155}i}{528} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 \\ -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & -\frac{\sqrt{77}i}{308} & -\frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} \\ 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{308} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 \\ 0 & 0 & \frac{\sqrt{462}}{132} & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & -\frac{2\sqrt{77}}{77} & 0 & 0 & \frac{\sqrt{770}}{308} & 0 & 0 & \frac{5\sqrt{1155}i}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{132} & \frac{\sqrt{77}i}{308} & 0 & -\frac{2\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & -\frac{5\sqrt{1155}i}{924} & 0 \\ 0 & \frac{\sqrt{1155}}{308} & 0 & -\frac{\sqrt{1155}i}{528} & -\frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & \frac{25\sqrt{77}i}{1232} & \frac{5\sqrt{462}}{3696} & 0 \\ \frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{1155}i}{528} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & -\frac{25\sqrt{77}i}{1232} & 0 & 0 & -\frac{5\sqrt{462}}{3696} \\ 0 & \frac{\sqrt{1155}i}{1232} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & \frac{\sqrt{770}}{308} & 0 & 0 & \frac{25\sqrt{77}i}{1232} & 0 & \frac{5\sqrt{77}}{616} & 0 & 0 \\ -\frac{\sqrt{1155}i}{1232} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & -\frac{25\sqrt{77}i}{1232} & 0 & \frac{5\sqrt{77}}{616} & 0 & 0 & 0 \\ -\frac{3\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & \frac{5\sqrt{1155}i}{924} & \frac{5\sqrt{462}}{3696} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{308} \\ 0 & \frac{3\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & -\frac{5\sqrt{1155}i}{924} & 0 & 0 & -\frac{5\sqrt{462}}{3696} & 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 \end{pmatrix}$ |
| 943 | symmetry | $\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{6,1}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}i}{448} & \frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{448} & \frac{\sqrt{105}}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{448} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{448} & 0 & 0 & -\frac{\sqrt{105}}{224} \\ 0 & \frac{\sqrt{42}i}{448} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{70}i}{448} & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 \\ -\frac{\sqrt{42}i}{448} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{112} & -\frac{\sqrt{70}i}{448} & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{42}i}{112} & \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} \\ 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{3\sqrt{42}}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{112} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & 0 & \frac{\sqrt{70}i}{112} \\ 0 & 0 & 0 & \frac{\sqrt{7}}{112} & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{112} & -\frac{\sqrt{70}i}{112} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{448} & \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{448} & \frac{5\sqrt{7}}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{448} & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{448} & 0 & 0 & -\frac{5\sqrt{7}}{224} \\ 0 & \frac{3\sqrt{70}i}{448} & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & 0 & \frac{5\sqrt{42}i}{448} & 0 & \frac{15\sqrt{42}}{224} & 0 & 0 \\ -\frac{3\sqrt{70}i}{448} & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{112} & -\frac{5\sqrt{42}i}{448} & 0 & \frac{15\sqrt{42}}{224} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} & 0 & \frac{\sqrt{70}i}{112} & \frac{5\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{112} \\ 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & -\frac{5\sqrt{7}}{224} & 0 & 0 & -\frac{5\sqrt{42}}{112} & 0 \end{bmatrix}$ |
| 944 | symmetry | $\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{6,2}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{448} & 0 & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & -\frac{3\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{448} & 0 & 0 \\ -\frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{42}}{448} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{112} & \frac{3\sqrt{70}i}{224} & 0 & -\frac{\sqrt{70}}{448} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{448} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{448} & 0 & 0 & \frac{\sqrt{105}}{224} & 0 \\ \frac{\sqrt{42}}{448} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{448} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{224} \\ 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & \frac{3\sqrt{70}i}{112} \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{224} & \frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{224} & -\frac{3\sqrt{70}i}{112} & 0 \\ -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} \\ 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 \\ 0 & -\frac{3\sqrt{70}i}{224} & 0 & -\frac{3\sqrt{70}}{448} & 0 & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{15\sqrt{42}i}{224} & 0 & \frac{5\sqrt{42}}{448} & 0 & 0 \\ \frac{3\sqrt{70}i}{224} & 0 & -\frac{3\sqrt{70}}{448} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & -\frac{15\sqrt{42}i}{224} & 0 & \frac{5\sqrt{42}}{448} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{448} & 0 & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{448} & 0 & 0 & -\frac{5\sqrt{7}}{224} & 0 \\ -\frac{\sqrt{70}}{448} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & \frac{5\sqrt{42}}{448} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{224} \\ 0 & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & \frac{3\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & -\frac{5\sqrt{7}}{224} & 0 & 0 & -\frac{5\sqrt{42}i}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{224} & -\frac{3\sqrt{70}i}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{224} & \frac{5\sqrt{42}i}{112} & 0 \end{bmatrix}$ |
| 945 | symmetry | $\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 946 | $\mathbb{T}^{(1,0;a)}_{6,1}(E_g,3)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{2310}}{462} & 0 & \frac{\sqrt{2310}i}{448} & \frac{83\sqrt{385}}{7392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{462} & 0 & -\frac{113\sqrt{154}i}{14784} & -\frac{9\sqrt{231}}{2464} & 0 \\ -\frac{\sqrt{2310}}{462} & 0 & -\frac{\sqrt{2310}i}{448} & 0 & 0 & -\frac{83\sqrt{385}}{7392} & 0 & 0 & -\frac{\sqrt{154}}{462} & 0 & \frac{113\sqrt{154}i}{14784} & 0 & 0 & \frac{9\sqrt{231}}{2464} \\ 0 & \frac{\sqrt{2310}i}{448} & 0 & \frac{17\sqrt{2310}}{7392} & 0 & 0 & -\frac{41\sqrt{385}}{3696} & 0 & 0 & -\frac{17\sqrt{154}i}{1344} & 0 & -\frac{\sqrt{154}}{7392} & 0 & 0 \\ -\frac{\sqrt{2310}i}{448} & 0 & \frac{17\sqrt{2310}}{7392} & 0 & 0 & 0 & 0 & \frac{41\sqrt{385}}{3696} & \frac{17\sqrt{154}i}{1344} & 0 & -\frac{\sqrt{154}}{7392} & 0 & 0 & 0 \\ \frac{83\sqrt{385}}{7392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1232} & 0 & -\frac{5\sqrt{2310}i}{1232} & -\frac{43\sqrt{231}}{7392} & 0 & 0 & 0 & 0 & \frac{17\sqrt{154}}{3696} \\ 0 & -\frac{83\sqrt{385}}{7392} & 0 & 0 & -\frac{\sqrt{2310}}{1232} & 0 & \frac{5\sqrt{2310}i}{1232} & 0 & 0 & \frac{43\sqrt{231}}{7392} & 0 & 0 & \frac{17\sqrt{154}}{3696} & 0 \\ 0 & 0 & -\frac{41\sqrt{385}}{3696} & 0 & 0 & -\frac{5\sqrt{2310}i}{1232} & 0 & 0 & 0 & 0 & \frac{19\sqrt{231}}{3696} & 0 & 0 & \frac{37\sqrt{154}i}{3696} \\ 0 & 0 & 0 & \frac{41\sqrt{385}}{3696} & \frac{5\sqrt{2310}i}{1232} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{231}}{3696} & -\frac{37\sqrt{154}i}{3696} & 0 \\ 0 & -\frac{\sqrt{154}}{462} & 0 & -\frac{17\sqrt{154}i}{1344} & -\frac{43\sqrt{231}}{7392} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & \frac{37\sqrt{2310}i}{14784} & \frac{5\sqrt{385}}{7392} & 0 \\ -\frac{\sqrt{154}}{462} & 0 & \frac{17\sqrt{154}i}{1344} & 0 & 0 & \frac{43\sqrt{231}}{7392} & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & -\frac{37\sqrt{2310}i}{14784} & 0 & 0 & -\frac{5\sqrt{385}}{7392} \\ 0 & -\frac{113\sqrt{154}i}{14784} & 0 & -\frac{\sqrt{154}}{7392} & 0 & 0 & \frac{19\sqrt{231}}{3696} & 0 & 0 & \frac{37\sqrt{2310}i}{14784} & 0 & -\frac{\sqrt{2310}}{7392} & 0 & 0 \\ \frac{113\sqrt{154}i}{14784} & 0 & -\frac{\sqrt{154}}{7392} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{231}}{3696} & -\frac{37\sqrt{2310}i}{14784} & 0 & -\frac{\sqrt{2310}}{7392} & 0 & 0 & 0 \\ -\frac{9\sqrt{231}}{2464} & 0 & 0 & 0 & 0 & \frac{17\sqrt{154}}{3696} & 0 & \frac{37\sqrt{154}i}{3696} & \frac{5\sqrt{385}}{7392} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2310}}{3696} \\ 0 & \frac{9\sqrt{231}}{2464} & 0 & 0 & \frac{17\sqrt{154}}{3696} & 0 & -\frac{37\sqrt{154}i}{3696} & 0 & 0 & -\frac{5\sqrt{385}}{7392} & 0 & 0 & -\frac{5\sqrt{2310}}{3696} & 0 \end{pmatrix}$ |
| | symmetry | $\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 947 | symmetry | $z$ |

$\mathbb{T}^{(1,0;a)}_{6,2}(E_g,3)$

$$\begin{pmatrix}
0 & \frac{17\sqrt{2310}i}{7392} & 0 & \frac{\sqrt{2310}}{448} & 0 & 0 & -\frac{41\sqrt{385}}{3696} & 0 & 0 & \frac{\sqrt{154}i}{7392} & 0 & \frac{17\sqrt{154}}{1344} & 0 & 0 \\
-\frac{17\sqrt{2310}i}{7392} & 0 & \frac{\sqrt{2310}}{448} & 0 & 0 & 0 & 0 & \frac{41\sqrt{385}}{3696} & -\frac{\sqrt{154}i}{7392} & 0 & \frac{17\sqrt{154}}{1344} & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{448} & 0 & -\frac{\sqrt{2310}i}{462} & -\frac{83\sqrt{385}}{7392} & 0 & 0 & 0 & 0 & \frac{113\sqrt{154}}{14784} & 0 & \frac{\sqrt{154}i}{462} & -\frac{9\sqrt{231}}{2464} & 0 \\
\frac{\sqrt{2310}}{448} & 0 & \frac{\sqrt{2310}i}{462} & 0 & 0 & \frac{83\sqrt{385}}{7392} & 0 & 0 & \frac{113\sqrt{154}}{14784} & 0 & -\frac{\sqrt{154}i}{462} & 0 & 0 & \frac{9\sqrt{231}}{2464} \\
0 & 0 & -\frac{83\sqrt{385}}{7392} & 0 & 0 & -\frac{\sqrt{2310}i}{1232} & 0 & \frac{5\sqrt{2310}}{1232} & 0 & 0 & -\frac{43\sqrt{231}}{7392} & 0 & 0 & -\frac{17\sqrt{154}i}{3696} \\
0 & 0 & 0 & \frac{83\sqrt{385}}{7392} & \frac{\sqrt{2310}i}{1232} & 0 & \frac{5\sqrt{2310}}{1232} & 0 & 0 & 0 & 0 & \frac{43\sqrt{231}}{7392} & \frac{17\sqrt{154}i}{3696} & 0 \\
-\frac{41\sqrt{385}}{3696} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2310}}{1232} & 0 & 0 & -\frac{19\sqrt{231}}{3696} & 0 & 0 & 0 & 0 & \frac{37\sqrt{154}}{3696} \\
0 & \frac{41\sqrt{385}}{3696} & 0 & 0 & \frac{5\sqrt{2310}}{1232} & 0 & 0 & 0 & 0 & \frac{19\sqrt{231}}{3696} & 0 & 0 & \frac{37\sqrt{154}}{3696} & 0 \\
0 & \frac{\sqrt{154}i}{7392} & 0 & \frac{113\sqrt{154}}{14784} & 0 & 0 & -\frac{19\sqrt{231}}{3696} & 0 & 0 & -\frac{\sqrt{2310}i}{7392} & 0 & \frac{37\sqrt{2310}}{14784} & 0 & 0 \\
-\frac{\sqrt{154}i}{7392} & 0 & \frac{113\sqrt{154}}{14784} & 0 & 0 & 0 & 0 & \frac{19\sqrt{231}}{3696} & \frac{\sqrt{2310}i}{7392} & 0 & \frac{37\sqrt{2310}}{14784} & 0 & 0 & 0 \\
0 & \frac{17\sqrt{154}}{1344} & 0 & \frac{\sqrt{154}i}{462} & -\frac{43\sqrt{231}}{7392} & 0 & 0 & 0 & 0 & \frac{37\sqrt{2310}}{14784} & 0 & \frac{\sqrt{2310}i}{462} & -\frac{5\sqrt{385}}{7392} & 0 \\
\frac{17\sqrt{154}}{1344} & 0 & -\frac{\sqrt{154}i}{462} & 0 & 0 & \frac{43\sqrt{231}}{7392} & 0 & 0 & \frac{37\sqrt{2310}}{14784} & 0 & -\frac{\sqrt{2310}i}{462} & 0 & 0 & \frac{5\sqrt{385}}{7392} \\
0 & 0 & -\frac{9\sqrt{231}}{2464} & 0 & 0 & -\frac{17\sqrt{154}i}{3696} & 0 & \frac{37\sqrt{154}}{3696} & 0 & 0 & -\frac{5\sqrt{385}}{7392} & 0 & 0 & -\frac{5\sqrt{2310}i}{3696} \\
0 & 0 & 0 & \frac{9\sqrt{231}}{2464} & \frac{17\sqrt{154}i}{3696} & 0 & \frac{37\sqrt{154}}{3696} & 0 & 0 & 0 & 0 & \frac{5\sqrt{385}}{7392} & \frac{5\sqrt{2310}i}{3696} & 0
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 948 | symmetry | $\mathbb{M}_1^{(a)}(A_g)$ $\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $y$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 949 | symmetry | $x$ |

$\mathbb{M}^{(a)}_{1,1}(E_g)$

$$\begin{pmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{1,2}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 \end{bmatrix}$ |
| 950 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A_g)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 951 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 952 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(B_g, 2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 953 | symmetry | $-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 954 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} \\ 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \end{bmatrix}$ |
| 955 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{3,1}(E_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} \\ 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \end{pmatrix}$ |
| 956 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 957 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 958 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A_g,2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 959 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 960 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 961 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 962 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{48} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{112} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{112} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 963 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{48} \\ 0 & 0 & \frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{13\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{112} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{112} & 0 & 0 \end{bmatrix}$ |
| 964 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 965 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{5,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} \\ 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 \end{bmatrix}$ |
| 966 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{5,1}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} \\ 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \\ -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 \end{bmatrix}$ |
| 967 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 968 | symmetry | $z$ |

$\mathbb{M}^{(a)}_{5,2}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 969 | symmetry | $\mathbb{M}_1^{(1,-1;a)}(A_g)$ $y$ |

$$\begin{bmatrix}
\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 \end{pmatrix}$ |
| 970 | symmetry | $x$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{pmatrix} 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{pmatrix}$ |
| 971 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A_g)$ | $\begin{bmatrix} -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{105}i}{210} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & -\frac{\sqrt{105}i}{210} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & \frac{\sqrt{105}i}{210} & \frac{\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & -\frac{\sqrt{70}}{35} \end{bmatrix}$ |
| 972 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{84} & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & -\frac{5\sqrt{7}i}{84} & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & \frac{5\sqrt{7}i}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 \\ \frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{7}}{84} & 0 & -\frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{70}}{42} & 0 \\ -\frac{5\sqrt{7}}{84} & 0 & \frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{70}}{42} \\ 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{84} & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & \frac{\sqrt{7}i}{42} & 0 \\ \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} \\ 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{42} & \frac{\sqrt{7}i}{42} & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 \end{pmatrix}$ |
| 973 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & -\frac{5\sqrt{7}i}{84} & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & \frac{5\sqrt{7}i}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 \\ 0 & \frac{5\sqrt{7}}{84} & 0 & \frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{70}}{42} & 0 \\ \frac{5\sqrt{7}}{84} & 0 & -\frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{70}}{42} \\ 0 & -\frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\ \frac{5\sqrt{7}i}{84} & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{105}i}{84} & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} \\ 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & \frac{\sqrt{7}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{84} & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & -\frac{\sqrt{7}i}{42} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 \end{pmatrix}$ |
| 974 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{28} & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{9\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 \\ \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & -\frac{9\sqrt{70}i}{280} & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{42}i}{56} & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & -\frac{3\sqrt{70}i}{280} & -\frac{\sqrt{105}}{210} & 0 \\ -\frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & \frac{\sqrt{105}}{210} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & \frac{\sqrt{70}i}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & -\frac{\sqrt{70}i}{70} & 0 \end{pmatrix}$ |
| 975 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 \\ \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & \frac{\sqrt{42}i}{42} & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{84} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & \frac{\sqrt{70}i}{70} & -\frac{\sqrt{105}}{210} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & \frac{\sqrt{105}}{210} \\ 0 & \frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & -\frac{9\sqrt{70}}{280} & 0 & 0 \\ -\frac{\sqrt{42}i}{84} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & -\frac{\sqrt{70}i}{70} & 0 & -\frac{9\sqrt{70}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & -\frac{\sqrt{42}i}{42} & -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 \end{bmatrix}$ |
| 976 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 \\ \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 \\ -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} \\ 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 \\ 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}i}{42} \\ 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & -\frac{\sqrt{70}i}{42} & 0 \\ 0 & \frac{\sqrt{70}}{168} & 0 & \frac{\sqrt{70}i}{84} & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & -\frac{\sqrt{42}i}{42} & -\frac{\sqrt{7}}{42} & 0 \\ \frac{\sqrt{70}}{168} & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{7}}{42} \\ 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 \\ \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{42} & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 \end{pmatrix}$ |
| 977 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 \\ -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & \frac{\sqrt{70}i}{168} & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{70}i}{84} \\ 0 & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & \frac{\sqrt{70}i}{84} & 0 \\ -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{42} \\ 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{42} & 0 \\ 0 & -\frac{\sqrt{70}i}{168} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 \\ \frac{\sqrt{70}i}{168} & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{84} & 0 & -\frac{\sqrt{70}i}{168} & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & -\frac{5\sqrt{42}i}{168} & \frac{\sqrt{7}}{42} & 0 \\ -\frac{\sqrt{70}}{84} & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{7}}{42} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & \frac{\sqrt{42}i}{42} & 0 \end{pmatrix}$ |
| 978 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{110} & 0 & 0 & \frac{3\sqrt{110}i}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{110} & -\frac{3\sqrt{110}i}{110} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{110} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & -\frac{3\sqrt{110}}{110} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & -\frac{\sqrt{110}i}{110} & 0 & \frac{\sqrt{110}}{110} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & \frac{\sqrt{110}i}{110} & 0 & \frac{\sqrt{110}}{110} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & \frac{\sqrt{110}i}{110} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{110} & 0 & 0 & -\frac{\sqrt{110}i}{110} & 0 & \frac{\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{165}}{110} & \frac{\sqrt{110}i}{110} & 0 & \frac{\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & \frac{\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{110} & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{110}i}{110} & 0 & -\frac{3\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{110}i}{110} & 0 & -\frac{3\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 979 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,2)$ | $\begin{bmatrix} \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & \frac{\sqrt{2310}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & -\frac{\sqrt{2310}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & -\frac{\sqrt{2310}i}{462} & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{385}}{154} & \frac{\sqrt{2310}i}{462} & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{462} & 0 & -\frac{\sqrt{2310}i}{462} & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{231} & 0 & -\frac{2\sqrt{154}i}{231} & 0 & 0 \\ \frac{\sqrt{2310}}{462} & 0 & \frac{\sqrt{2310}i}{462} & 0 & 0 & \frac{\sqrt{385}}{66} & 0 & 0 & -\frac{2\sqrt{154}}{231} & 0 & \frac{2\sqrt{154}i}{231} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}i}{462} & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & \frac{2\sqrt{154}i}{231} & 0 & -\frac{2\sqrt{154}}{231} & 0 & 0 \\ -\frac{\sqrt{2310}i}{462} & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{66} & -\frac{2\sqrt{154}i}{231} & 0 & -\frac{2\sqrt{154}}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{231} & 0 & \frac{2\sqrt{154}i}{231} & \frac{\sqrt{385}}{462} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{231} & 0 & -\frac{2\sqrt{154}i}{231} & 0 & 0 & -\frac{\sqrt{385}}{462} & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}i}{231} & 0 & -\frac{2\sqrt{154}}{231} & 0 & 0 & \frac{\sqrt{385}}{462} & 0 & 0 & \frac{\sqrt{2310}i}{462} \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{154}i}{231} & 0 & -\frac{2\sqrt{154}}{231} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{462} & -\frac{\sqrt{2310}i}{462} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & \frac{\sqrt{2310}i}{462} & \frac{\sqrt{385}}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & -\frac{\sqrt{2310}i}{462} & 0 & 0 & -\frac{\sqrt{385}}{77} \end{bmatrix}$ |
| 980 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{110} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & -\frac{3\sqrt{110}}{110} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & -\frac{3\sqrt{110}i}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{110} & \frac{3\sqrt{110}i}{110} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & \frac{\sqrt{110}i}{110} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}i}{110} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & -\frac{\sqrt{110}i}{110} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & \frac{\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{110} & 0 & 0 & \frac{\sqrt{110}}{110} & 0 & -\frac{\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & \frac{\sqrt{110}i}{110} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{110} & -\frac{\sqrt{110}i}{110} & 0 & -\frac{\sqrt{110}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{110}}{110} & 0 & -\frac{3\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{110}}{110} & 0 & \frac{3\sqrt{110}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 981 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}i}{220} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{110} & \frac{\sqrt{330}i}{220} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{22}i}{44} & -\frac{3\sqrt{55}}{110} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{220} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & \frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 \\ 0 & -\frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 \\ \frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 & \frac{\sqrt{330}i}{220} & \frac{\sqrt{55}}{110} & 0 \\ -\frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 & -\frac{\sqrt{330}i}{220} & 0 & 0 & -\frac{\sqrt{55}}{110} \\ 0 & 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & \frac{\sqrt{22}i}{44} \\ 0 & 0 & 0 & \frac{3\sqrt{55}}{110} & -\frac{\sqrt{330}i}{132} & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & -\frac{\sqrt{22}i}{44} & 0 \\ \frac{3\sqrt{55}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & \frac{\sqrt{330}i}{220} & -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & -\frac{\sqrt{330}i}{220} & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 \\ 0 & -\frac{\sqrt{330}i}{220} & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & \frac{\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 \\ \frac{\sqrt{330}i}{220} & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{110} & -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 \end{bmatrix}$ |
| 982 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & \frac{3\sqrt{55}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{220} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{330}}{220} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}i}{220} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{110} & \frac{\sqrt{330}i}{220} & 0 \\ 0 & -\frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{220} & 0 & -\frac{\sqrt{330}i}{220} & -\frac{\sqrt{55}}{110} & 0 \\ -\frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{220} & 0 & \frac{\sqrt{330}i}{220} & 0 & 0 & \frac{\sqrt{55}}{110} \\ 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 \\ -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 \\ \frac{3\sqrt{55}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{220} & 0 & \frac{\sqrt{330}i}{132} & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} \\ 0 & -\frac{3\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{330}}{220} & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & \frac{\sqrt{22}}{44} & 0 \\ 0 & 0 & \frac{3\sqrt{55}}{110} & 0 & 0 & -\frac{\sqrt{330}i}{220} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & \frac{\sqrt{22}i}{44} \\ 0 & 0 & 0 & -\frac{3\sqrt{55}}{110} & \frac{\sqrt{330}i}{220} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & -\frac{\sqrt{22}i}{44} & 0 \\ 0 & \frac{\sqrt{330}}{220} & 0 & -\frac{\sqrt{330}i}{220} & -\frac{\sqrt{55}}{110} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 \\ \frac{\sqrt{330}}{220} & 0 & \frac{\sqrt{330}i}{220} & 0 & 0 & \frac{\sqrt{55}}{110} & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 \end{bmatrix}$ |
| 983 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & -\frac{3\sqrt{385}i}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & \frac{5\sqrt{231}i}{3696} & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 \\ \frac{3\sqrt{385}i}{1232} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & -\frac{5\sqrt{231}i}{3696} & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1232} & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & -\frac{65\sqrt{231}i}{3696} & -\frac{\sqrt{154}}{88} & 0 \\ 0 & 0 & \frac{3\sqrt{385}i}{1232} & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & \frac{65\sqrt{231}i}{3696} & 0 & 0 & \frac{\sqrt{154}}{88} \\ 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{\sqrt{385}i}{264} & 0 & \frac{\sqrt{385}}{132} & 0 & 0 & -\frac{5\sqrt{154}}{1848} & 0 & 0 & \frac{5\sqrt{231}i}{1848} \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & \frac{\sqrt{385}i}{264} & 0 & \frac{\sqrt{385}}{132} & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} & -\frac{5\sqrt{231}i}{1848} & 0 \\ \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{132} & 0 & \frac{\sqrt{385}i}{66} & \frac{19\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{924} \\ 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & \frac{\sqrt{385}}{132} & 0 & -\frac{\sqrt{385}i}{66} & 0 & 0 & -\frac{19\sqrt{154}}{1848} & 0 & 0 & -\frac{\sqrt{231}}{924} & 0 \\ 0 & \frac{5\sqrt{231}i}{3696} & 0 & -\frac{\sqrt{231}}{231} & 0 & 0 & \frac{19\sqrt{154}}{1848} & 0 & 0 & -\frac{23\sqrt{385}i}{3696} & 0 & \frac{\sqrt{385}}{462} & 0 & 0 \\ -\frac{5\sqrt{231}i}{3696} & 0 & -\frac{\sqrt{231}}{231} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{154}}{1848} & \frac{23\sqrt{385}i}{3696} & 0 & \frac{\sqrt{385}}{462} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{231}}{462} & 0 & -\frac{65\sqrt{231}i}{3696} & -\frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{462} & 0 & \frac{17\sqrt{385}i}{3696} & \frac{\sqrt{2310}}{616} & 0 \\ -\frac{5\sqrt{231}}{462} & 0 & \frac{65\sqrt{231}i}{3696} & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & \frac{\sqrt{385}}{462} & 0 & -\frac{17\sqrt{385}i}{3696} & 0 & 0 & -\frac{\sqrt{2310}}{616} \\ 0 & 0 & -\frac{\sqrt{154}}{88} & 0 & 0 & \frac{5\sqrt{231}i}{1848} & 0 & -\frac{\sqrt{231}}{924} & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{3\sqrt{385}i}{616} \\ 0 & 0 & 0 & \frac{\sqrt{154}}{88} & -\frac{5\sqrt{231}i}{1848} & 0 & -\frac{\sqrt{231}}{924} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & \frac{3\sqrt{385}i}{616} & 0 \end{pmatrix}$ |
| 984 | symmetry | $x^5-5x^3y^2-5x^3z^2+\frac{15xy^4}{8}+\frac{15xy^2z^2}{4}+\frac{15xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{385}}{1232} & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & -\frac{65\sqrt{231}}{3696} & 0 & -\frac{\sqrt{231}i}{231} & \frac{\sqrt{154}}{88} & 0 \\ \frac{3\sqrt{385}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{65\sqrt{231}}{3696} & 0 & \frac{\sqrt{231}i}{231} & 0 & 0 & -\frac{\sqrt{154}}{88} \\ 0 & 0 & 0 & \frac{3\sqrt{385}}{1232} & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{5\sqrt{231}i}{462} & 0 & \frac{5\sqrt{231}}{3696} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{385}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & \frac{5\sqrt{231}i}{462} & 0 & \frac{5\sqrt{231}}{3696} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{264} & 0 & \frac{\sqrt{385}i}{132} & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} \\ 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & \frac{\sqrt{385}}{264} & 0 & -\frac{\sqrt{385}i}{132} & 0 & 0 & -\frac{5\sqrt{154}}{1848} & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & \frac{\sqrt{385}i}{132} & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & \frac{19\sqrt{154}}{1848} & 0 & 0 & \frac{\sqrt{231}i}{924} \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & -\frac{\sqrt{385}i}{132} & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{154}}{1848} & -\frac{\sqrt{231}i}{924} & 0 \\ 0 & -\frac{65\sqrt{231}}{3696} & 0 & -\frac{5\sqrt{231}i}{462} & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{3696} & 0 & -\frac{\sqrt{385}i}{462} & \frac{\sqrt{2310}}{616} & 0 \\ -\frac{65\sqrt{231}}{3696} & 0 & \frac{5\sqrt{231}i}{462} & 0 & 0 & -\frac{5\sqrt{154}}{1848} & 0 & 0 & -\frac{17\sqrt{385}}{3696} & 0 & \frac{\sqrt{385}i}{462} & 0 & 0 & -\frac{\sqrt{2310}}{616} \\ 0 & -\frac{\sqrt{231}i}{231} & 0 & \frac{5\sqrt{231}}{3696} & 0 & 0 & \frac{19\sqrt{154}}{1848} & 0 & 0 & -\frac{\sqrt{385}i}{462} & 0 & \frac{23\sqrt{385}}{3696} & 0 & 0 \\ \frac{\sqrt{231}i}{231} & 0 & \frac{5\sqrt{231}}{3696} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{154}}{1848} & \frac{\sqrt{385}i}{462} & 0 & \frac{23\sqrt{385}}{3696} & 0 & 0 & 0 \\ \frac{\sqrt{154}}{88} & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & \frac{\sqrt{231}i}{924} & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{616} \\ 0 & -\frac{\sqrt{154}}{88} & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & -\frac{\sqrt{231}i}{924} & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & \frac{3\sqrt{385}}{616} & 0 \end{pmatrix}$ |
| 985 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & -\frac{3\sqrt{11}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & \frac{7\sqrt{165}i}{880} & 0 & \frac{\sqrt{165}}{110} & 0 & 0 \\ \frac{3\sqrt{11}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & -\frac{7\sqrt{165}i}{880} & 0 & \frac{\sqrt{165}}{110} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}i}{176} & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{55} & 0 & \frac{\sqrt{165}i}{176} & \frac{9\sqrt{110}}{440} & 0 \\ 0 & 0 & \frac{3\sqrt{11}i}{176} & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{\sqrt{165}}{55} & 0 & -\frac{\sqrt{165}i}{176} & 0 & 0 & -\frac{9\sqrt{110}}{440} \\ 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{3\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & -\frac{7\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{165}i}{440} \\ 0 & 0 & 0 & \frac{\sqrt{66}}{88} & -\frac{3\sqrt{11}i}{88} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & \frac{7\sqrt{110}}{440} & \frac{\sqrt{165}i}{440} & 0 \\ \frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & \frac{\sqrt{11}i}{22} & \frac{\sqrt{110}}{440} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{220} \\ 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & -\frac{\sqrt{110}}{440} & 0 & 0 & \frac{\sqrt{165}}{220} & 0 \\ 0 & \frac{7\sqrt{165}i}{880} & 0 & -\frac{\sqrt{165}}{55} & 0 & 0 & \frac{\sqrt{110}}{440} & 0 & 0 & \frac{3\sqrt{11}i}{176} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 \\ -\frac{7\sqrt{165}i}{880} & 0 & -\frac{\sqrt{165}}{55} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{440} & -\frac{3\sqrt{11}i}{176} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{110} & 0 & \frac{\sqrt{165}i}{176} & -\frac{7\sqrt{110}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & -\frac{5\sqrt{11}i}{176} & \frac{\sqrt{66}}{88} & 0 \\ \frac{\sqrt{165}}{110} & 0 & -\frac{\sqrt{165}i}{176} & 0 & 0 & \frac{7\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & \frac{5\sqrt{11}i}{176} & 0 & 0 & -\frac{\sqrt{66}}{88} \\ 0 & 0 & \frac{9\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{165}i}{440} & 0 & \frac{\sqrt{165}}{220} & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{3\sqrt{11}i}{88} \\ 0 & 0 & 0 & -\frac{9\sqrt{110}}{440} & \frac{\sqrt{165}i}{440} & 0 & \frac{\sqrt{165}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & \frac{3\sqrt{11}i}{88} & 0 \end{pmatrix}$ |
| 986 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{11}}{176} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{165}i}{55} & -\frac{9\sqrt{110}}{440} & 0 \\ \frac{3\sqrt{11}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{165}i}{55} & 0 & 0 & \frac{9\sqrt{110}}{440} \\ 0 & 0 & 0 & \frac{3\sqrt{11}}{176} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{165}i}{110} & 0 & \frac{7\sqrt{165}}{880} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & -\frac{\sqrt{165}i}{110} & 0 & \frac{7\sqrt{165}}{880} & 0 & 0 & 0 \\ -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{88} & 0 & \frac{\sqrt{11}i}{44} & \frac{7\sqrt{110}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{440} \\ 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{3\sqrt{11}}{88} & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & -\frac{7\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{165}}{440} & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{165}i}{220} \\ 0 & 0 & 0 & \frac{\sqrt{66}}{88} & -\frac{\sqrt{11}i}{44} & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{440} & \frac{\sqrt{165}i}{220} & 0 \\ 0 & \frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{165}i}{110} & \frac{7\sqrt{110}}{440} & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{176} & 0 & \frac{\sqrt{11}i}{22} & \frac{\sqrt{66}}{88} & 0 \\ \frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{165}i}{110} & 0 & 0 & -\frac{7\sqrt{110}}{440} & 0 & 0 & \frac{5\sqrt{11}}{176} & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & -\frac{\sqrt{66}}{88} \\ 0 & -\frac{\sqrt{165}i}{55} & 0 & \frac{7\sqrt{165}}{880} & 0 & 0 & \frac{\sqrt{110}}{440} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & -\frac{3\sqrt{11}}{176} & 0 & 0 \\ \frac{\sqrt{165}i}{55} & 0 & \frac{7\sqrt{165}}{880} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{440} & -\frac{\sqrt{11}i}{22} & 0 & -\frac{3\sqrt{11}}{176} & 0 & 0 & 0 \\ -\frac{9\sqrt{110}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{440} & 0 & -\frac{\sqrt{165}i}{220} & \frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{88} \\ 0 & \frac{9\sqrt{110}}{440} & 0 & 0 & -\frac{\sqrt{165}}{440} & 0 & \frac{\sqrt{165}i}{220} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & \frac{3\sqrt{11}}{88} & 0 \end{bmatrix}$ |
| 987 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & \frac{\sqrt{33}}{88} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{440} & 0 & \frac{\sqrt{55}i}{55} & \frac{\sqrt{330}}{220} & 0 \\ \frac{\sqrt{33}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{55}}{440} & 0 & -\frac{\sqrt{55}i}{55} & 0 & 0 & -\frac{\sqrt{330}}{220} \\ 0 & 0 & 0 & \frac{\sqrt{33}}{88} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{55}i}{55} & 0 & -\frac{13\sqrt{55}}{440} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & -\frac{\sqrt{55}i}{55} & 0 & -\frac{13\sqrt{55}}{440} & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{33}}{132} & 0 & -\frac{\sqrt{33}i}{33} & \frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{220} \\ 0 & \frac{\sqrt{22}}{44} & 0 & 0 & -\frac{7\sqrt{33}}{132} & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & \frac{\sqrt{55}}{220} & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}}{44} & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 \\ 0 & \frac{\sqrt{55}}{440} & 0 & \frac{\sqrt{55}i}{55} & \frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & 0 & \frac{\sqrt{22}}{44} & 0 \\ \frac{\sqrt{55}}{440} & 0 & -\frac{\sqrt{55}i}{55} & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & -\frac{\sqrt{33}}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & \frac{\sqrt{55}i}{55} & 0 & -\frac{13\sqrt{55}}{440} & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 \\ -\frac{\sqrt{55}i}{55} & 0 & -\frac{13\sqrt{55}}{440} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 \\ \frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} \\ 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{33}}{44} & 0 \end{bmatrix}$ |
| 988 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{pmatrix} 0 & \frac{\sqrt{33}i}{88} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{13\sqrt{55}i}{440} & 0 & -\frac{\sqrt{55}}{55} & 0 & 0 \\ -\frac{\sqrt{33}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & -\frac{13\sqrt{55}i}{440} & 0 & -\frac{\sqrt{55}}{55} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}i}{88} & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{55} & 0 & -\frac{\sqrt{55}i}{440} & \frac{\sqrt{330}}{220} & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{88} & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & -\frac{\sqrt{55}}{55} & 0 & \frac{\sqrt{55}i}{440} & 0 & 0 & -\frac{\sqrt{330}}{220} \\ 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & -\frac{7\sqrt{33}i}{132} & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & \frac{\sqrt{330}}{220} & 0 & 0 & -\frac{\sqrt{55}i}{220} \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & \frac{7\sqrt{33}i}{132} & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{220} & \frac{\sqrt{55}i}{220} & 0 \\ -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{55}i}{440} & 0 & -\frac{\sqrt{55}}{55} & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & \frac{5\sqrt{33}i}{264} & 0 & 0 & 0 & 0 \\ -\frac{13\sqrt{55}i}{440} & 0 & -\frac{\sqrt{55}}{55} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & -\frac{5\sqrt{33}i}{264} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{55} & 0 & -\frac{\sqrt{55}i}{440} & \frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{88} & -\frac{\sqrt{22}}{44} & 0 \\ -\frac{\sqrt{55}}{55} & 0 & \frac{\sqrt{55}i}{440} & 0 & 0 & -\frac{\sqrt{330}}{220} & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{88} & 0 & 0 & \frac{\sqrt{22}}{44} \\ 0 & 0 & \frac{\sqrt{330}}{220} & 0 & 0 & -\frac{\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & \frac{\sqrt{33}i}{44} \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{220} & \frac{\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & -\frac{\sqrt{33}i}{44} & 0 \end{pmatrix}$ |
| 989 | symmetry | $-\frac{3\sqrt{231}x^5yz}{4}+\frac{5\sqrt{231}x^3yz^3}{2}+\frac{3\sqrt{231}xy^5z}{4}-\frac{5\sqrt{231}xy^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}i}{364} & 0 & -\frac{\sqrt{273}}{364} & 0 & 0 & \frac{\sqrt{2730}}{364} & 0 & 0 & -\frac{\sqrt{455}i}{182} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{273}i}{364} & 0 & -\frac{\sqrt{273}}{364} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2730}}{364} & \frac{\sqrt{455}i}{182} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{364} & 0 & \frac{\sqrt{273}i}{364} & \frac{\sqrt{2730}}{364} & 0 & 0 & 0 & 0 & \frac{\sqrt{455}}{182} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{364} & 0 & -\frac{\sqrt{273}i}{364} & 0 & 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{\sqrt{455}}{182} & 0 \\ 0 & -\frac{\sqrt{273}i}{364} & 0 & -\frac{\sqrt{273}}{364} & 0 & 0 & \frac{3\sqrt{182}}{182} & 0 & 0 & -\frac{3\sqrt{455}i}{364} & 0 & \frac{3\sqrt{455}}{364} & 0 & 0 \\ \frac{\sqrt{273}i}{364} & 0 & -\frac{\sqrt{273}}{364} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{182}}{182} & \frac{3\sqrt{455}i}{364} & 0 & \frac{3\sqrt{455}}{364} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{273}}{364} & 0 & \frac{\sqrt{273}i}{364} & \frac{3\sqrt{182}}{182} & 0 & 0 & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & \frac{3\sqrt{455}i}{364} & 0 & 0 \\ -\frac{\sqrt{273}}{364} & 0 & -\frac{\sqrt{273}i}{364} & 0 & 0 & -\frac{3\sqrt{182}}{182} & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & -\frac{3\sqrt{455}i}{364} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2730}}{364} & 0 & 0 & -\frac{3\sqrt{455}i}{364} & 0 & \frac{3\sqrt{455}}{364} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2730}}{364} & \frac{3\sqrt{455}i}{364} & 0 & \frac{3\sqrt{455}}{364} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2730}}{364} & 0 & 0 & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & \frac{3\sqrt{455}i}{364} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & -\frac{3\sqrt{455}i}{364} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{455}i}{182} & 0 & \frac{\sqrt{455}}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{455}i}{182} & 0 & \frac{\sqrt{455}}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 990 | symmetry | $-\frac{35x^6z}{16}-\frac{105x^4y^2z}{16}+\frac{105x^4z^3}{8}-\frac{105x^2y^4z}{16}+\frac{105x^2y^2z^3}{4}-\frac{21x^2z^5}{2}-\frac{35y^6z}{16}+\frac{105y^4z^3}{8}-\frac{21y^2z^5}{2}+z^7$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A_g,2)$ | $\begin{pmatrix} -\frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{572} & 0 & -\frac{\sqrt{143}i}{572} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{858}}{1716} & 0 & 0 & -\frac{\sqrt{143}}{572} & 0 & \frac{\sqrt{143}i}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & \frac{\sqrt{143}i}{572} & 0 & -\frac{\sqrt{143}}{572} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{858}}{1716} & -\frac{\sqrt{143}i}{572} & 0 & -\frac{\sqrt{143}}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{143}}{572} & 0 & \frac{\sqrt{143}i}{572} & \frac{\sqrt{858}}{286} & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{572} & 0 & \frac{\sqrt{2145}i}{572} & 0 & 0 \\ -\frac{\sqrt{143}}{572} & 0 & -\frac{\sqrt{143}i}{572} & 0 & 0 & -\frac{\sqrt{858}}{286} & 0 & 0 & \frac{\sqrt{2145}}{572} & 0 & -\frac{\sqrt{2145}i}{572} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{143}i}{572} & 0 & -\frac{\sqrt{143}}{572} & 0 & 0 & \frac{\sqrt{858}}{286} & 0 & 0 & -\frac{\sqrt{2145}i}{572} & 0 & \frac{\sqrt{2145}}{572} & 0 & 0 \\ \frac{\sqrt{143}i}{572} & 0 & -\frac{\sqrt{143}}{572} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{286} & \frac{\sqrt{2145}i}{572} & 0 & \frac{\sqrt{2145}}{572} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{572} & 0 & -\frac{\sqrt{2145}i}{572} & -\frac{5\sqrt{858}}{572} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{286} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{572} & 0 & \frac{\sqrt{2145}i}{572} & 0 & 0 & \frac{5\sqrt{858}}{572} & 0 & 0 & -\frac{5\sqrt{143}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{572} & 0 & \frac{\sqrt{2145}}{572} & 0 & 0 & -\frac{5\sqrt{858}}{572} & 0 & 0 & \frac{5\sqrt{143}i}{286} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{572} & 0 & \frac{\sqrt{2145}}{572} & 0 & 0 & 0 & 0 & \frac{5\sqrt{858}}{572} & -\frac{5\sqrt{143}i}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{286} & 0 & \frac{5\sqrt{143}i}{286} & \frac{5\sqrt{858}}{429} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{286} & 0 & -\frac{5\sqrt{143}i}{286} & 0 & 0 & -\frac{5\sqrt{858}}{429} \end{pmatrix}$ |
| 991 | symmetry | $-\frac{3\sqrt{231}x^6z}{16}+\frac{15\sqrt{231}x^4y^2z}{16}+\frac{5\sqrt{231}x^4z^3}{8}+\frac{15\sqrt{231}x^2y^4z}{16}-\frac{15\sqrt{231}x^2y^2z^3}{4}-\frac{3\sqrt{231}y^6z}{16}+\frac{5\sqrt{231}y^4z^3}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A_g,3)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{364} & 0 & \frac{\sqrt{273}i}{364} & \frac{\sqrt{2730}}{364} & 0 & 0 & 0 & 0 & \frac{\sqrt{455}}{182} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{364} & 0 & -\frac{\sqrt{273}i}{364} & 0 & 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{\sqrt{455}}{182} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{273}i}{364} & 0 & \frac{\sqrt{273}}{364} & 0 & 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{\sqrt{455}i}{182} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{273}i}{364} & 0 & \frac{\sqrt{273}}{364} & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}}{364} & -\frac{\sqrt{455}i}{182} & 0 \\ 0 & -\frac{\sqrt{273}}{364} & 0 & \frac{\sqrt{273}i}{364} & \frac{3\sqrt{182}}{182} & 0 & 0 & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & \frac{3\sqrt{455}i}{364} & 0 & 0 \\ -\frac{\sqrt{273}}{364} & 0 & -\frac{\sqrt{273}i}{364} & 0 & 0 & -\frac{3\sqrt{182}}{182} & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & -\frac{3\sqrt{455}i}{364} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{273}i}{364} & 0 & \frac{\sqrt{273}}{364} & 0 & 0 & -\frac{3\sqrt{182}}{182} & 0 & 0 & \frac{3\sqrt{455}i}{364} & 0 & -\frac{3\sqrt{455}}{364} & 0 & 0 \\ -\frac{\sqrt{273}i}{364} & 0 & \frac{\sqrt{273}}{364} & 0 & 0 & 0 & 0 & \frac{3\sqrt{182}}{182} & -\frac{3\sqrt{455}i}{364} & 0 & -\frac{3\sqrt{455}}{364} & 0 & 0 & 0 \\ \frac{\sqrt{2730}}{364} & 0 & 0 & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & \frac{3\sqrt{455}i}{364} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{3\sqrt{455}}{364} & 0 & -\frac{3\sqrt{455}i}{364} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2730}}{364} & 0 & 0 & \frac{3\sqrt{455}i}{364} & 0 & -\frac{3\sqrt{455}}{364} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2730}}{364} & -\frac{3\sqrt{455}i}{364} & 0 & -\frac{3\sqrt{455}}{364} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{455}}{182} & 0 & \frac{\sqrt{455}i}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{455}}{182} & 0 & -\frac{\sqrt{455}i}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 992 | symmetry | $\frac{3\sqrt{91}x^5yz}{2} - \frac{5\sqrt{91}x^3y^3z}{2} - \frac{5\sqrt{91}x^3yz^3}{2} + \frac{3\sqrt{91}xy^5z}{2} - \frac{5\sqrt{91}xy^3z^3}{2} + \frac{3\sqrt{91}xyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(B_g,1)$ | $\begin{pmatrix} 0 & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & -\frac{\sqrt{1155}i}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{168} & \frac{3\sqrt{77}i}{154} & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{616} & \frac{\sqrt{1155}i}{924} & 0 \\ \frac{\sqrt{462}}{168} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & \frac{5\sqrt{77}i}{308} & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{924} \\ 0 & -\frac{\sqrt{462}}{168} & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & -\frac{\sqrt{1155}}{924} & 0 \\ 0 & -\frac{3\sqrt{77}i}{154} & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{77}i}{154} & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{77}}{308} & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{1155}i}{308} & \frac{\sqrt{770}}{154} & 0 \\ \frac{5\sqrt{77}}{308} & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & 0 & -\frac{\sqrt{770}}{154} \\ 0 & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & 0 & \frac{5\sqrt{462}}{616} & 0 & 0 & -\frac{5\sqrt{77}i}{308} \\ 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{616} & \frac{5\sqrt{77}i}{308} & 0 \\ \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{308} & \frac{5\sqrt{462}}{616} & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{308} \\ 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{308} & 0 & 0 & -\frac{5\sqrt{462}}{616} & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 \\ 0 & -\frac{\sqrt{1155}i}{924} & 0 & -\frac{\sqrt{1155}}{924} & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 \\ \frac{\sqrt{1155}i}{924} & 0 & -\frac{\sqrt{1155}}{924} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{154} & \frac{5\sqrt{77}i}{308} & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & 0 \end{pmatrix}$ |
| 993 | symmetry | $-\frac{3\sqrt{77}x^5yz}{4}+5\sqrt{77}x^3y^3z-\frac{5\sqrt{77}x^3yz^3}{2}-\frac{3\sqrt{77}xy^5z}{4}-\frac{5\sqrt{77}xy^3z^3}{2}+\frac{3\sqrt{77}xyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(B_g,2)$ | $\begin{matrix} 0 & 0 & -\frac{\sqrt{546}}{168} & 0 & 0 & \frac{3\sqrt{91}i}{182} & 0 & -\frac{\sqrt{91}}{52} & 0 & 0 & \frac{\sqrt{910}}{728} & 0 & 0 & -\frac{\sqrt{1365}i}{1092} \\ 0 & 0 & 0 & \frac{\sqrt{546}}{168} & -\frac{3\sqrt{91}i}{182} & 0 & -\frac{\sqrt{91}}{52} & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}}{728} & \frac{\sqrt{1365}i}{1092} & 0 \\ -\frac{\sqrt{546}}{168} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{91}}{182} & 0 & -\frac{\sqrt{91}i}{52} & -\frac{\sqrt{910}}{728} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1365}}{1092} \\ 0 & \frac{\sqrt{546}}{168} & 0 & 0 & -\frac{3\sqrt{91}}{182} & 0 & \frac{\sqrt{91}i}{52} & 0 & 0 & \frac{\sqrt{910}}{728} & 0 & 0 & -\frac{\sqrt{1365}}{1092} & 0 \\ 0 & \frac{3\sqrt{91}i}{182} & 0 & -\frac{3\sqrt{91}}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{91}i}{182} & 0 & -\frac{3\sqrt{91}}{182} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{91}}{52} & 0 & -\frac{\sqrt{91}i}{52} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1365}}{364} & 0 & \frac{\sqrt{1365}i}{364} & \frac{\sqrt{910}}{182} & 0 \\ -\frac{\sqrt{91}}{52} & 0 & \frac{\sqrt{91}i}{52} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1365}}{364} & 0 & -\frac{\sqrt{1365}i}{364} & 0 & 0 & -\frac{\sqrt{910}}{182} \\ 0 & 0 & -\frac{\sqrt{910}}{728} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1365}}{364} & 0 & 0 & \frac{5\sqrt{546}}{728} & 0 & 0 & -\frac{5\sqrt{91}i}{364} \\ 0 & 0 & 0 & \frac{\sqrt{910}}{728} & 0 & 0 & -\frac{\sqrt{1365}}{364} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{546}}{728} & \frac{5\sqrt{91}i}{364} & 0 \\ \frac{\sqrt{910}}{728} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1365}i}{364} & \frac{5\sqrt{546}}{728} & 0 & 0 & 0 & 0 & \frac{5\sqrt{91}}{364} \\ 0 & -\frac{\sqrt{910}}{728} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1365}i}{364} & 0 & 0 & -\frac{5\sqrt{546}}{728} & 0 & 0 & \frac{5\sqrt{91}}{364} & 0 \\ 0 & -\frac{\sqrt{1365}i}{1092} & 0 & -\frac{\sqrt{1365}}{1092} & 0 & 0 & \frac{\sqrt{910}}{182} & 0 & 0 & -\frac{5\sqrt{91}i}{364} & 0 & \frac{5\sqrt{91}}{364} & 0 & 0 \\ \frac{\sqrt{1365}i}{1092} & 0 & -\frac{\sqrt{1365}}{1092} & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}}{182} & \frac{5\sqrt{91}i}{364} & 0 & \frac{5\sqrt{91}}{364} & 0 & 0 & 0 \end{matrix}$ |
| 994 | symmetry | $\frac{\sqrt{6006}x^6z}{32} - \frac{15\sqrt{6006}x^4y^2z}{32} + \frac{15\sqrt{6006}x^2y^4z}{32} - \frac{\sqrt{6006}y^6z}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(B_g,3)$ | $\begin{bmatrix} \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{28} & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 995 | symmetry | $\frac{15\sqrt{42}x^6z}{32}+\frac{15\sqrt{42}x^4y^2z}{32}-\frac{5\sqrt{42}x^4z^3}{2}-\frac{15\sqrt{42}x^2y^4z}{32}+\frac{3\sqrt{42}x^2z^5}{2}-\frac{15\sqrt{42}y^6z}{32}+\frac{5\sqrt{42}y^4z^3}{2}-\frac{3\sqrt{42}y^2z^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(B_g,4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{4004} & 0 & -\frac{\sqrt{6006}i}{4004} & -\frac{\sqrt{15015}}{2002} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2002} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{4004} & 0 & \frac{\sqrt{6006}i}{4004} & 0 & 0 & \frac{\sqrt{15015}}{2002} & 0 & 0 & -\frac{\sqrt{10010}}{2002} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}i}{4004} & 0 & \frac{\sqrt{6006}}{4004} & 0 & 0 & -\frac{\sqrt{15015}}{2002} & 0 & 0 & \frac{\sqrt{10010}i}{2002} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}i}{4004} & 0 & \frac{\sqrt{6006}}{4004} & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{2002} & -\frac{\sqrt{10010}i}{2002} & 0 \\ 0 & \frac{\sqrt{6006}}{4004} & 0 & \frac{\sqrt{6006}i}{4004} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{2002} & 0 & \frac{3\sqrt{10010}i}{2002} & \frac{2\sqrt{15015}}{1001} & 0 \\ \frac{\sqrt{6006}}{4004} & 0 & -\frac{\sqrt{6006}i}{4004} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{2002} & 0 & -\frac{3\sqrt{10010}i}{2002} & 0 & 0 & -\frac{2\sqrt{15015}}{1001} \\ 0 & -\frac{\sqrt{6006}i}{4004} & 0 & \frac{\sqrt{6006}}{4004} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6006}i}{4004} & 0 & \frac{\sqrt{6006}}{4004} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15015}}{2002} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{2002} & 0 & 0 & \frac{15\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{2002} \\ 0 & \frac{\sqrt{15015}}{2002} & 0 & 0 & -\frac{3\sqrt{10010}}{2002} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{2002} & 0 & 0 & \frac{5\sqrt{6006}}{2002} & 0 \\ 0 & 0 & -\frac{\sqrt{15015}}{2002} & 0 & 0 & \frac{3\sqrt{10010}i}{2002} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{2002} & 0 & 0 & \frac{5\sqrt{6006}i}{2002} \\ 0 & 0 & 0 & \frac{\sqrt{15015}}{2002} & -\frac{3\sqrt{10010}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{2002} & -\frac{5\sqrt{6006}i}{2002} & 0 \\ 0 & -\frac{\sqrt{10010}}{2002} & 0 & \frac{\sqrt{10010}i}{2002} & \frac{2\sqrt{15015}}{1001} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{2002} & 0 & \frac{5\sqrt{6006}i}{2002} & 0 & 0 \\ -\frac{\sqrt{10010}}{2002} & 0 & -\frac{\sqrt{10010}i}{2002} & 0 & 0 & -\frac{2\sqrt{15015}}{1001} & 0 & 0 & \frac{5\sqrt{6006}}{2002} & 0 & -\frac{5\sqrt{6006}i}{2002} & 0 & 0 & 0 \end{pmatrix}$ |
| 996 | symmetry | $-\frac{35x^6y}{16}+\frac{105x^4y^3}{8}-\frac{105x^4yz^2}{16}-\frac{21x^2y^5}{2}+\frac{105x^2y^3z^2}{4}-\frac{105x^2yz^4}{16}+y^7-\frac{21y^5z^2}{2}+\frac{105y^3z^4}{8}-\frac{35yz^6}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{113\sqrt{858}i}{13728} & 0 & -\frac{3\sqrt{858}}{416} & 0 & 0 & -\frac{7\sqrt{143}}{1144} & 0 & 0 & \frac{7\sqrt{1430}i}{4576} & 0 & -\frac{7\sqrt{1430}}{4576} & 0 & 0 \\ -\frac{113\sqrt{858}i}{13728} & 0 & -\frac{3\sqrt{858}}{416} & 0 & 0 & 0 & 0 & \frac{7\sqrt{143}}{1144} & -\frac{7\sqrt{1430}i}{4576} & 0 & -\frac{7\sqrt{1430}}{4576} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{858}}{416} & 0 & -\frac{59\sqrt{858}i}{6864} & -\frac{19\sqrt{143}}{2288} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1430}}{4576} & 0 & -\frac{7\sqrt{1430}i}{2288} & -\frac{3\sqrt{2145}}{2288} & 0 \\ -\frac{3\sqrt{858}}{416} & 0 & \frac{59\sqrt{858}i}{6864} & 0 & 0 & \frac{19\sqrt{143}}{2288} & 0 & 0 & -\frac{5\sqrt{1430}}{4576} & 0 & \frac{7\sqrt{1430}i}{2288} & 0 & 0 & \frac{3\sqrt{2145}}{2288} \\ 0 & 0 & -\frac{19\sqrt{143}}{2288} & 0 & 0 & \frac{\sqrt{858}i}{176} & 0 & -\frac{3\sqrt{858}}{1144} & 0 & 0 & -\frac{7\sqrt{2145}}{2288} & 0 & 0 & \frac{7\sqrt{1430}i}{2288} \\ 0 & 0 & 0 & \frac{19\sqrt{143}}{2288} & -\frac{\sqrt{858}i}{176} & 0 & -\frac{3\sqrt{858}}{1144} & 0 & 0 & 0 & 0 & \frac{7\sqrt{2145}}{2288} & -\frac{7\sqrt{1430}i}{2288} & 0 \\ -\frac{7\sqrt{143}}{1144} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{858}}{1144} & 0 & -\frac{\sqrt{858}i}{286} & -\frac{\sqrt{2145}}{1144} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{1144} \\ 0 & \frac{7\sqrt{143}}{1144} & 0 & 0 & -\frac{3\sqrt{858}}{1144} & 0 & \frac{\sqrt{858}i}{286} & 0 & 0 & \frac{\sqrt{2145}}{1144} & 0 & 0 & -\frac{\sqrt{1430}}{1144} & 0 \\ 0 & \frac{7\sqrt{1430}i}{4576} & 0 & -\frac{5\sqrt{1430}}{4576} & 0 & 0 & -\frac{\sqrt{2145}}{1144} & 0 & 0 & \frac{5\sqrt{858}i}{4576} & 0 & -\frac{5\sqrt{858}}{4576} & 0 & 0 \\ -\frac{7\sqrt{1430}i}{4576} & 0 & -\frac{5\sqrt{1430}}{4576} & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{1144} & -\frac{5\sqrt{858}i}{4576} & 0 & -\frac{5\sqrt{858}}{4576} & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{1430}}{4576} & 0 & -\frac{7\sqrt{1430}i}{2288} & -\frac{7\sqrt{2145}}{2288} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{858}}{4576} & 0 & -\frac{15\sqrt{858}i}{2288} & -\frac{25\sqrt{143}}{2288} & 0 \\ -\frac{7\sqrt{1430}}{4576} & 0 & \frac{7\sqrt{1430}i}{2288} & 0 & 0 & \frac{7\sqrt{2145}}{2288} & 0 & 0 & -\frac{5\sqrt{858}}{4576} & 0 & \frac{15\sqrt{858}i}{2288} & 0 & 0 & \frac{25\sqrt{143}}{2288} \\ 0 & 0 & -\frac{3\sqrt{2145}}{2288} & 0 & 0 & \frac{7\sqrt{1430}i}{2288} & 0 & -\frac{\sqrt{1430}}{1144} & 0 & 0 & -\frac{25\sqrt{143}}{2288} & 0 & 0 & \frac{25\sqrt{858}i}{6864} \\ 0 & 0 & 0 & \frac{3\sqrt{2145}}{2288} & -\frac{7\sqrt{1430}i}{2288} & 0 & -\frac{\sqrt{1430}}{1144} & 0 & 0 & 0 & 0 & \frac{25\sqrt{143}}{2288} & -\frac{25\sqrt{858}i}{6864} & 0 \end{pmatrix}$ |
| 997 | symmetry | $x^7 - \frac{21x^5y^2}{2} - \frac{21x^5z^2}{2} + \frac{105x^3y^4}{8} + \frac{105x^3y^2z^2}{4} + \frac{105x^3z^4}{8} - \frac{35xy^6}{16} - \frac{105xy^4z^2}{16} - \frac{105xy^2z^4}{16} - \frac{35xz^6}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{59\sqrt{858}}{6864} & 0 & \frac{3\sqrt{858}i}{416} & -\frac{19\sqrt{143}}{2288} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{1430}}{2288} & 0 & -\frac{5\sqrt{1430}i}{4576} & \frac{3\sqrt{2145}}{2288} & 0 \\ \frac{59\sqrt{858}}{6864} & 0 & -\frac{3\sqrt{858}i}{416} & 0 & 0 & \frac{19\sqrt{143}}{2288} & 0 & 0 & -\frac{7\sqrt{1430}}{2288} & 0 & \frac{5\sqrt{1430}i}{4576} & 0 & 0 & -\frac{3\sqrt{2145}}{2288} \\ 0 & \frac{3\sqrt{858}i}{416} & 0 & -\frac{113\sqrt{858}}{13728} & 0 & 0 & \frac{7\sqrt{143}}{1144} & 0 & 0 & -\frac{7\sqrt{1430}i}{4576} & 0 & \frac{7\sqrt{1430}}{4576} & 0 & 0 \\ -\frac{3\sqrt{858}i}{416} & 0 & -\frac{113\sqrt{858}}{13728} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{143}}{1144} & \frac{7\sqrt{1430}i}{4576} & 0 & \frac{7\sqrt{1430}}{4576} & 0 & 0 & 0 \\ -\frac{19\sqrt{143}}{2288} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{176} & 0 & -\frac{3\sqrt{858}i}{1144} & \frac{7\sqrt{2145}}{2288} & 0 & 0 & 0 & 0 & \frac{7\sqrt{1430}}{2288} \\ 0 & \frac{19\sqrt{143}}{2288} & 0 & 0 & -\frac{\sqrt{858}}{176} & 0 & \frac{3\sqrt{858}i}{1144} & 0 & 0 & -\frac{7\sqrt{2145}}{2288} & 0 & 0 & \frac{7\sqrt{1430}}{2288} & 0 \\ 0 & 0 & \frac{7\sqrt{143}}{1144} & 0 & 0 & -\frac{3\sqrt{858}i}{1144} & 0 & \frac{\sqrt{858}}{286} & 0 & 0 & -\frac{\sqrt{2145}}{1144} & 0 & 0 & \frac{\sqrt{1430}i}{1144} \\ 0 & 0 & 0 & -\frac{7\sqrt{143}}{1144} & \frac{3\sqrt{858}i}{1144} & 0 & \frac{\sqrt{858}}{286} & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{1144} & -\frac{\sqrt{1430}i}{1144} & 0 \\ 0 & -\frac{7\sqrt{1430}}{2288} & 0 & -\frac{7\sqrt{1430}i}{4576} & \frac{7\sqrt{2145}}{2288} & 0 & 0 & 0 & 0 & \frac{15\sqrt{858}}{2288} & 0 & \frac{5\sqrt{858}i}{4576} & -\frac{25\sqrt{143}}{2288} & 0 \\ -\frac{7\sqrt{1430}}{2288} & 0 & \frac{7\sqrt{1430}i}{4576} & 0 & 0 & -\frac{7\sqrt{2145}}{2288} & 0 & 0 & \frac{15\sqrt{858}}{2288} & 0 & -\frac{5\sqrt{858}i}{4576} & 0 & 0 & \frac{25\sqrt{143}}{2288} \\ 0 & -\frac{5\sqrt{1430}i}{4576} & 0 & \frac{7\sqrt{1430}}{4576} & 0 & 0 & -\frac{\sqrt{2145}}{1144} & 0 & 0 & \frac{5\sqrt{858}i}{4576} & 0 & -\frac{5\sqrt{858}}{4576} & 0 & 0 \\ \frac{5\sqrt{1430}i}{4576} & 0 & \frac{7\sqrt{1430}}{4576} & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{1144} & -\frac{5\sqrt{858}i}{4576} & 0 & -\frac{5\sqrt{858}}{4576} & 0 & 0 & 0 \\ \frac{3\sqrt{2145}}{2288} & 0 & 0 & 0 & 0 & \frac{7\sqrt{1430}}{2288} & 0 & \frac{\sqrt{1430}i}{1144} & -\frac{25\sqrt{143}}{2288} & 0 & 0 & 0 & 0 & -\frac{25\sqrt{858}}{6864} \\ 0 & -\frac{3\sqrt{2145}}{2288} & 0 & 0 & \frac{7\sqrt{1430}}{2288} & 0 & -\frac{\sqrt{1430}i}{1144} & 0 & 0 & \frac{25\sqrt{143}}{2288} & 0 & 0 & -\frac{25\sqrt{858}}{6864} & 0 \end{pmatrix}$ |
| 998 | symmetry | $-\frac{3\sqrt{231}x^6y}{16}+\frac{5\sqrt{231}x^4y^3}{8}+\frac{15\sqrt{231}x^4yz^2}{16}-\frac{15\sqrt{231}x^2y^3z^2}{4}+\frac{15\sqrt{231}x^2yz^4}{16}+\frac{5\sqrt{231}y^3z^4}{8}-\frac{3\sqrt{231}yz^6}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{15\sqrt{182}i}{2912} & 0 & -\frac{29\sqrt{182}}{2912} & 0 & 0 & \frac{\sqrt{273}}{104} & 0 & 0 & -\frac{\sqrt{2730}i}{416} & 0 & \frac{\sqrt{2730}}{416} & 0 & 0 \\ -\frac{15\sqrt{182}i}{2912} & 0 & -\frac{29\sqrt{182}}{2912} & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{104} & \frac{\sqrt{2730}i}{416} & 0 & \frac{\sqrt{2730}}{416} & 0 & 0 & 0 \\ 0 & -\frac{29\sqrt{182}}{2912} & 0 & -\frac{9\sqrt{182}i}{1456} & \frac{11\sqrt{273}}{1456} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2730}}{2912} & 0 & \frac{3\sqrt{2730}i}{1456} & \frac{\sqrt{455}}{1456} & 0 \\ -\frac{29\sqrt{182}}{2912} & 0 & \frac{9\sqrt{182}i}{1456} & 0 & 0 & -\frac{11\sqrt{273}}{1456} & 0 & 0 & \frac{5\sqrt{2730}}{2912} & 0 & -\frac{3\sqrt{2730}i}{1456} & 0 & 0 & -\frac{\sqrt{455}}{1456} \\ 0 & 0 & \frac{11\sqrt{273}}{1456} & 0 & 0 & -\frac{15\sqrt{182}i}{1456} & 0 & \frac{9\sqrt{182}}{728} & 0 & 0 & -\frac{3\sqrt{455}}{1456} & 0 & 0 & \frac{\sqrt{2730}i}{1456} \\ 0 & 0 & 0 & -\frac{11\sqrt{273}}{1456} & \frac{15\sqrt{182}i}{1456} & 0 & \frac{9\sqrt{182}}{728} & 0 & 0 & 0 & 0 & \frac{3\sqrt{455}}{1456} & -\frac{\sqrt{2730}i}{1456} & 0 \\ \frac{\sqrt{273}}{104} & 0 & 0 & 0 & 0 & \frac{9\sqrt{182}}{728} & 0 & \frac{3\sqrt{182}i}{182} & \frac{3\sqrt{455}}{728} & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}}{728} \\ 0 & -\frac{\sqrt{273}}{104} & 0 & 0 & \frac{9\sqrt{182}}{728} & 0 & -\frac{3\sqrt{182}i}{182} & 0 & 0 & -\frac{3\sqrt{455}}{728} & 0 & 0 & \frac{\sqrt{2730}}{728} & 0 \\ 0 & -\frac{\sqrt{2730}i}{416} & 0 & \frac{5\sqrt{2730}}{2912} & 0 & 0 & \frac{3\sqrt{455}}{728} & 0 & 0 & -\frac{15\sqrt{182}i}{2912} & 0 & \frac{15\sqrt{182}}{2912} & 0 & 0 \\ \frac{\sqrt{2730}i}{416} & 0 & \frac{5\sqrt{2730}}{2912} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{455}}{728} & \frac{15\sqrt{182}i}{2912} & 0 & \frac{15\sqrt{182}}{2912} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2730}}{416} & 0 & \frac{3\sqrt{2730}i}{1456} & -\frac{3\sqrt{455}}{1456} & 0 & 0 & 0 & 0 & \frac{15\sqrt{182}}{2912} & 0 & -\frac{15\sqrt{182}i}{1456} & -\frac{15\sqrt{273}}{1456} & 0 \\ \frac{\sqrt{2730}}{416} & 0 & -\frac{3\sqrt{2730}i}{1456} & 0 & 0 & \frac{3\sqrt{455}}{1456} & 0 & 0 & \frac{15\sqrt{182}}{2912} & 0 & \frac{15\sqrt{182}i}{1456} & 0 & 0 & \frac{15\sqrt{273}}{1456} \\ 0 & 0 & \frac{\sqrt{455}}{1456} & 0 & 0 & \frac{\sqrt{2730}i}{1456} & 0 & \frac{\sqrt{2730}}{728} & 0 & 0 & -\frac{15\sqrt{273}}{1456} & 0 & 0 & \frac{15\sqrt{182}i}{1456} \\ 0 & 0 & 0 & -\frac{\sqrt{455}}{1456} & -\frac{\sqrt{2730}i}{1456} & 0 & \frac{\sqrt{2730}}{728} & 0 & 0 & 0 & 0 & \frac{15\sqrt{273}}{1456} & -\frac{15\sqrt{182}i}{1456} & 0 \end{pmatrix}$ |
| 999 | symmetry | $\frac{5\sqrt{231}x^3y^4}{8} - \frac{15\sqrt{231}x^3y^2z^2}{4} + \frac{5\sqrt{231}x^3z^4}{8} - \frac{3\sqrt{231}xy^6}{16} + \frac{15\sqrt{231}xy^4z^2}{16} + \frac{15\sqrt{231}xy^2z^4}{16} - \frac{3\sqrt{231}xz^6}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{9\sqrt{182}}{1456} & 0 & \frac{29\sqrt{182}i}{2912} & \frac{11\sqrt{273}}{1456} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2730}}{1456} & 0 & \frac{5\sqrt{2730}i}{2912} & -\frac{\sqrt{455}}{1456} & 0 \\ \frac{9\sqrt{182}}{1456} & 0 & -\frac{29\sqrt{182}i}{2912} & 0 & 0 & -\frac{11\sqrt{273}}{1456} & 0 & 0 & \frac{3\sqrt{2730}}{1456} & 0 & -\frac{5\sqrt{2730}i}{2912} & 0 & 0 & \frac{\sqrt{455}}{1456} \\ 0 & \frac{29\sqrt{182}i}{2912} & 0 & -\frac{15\sqrt{182}}{2912} & 0 & 0 & -\frac{\sqrt{273}}{104} & 0 & 0 & \frac{\sqrt{2730}i}{416} & 0 & -\frac{\sqrt{2730}}{416} & 0 & 0 \\ -\frac{29\sqrt{182}i}{2912} & 0 & -\frac{15\sqrt{182}}{2912} & 0 & 0 & 0 & 0 & \frac{\sqrt{273}}{104} & -\frac{\sqrt{2730}i}{416} & 0 & -\frac{\sqrt{2730}}{416} & 0 & 0 & 0 \\ \frac{11\sqrt{273}}{1456} & 0 & 0 & 0 & 0 & \frac{15\sqrt{182}}{1456} & 0 & \frac{9\sqrt{182}i}{728} & \frac{3\sqrt{455}}{1456} & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}}{1456} \\ 0 & -\frac{11\sqrt{273}}{1456} & 0 & 0 & \frac{15\sqrt{182}}{1456} & 0 & -\frac{9\sqrt{182}i}{728} & 0 & 0 & -\frac{3\sqrt{455}}{1456} & 0 & 0 & \frac{\sqrt{2730}}{1456} & 0 \\ 0 & 0 & -\frac{\sqrt{273}}{104} & 0 & 0 & \frac{9\sqrt{182}i}{728} & 0 & -\frac{3\sqrt{182}}{182} & 0 & 0 & \frac{3\sqrt{455}}{728} & 0 & 0 & -\frac{\sqrt{2730}i}{728} \\ 0 & 0 & 0 & \frac{\sqrt{273}}{104} & -\frac{9\sqrt{182}i}{728} & 0 & -\frac{3\sqrt{182}}{182} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{455}}{728} & \frac{\sqrt{2730}i}{728} & 0 \\ 0 & \frac{3\sqrt{2730}}{1456} & 0 & \frac{\sqrt{2730}i}{416} & \frac{3\sqrt{455}}{1456} & 0 & 0 & 0 & 0 & \frac{15\sqrt{182}}{1456} & 0 & -\frac{15\sqrt{182}i}{2912} & -\frac{15\sqrt{273}}{1456} & 0 \\ \frac{3\sqrt{2730}}{1456} & 0 & -\frac{\sqrt{2730}i}{416} & 0 & 0 & -\frac{3\sqrt{455}}{1456} & 0 & 0 & \frac{15\sqrt{182}}{1456} & 0 & \frac{15\sqrt{182}i}{2912} & 0 & 0 & \frac{15\sqrt{273}}{1456} \\ 0 & \frac{5\sqrt{2730}i}{2912} & 0 & -\frac{\sqrt{2730}}{416} & 0 & 0 & \frac{3\sqrt{455}}{728} & 0 & 0 & -\frac{15\sqrt{182}i}{2912} & 0 & \frac{15\sqrt{182}}{2912} & 0 & 0 \\ -\frac{5\sqrt{2730}i}{2912} & 0 & -\frac{\sqrt{2730}}{416} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{455}}{728} & \frac{15\sqrt{182}i}{2912} & 0 & \frac{15\sqrt{182}}{2912} & 0 & 0 & 0 \\ -\frac{\sqrt{455}}{1456} & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}}{1456} & 0 & -\frac{\sqrt{2730}i}{728} & -\frac{15\sqrt{273}}{1456} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{182}}{1456} \\ 0 & \frac{\sqrt{455}}{1456} & 0 & 0 & \frac{\sqrt{2730}}{1456} & 0 & \frac{\sqrt{2730}i}{728} & 0 & 0 & \frac{15\sqrt{273}}{1456} & 0 & 0 & -\frac{15\sqrt{182}}{1456} & 0 \end{pmatrix}$ |
| 1000 | symmetry | $\frac{\sqrt{6006}xy^6}{32} - \frac{15\sqrt{6006}xy^4z^2}{32} + \frac{15\sqrt{6006}xy^2z^4}{32} - \frac{\sqrt{6006}xz^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{7}i}{224} & -\frac{3\sqrt{42}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{224} & -\frac{3\sqrt{70}}{224} & 0 \\ 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & 0 & \frac{3\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}i}{224} & 0 & 0 & \frac{3\sqrt{70}}{224} \\ 0 & -\frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{7}}{224} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 \\ \frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & \frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & 0 \\ -\frac{3\sqrt{42}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{112} & 0 & -\frac{3\sqrt{7}i}{56} & -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} \\ 0 & \frac{3\sqrt{42}}{224} & 0 & 0 & -\frac{3\sqrt{7}}{112} & 0 & \frac{3\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} & \frac{\sqrt{105}i}{56} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{224} & -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}i}{224} & -\frac{5\sqrt{42}}{224} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{224} & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & \frac{15\sqrt{7}i}{224} & 0 & 0 & \frac{5\sqrt{42}}{224} \\ 0 & -\frac{3\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & \frac{3\sqrt{70}}{112} & 0 & 0 & -\frac{15\sqrt{7}i}{224} & 0 & \frac{15\sqrt{7}}{224} & 0 & 0 \\ \frac{3\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} & \frac{15\sqrt{7}i}{224} & 0 & \frac{15\sqrt{7}}{224} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{105}i}{56} & -\frac{5\sqrt{42}}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{112} \\ 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{105}i}{56} & 0 & 0 & \frac{5\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}}{112} & 0 \end{bmatrix}$ |
| 1001 | symmetry | $-\frac{\sqrt{6006}x^6y}{32}+\frac{15\sqrt{6006}x^4yz^2}{32}-\frac{15\sqrt{6006}x^2yz^4}{32}+\frac{\sqrt{6006}yz^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)$ | $\begin{pmatrix} 0 & \frac{\sqrt{7}i}{224} & 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 \\ -\frac{\sqrt{7}i}{224} & 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & \frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{224} & 0 & 0 & \frac{3\sqrt{42}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 \\ -\frac{3\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{224} & 0 & 0 & \frac{3\sqrt{105}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{224} \\ 0 & 0 & \frac{3\sqrt{42}}{224} & 0 & 0 & -\frac{3\sqrt{7}i}{112} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}i}{112} \\ 0 & 0 & 0 & -\frac{3\sqrt{42}}{224} & \frac{3\sqrt{7}i}{112} & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{224} & -\frac{\sqrt{105}i}{112} & 0 \\ \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & -\frac{3\sqrt{70}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\ 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 \\ 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{105}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{112} & 0 & 0 & \frac{15\sqrt{7}i}{224} & 0 & -\frac{15\sqrt{7}}{224} & 0 & 0 \\ \frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{105}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & -\frac{15\sqrt{7}i}{224} & 0 & -\frac{15\sqrt{7}}{224} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{224} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{224} & 0 & 0 & \frac{5\sqrt{42}}{224} & 0 \\ \frac{\sqrt{105}}{224} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{224} & 0 & 0 & -\frac{15\sqrt{7}}{224} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{224} \\ 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & \frac{5\sqrt{42}}{224} & 0 & 0 & -\frac{5\sqrt{7}i}{112} \\ 0 & 0 & 0 & \frac{3\sqrt{70}}{224} & -\frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{224} & \frac{5\sqrt{7}i}{112} & 0 \end{pmatrix}$ |
| 1002 | symmetry | $\frac{3\sqrt{42}x^5y^2}{2} - \frac{3\sqrt{42}x^5z^2}{2} - \frac{5\sqrt{42}x^3y^4}{2} + \frac{5\sqrt{42}x^3z^4}{2} + \frac{15\sqrt{42}xy^6}{32} + \frac{15\sqrt{42}xy^4z^2}{32} - \frac{15\sqrt{42}xy^2z^4}{32} - \frac{15\sqrt{42}xz^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1003 | symmetry | $-\frac{15\sqrt{42}x^6y}{32}+\frac{5\sqrt{42}x^4y^3}{2}-\frac{15\sqrt{42}x^4yz^2}{32}-\frac{3\sqrt{42}x^2y^5}{2}+\frac{15\sqrt{42}x^2yz^4}{32}+\frac{3\sqrt{42}y^5z^2}{2}-\frac{5\sqrt{42}y^3z^4}{2}+\frac{15\sqrt{42}yz^6}{32}$ |

$\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)$

$$\begin{bmatrix}
0 & -\frac{15\sqrt{1001}}{2002} & 0 & -\frac{23\sqrt{1001}i}{2912} & -\frac{\sqrt{6006}}{2464} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{2002} & 0 & \frac{3\sqrt{15015}i}{32032} & \frac{19\sqrt{10010}}{32032} & 0 \\
-\frac{15\sqrt{1001}}{2002} & 0 & \frac{23\sqrt{1001}i}{2912} & 0 & 0 & \frac{\sqrt{6006}}{2464} & 0 & 0 & -\frac{\sqrt{15015}}{2002} & 0 & -\frac{3\sqrt{15015}i}{32032} & 0 & 0 & -\frac{19\sqrt{10010}}{32032} \\
0 & -\frac{23\sqrt{1001}i}{2912} & 0 & \frac{255\sqrt{1001}}{32032} & 0 & 0 & -\frac{\sqrt{6006}}{16016} & 0 & 0 & \frac{\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 \\
\frac{23\sqrt{1001}i}{2912} & 0 & \frac{255\sqrt{1001}}{32032} & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{16016} & -\frac{\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 & 0 \\
-\frac{\sqrt{6006}}{2464} & 0 & 0 & 0 & 0 & -\frac{45\sqrt{1001}}{16016} & 0 & \frac{3\sqrt{1001}i}{8008} & \frac{51\sqrt{10010}}{32032} & 0 & 0 & 0 & 0 & \frac{17\sqrt{15015}}{16016} \\
0 & \frac{\sqrt{6006}}{2464} & 0 & 0 & -\frac{45\sqrt{1001}}{16016} & 0 & -\frac{3\sqrt{1001}i}{8008} & 0 & 0 & -\frac{51\sqrt{10010}}{32032} & 0 & 0 & \frac{17\sqrt{15015}}{16016} & 0 \\
0 & 0 & -\frac{\sqrt{6006}}{16016} & 0 & 0 & \frac{3\sqrt{1001}i}{8008} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{16016} & 0 & 0 & \frac{\sqrt{15015}i}{8008} \\
0 & 0 & 0 & \frac{\sqrt{6006}}{16016} & -\frac{3\sqrt{1001}i}{8008} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{16016} & -\frac{\sqrt{15015}i}{8008} & 0 \\
0 & -\frac{\sqrt{15015}}{2002} & 0 & \frac{\sqrt{15015}i}{32032} & \frac{51\sqrt{10010}}{32032} & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{2002} & 0 & \frac{15\sqrt{1001}i}{32032} & -\frac{75\sqrt{6006}}{32032} & 0 \\
-\frac{\sqrt{15015}}{2002} & 0 & -\frac{\sqrt{15015}i}{32032} & 0 & 0 & -\frac{51\sqrt{10010}}{32032} & 0 & 0 & \frac{15\sqrt{1001}}{2002} & 0 & -\frac{15\sqrt{1001}i}{32032} & 0 & 0 & \frac{75\sqrt{6006}}{32032} \\
0 & \frac{3\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 & -\frac{3\sqrt{10010}}{16016} & 0 & 0 & \frac{15\sqrt{1001}i}{32032} & 0 & -\frac{15\sqrt{1001}}{32032} & 0 & 0 \\
-\frac{3\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{16016} & -\frac{15\sqrt{1001}i}{32032} & 0 & -\frac{15\sqrt{1001}}{32032} & 0 & 0 & 0 \\
\frac{19\sqrt{10010}}{32032} & 0 & 0 & 0 & 0 & \frac{17\sqrt{15015}}{16016} & 0 & \frac{\sqrt{15015}i}{8008} & -\frac{75\sqrt{6006}}{32032} & 0 & 0 & 0 & 0 & -\frac{75\sqrt{1001}}{16016} \\
0 & -\frac{19\sqrt{10010}}{32032} & 0 & 0 & \frac{17\sqrt{15015}}{16016} & 0 & -\frac{\sqrt{15015}i}{8008} & 0 & 0 & \frac{75\sqrt{6006}}{32032} & 0 & 0 & -\frac{75\sqrt{1001}}{16016} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1004 | $\mathbb{M}^{(1,-1;a)}_{7,2}(E_g,4)$ | $\begin{pmatrix} 0 & \frac{255\sqrt{1001}i}{32032} & 0 & -\frac{23\sqrt{1001}}{2912} & 0 & 0 & -\frac{\sqrt{6006}}{16016} & 0 & 0 & \frac{\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 \\ -\frac{255\sqrt{1001}i}{32032} & 0 & -\frac{23\sqrt{1001}}{2912} & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{16016} & -\frac{\sqrt{15015}i}{32032} & 0 & -\frac{\sqrt{15015}}{32032} & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{1001}}{2912} & 0 & -\frac{15\sqrt{1001}i}{2002} & \frac{\sqrt{6006}}{2464} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15015}}{32032} & 0 & \frac{\sqrt{15015}i}{2002} & \frac{19\sqrt{10010}}{32032} & 0 \\ -\frac{23\sqrt{1001}}{2912} & 0 & \frac{15\sqrt{1001}i}{2002} & 0 & 0 & -\frac{\sqrt{6006}}{2464} & 0 & 0 & -\frac{3\sqrt{15015}}{32032} & 0 & -\frac{\sqrt{15015}i}{2002} & 0 & 0 & -\frac{19\sqrt{10010}}{32032} \\ 0 & 0 & \frac{\sqrt{6006}}{2464} & 0 & 0 & -\frac{45\sqrt{1001}i}{16016} & 0 & -\frac{3\sqrt{1001}}{8008} & 0 & 0 & \frac{51\sqrt{10010}}{32032} & 0 & 0 & -\frac{17\sqrt{15015}i}{16016} \\ 0 & 0 & 0 & -\frac{\sqrt{6006}}{2464} & \frac{45\sqrt{1001}i}{16016} & 0 & -\frac{3\sqrt{1001}}{8008} & 0 & 0 & 0 & 0 & -\frac{51\sqrt{10010}}{32032} & \frac{17\sqrt{15015}i}{16016} & 0 \\ -\frac{\sqrt{6006}}{16016} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{8008} & 0 & 0 & \frac{3\sqrt{10010}}{16016} & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{8008} \\ 0 & \frac{\sqrt{6006}}{16016} & 0 & 0 & -\frac{3\sqrt{1001}}{8008} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{16016} & 0 & 0 & \frac{\sqrt{15015}}{8008} & 0 \\ 0 & \frac{\sqrt{15015}i}{32032} & 0 & -\frac{3\sqrt{15015}}{32032} & 0 & 0 & \frac{3\sqrt{10010}}{16016} & 0 & 0 & -\frac{15\sqrt{1001}i}{32032} & 0 & \frac{15\sqrt{1001}}{32032} & 0 & 0 \\ -\frac{\sqrt{15015}i}{32032} & 0 & -\frac{3\sqrt{15015}}{32032} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10010}}{16016} & \frac{15\sqrt{1001}i}{32032} & 0 & \frac{15\sqrt{1001}}{32032} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15015}}{32032} & 0 & \frac{\sqrt{15015}i}{2002} & \frac{51\sqrt{10010}}{32032} & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{32032} & 0 & \frac{15\sqrt{1001}i}{2002} & \frac{75\sqrt{6006}}{32032} & 0 \\ -\frac{\sqrt{15015}}{32032} & 0 & -\frac{\sqrt{15015}i}{2002} & 0 & 0 & -\frac{51\sqrt{10010}}{32032} & 0 & 0 & \frac{15\sqrt{1001}}{32032} & 0 & -\frac{15\sqrt{1001}i}{2002} & 0 & 0 & -\frac{75\sqrt{6006}}{32032} \\ 0 & 0 & \frac{19\sqrt{10010}}{32032} & 0 & 0 & -\frac{17\sqrt{15015}i}{16016} & 0 & \frac{\sqrt{15015}}{8008} & 0 & 0 & \frac{75\sqrt{6006}}{32032} & 0 & 0 & -\frac{75\sqrt{1001}i}{16016} \\ 0 & 0 & 0 & -\frac{19\sqrt{10010}}{32032} & \frac{17\sqrt{15015}i}{16016} & 0 & \frac{\sqrt{15015}}{8008} & 0 & 0 & 0 & 0 & -\frac{75\sqrt{6006}}{32032} & \frac{75\sqrt{1001}i}{16016} & 0 \end{pmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1005 | symmetry | $y$ |

$\mathbb{M}_1^{(1,1;a)}(A_g)$

$$\begin{pmatrix}
-\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\
\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 \\
-\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{42}i}{56} & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{\sqrt{70}}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & -\frac{\sqrt{70}i}{140} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & \frac{\sqrt{70}i}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & -\frac{\sqrt{70}i}{140} & \frac{2\sqrt{105}}{105} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & -\frac{2\sqrt{105}}{105}
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1006 | symmetry | $x$ |

$\mathbb{M}^{(1,1;a)}_{1,1}(E_g)$

$$\begin{pmatrix}
0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\
\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & \frac{\sqrt{7}i}{14} & 0 \\
-\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\
0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\
0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\
\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & -\frac{3\sqrt{105}i}{140} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{105}i}{140} & \frac{\sqrt{70}}{140} & 0 \\
\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{70}}{140} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{105}i}{105} \\
0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & -\frac{\sqrt{105}i}{105} & 0
\end{pmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{pmatrix} 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{7}i}{28} & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{105}i}{70} & \frac{\sqrt{70}}{140} & 0 \\ -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{\sqrt{70}}{140} \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & \frac{\sqrt{105}i}{70} & 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 \end{pmatrix}$ |
| 1007 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(A_g)$ | $\begin{bmatrix} \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{924} & 0 & -\frac{5\sqrt{462}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & -\frac{5\sqrt{462}}{924} & 0 & \frac{5\sqrt{462}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & \frac{5\sqrt{462}i}{924} & 0 & -\frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & -\frac{5\sqrt{462}i}{924} & 0 & -\frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{462}}{924} & 0 & \frac{5\sqrt{462}i}{924} & -\frac{\sqrt{77}}{33} & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{231} & 0 & \frac{\sqrt{770}i}{231} & 0 & 0 \\ -\frac{5\sqrt{462}}{924} & 0 & -\frac{5\sqrt{462}i}{924} & 0 & 0 & \frac{\sqrt{77}}{33} & 0 & 0 & \frac{\sqrt{770}}{231} & 0 & -\frac{\sqrt{770}i}{231} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{462}i}{924} & 0 & -\frac{5\sqrt{462}}{924} & 0 & 0 & -\frac{\sqrt{77}}{33} & 0 & 0 & -\frac{\sqrt{770}i}{231} & 0 & \frac{\sqrt{770}}{231} & 0 & 0 \\ \frac{5\sqrt{462}i}{924} & 0 & -\frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{33} & \frac{\sqrt{770}i}{231} & 0 & \frac{\sqrt{770}}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{231} & 0 & -\frac{\sqrt{770}i}{231} & \frac{\sqrt{77}}{231} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{231} & 0 & \frac{\sqrt{770}i}{231} & 0 & 0 & -\frac{\sqrt{77}}{231} & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{231} & 0 & \frac{\sqrt{770}}{231} & 0 & 0 & \frac{\sqrt{77}}{231} & 0 & 0 & -\frac{5\sqrt{462}i}{924} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{231} & 0 & \frac{\sqrt{770}}{231} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{231} & \frac{5\sqrt{462}i}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & -\frac{5\sqrt{462}i}{924} & \frac{2\sqrt{77}}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & \frac{5\sqrt{462}i}{924} & 0 & 0 & -\frac{2\sqrt{77}}{77} \end{bmatrix}$ |
| 1008 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & -\frac{\sqrt{462}i}{88} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & \frac{\sqrt{462}i}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & -\frac{\sqrt{770}i}{616} & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 \\ 0 & -\frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 \\ \frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{770}}{616} & 0 & -\frac{\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & -\frac{\sqrt{462}i}{616} & -\frac{\sqrt{77}}{154} & 0 \\ -\frac{\sqrt{770}}{616} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & \frac{\sqrt{462}i}{616} & 0 & 0 & \frac{\sqrt{77}}{154} \\ 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{616} & 0 & 0 & \frac{\sqrt{1155}}{231} & 0 & 0 & \frac{\sqrt{770}i}{616} \\ 0 & 0 & 0 & -\frac{3\sqrt{77}}{154} & -\frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{462}}{616} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{231} & -\frac{\sqrt{770}i}{616} & 0 \\ -\frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{462}i}{616} & \frac{\sqrt{1155}}{231} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{616} \\ 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & \frac{\sqrt{462}}{168} & 0 & \frac{\sqrt{462}i}{616} & 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{\sqrt{770}}{616} & 0 \\ 0 & -\frac{\sqrt{462}i}{88} & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 \\ \frac{\sqrt{462}i}{88} & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & -\frac{\sqrt{770}i}{616} & 0 & -\frac{\sqrt{770}}{616} & 0 & 0 & 0 \end{bmatrix}$ |
| 1009 | symmetry | $\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(B_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & -\frac{\sqrt{770}i}{616} & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{616} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & \frac{\sqrt{462}i}{88} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{154} & -\frac{\sqrt{462}i}{88} & 0 \\ 0 & \frac{\sqrt{770}}{616} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & -\frac{\sqrt{462}i}{616} & -\frac{\sqrt{77}}{154} & 0 \\ \frac{\sqrt{770}}{616} & 0 & -\frac{\sqrt{770}i}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & \frac{\sqrt{462}i}{616} & 0 & 0 & \frac{\sqrt{77}}{154} \\ 0 & -\frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{168} & 0 & -\frac{\sqrt{462}}{168} & 0 & 0 \\ \frac{\sqrt{770}i}{616} & 0 & \frac{\sqrt{770}}{616} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{168} & 0 & -\frac{\sqrt{462}}{168} & 0 & 0 & 0 \\ \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & -\frac{\sqrt{462}i}{168} & \frac{\sqrt{1155}}{231} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{616} \\ 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & \frac{\sqrt{462}}{616} & 0 & \frac{\sqrt{462}i}{168} & 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{\sqrt{770}}{616} & 0 \\ 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & -\frac{\sqrt{462}i}{616} & 0 & -\frac{\sqrt{462}}{168} & 0 & 0 & -\frac{\sqrt{1155}}{231} & 0 & 0 & -\frac{\sqrt{770}i}{616} \\ 0 & 0 & 0 & -\frac{3\sqrt{77}}{154} & \frac{\sqrt{462}i}{616} & 0 & -\frac{\sqrt{462}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{231} & \frac{\sqrt{770}i}{616} & 0 \\ 0 & -\frac{\sqrt{462}}{88} & 0 & \frac{\sqrt{462}i}{88} & -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{616} & 0 & -\frac{\sqrt{770}i}{616} & 0 & 0 \\ -\frac{\sqrt{462}}{88} & 0 & -\frac{\sqrt{462}i}{88} & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & -\frac{\sqrt{770}}{616} & 0 & \frac{\sqrt{770}i}{616} & 0 & 0 & 0 \end{bmatrix}$ |
| 1010 | symmetry | $-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & -\frac{3\sqrt{77}i}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{1848} & 0 & \frac{5\sqrt{1155}}{924} & 0 & 0 \\ \frac{3\sqrt{77}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{1232} & -\frac{\sqrt{1155}i}{1848} & 0 & \frac{5\sqrt{1155}}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{77}i}{616} & \frac{5\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & -\frac{13\sqrt{1155}i}{1848} & \frac{\sqrt{770}}{176} & 0 \\ 0 & 0 & \frac{3\sqrt{77}i}{616} & 0 & 0 & -\frac{5\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & \frac{13\sqrt{1155}i}{1848} & 0 & 0 & -\frac{\sqrt{770}}{176} \\ 0 & 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{77}i}{132} & 0 & -\frac{5\sqrt{77}}{264} & 0 & 0 & \frac{5\sqrt{770}}{3696} & 0 & 0 & \frac{\sqrt{1155}i}{924} \\ 0 & 0 & 0 & -\frac{5\sqrt{462}}{1232} & \frac{\sqrt{77}i}{132} & 0 & -\frac{5\sqrt{77}}{264} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{3696} & -\frac{\sqrt{1155}i}{924} & 0 \\ -\frac{5\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{264} & 0 & \frac{\sqrt{77}i}{33} & -\frac{19\sqrt{770}}{3696} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1848} \\ 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & -\frac{5\sqrt{77}}{264} & 0 & -\frac{\sqrt{77}i}{33} & 0 & 0 & \frac{19\sqrt{770}}{3696} & 0 & 0 & \frac{\sqrt{1155}}{1848} & 0 \\ 0 & \frac{\sqrt{1155}i}{1848} & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & -\frac{19\sqrt{770}}{3696} & 0 & 0 & -\frac{23\sqrt{77}i}{1848} & 0 & -\frac{5\sqrt{77}}{924} & 0 & 0 \\ -\frac{\sqrt{1155}i}{1848} & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & \frac{19\sqrt{770}}{3696} & \frac{23\sqrt{77}i}{1848} & 0 & -\frac{5\sqrt{77}}{924} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{1155}}{924} & 0 & -\frac{13\sqrt{1155}i}{1848} & \frac{5\sqrt{770}}{3696} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{924} & 0 & \frac{17\sqrt{77}i}{1848} & -\frac{5\sqrt{462}}{1232} & 0 \\ \frac{5\sqrt{1155}}{924} & 0 & \frac{13\sqrt{1155}i}{1848} & 0 & 0 & -\frac{5\sqrt{770}}{3696} & 0 & 0 & -\frac{5\sqrt{77}}{924} & 0 & -\frac{17\sqrt{77}i}{1848} & 0 & 0 & \frac{5\sqrt{462}}{1232} \\ 0 & 0 & \frac{\sqrt{770}}{176} & 0 & 0 & \frac{\sqrt{1155}i}{924} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 & -\frac{5\sqrt{462}}{1232} & 0 & 0 & -\frac{3\sqrt{77}i}{308} \\ 0 & 0 & 0 & -\frac{\sqrt{770}}{176} & -\frac{\sqrt{1155}i}{924} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{1232} & \frac{3\sqrt{77}i}{308} & 0 \end{pmatrix}$ |
| 1011 | symmetry | $x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{1155}}{1848} & 0 & \frac{\sqrt{1155}i}{462} & -\frac{\sqrt{770}}{176} & 0 \\ \frac{3\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1232} & 0 & 0 & -\frac{13\sqrt{1155}}{1848} & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & \frac{\sqrt{770}}{176} \\ 0 & 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & \frac{5\sqrt{1155}i}{924} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1232} & -\frac{5\sqrt{1155}i}{924} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 & 0 \\ \frac{5\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{132} & 0 & -\frac{5\sqrt{77}i}{264} & -\frac{5\sqrt{770}}{3696} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} \\ 0 & -\frac{5\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{77}}{132} & 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & \frac{5\sqrt{770}}{3696} & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 \\ 0 & 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & -\frac{\sqrt{77}}{33} & 0 & 0 & -\frac{19\sqrt{770}}{3696} & 0 & 0 & -\frac{\sqrt{1155}i}{1848} \\ 0 & 0 & 0 & -\frac{5\sqrt{462}}{1232} & \frac{5\sqrt{77}i}{264} & 0 & -\frac{\sqrt{77}}{33} & 0 & 0 & 0 & 0 & \frac{19\sqrt{770}}{3696} & \frac{\sqrt{1155}i}{1848} & 0 \\ 0 & -\frac{13\sqrt{1155}}{1848} & 0 & \frac{5\sqrt{1155}i}{924} & -\frac{5\sqrt{770}}{3696} & 0 & 0 & 0 & 0 & -\frac{17\sqrt{77}}{1848} & 0 & \frac{5\sqrt{77}i}{924} & -\frac{5\sqrt{462}}{1232} & 0 \\ -\frac{13\sqrt{1155}}{1848} & 0 & -\frac{5\sqrt{1155}i}{924} & 0 & 0 & \frac{5\sqrt{770}}{3696} & 0 & 0 & -\frac{17\sqrt{77}}{1848} & 0 & -\frac{5\sqrt{77}i}{924} & 0 & 0 & \frac{5\sqrt{462}}{1232} \\ 0 & \frac{\sqrt{1155}i}{462} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 & -\frac{19\sqrt{770}}{3696} & 0 & 0 & \frac{5\sqrt{77}i}{924} & 0 & \frac{23\sqrt{77}}{1848} & 0 & 0 \\ -\frac{\sqrt{1155}i}{462} & 0 & \frac{\sqrt{1155}}{1848} & 0 & 0 & 0 & 0 & \frac{19\sqrt{770}}{3696} & -\frac{5\sqrt{77}i}{924} & 0 & \frac{23\sqrt{77}}{1848} & 0 & 0 & 0 \\ -\frac{\sqrt{770}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & -\frac{\sqrt{1155}i}{1848} & -\frac{5\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{308} \\ 0 & \frac{\sqrt{770}}{176} & 0 & 0 & \frac{\sqrt{1155}}{924} & 0 & \frac{\sqrt{1155}i}{1848} & 0 & 0 & \frac{5\sqrt{462}}{1232} & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 \end{pmatrix}$ |
| 1012 | symmetry | $\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{616} & 0 & -\frac{2\sqrt{77}i}{77} & \frac{\sqrt{462}}{176} & 0 \\ \frac{\sqrt{1155}}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{77}}{616} & 0 & \frac{2\sqrt{77}i}{77} & 0 & 0 & -\frac{\sqrt{462}}{176} \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{77}i}{308} & 0 & -\frac{13\sqrt{77}}{616} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{\sqrt{77}i}{308} & 0 & -\frac{13\sqrt{77}}{616} & 0 & 0 & 0 \\ \frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{132} & 0 & \frac{\sqrt{1155}i}{264} & -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{308} \\ 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & -\frac{\sqrt{1155}}{132} & 0 & -\frac{\sqrt{1155}i}{264} & 0 & 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{77}}{308} & 0 \\ 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}}{3696} & 0 & 0 & -\frac{3\sqrt{77}i}{616} \\ 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{462}}{3696} & \frac{3\sqrt{77}i}{616} & 0 \\ 0 & \frac{\sqrt{77}}{616} & 0 & \frac{\sqrt{77}i}{308} & -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{1155}i}{308} & -\frac{5\sqrt{770}}{1232} & 0 \\ \frac{\sqrt{77}}{616} & 0 & -\frac{\sqrt{77}i}{308} & 0 & 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & 0 & \frac{5\sqrt{770}}{1232} \\ 0 & -\frac{2\sqrt{77}i}{77} & 0 & -\frac{13\sqrt{77}}{616} & 0 & 0 & -\frac{13\sqrt{462}}{3696} & 0 & 0 & \frac{\sqrt{1155}i}{308} & 0 & \frac{5\sqrt{1155}}{1848} & 0 & 0 \\ \frac{2\sqrt{77}i}{77} & 0 & -\frac{13\sqrt{77}}{616} & 0 & 0 & 0 & 0 & \frac{13\sqrt{462}}{3696} & -\frac{\sqrt{1155}i}{308} & 0 & \frac{5\sqrt{1155}}{1848} & 0 & 0 & 0 \\ \frac{\sqrt{462}}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{308} & 0 & -\frac{3\sqrt{77}i}{616} & -\frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} \\ 0 & -\frac{\sqrt{462}}{176} & 0 & 0 & \frac{\sqrt{77}}{308} & 0 & \frac{3\sqrt{77}i}{616} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 \end{pmatrix}$ |
| 1013 | symmetry | $-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & \frac{\sqrt{1155}i}{616} & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{13\sqrt{77}i}{616} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 \\ -\frac{\sqrt{1155}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{13\sqrt{77}i}{616} & 0 & -\frac{\sqrt{77}}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}i}{616} & -\frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{77} & 0 & -\frac{\sqrt{77}i}{616} & \frac{\sqrt{462}}{176} & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{616} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{2\sqrt{77}}{77} & 0 & \frac{\sqrt{77}i}{616} & 0 & 0 & -\frac{\sqrt{462}}{176} \\ 0 & 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & -\frac{\sqrt{1155}i}{132} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & -\frac{9\sqrt{462}}{1232} & 0 & 0 & -\frac{\sqrt{77}i}{308} \\ 0 & 0 & 0 & \frac{5\sqrt{770}}{1232} & \frac{\sqrt{1155}i}{132} & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 & 0 & \frac{9\sqrt{462}}{1232} & \frac{\sqrt{77}i}{308} & 0 \\ \frac{5\sqrt{770}}{1232} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & \frac{13\sqrt{462}}{3696} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{616} \\ 0 & -\frac{5\sqrt{770}}{1232} & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}}{3696} & 0 & 0 & -\frac{3\sqrt{77}}{616} & 0 \\ 0 & \frac{13\sqrt{77}i}{616} & 0 & \frac{2\sqrt{77}}{77} & 0 & 0 & \frac{13\sqrt{462}}{3696} & 0 & 0 & \frac{5\sqrt{1155}i}{1848} & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 \\ -\frac{13\sqrt{77}i}{616} & 0 & \frac{2\sqrt{77}}{77} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}}{3696} & -\frac{5\sqrt{1155}i}{1848} & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{77}}{308} & 0 & -\frac{\sqrt{77}i}{616} & -\frac{9\sqrt{462}}{1232} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & -\frac{\sqrt{1155}i}{616} & \frac{5\sqrt{770}}{1232} & 0 \\ -\frac{\sqrt{77}}{308} & 0 & \frac{\sqrt{77}i}{616} & 0 & 0 & \frac{9\sqrt{462}}{1232} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{1155}i}{616} & 0 & 0 & -\frac{5\sqrt{770}}{1232} \\ 0 & 0 & \frac{\sqrt{462}}{176} & 0 & 0 & -\frac{\sqrt{77}i}{308} & 0 & -\frac{3\sqrt{77}}{616} & 0 & 0 & \frac{5\sqrt{770}}{1232} & 0 & 0 & \frac{\sqrt{1155}i}{308} \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{176} & \frac{\sqrt{77}i}{308} & 0 & -\frac{3\sqrt{77}}{616} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1232} & -\frac{\sqrt{1155}i}{308} & 0 \end{bmatrix}$ |
| 1014 | symmetry | $\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A_g,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}i}{312} & 0 & \frac{\sqrt{4290}}{312} & 0 & 0 & \frac{5\sqrt{429}}{858} & 0 & 0 & \frac{\sqrt{286}i}{572} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}i}{312} & 0 & \frac{\sqrt{4290}}{312} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}}{858} & -\frac{\sqrt{286}i}{572} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{312} & 0 & -\frac{\sqrt{4290}i}{312} & \frac{5\sqrt{429}}{858} & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}}{572} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{312} & 0 & \frac{\sqrt{4290}i}{312} & 0 & 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{\sqrt{286}}{572} & 0 \\ 0 & \frac{\sqrt{4290}i}{312} & 0 & \frac{\sqrt{4290}}{312} & 0 & 0 & \frac{\sqrt{715}}{143} & 0 & 0 & \frac{3\sqrt{286}i}{1144} & 0 & -\frac{3\sqrt{286}}{1144} & 0 & 0 \\ -\frac{\sqrt{4290}i}{312} & 0 & \frac{\sqrt{4290}}{312} & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{143} & -\frac{3\sqrt{286}i}{1144} & 0 & -\frac{3\sqrt{286}}{1144} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{4290}}{312} & 0 & -\frac{\sqrt{4290}i}{312} & \frac{\sqrt{715}}{143} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & 0 \\ \frac{\sqrt{4290}}{312} & 0 & \frac{\sqrt{4290}i}{312} & 0 & 0 & -\frac{\sqrt{715}}{143} & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & \frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{429}}{858} & 0 & 0 & \frac{3\sqrt{286}i}{1144} & 0 & -\frac{3\sqrt{286}}{1144} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{429}}{858} & -\frac{3\sqrt{286}i}{1144} & 0 & -\frac{3\sqrt{286}}{1144} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{429}}{858} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & \frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{286}i}{572} & 0 & -\frac{\sqrt{286}}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{286}i}{572} & 0 & -\frac{\sqrt{286}}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 1015 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A_g,2)$ | $\begin{pmatrix} -\frac{\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{3432} & 0 & \frac{\sqrt{6006}i}{3432} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1001}}{2002} & 0 & 0 & \frac{\sqrt{6006}}{3432} & 0 & -\frac{\sqrt{6006}i}{3432} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1001}}{2002} & 0 & 0 & -\frac{\sqrt{6006}i}{3432} & 0 & \frac{\sqrt{6006}}{3432} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1001}}{2002} & \frac{\sqrt{6006}i}{3432} & 0 & \frac{\sqrt{6006}}{3432} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6006}}{3432} & 0 & -\frac{\sqrt{6006}i}{3432} & \frac{3\sqrt{1001}}{1001} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{1144} & 0 & -\frac{\sqrt{10010}i}{1144} & 0 & 0 \\ \frac{\sqrt{6006}}{3432} & 0 & \frac{\sqrt{6006}i}{3432} & 0 & 0 & -\frac{3\sqrt{1001}}{1001} & 0 & 0 & -\frac{\sqrt{10010}}{1144} & 0 & \frac{\sqrt{10010}i}{1144} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6006}i}{3432} & 0 & \frac{\sqrt{6006}}{3432} & 0 & 0 & \frac{3\sqrt{1001}}{1001} & 0 & 0 & \frac{\sqrt{10010}i}{1144} & 0 & -\frac{\sqrt{10010}}{1144} & 0 & 0 \\ -\frac{\sqrt{6006}i}{3432} & 0 & \frac{\sqrt{6006}}{3432} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{1001} & -\frac{\sqrt{10010}i}{1144} & 0 & -\frac{\sqrt{10010}}{1144} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{1144} & 0 & \frac{\sqrt{10010}i}{1144} & -\frac{15\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{1716} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{1144} & 0 & -\frac{\sqrt{10010}i}{1144} & 0 & 0 & \frac{15\sqrt{1001}}{2002} & 0 & 0 & \frac{5\sqrt{6006}}{1716} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}i}{1144} & 0 & -\frac{\sqrt{10010}}{1144} & 0 & 0 & -\frac{15\sqrt{1001}}{2002} & 0 & 0 & -\frac{5\sqrt{6006}i}{1716} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10010}i}{1144} & 0 & -\frac{\sqrt{10010}}{1144} & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{2002} & \frac{5\sqrt{6006}i}{1716} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{1716} & 0 & -\frac{5\sqrt{6006}i}{1716} & \frac{10\sqrt{1001}}{1001} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{1716} & 0 & \frac{5\sqrt{6006}i}{1716} & 0 & 0 & -\frac{10\sqrt{1001}}{1001} \end{pmatrix}$ |
| 1016 | symmetry | $\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{312} & 0 & -\frac{\sqrt{4290}i}{312} & \frac{5\sqrt{429}}{858} & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}}{572} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{312} & 0 & \frac{\sqrt{4290}i}{312} & 0 & 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{\sqrt{286}}{572} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}i}{312} & 0 & -\frac{\sqrt{4290}}{312} & 0 & 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{\sqrt{286}i}{572} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{4290}i}{312} & 0 & -\frac{\sqrt{4290}}{312} & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{858} & \frac{\sqrt{286}i}{572} & 0 \\ 0 & \frac{\sqrt{4290}}{312} & 0 & -\frac{\sqrt{4290}i}{312} & \frac{\sqrt{715}}{143} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & 0 \\ \frac{\sqrt{4290}}{312} & 0 & \frac{\sqrt{4290}i}{312} & 0 & 0 & -\frac{\sqrt{715}}{143} & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & \frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{4290}i}{312} & 0 & -\frac{\sqrt{4290}}{312} & 0 & 0 & -\frac{\sqrt{715}}{143} & 0 & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & \frac{3\sqrt{286}}{1144} & 0 & 0 \\ \frac{\sqrt{4290}i}{312} & 0 & -\frac{\sqrt{4290}}{312} & 0 & 0 & 0 & 0 & \frac{\sqrt{715}}{143} & \frac{3\sqrt{286}i}{1144} & 0 & \frac{3\sqrt{286}}{1144} & 0 & 0 & 0 \\ \frac{5\sqrt{429}}{858} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{3\sqrt{286}}{1144} & 0 & \frac{3\sqrt{286}i}{1144} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{429}}{858} & 0 & 0 & -\frac{3\sqrt{286}i}{1144} & 0 & \frac{3\sqrt{286}}{1144} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{429}}{858} & \frac{3\sqrt{286}i}{1144} & 0 & \frac{3\sqrt{286}}{1144} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{286}}{572} & 0 & -\frac{\sqrt{286}i}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{286}}{572} & 0 & \frac{\sqrt{286}i}{572} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1017 | symmetry | $\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(B_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{\sqrt{858}i}{286} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}}{429} & \frac{\sqrt{858}i}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{3432} & 0 & -\frac{\sqrt{1430}i}{3432} & \frac{2\sqrt{143}}{429} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{286} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{3432} & 0 & \frac{\sqrt{1430}i}{3432} & 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{\sqrt{858}}{286} & 0 \\ 0 & -\frac{\sqrt{1430}i}{3432} & 0 & \frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{264} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 \\ \frac{\sqrt{1430}i}{3432} & 0 & \frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{264} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1430}}{3432} & 0 & -\frac{\sqrt{1430}i}{3432} & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{858}}{3432} & 0 & \frac{23\sqrt{858}i}{3432} & -\frac{8\sqrt{143}}{429} & 0 \\ -\frac{\sqrt{1430}}{3432} & 0 & \frac{\sqrt{1430}i}{3432} & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{858}}{3432} & 0 & -\frac{23\sqrt{858}i}{3432} & 0 & 0 & \frac{8\sqrt{143}}{429} \\ 0 & 0 & \frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{\sqrt{858}i}{264} & 0 & -\frac{23\sqrt{858}}{3432} & 0 & 0 & -\frac{2\sqrt{2145}}{429} & 0 & 0 & -\frac{5\sqrt{1430}i}{1716} \\ 0 & 0 & 0 & -\frac{2\sqrt{143}}{429} & \frac{\sqrt{858}i}{264} & 0 & -\frac{23\sqrt{858}}{3432} & 0 & 0 & 0 & 0 & \frac{2\sqrt{2145}}{429} & \frac{5\sqrt{1430}i}{1716} & 0 \\ -\frac{2\sqrt{143}}{429} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{264} & 0 & \frac{23\sqrt{858}i}{3432} & -\frac{2\sqrt{2145}}{429} & 0 & 0 & 0 & 0 & \frac{5\sqrt{1430}}{1716} \\ 0 & \frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{\sqrt{858}}{264} & 0 & -\frac{23\sqrt{858}i}{3432} & 0 & 0 & \frac{2\sqrt{2145}}{429} & 0 & 0 & \frac{5\sqrt{1430}}{1716} & 0 \\ 0 & -\frac{\sqrt{858}i}{286} & 0 & -\frac{\sqrt{858}}{286} & 0 & 0 & -\frac{8\sqrt{143}}{429} & 0 & 0 & -\frac{5\sqrt{1430}i}{1716} & 0 & \frac{5\sqrt{1430}}{1716} & 0 & 0 \\ \frac{\sqrt{858}i}{286} & 0 & -\frac{\sqrt{858}}{286} & 0 & 0 & 0 & 0 & \frac{8\sqrt{143}}{429} & \frac{5\sqrt{1430}i}{1716} & 0 & \frac{5\sqrt{1430}}{1716} & 0 & 0 & 0 \end{bmatrix}$ |
| 1018 | symmetry | $-\frac{\sqrt{105}x^4z}{4}+\frac{\sqrt{105}x^2z^3}{2}+\frac{\sqrt{105}y^4z}{4}-\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(B_g,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{3432} & 0 & \frac{\sqrt{1430}i}{3432} & -\frac{2\sqrt{143}}{429} & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{286} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{3432} & 0 & -\frac{\sqrt{1430}i}{3432} & 0 & 0 & \frac{2\sqrt{143}}{429} & 0 & 0 & \frac{\sqrt{858}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{\sqrt{858}i}{286} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}}{429} & \frac{\sqrt{858}i}{286} & 0 \\ 0 & -\frac{\sqrt{1430}}{3432} & 0 & -\frac{\sqrt{1430}i}{3432} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{858}}{3432} & 0 & -\frac{23\sqrt{858}i}{3432} & \frac{8\sqrt{143}}{429} & 0 \\ -\frac{\sqrt{1430}}{3432} & 0 & \frac{\sqrt{1430}i}{3432} & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{858}}{3432} & 0 & \frac{23\sqrt{858}i}{3432} & 0 & 0 & -\frac{8\sqrt{143}}{429} \\ 0 & \frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{264} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 \\ -\frac{\sqrt{1430}i}{3432} & 0 & -\frac{\sqrt{1430}}{3432} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{264} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 & 0 \\ -\frac{2\sqrt{143}}{429} & 0 & 0 & 0 & 0 & \frac{23\sqrt{858}}{3432} & 0 & -\frac{\sqrt{858}i}{264} & \frac{2\sqrt{2145}}{429} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1430}}{1716} \\ 0 & \frac{2\sqrt{143}}{429} & 0 & 0 & \frac{23\sqrt{858}}{3432} & 0 & \frac{\sqrt{858}i}{264} & 0 & 0 & -\frac{2\sqrt{2145}}{429} & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & 0 \\ 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & 0 & -\frac{23\sqrt{858}i}{3432} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 & -\frac{2\sqrt{2145}}{429} & 0 & 0 & -\frac{5\sqrt{1430}i}{1716} \\ 0 & 0 & 0 & \frac{2\sqrt{143}}{429} & \frac{23\sqrt{858}i}{3432} & 0 & -\frac{\sqrt{858}}{264} & 0 & 0 & 0 & 0 & \frac{2\sqrt{2145}}{429} & \frac{5\sqrt{1430}i}{1716} & 0 \\ 0 & \frac{\sqrt{858}}{286} & 0 & -\frac{\sqrt{858}i}{286} & \frac{8\sqrt{143}}{429} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & 0 & -\frac{5\sqrt{1430}i}{1716} & 0 & 0 \\ \frac{\sqrt{858}}{286} & 0 & \frac{\sqrt{858}i}{286} & 0 & 0 & -\frac{8\sqrt{143}}{429} & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & 0 & \frac{5\sqrt{1430}i}{1716} & 0 & 0 & 0 \end{pmatrix}$ |
| 1019 | symmetry | $\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{113\sqrt{1001}i}{16016} & 0 & \frac{3\sqrt{1001}}{416} & 0 & 0 & \frac{7\sqrt{6006}}{6864} & 0 & 0 & \frac{\sqrt{15015}i}{2288} & 0 & \frac{7\sqrt{15015}}{13728} & 0 & 0 \\ -\frac{113\sqrt{1001}i}{16016} & 0 & \frac{3\sqrt{1001}}{416} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6006}}{6864} & -\frac{\sqrt{15015}i}{2288} & 0 & \frac{7\sqrt{15015}}{13728} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{1001}}{416} & 0 & -\frac{59\sqrt{1001}i}{8008} & \frac{19\sqrt{6006}}{13728} & 0 & 0 & 0 & 0 & \frac{5\sqrt{15015}}{13728} & 0 & -\frac{\sqrt{15015}i}{1144} & \frac{3\sqrt{10010}}{4576} & 0 \\ \frac{3\sqrt{1001}}{416} & 0 & \frac{59\sqrt{1001}i}{8008} & 0 & 0 & -\frac{19\sqrt{6006}}{13728} & 0 & 0 & \frac{5\sqrt{15015}}{13728} & 0 & \frac{\sqrt{15015}i}{1144} & 0 & 0 & -\frac{3\sqrt{10010}}{4576} \\ 0 & 0 & \frac{19\sqrt{6006}}{13728} & 0 & 0 & \frac{3\sqrt{1001}i}{616} & 0 & \frac{3\sqrt{1001}}{1144} & 0 & 0 & \frac{7\sqrt{10010}}{4576} & 0 & 0 & \frac{\sqrt{15015}i}{1144} \\ 0 & 0 & 0 & -\frac{19\sqrt{6006}}{13728} & -\frac{3\sqrt{1001}i}{616} & 0 & \frac{3\sqrt{1001}}{1144} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{10010}}{4576} & -\frac{\sqrt{15015}i}{1144} & 0 \\ \frac{7\sqrt{6006}}{6864} & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{1144} & 0 & -\frac{3\sqrt{1001}i}{1001} & \frac{\sqrt{10010}}{2288} & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{3432} \\ 0 & -\frac{7\sqrt{6006}}{6864} & 0 & 0 & \frac{3\sqrt{1001}}{1144} & 0 & \frac{3\sqrt{1001}i}{1001} & 0 & 0 & -\frac{\sqrt{10010}}{2288} & 0 & 0 & \frac{\sqrt{15015}}{3432} & 0 \\ 0 & \frac{\sqrt{15015}i}{2288} & 0 & \frac{5\sqrt{15015}}{13728} & 0 & 0 & \frac{\sqrt{10010}}{2288} & 0 & 0 & \frac{15\sqrt{1001}i}{16016} & 0 & \frac{5\sqrt{1001}}{4576} & 0 & 0 \\ -\frac{\sqrt{15015}i}{2288} & 0 & \frac{5\sqrt{15015}}{13728} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2288} & -\frac{15\sqrt{1001}i}{16016} & 0 & \frac{5\sqrt{1001}}{4576} & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{15015}}{13728} & 0 & -\frac{\sqrt{15015}i}{1144} & \frac{7\sqrt{10010}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{1001}}{4576} & 0 & -\frac{45\sqrt{1001}i}{8008} & \frac{25\sqrt{6006}}{13728} & 0 \\ \frac{7\sqrt{15015}}{13728} & 0 & \frac{\sqrt{15015}i}{1144} & 0 & 0 & -\frac{7\sqrt{10010}}{4576} & 0 & 0 & \frac{5\sqrt{1001}}{4576} & 0 & \frac{45\sqrt{1001}i}{8008} & 0 & 0 & -\frac{25\sqrt{6006}}{13728} \\ 0 & 0 & \frac{3\sqrt{10010}}{4576} & 0 & 0 & \frac{\sqrt{15015}i}{1144} & 0 & \frac{\sqrt{15015}}{3432} & 0 & 0 & \frac{25\sqrt{6006}}{13728} & 0 & 0 & \frac{25\sqrt{1001}i}{8008} \\ 0 & 0 & 0 & -\frac{3\sqrt{10010}}{4576} & -\frac{\sqrt{15015}i}{1144} & 0 & \frac{\sqrt{15015}}{3432} & 0 & 0 & 0 & 0 & -\frac{25\sqrt{6006}}{13728} & -\frac{25\sqrt{1001}i}{8008} & 0 \end{bmatrix}$ |
| 1020 | symmetry | $x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)$ | $\begin{pmatrix} 0 & \frac{59\sqrt{1001}}{8008} & 0 & -\frac{3\sqrt{1001}i}{416} & \frac{19\sqrt{6006}}{13728} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{1144} & 0 & \frac{5\sqrt{15015}i}{13728} & -\frac{3\sqrt{10010}}{4576} & 0 \\ \frac{59\sqrt{1001}}{8008} & 0 & \frac{3\sqrt{1001}i}{416} & 0 & 0 & -\frac{19\sqrt{6006}}{13728} & 0 & 0 & -\frac{\sqrt{15015}}{1144} & 0 & -\frac{5\sqrt{15015}i}{13728} & 0 & 0 & \frac{3\sqrt{10010}}{4576} \\ 0 & -\frac{3\sqrt{1001}i}{416} & 0 & -\frac{113\sqrt{1001}}{16016} & 0 & 0 & -\frac{7\sqrt{6006}}{6864} & 0 & 0 & \frac{7\sqrt{15015}i}{13728} & 0 & \frac{\sqrt{15015}}{2288} & 0 & 0 \\ \frac{3\sqrt{1001}i}{416} & 0 & -\frac{113\sqrt{1001}}{16016} & 0 & 0 & 0 & 0 & \frac{7\sqrt{6006}}{6864} & -\frac{7\sqrt{15015}i}{13728} & 0 & \frac{\sqrt{15015}}{2288} & 0 & 0 & 0 \\ \frac{19\sqrt{6006}}{13728} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{616} & 0 & \frac{3\sqrt{1001}i}{1144} & -\frac{7\sqrt{10010}}{4576} & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{1144} \\ 0 & -\frac{19\sqrt{6006}}{13728} & 0 & 0 & -\frac{3\sqrt{1001}}{616} & 0 & -\frac{3\sqrt{1001}i}{1144} & 0 & 0 & \frac{7\sqrt{10010}}{4576} & 0 & 0 & \frac{\sqrt{15015}}{1144} & 0 \\ 0 & 0 & -\frac{7\sqrt{6006}}{6864} & 0 & 0 & \frac{3\sqrt{1001}i}{1144} & 0 & \frac{3\sqrt{1001}}{1001} & 0 & 0 & \frac{\sqrt{10010}}{2288} & 0 & 0 & -\frac{\sqrt{15015}i}{3432} \\ 0 & 0 & 0 & \frac{7\sqrt{6006}}{6864} & -\frac{3\sqrt{1001}i}{1144} & 0 & \frac{3\sqrt{1001}}{1001} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2288} & \frac{\sqrt{15015}i}{3432} & 0 \\ 0 & -\frac{\sqrt{15015}}{1144} & 0 & \frac{7\sqrt{15015}i}{13728} & -\frac{7\sqrt{10010}}{4576} & 0 & 0 & 0 & 0 & \frac{45\sqrt{1001}}{8008} & 0 & -\frac{5\sqrt{1001}i}{4576} & \frac{25\sqrt{6006}}{13728} & 0 \\ -\frac{\sqrt{15015}}{1144} & 0 & -\frac{7\sqrt{15015}i}{13728} & 0 & 0 & \frac{7\sqrt{10010}}{4576} & 0 & 0 & \frac{45\sqrt{1001}}{8008} & 0 & \frac{5\sqrt{1001}i}{4576} & 0 & 0 & -\frac{25\sqrt{6006}}{13728} \\ 0 & \frac{5\sqrt{15015}i}{13728} & 0 & \frac{\sqrt{15015}}{2288} & 0 & 0 & \frac{\sqrt{10010}}{2288} & 0 & 0 & -\frac{5\sqrt{1001}i}{4576} & 0 & -\frac{15\sqrt{1001}}{16016} & 0 & 0 \\ -\frac{5\sqrt{15015}i}{13728} & 0 & \frac{\sqrt{15015}}{2288} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2288} & \frac{5\sqrt{1001}i}{4576} & 0 & -\frac{15\sqrt{1001}}{16016} & 0 & 0 & 0 \\ -\frac{3\sqrt{10010}}{4576} & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{1144} & 0 & -\frac{\sqrt{15015}i}{3432} & \frac{25\sqrt{6006}}{13728} & 0 & 0 & 0 & 0 & -\frac{25\sqrt{1001}}{8008} \\ 0 & \frac{3\sqrt{10010}}{4576} & 0 & 0 & \frac{\sqrt{15015}}{1144} & 0 & \frac{\sqrt{15015}i}{3432} & 0 & 0 & -\frac{25\sqrt{6006}}{13728} & 0 & 0 & -\frac{25\sqrt{1001}}{8008} & 0 \end{pmatrix}$ |
| 1021 | symmetry | $\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{5\sqrt{715}i}{2288} & 0 & \frac{\sqrt{715}}{416} & 0 & 0 & \frac{\sqrt{4290}}{6864} & 0 & 0 & -\frac{35\sqrt{429}i}{6864} & 0 & -\frac{47\sqrt{429}}{13728} & 0 & 0 \\ -\frac{5\sqrt{715}i}{2288} & 0 & \frac{\sqrt{715}}{416} & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{6864} & \frac{35\sqrt{429}i}{6864} & 0 & -\frac{47\sqrt{429}}{13728} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{715}}{416} & 0 & -\frac{3\sqrt{715}i}{1144} & \frac{3\sqrt{4290}}{4576} & 0 & 0 & 0 & 0 & -\frac{31\sqrt{429}}{4576} & 0 & \frac{5\sqrt{429}i}{1144} & -\frac{27\sqrt{286}}{4576} & 0 \\ \frac{\sqrt{715}}{416} & 0 & \frac{3\sqrt{715}i}{1144} & 0 & 0 & -\frac{3\sqrt{4290}}{4576} & 0 & 0 & -\frac{31\sqrt{429}}{4576} & 0 & -\frac{5\sqrt{429}i}{1144} & 0 & 0 & \frac{27\sqrt{286}}{4576} \\ 0 & 0 & \frac{3\sqrt{4290}}{4576} & 0 & 0 & -\frac{5\sqrt{715}i}{1144} & 0 & -\frac{7\sqrt{715}}{1144} & 0 & 0 & -\frac{23\sqrt{286}}{4576} & 0 & 0 & \frac{5\sqrt{429}i}{3432} \\ 0 & 0 & 0 & -\frac{3\sqrt{4290}}{4576} & \frac{5\sqrt{715}i}{1144} & 0 & -\frac{7\sqrt{715}}{1144} & 0 & 0 & 0 & 0 & \frac{23\sqrt{286}}{4576} & -\frac{5\sqrt{429}i}{3432} & 0 \\ \frac{\sqrt{4290}}{6864} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{715}}{1144} & 0 & \frac{\sqrt{715}i}{143} & -\frac{29\sqrt{286}}{2288} & 0 & 0 & 0 & 0 & \frac{23\sqrt{429}}{3432} \\ 0 & -\frac{\sqrt{4290}}{6864} & 0 & 0 & -\frac{7\sqrt{715}}{1144} & 0 & -\frac{\sqrt{715}i}{143} & 0 & 0 & \frac{29\sqrt{286}}{2288} & 0 & 0 & \frac{23\sqrt{429}}{3432} & 0 \\ 0 & -\frac{35\sqrt{429}i}{6864} & 0 & -\frac{31\sqrt{429}}{4576} & 0 & 0 & -\frac{29\sqrt{286}}{2288} & 0 & 0 & -\frac{5\sqrt{715}i}{2288} & 0 & \frac{23\sqrt{715}}{4576} & 0 & 0 \\ \frac{35\sqrt{429}i}{6864} & 0 & -\frac{31\sqrt{429}}{4576} & 0 & 0 & 0 & 0 & \frac{29\sqrt{286}}{2288} & \frac{5\sqrt{715}i}{2288} & 0 & \frac{23\sqrt{715}}{4576} & 0 & 0 & 0 \\ 0 & -\frac{47\sqrt{429}}{13728} & 0 & \frac{5\sqrt{429}i}{1144} & -\frac{23\sqrt{286}}{4576} & 0 & 0 & 0 & 0 & \frac{23\sqrt{715}}{4576} & 0 & -\frac{5\sqrt{715}i}{1144} & \frac{35\sqrt{4290}}{13728} & 0 \\ -\frac{47\sqrt{429}}{13728} & 0 & -\frac{5\sqrt{429}i}{1144} & 0 & 0 & \frac{23\sqrt{286}}{4576} & 0 & 0 & \frac{23\sqrt{715}}{4576} & 0 & \frac{5\sqrt{715}i}{1144} & 0 & 0 & -\frac{35\sqrt{4290}}{13728} \\ 0 & 0 & -\frac{27\sqrt{286}}{4576} & 0 & 0 & \frac{5\sqrt{429}i}{3432} & 0 & \frac{23\sqrt{429}}{3432} & 0 & 0 & \frac{35\sqrt{4290}}{13728} & 0 & 0 & \frac{5\sqrt{715}i}{1144} \\ 0 & 0 & 0 & \frac{27\sqrt{286}}{4576} & -\frac{5\sqrt{429}i}{3432} & 0 & \frac{23\sqrt{429}}{3432} & 0 & 0 & 0 & 0 & -\frac{35\sqrt{4290}}{13728} & -\frac{5\sqrt{715}i}{1144} & 0 \end{pmatrix}$ |
| 1022 | symmetry | $\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)$ | $\begin{pmatrix} 0 & \frac{3\sqrt{715}}{1144} & 0 & -\frac{\sqrt{715}i}{416} & \frac{3\sqrt{4290}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{1144} & 0 & -\frac{31\sqrt{429}i}{4576} & \frac{27\sqrt{286}}{4576} & 0 \\ \frac{3\sqrt{715}}{1144} & 0 & \frac{\sqrt{715}i}{416} & 0 & 0 & -\frac{3\sqrt{4290}}{4576} & 0 & 0 & \frac{5\sqrt{429}}{1144} & 0 & \frac{31\sqrt{429}i}{4576} & 0 & 0 & -\frac{27\sqrt{286}}{4576} \\ 0 & -\frac{\sqrt{715}i}{416} & 0 & -\frac{5\sqrt{715}}{2288} & 0 & 0 & -\frac{\sqrt{4290}}{6864} & 0 & 0 & -\frac{47\sqrt{429}i}{13728} & 0 & -\frac{35\sqrt{429}}{6864} & 0 & 0 \\ \frac{\sqrt{715}i}{416} & 0 & -\frac{5\sqrt{715}}{2288} & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{6864} & \frac{47\sqrt{429}i}{13728} & 0 & -\frac{35\sqrt{429}}{6864} & 0 & 0 & 0 \\ \frac{3\sqrt{4290}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{715}}{1144} & 0 & -\frac{7\sqrt{715}i}{1144} & \frac{23\sqrt{286}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{3432} \\ 0 & -\frac{3\sqrt{4290}}{4576} & 0 & 0 & \frac{5\sqrt{715}}{1144} & 0 & \frac{7\sqrt{715}i}{1144} & 0 & 0 & -\frac{23\sqrt{286}}{4576} & 0 & 0 & \frac{5\sqrt{429}}{3432} & 0 \\ 0 & 0 & -\frac{\sqrt{4290}}{6864} & 0 & 0 & -\frac{7\sqrt{715}i}{1144} & 0 & -\frac{\sqrt{715}}{143} & 0 & 0 & -\frac{29\sqrt{286}}{2288} & 0 & 0 & -\frac{23\sqrt{429}i}{3432} \\ 0 & 0 & 0 & \frac{\sqrt{4290}}{6864} & \frac{7\sqrt{715}i}{1144} & 0 & -\frac{\sqrt{715}}{143} & 0 & 0 & 0 & 0 & \frac{29\sqrt{286}}{2288} & \frac{23\sqrt{429}i}{3432} & 0 \\ 0 & \frac{5\sqrt{429}}{1144} & 0 & -\frac{47\sqrt{429}i}{13728} & \frac{23\sqrt{286}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{715}}{1144} & 0 & -\frac{23\sqrt{715}i}{4576} & \frac{35\sqrt{4290}}{13728} & 0 \\ \frac{5\sqrt{429}}{1144} & 0 & \frac{47\sqrt{429}i}{13728} & 0 & 0 & -\frac{23\sqrt{286}}{4576} & 0 & 0 & \frac{5\sqrt{715}}{1144} & 0 & \frac{23\sqrt{715}i}{4576} & 0 & 0 & -\frac{35\sqrt{4290}}{13728} \\ 0 & -\frac{31\sqrt{429}i}{4576} & 0 & -\frac{35\sqrt{429}}{6864} & 0 & 0 & -\frac{29\sqrt{286}}{2288} & 0 & 0 & -\frac{23\sqrt{715}i}{4576} & 0 & \frac{5\sqrt{715}}{2288} & 0 & 0 \\ \frac{31\sqrt{429}i}{4576} & 0 & -\frac{35\sqrt{429}}{6864} & 0 & 0 & 0 & 0 & \frac{29\sqrt{286}}{2288} & \frac{23\sqrt{715}i}{4576} & 0 & \frac{5\sqrt{715}}{2288} & 0 & 0 & 0 \\ \frac{27\sqrt{286}}{4576} & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{3432} & 0 & -\frac{23\sqrt{429}i}{3432} & \frac{35\sqrt{4290}}{13728} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{715}}{1144} \\ 0 & -\frac{27\sqrt{286}}{4576} & 0 & 0 & \frac{5\sqrt{429}}{3432} & 0 & \frac{23\sqrt{429}i}{3432} & 0 & 0 & -\frac{35\sqrt{4290}}{13728} & 0 & 0 & -\frac{5\sqrt{715}}{1144} & 0 \end{pmatrix}$ |
| 1023 | symmetry | $\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & -\frac{2\sqrt{2145}}{429} & 0 & \frac{\sqrt{2145}i}{208} & -\frac{\sqrt{1430}}{528} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & \frac{19\sqrt{143}i}{6864} & -\frac{3\sqrt{858}}{2288} & 0 \\ -\frac{2\sqrt{2145}}{429} & 0 & -\frac{\sqrt{2145}i}{208} & 0 & 0 & \frac{\sqrt{1430}}{528} & 0 & 0 & -\frac{2\sqrt{143}}{429} & 0 & -\frac{19\sqrt{143}i}{6864} & 0 & 0 & \frac{3\sqrt{858}}{2288} \\ 0 & \frac{\sqrt{2145}i}{208} & 0 & \frac{17\sqrt{2145}}{3432} & 0 & 0 & \frac{5\sqrt{1430}}{1716} & 0 & 0 & \frac{41\sqrt{143}i}{6864} & 0 & -\frac{\sqrt{143}}{3432} & 0 & 0 \\ -\frac{\sqrt{2145}i}{208} & 0 & \frac{17\sqrt{2145}}{3432} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & -\frac{41\sqrt{143}i}{6864} & 0 & -\frac{\sqrt{143}}{3432} & 0 & 0 & 0 \\ -\frac{\sqrt{1430}}{528} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{572} & 0 & \frac{\sqrt{2145}i}{572} & -\frac{31\sqrt{858}}{6864} & 0 & 0 & 0 & 0 & \frac{17\sqrt{143}}{1716} \\ 0 & \frac{\sqrt{1430}}{528} & 0 & 0 & -\frac{\sqrt{2145}}{572} & 0 & -\frac{\sqrt{2145}i}{572} & 0 & 0 & \frac{31\sqrt{858}}{6864} & 0 & 0 & \frac{17\sqrt{143}}{1716} & 0 \\ 0 & 0 & \frac{5\sqrt{1430}}{1716} & 0 & 0 & \frac{\sqrt{2145}i}{572} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & -\frac{\sqrt{143}i}{156} \\ 0 & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & -\frac{\sqrt{2145}i}{572} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716} & \frac{\sqrt{143}i}{156} & 0 \\ 0 & -\frac{2\sqrt{143}}{429} & 0 & \frac{41\sqrt{143}i}{6864} & -\frac{31\sqrt{858}}{6864} & 0 & 0 & 0 & 0 & \frac{2\sqrt{2145}}{429} & 0 & -\frac{\sqrt{2145}i}{624} & \frac{35\sqrt{1430}}{6864} & 0 \\ -\frac{2\sqrt{143}}{429} & 0 & -\frac{41\sqrt{143}i}{6864} & 0 & 0 & \frac{31\sqrt{858}}{6864} & 0 & 0 & \frac{2\sqrt{2145}}{429} & 0 & \frac{\sqrt{2145}i}{624} & 0 & 0 & -\frac{35\sqrt{1430}}{6864} \\ 0 & \frac{19\sqrt{143}i}{6864} & 0 & -\frac{\sqrt{143}}{3432} & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & -\frac{\sqrt{2145}i}{624} & 0 & -\frac{\sqrt{2145}}{3432} & 0 & 0 \\ -\frac{19\sqrt{143}i}{6864} & 0 & -\frac{\sqrt{143}}{3432} & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716} & \frac{\sqrt{2145}i}{624} & 0 & -\frac{\sqrt{2145}}{3432} & 0 & 0 & 0 \\ -\frac{3\sqrt{858}}{2288} & 0 & 0 & 0 & 0 & \frac{17\sqrt{143}}{1716} & 0 & -\frac{\sqrt{143}i}{156} & \frac{35\sqrt{1430}}{6864} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2145}}{1716} \\ 0 & \frac{3\sqrt{858}}{2288} & 0 & 0 & \frac{17\sqrt{143}}{1716} & 0 & \frac{\sqrt{143}i}{156} & 0 & 0 & -\frac{35\sqrt{1430}}{6864} & 0 & 0 & -\frac{5\sqrt{2145}}{1716} & 0 \end{bmatrix}$ |
| 1024 | symmetry | $\frac{\sqrt{105}x^4y}{4} - \frac{\sqrt{105}x^2y^3}{2} + \frac{\sqrt{105}y^3z^2}{2} - \frac{\sqrt{105}yz^4}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)$ | $\begin{pmatrix} 0 & \frac{17\sqrt{2145}i}{3432} & 0 & \frac{\sqrt{2145}}{208} & 0 & 0 & \frac{5\sqrt{1430}}{1716} & 0 & 0 & \frac{\sqrt{143}i}{3432} & 0 & -\frac{41\sqrt{143}}{6864} & 0 & 0 \\ -\frac{17\sqrt{2145}i}{3432} & 0 & \frac{\sqrt{2145}}{208} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{1430}}{1716} & -\frac{\sqrt{143}i}{3432} & 0 & -\frac{41\sqrt{143}}{6864} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2145}}{208} & 0 & -\frac{2\sqrt{2145}i}{429} & \frac{\sqrt{1430}}{528} & 0 & 0 & 0 & 0 & -\frac{19\sqrt{143}}{6864} & 0 & \frac{2\sqrt{143}i}{429} & -\frac{3\sqrt{858}}{2288} & 0 \\ \frac{\sqrt{2145}}{208} & 0 & \frac{2\sqrt{2145}i}{429} & 0 & 0 & -\frac{\sqrt{1430}}{528} & 0 & 0 & -\frac{19\sqrt{143}}{6864} & 0 & -\frac{2\sqrt{143}i}{429} & 0 & 0 & \frac{3\sqrt{858}}{2288} \\ 0 & 0 & \frac{\sqrt{1430}}{528} & 0 & 0 & -\frac{\sqrt{2145}i}{572} & 0 & -\frac{\sqrt{2145}}{572} & 0 & 0 & -\frac{31\sqrt{858}}{6864} & 0 & 0 & -\frac{17\sqrt{143}i}{1716} \\ 0 & 0 & 0 & -\frac{\sqrt{1430}}{528} & \frac{\sqrt{2145}i}{572} & 0 & -\frac{\sqrt{2145}}{572} & 0 & 0 & 0 & 0 & \frac{31\sqrt{858}}{6864} & \frac{17\sqrt{143}i}{1716} & 0 \\ \frac{5\sqrt{1430}}{1716} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{572} & 0 & 0 & \frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{156} \\ 0 & -\frac{5\sqrt{1430}}{1716} & 0 & 0 & -\frac{\sqrt{2145}}{572} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & -\frac{\sqrt{143}}{156} & 0 \\ 0 & \frac{\sqrt{143}i}{3432} & 0 & -\frac{19\sqrt{143}}{6864} & 0 & 0 & \frac{\sqrt{858}}{1716} & 0 & 0 & -\frac{\sqrt{2145}i}{3432} & 0 & -\frac{\sqrt{2145}}{624} & 0 & 0 \\ -\frac{\sqrt{143}i}{3432} & 0 & -\frac{19\sqrt{143}}{6864} & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & \frac{\sqrt{2145}i}{3432} & 0 & -\frac{\sqrt{2145}}{624} & 0 & 0 & 0 \\ 0 & -\frac{41\sqrt{143}}{6864} & 0 & \frac{2\sqrt{143}i}{429} & -\frac{31\sqrt{858}}{6864} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{624} & 0 & \frac{2\sqrt{2145}i}{429} & -\frac{35\sqrt{1430}}{6864} & 0 \\ -\frac{41\sqrt{143}}{6864} & 0 & -\frac{2\sqrt{143}i}{429} & 0 & 0 & \frac{31\sqrt{858}}{6864} & 0 & 0 & -\frac{\sqrt{2145}}{624} & 0 & -\frac{2\sqrt{2145}i}{429} & 0 & 0 & \frac{35\sqrt{1430}}{6864} \\ 0 & 0 & -\frac{3\sqrt{858}}{2288} & 0 & 0 & -\frac{17\sqrt{143}i}{1716} & 0 & -\frac{\sqrt{143}}{156} & 0 & 0 & -\frac{35\sqrt{1430}}{6864} & 0 & 0 & -\frac{5\sqrt{2145}i}{1716} \\ 0 & 0 & 0 & \frac{3\sqrt{858}}{2288} & \frac{17\sqrt{143}i}{1716} & 0 & -\frac{\sqrt{143}}{156} & 0 & 0 & 0 & 0 & \frac{35\sqrt{1430}}{6864} & \frac{5\sqrt{2145}i}{1716} & 0 \end{pmatrix}$ |